

Contents

Every line is a link. The page numbers are read back from the rendered book, never typed by hand.

How to use this manual	2
FOUNDATIONS • THE LANGUAGE OF DISCRETE MATHS	6
F.1 • Numbers, Symbols and Notation	7
F.2 • Reading and Writing Mathematics	13
PART ONE • LOGIC	19
1.1 • Propositions and Truth Tables	20
1.2 • Implication, Converse, Inverse, Contrapositive	27
1.3 • Logical Equivalence and the Laws	32
1.4 • Predicate Logic and Quantifiers	38
1.5 • Rules of Inference and Valid Arguments	46
PART TWO • SETS	51
2.1 • Sets, Membership and the Nine Types	52
2.2 • Set Operations and Venn Diagrams	58
2.3 • Bit Strings, Power Sets and Cartesian Products	64
PART THREE • RELATIONS	74
3.1 • Relations and Their Representations	75
3.2 • The Five Properties of a Relation	79
3.3 • Equivalence Relations and Partitions	86
3.4 • Partial and Total Orders	92
PART FOUR • FUNCTIONS	98
4.1 • Functions, Domain and Range	99
4.2 • The Five Correspondence Types	101
4.3 • Composition and Inverse Functions	105
PART FIVE • COUNTING	111
5.1 • The Product and Sum Rules	112
5.2 • Permutations and Combinations	121
5.3 • Inclusion, Exclusion and the Pigeonhole Principle	130
5.4 • The Binomial Theorem	138
PART SIX • GRAPH THEORY	144
6.1 • Graphs, Vertices, Edges and Degree	145
6.2 • The Types of Undirected Graph	155
6.3 • The Handshaking Theorem	163
6.4 • Directed Graphs, In-degree and Out-degree	171
6.5 • Adjacency, Path and Relation Matrices	179
6.6 • Isomorphism, Euler and Hamilton	187
6.7 • Planarity, Euler's Formula and Kuratowski	194
6.8 • Shortest Routes and the Travelling Salesman	202
PART SEVEN • TREES	209
7.1 • Trees and Rooted-Tree Terminology	210
7.2 • m-ary and Balanced Trees	216

7.3 · Traversals and Expression Trees	227
PART EIGHT · BOOLEAN ALGEBRA	236
8.1 · Boolean Algebra and Its Laws	237
PART NINE · PROOF AND RECURSION	245
9.1 · Proof Techniques	246
9.2 · Mathematical Induction	257
9.3 · Sequences, Series and Progressions	265
9.4 · Recurrence Relations	272
PART TEN · STATISTICS AND PROBABILITY	282
10.1 · Descriptive Statistics	283
10.2 · Discrete Probability	288
PART ELEVEN · C++ FOR DISCRETE STRUCTURES	299
11.1 · Enough C++ to Answer the Question	300
11.2 · The Four Examined Programs	300
REFERENCE · FORMULAS, NOTATION AND EXAM CRAFT	301
R.1 · Formula and Notation Reference	302
R.2 · Exam Day Strategy	302
R.3 · Trap Index and Cram Sheet	302
PAST PAPERS · ALL FIVE, SOLVED IN FULL	303
2025/2026 Solved in Full	304
2024/2025 Solved in Full	304
2023/2024 Solved in Full	304
2021/2022 Solved in Full	304
2020/2021 Solved in Full	304
MOCK EXAMINATION ONE	305
Question Paper	306
Answers	306
MOCK EXAMINATION TWO	307
Question Paper	308
Answers	308
MOCK EXAMINATION THREE	309
Question Paper	310
Answers	310

BEFORE YOU START

How to use this manual

Read this page once. It will save you hours, and it explains one thing about this course that nothing else will tell you.

The thing you need to know first

This course has an official course text of 161 pages. **It does not contain the exam.** That is not a complaint, it is a measurement. Every past paper leans heavily on graph theory, trees, the properties of relations, ordered sets and Boolean algebra. The official text covers none of them. Searched end to end, it contains the word "vertex" zero times, "reflexive" zero

times, "adjacency" zero times, "Handshaking" zero times, "planar" zero times. Its five uses of the word "graph" are all about plotting a function, and its twelve uses of "tree" all belong to one worked example about trees growing in a street.

Meanwhile, on the 2025/2026 paper, roughly two marks in every five come from exactly that missing material. A student who revises only the course text walks into the hall having never seen two fifths of the paper.

So this manual teaches **both**: everything the course text covers, and everything the papers demand that it does not. Wherever a topic falls in the second group, it is flagged like this, once, at the top of the section:

NOT IN THE COURSE TEXT

The official 161-page text does not cover this topic at all, and the exam tests it every year. Everything you need is taught here from zero. If you go back to the course text to revise this, you will not find it. That is the text's gap, not yours.

How every past question is presented

Past questions are the whole point of a manual like this, and they are easy to present dishonestly. If a book tidies a question into its own neat prose, it has done the hardest half of the work for you, namely working out what is being asked, and left you only the arithmetic. Worse, it hides the examiner's habits of phrasing, which are exactly what you are training to recognise, and it quietly repairs the paper's own mistakes, so you never learn that papers contain mistakes.

This manual therefore never restates a past question. Every one arrives in three parts, in this order.

AS PRINTED

EXAMPLE ONLY

The examiner's exact words appear in a box like this one, carrying the paper's own numbering, spelling, symbols and mark text, down to the last comma. **(2 marks)**

NOTE ON THE PAPER

If a question contains a typo, a contradiction or an ambiguity, it stays in the quote and is flagged here, underneath. It is never silently corrected, because when you meet it in the hall it will not have been corrected for you either. Handling a broken question is part of the skill.

BREAK IT DOWN

GIVEN	what the question hands you, listed plainly.
ASKED	what it actually wants. The instruction verb decides this, and it is where marks are most often thrown away: "state" is not "prove", "differentiate" is not "define", and "determine which is true" is not "convert".
MARKS SAY	what the mark count tells you about the depth expected, and roughly how long to spend.
TEACHES	the section of this manual that covers it.
TRAP	the specific way people lose marks on this one.

Only then comes the worked answer. Questions this manual wrote itself, for practice, are labelled as practice and never wear an "As printed" bar, because they were never printed by anybody.

The boxes, and what each one is for

Navigate by shape and colour. Each box does exactly one job.

TEACH

One idea, in plain language, built from what you already know. Never two ideas in one box.

● MUST MEMORISE

THE RESERVED COLOUR

Crimson appears nowhere else in this book. When you see it, the content is going into your memory, not just past your eyes. Each one carries a hook to make that easier.

◆ **Memory hook:** crimson means learn this by heart.

WORKED EXAMPLE

PQ 25/26 Q1C

A full solution with the reasoning shown on every line. The chip on the right names where the question came from, so you always know whether you are looking at a real exam question and which paper it is from. Numeric ones open by listing what you were given and what you must find, before any formula appears.

⚠ TRAP

The mistake made every year. Read these twice.

RECALL AND CHECK

A question with its answer hidden below a dashed rule. Cover the answer, say yours out loud, then look. Reading the answer without trying first teaches you almost nothing, however productive it feels.

FILL IN THE GAP

Cloze practice. Write the missing words on paper, not in your head.

C++

A program, with its real output. Every listing in this manual was compiled and run, and the output shown is what the compiler actually printed, not what it ought to have printed.

How to work through it

Work with a pen and blank paper. When a worked example ends, cover it and redo it from the question. When you reach a past paper, sit it under exam conditions before you read a single answer: three hours, four questions, no notes. A paper you have read the answers to teaches you far less than one you have struggled through and then marked.

The parts are ordered so that nothing is used before it is taught. If you meet a symbol you do not recognise, it was introduced earlier, and the Formula and Notation Reference at the back lists every one.

What the exam looks like

The format changed part way through the papers we hold, so be careful with the older ones.

SESSION	CODE PRINTED	INSTRUCTION	TIME	UNITS	MARKS PER QUESTION
2020/2021	CSC311	Attempt any THREE	2 hours	2	23.5
2021/2022	CSC416	Attempt any THREE	2 hours	2	23.5
2023/2024	CSC 242	not on our copy	not stated	not stated	23.5
2024/2025	CSC416	Attempt any FOUR	3 hours	3	17.5
2025/2026	CSC 242	Answer any Four (4)	3 hours	3	17.5

It is one course that has been renamed twice, so all five papers are valid practice. The current shape is the last row: **six questions, answer any four, 17.5 marks each, 70 marks in three hours**. That is about 44 minutes a question. This manual solves all five papers in full, then gives you three more in the current shape to sit.

⚠ TRAP

BEFORE YOU START REVISING

You choose four questions out of six, so it is tempting to plan on dodging a topic. Look at what that costs on the 2025/2026 paper: graphs appear in Q1, Q4, Q5 and Q6, and counting appears in Q3, Q4 and Q6. There is no set of four questions you can answer while avoiding graph theory, and no honest way to avoid counting either.

The choice protects you against a bad day on one topic. It does not let you leave one out.

FOUNDATIONS

The Language of Discrete Maths

Every symbol this course uses, introduced once, plainly, before anything is built on it. If you have ever stared at a line of mathematics and not known how to say it out loud, start here.

$\in$	$\notin$	$\subseteq$	$\subset$	$\cup$	$\cap$	$\emptyset$	Σ	$\forall$
is in	is not in	subset	proper	union	meet	empty	sum	all
$\exists$	$\neg$	$\wedge$	$\vee$	$\rightarrow$	$\leftrightarrow$	$\equiv$	$\times$	Σ

Numbers, Symbols and Notation

Discrete mathematics has a small vocabulary. Learn it once here and nothing later in this book will look like a foreign language.

By the end of this section you will know what "discrete" means, be able to name and write the standard number sets, read and write the membership and subset symbols out loud, and use factorial and summation notation without hesitating.

What "discrete" means, and why the course exists

TEACH

THE ONE WORD IN THE TITLE

Discrete means separated, made of individual pieces you can count, with nothing in between. Its opposite is **continuous**, which flows without gaps.

The whole numbers 1, 2, 3 are discrete: between 2 and 3 there is no other whole number. Temperature is continuous: between 20 degrees and 21 there are infinitely many values, and you can always find another.

A computer is a discrete machine. It stores whole bits, executes whole instructions, and holds finitely many values. That is the whole reason this course exists: **the mathematics of computing is the mathematics of things you can count and separate**, not the mathematics of smooth curves.

TEACH

WHERE EACH PART GOES IN COMPUTING

Keep this map in your head. It also answers a question the examiner likes to ask directly (2025/2026 Q2a asks for five justifications for including discrete structures in a computer science curriculum).

TOPIC	WHAT IT IS FOR IN COMPUTING
Logic	Circuit design, conditions in programs, database queries, proving a program correct, artificial intelligence
Sets	Data types, database tables, collections, search spaces
Relations	Relational databases, dependency between tasks, equivalence of states
Functions	Every subprogram, hashing, encryption, compression
Counting	Algorithm complexity, password strength, probability of collisions
Graphs	Networks, routing, the web, social networks, scheduling, compilers
Trees	File systems, search structures, parsing, decision procedures
Proof and induction	Showing an algorithm always works, not just on the cases you tried
Recurrence	The running time of a recursive algorithm

(a) Provide five Justification the inclusion of Discrete structure to Computer Science students curriculum. **(5 marks)**

NOTE ON THE PAPER

A word is missing and one is capitalised mid-sentence. The sentence should read "Provide five justifications **FOR** the inclusion of discrete structures **IN** the computer science students' curriculum". The meaning is not in doubt: give five reasons the subject belongs on the degree. Answer it as the five-reason question it is.

BREAK IT DOWN

GIVEN	Nothing. This is an opinion question with a right answer.
ASKED	Provide five justifications. The number is explicit, so give exactly five, numbered. Four costs a mark; six wastes time.
MARKS SAY	5 marks, one per justification. Each wants a claim <i>and</i> a concrete example of where it bites. A bare list of topic names is not a justification.
TEACHES	This section, F.1, using the table above.
TRAP	Listing five topics ("logic, sets, graphs, trees, counting") instead of five reasons . The question asks why the subject is included, not what is in it.

Five justifications, each a claim with a concrete example attached.

1. **Computers are discrete machines, so discrete mathematics is their native mathematics.** A machine stores whole bits and executes whole instructions; it has no smooth quantities. Continuous mathematics describes a falling stone, but nothing inside a processor. *Example:* an integer overflow is a fact about a finite set of representable values, and only discrete reasoning predicts it.
2. **It supplies the language in which algorithms are specified and their cost measured.** Counting techniques give the running time of a procedure before it is ever run. *Example:* knowing that comparing every pair of n records takes $n(n-1)/2$ comparisons tells you the method will not scale, without writing a line of code.
3. **It is the basis of correctness, not merely of testing.** Logic and proof by induction establish that a program works for *every* input, where testing only covers the cases you thought of. *Example:* proving a loop invariant holds shows a sorting routine is right for all inputs, including the ones nobody tried.
4. **Its structures are the data structures of the subject.** Sets, relations, functions, graphs and trees are not analogies for computing objects; they are those objects. *Example:* a relational database is built directly on relations, and a file system is a tree.
5. **It underpins the applied areas the degree leads to.** Networking, cryptography, compilers and artificial intelligence all rest on it. *Example:* routing traffic is a shortest-path problem on a graph, and public-key cryptography rests on number theory over the integers.

How to write it under time pressure: number them 1 to 5, give each a bold opening claim and one sentence of example. Five short numbered paragraphs read as five justifications; one long paragraph reads as one.

The number sets

TEACH

NAMES FOR COLLECTIONS OF NUMBERS

A **set** is any collection of distinct objects, gathered together and treated as one thing. Part Two studies sets properly; for now you only need the five standard ones, because every question in this course lives in one of them.

They are written with a special doubled letter so nobody confuses the set of all integers with a variable called Z .

SYMBOL	NAME	WHAT IS IN IT
$\mathbb{N}$	the natural numbers	0, 1, 2, 3, ... the counting numbers
$\mathbb{Z}$	the integers	... -2, -1, 0, 1, 2, ... whole numbers, positive and negative
$\mathbb{Q}$	the rational numbers	anything writable as a fraction of two integers, such as $\frac{3}{4}$ or -7
$\mathbb{R}$	the real numbers	every point on the number line, including π and the square root of 2
$\mathbb{C}$	the complex numbers	reals extended with the square root of -1 (not examined here)

◆ **Memory hook:** Natural, **Z**ahlen (German for numbers, which is where $\mathbb{Z}$ comes from), **Q**uotient, **R**eal, **C**omplex. Each doubled stroke says "this is a whole universe of numbers, not a variable".

⚠ TRAP

DOES $\mathbb{N}$ CONTAIN 0?

Textbooks disagree. Some start the natural numbers at 0, some at 1. In computing, 0 is almost always included, and this manual includes it.

What matters in an exam: **say which convention you are using**. One short sentence ("taking $\mathbb{N}$ to include 0") protects you completely, and a marker cannot penalise a stated convention. What loses marks is leaving it ambiguous and then relying on it.

Membership: the two symbols you will use most

TEACH

∈ AND ∉

To say that an object is in a set we write $\in$. The statement $3 \in \mathbb{Z}$ is read "**3 is an element of the integers**", or more casually "3 is in $\mathbb{Z}$ ". The symbol is a stylised Greek epsilon, for the Greek word meaning "is".

To say an object is not in a set, strike it through: $\notin$. So $\frac{1}{2} \notin \mathbb{Z}$ reads "**one half is not an element of the integers**", which is true, because one half is not a whole number.

The left side is always an **object** and the right side is always a **set**. That direction never changes, and it is the single most common slip in first attempts.

● MUST MEMORISE

READ THESE OUT LOUD

WRITTEN	SAID	TRUE?
$5 \in \mathbb{N}$	5 is an element of the natural numbers	yes
$-5 \in \mathbb{N}$	minus 5 is an element of the natural numbers	no, $\mathbb{N}$ has no negatives
$-5 \in \mathbb{Z}$	minus 5 is an element of the integers	yes
$\pi \notin \mathbb{Q}$	pi is not an element of the rationals	yes, pi is not a fraction
$\pi \in \mathbb{R}$	pi is an element of the reals	yes

◆ **Memory hook:** $\in$ opens towards the set, like a mouth swallowing the element.

Two ways to describe a set

The 2025/2026 paper asks for exactly this, in two marks, so learn both names.

● MUST MEMORISE

ROSTER AND SET-BUILDER

1. The roster method, also called listing or the tabular method. Write the members out between braces, separated by commas.

$$A = \{2, 4, 6, 8\}$$

2. Set-builder notation, also called the rule method. State the property that decides membership.

$$A = \{x \mid x \text{ is an even number and } 0 < x < 10\}$$

The vertical bar is read "**such that**". Some books use a colon instead, which means the same thing. The whole line is read: "A is the set of all x such that x is even and x is between 0 and 10".

◆ **Memory hook:** the roster method names the members, the way a class register does. Set-builder gives the entry rule, the way a door policy does.

WORKED EXAMPLE

METHOD

Write each set both ways.

(a) The odd numbers under 10.

Roster: $\{1, 3, 5, 7, 9\}$.

Set-builder: $\{x \mid x \in \mathbb{N}, x \text{ is odd}, x < 10\}$.

(b) The days of the week starting with T.

Roster: $\{\text{Tuesday, Thursday}\}$.

Set-builder: $\{d \mid d \text{ is a day of the week and } d \text{ begins with T}\}$.

(c) The integers whose square is 4.

Set-builder: $\{x \mid x \in \mathbb{Z} \text{ and } x^2 = 4\}$.

Roster: $\{-2, 2\}$. Notice the roster form does the work of solving the equation for you, which is exactly why an examiner asks for it.

Now redo it: write "the prime numbers below 12" both ways, then check: $\{2, 3, 5, 7, 11\}$.

⚠ TRAP**SETS DO NOT REPEAT AND DO NOT CARE ABOUT ORDER**

$\{1, 2, 2, 3\}$ is the same set as $\{1, 2, 3\}$. Writing an element twice adds nothing, because a set records only *whether* a thing is in.

$\{3, 1, 2\}$ is also the same set as $\{1, 2, 3\}$. Order carries no information either.

This matters later. When order *does* matter you are no longer dealing with a set but with an **ordered pair** or a sequence, and the brackets change from braces to parentheses: $(1, 2)$ is not $(2, 1)$. Part Three is built entirely on that distinction.

Factorial

● MUST MEMORISE**N FACTORIAL**

For a whole number n , $n!$ (said "n factorial") means multiply every whole number from n down to 1.

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$$

n a whole number, 0 or more. $n!$ the product of every whole number from n down to 1. By definition $0! = 1$.

The first few, which you should know on sight:

N	0	1	2	3	4	5	6	7
N!	1	1	2	6	24	120	720	5040

◆ **Memory hook:** $0! = 1$ looks wrong but is forced: there is exactly **one** way to arrange nothing, namely do nothing. Every counting formula in Part Five breaks if you set it to 0.

TEACH**WHY FACTORIALS GROW SO FAST**

$10!$ is already 3 628 800, and $20!$ is bigger than a billion billion. This is not a curiosity: it is the reason some problems are hopeless to solve by trying every possibility. The travelling-salesman question in Part Six turns on exactly this.

Useful cancelling trick, used constantly in Part Five:

$$10! / 8! = (10 \times 9 \times 8!) / 8! = 10 \times 9 = 90$$

Never expand both factorials and divide. Cancel first. It turns an impossible arithmetic problem into a two-second one.

Summation

● MUST MEMORISE

SIGMA NOTATION

The Greek capital sigma, Σ , is shorthand for "add up". It is read "the sum of".

$$\sum_{i=1}^n a_i = a_1 + a_2 + \dots + a_n$$

i the index, a counter. **i=1** below the sigma, where the counter starts. **n** above the sigma, where it stops. **a_i** the thing being added, written in terms of i.

So $\sum_{i=1}^4 i^2$ means $1^2 + 2^2 + 3^2 + 4^2 = 1 + 4 + 9 + 16 = 30$.

♦ **Memory hook:** sigma is the Greek S, for Sum. Read it left to right as a sentence: "sum, from i equals 1, up to n, of a-sub-i".

RECALL AND CHECK

1. Say out loud what $7 \notin \mathbb{Q}$ would mean, and say whether it is true.
2. What is $6! / 4!$?
3. Write $\{x \mid x \in \mathbb{Z} \text{ and } -2 \leq x \leq 2\}$ by the roster method.
4. Evaluate $\sum_{i=1}^3 (2i + 1)$.

ANSWERS

1. "7 is not an element of the rationals". It is false: $7 = 7/1$, so 7 is rational.
2. $6!/4! = 6 \times 5 = 30$. Cancel, do not expand.
3. $\{-2, -1, 0, 1, 2\}$.
4. $(2+1) + (4+1) + (6+1) = 3 + 5 + 7 = 15$.

FILL IN THE GAP

1. The symbol $\in$ is read _____.
2. $\mathbb{Z}$ is the set of _____.
3. The two mathematical ways to describe a set are the _____ method and _____ notation.
4. $0!$ equals _____.
5. In set-builder notation the vertical bar is read _____.

ANSWERS

1. is an element of. 2. the integers. 3. roster (listing), set-builder (rule). 4. 1. 5. such that.

FOUNDATIONS • F.2

Reading and Writing Mathematics

Mathematical English is a dialect. A handful of phrases mean something narrower than they do in ordinary speech, and marks are lost every year by reading them casually.

By the end of this section you will know exactly what "if", "only if", "if and only if", "necessary" and "sufficient" translate to in symbols, what the instruction verbs on an exam paper demand, and how to lay out an answer so a marker can give you every mark you earned.

The five phrases that carry the marks

TEACH

TWO ARROWS, BEFORE WE USE THEM

Two symbols appear all through this section, so they are introduced here and studied properly in Part One.

$\rightarrow$ is the **implication**, also called the conditional. $p \rightarrow q$ is read "if p then q", and it is a single statement claiming that whenever p holds, q holds too.

$\leftrightarrow$ is the **biconditional**. $p \leftrightarrow q$ is read "p if and only if q", and it claims the two always hold together: each one implies the other. It points both ways because it works both ways.

Here p and q stand for whole statements. Section 1.1 gives them their exact truth tables; for now these readings are enough.

TEACH

NECESSARY AND SUFFICIENT

Take a statement of the form "if p then q", written $p \rightarrow q$. English dresses it up in several ways, and they are not interchangeable.

p is sufficient for q. Having p is enough to guarantee q. Being a dog is sufficient for being an animal.

q is necessary for p. Without q you cannot have p. Being an animal is necessary for being a dog.

So one implication, $p \rightarrow q$, says two things at once: p is enough, and q is required. Which side is which is the whole question, and it is the thing the examiner tests.

ENGLISH	SYMBOLS
if p, then q	$p \rightarrow q$
p implies q	$p \rightarrow q$
q if p	$p \rightarrow q$
p only if q	$p \rightarrow q$
p is sufficient for q	$p \rightarrow q$
q is necessary for p	$p \rightarrow q$
q whenever p	$p \rightarrow q$
q unless not p	$p \rightarrow q$
p if and only if q	$p \leftrightarrow q$
p is necessary and sufficient for q	$p \leftrightarrow q$

◆ **Memory hook:** the two dangerous ones are "**p if q**", which is $q \rightarrow p$, and "**p only if q**", which is $p \rightarrow q$. "If" alone introduces the *hypothesis*; "only if" introduces the *conclusion*. They point opposite ways.

⚠ TRAP

"WHENEVER" AND "ONLY IF" REVERSE PEOPLE EVERY YEAR

The 2024/2025 paper uses both, deliberately.

"I go to the beach whenever it is a sunny summer day." The "whenever" clause is the *hypothesis*. Let p be "it is a sunny summer day" and q "I go to the beach". This is $p \rightarrow q$, not $q \rightarrow p$. The sunny day triggers the beach trip, not the other way round.

"When I stay up late, it is necessary that I sleep until noon." Here "necessary" marks the *conclusion*. Let p be "I stay up late" and q "I sleep until noon". This is $p \rightarrow q$.

Method that never fails: rewrite the sentence as a plain "if ... then ..." **before** you touch symbols. Do that in the margin, then translate. Two extra seconds, and the whole question stops being about English.

Translate each into symbols, with p = "it is raining", q = "the match is cancelled".

(a) "The match is cancelled if it is raining."

Rewrite: if it is raining, then the match is cancelled. So $p \rightarrow q$.

(b) "The match is cancelled only if it is raining."

"Only if" marks the conclusion, so rain is required for cancellation. Rewrite: if the match is cancelled, then it is raining. So $q \rightarrow p$. Note that this is the reverse of (a), from a sentence that differs by one word.

(c) "Rain is necessary and sufficient for cancellation."

Both directions at once: $p \leftrightarrow q$.

(d) "The match is cancelled whenever it rains."

"Whenever" marks the hypothesis: $p \rightarrow q$, the same as (a).

Now redo it: translate "passing the exam is necessary for graduating". Answer below the fold of your hand: let g be graduating and e be passing, then $g \rightarrow e$.

What the instruction verbs demand

TEACH

READ THE VERB BEFORE YOU START WRITING

Marks are allocated to the verb, not to the topic. A perfect definition earns nothing when the question said "prove". This table is drawn from the verbs the five papers actually use.

VERB	WHAT EARNS THE MARKS
Define	One precise sentence. No example needed unless asked.
State	The claim itself, in full, with its conditions. Not a worked instance of it.
Explain	The idea in your own words, plus why it is so. Usually an example is expected.
Differentiate between	A stated point of contrast, both sides, on the same axis. A list of two definitions side by side does not differentiate.
Show that / Prove	An argument that settles every case, not a check of one or two.
Determine whether	A verdict plus the reason. A bare yes or no scores almost nothing.
Construct / Draw	The object itself, labelled.
Compute / Calculate / Find	The number, with the working that produced it.
With the aid of examples	The example is compulsory and carries marks of its own.

⚠ TRAP

"DIFFERENTIATE" IS NOT "DEFINE TWICE"

The 2025/2026 paper opens with ten marks of "differentiate between the following", five pairs at two marks each. The commonest way to lose half of those ten marks is to write two definitions and stop.

A contrast needs a shared axis. Compare this pair of answers on the two ways of describing a set, from F.1:

Scores poorly: "The roster method lists the elements. Set-builder notation uses a rule."

Scores fully: "Both name exactly the same collection, and either can describe any set that the other can. They differ in what they make easy. The roster method writes the members out, so it is direct but only practical for a small finite set: $\{2, 4, 6, 8\}$. Set-builder states the property that decides membership, so it handles infinite sets and sets defined by a condition: $\{x \mid x \text{ is even and } 0 < x < 10\}$. The roster shows you *what* is in; set-builder shows you *why*."

Same facts. The second one answers the verb, because every sentence contrasts the two on a stated axis: what they name, what they make easy, what they show.

How to lay out an answer

● MUST MEMORISE

THE FOUR-LINE ANSWER

For anything with a number in it, write these four things, in this order, every time.

- 1. Given.** The values the question handed you, listed.
- 2. Find.** What is being asked, in your own words.
- 3. Formula or rule.** Stated before it is used, with what each letter means.
- 4. Substitute, work, and answer.** With units or a unit of counting.

This is worth real marks even when your arithmetic goes wrong, because a marker can see the method and award it. An unexplained number cannot be given method marks, however correct it is.

◆ **Memory hook:** Given, Find, Formula, Substitute. Every worked example in this book opens that way, so you will have copied the habit a hundred times before the exam.

TEACH

THREE WORDS THIS COURSE USES PRECISELY

Definition. A naming. It cannot be true or false, only used or misused. " $n!$ means the product of every whole number from n down to 1" is a definition.

Theorem. A claim that has been proved. It is true, and something else depends on it. "There is no largest prime number" is a theorem: it is not a naming, it asserts something, and it has been established beyond doubt.

Proof. An argument that settles a claim for *every* case it covers. Checking three examples is not a proof, it is evidence. Part Nine builds proofs properly.

An exam that says "prove" and receives three examples gets the marks for three examples, which is usually zero.

RECALL AND CHECK

1. Translate "q is necessary for p" into symbols.
2. Translate "p only if q" into symbols.
3. A question says "determine whether 51 is prime". You write "No". How many of the 2 marks should you expect?
4. What is the difference between a definition and a theorem?

ANSWERS

1. $p \rightarrow q$. The necessary thing is the conclusion.
2. $p \rightarrow q$. "Only if" also marks the conclusion.
3. About one, at best. "Determine whether" wants the verdict *and* the reason: exhibit what settles it, for instance "no, because $51 = 3 \times 17$ ".
4. A definition names something and cannot be true or false. A theorem makes a claim that has been proved true.

FILL IN THE GAP

1. "p is sufficient for q" in symbols is _____.
2. "p if and only if q" in symbols is _____.
3. An argument that settles a claim for every case is called a _____.
4. The four-line answer runs Given, _____, Formula, _____.

ANSWERS

1. $p \rightarrow q$. 2. $p \leftrightarrow q$. 3. proof. 4. Find, Substitute.

PART ONE

Logic: The Grammar of Reasoning

Before you can prove anything, you need a language precise enough that a statement cannot wriggle. Logic is that language, and it is the one part of this course the examiner returns to on every single paper.

p	q	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Propositions and Truth Tables

Everything in this part is built from one idea: a sentence that is definitely true or definitely false, and nothing in between.

By the end of this section you will be able to say whether a sentence is a proposition, write an English sentence in symbols, build a truth table for any compound proposition however tangled, and name a proposition as a tautology, a contradiction or a contingency.

What a proposition is

TEACH

THE STARTING POINT

A **proposition** is a declarative sentence that is either true or false, but not both.

Three phrases in that definition are doing work. **Declarative** means it declares something, so it is not a question or a command. **Either true or false** means it has a definite truth value, even if nobody present knows which. **Not both** means it cannot be true in one breath and false in the next.

SENTENCE	PROPOSITION?	WHY
Lagos is in Nigeria.	Yes, true	It declares something, and that something is the case.
$2 + 2 = 5$	Yes, false	Being false does not stop it being a proposition. It only needs a truth value.
What time is the exam?	No	A question. It declares nothing, so there is nothing to be true.
Close the door.	No	A command, for the same reason.
$x + 1 = 3$	Not yet	Its truth depends on x . Until x is fixed it has no truth value. Section 1.4 turns sentences like this into propositions.
This sentence is false.	No	If it is true it is false, and if it is false it is true. It breaks the "not both" rule.

● MUST MEMORISE

DEFINITION

A **proposition** is a declarative sentence that is either true or false, but not both.

Propositions are written with letters, usually p , q , r , s . A letter standing for a proposition is a **propositional variable**. The two truth values are written **T** for true and **F** for false.

◆ **Memory hook:** if you can sensibly answer "is that true?", it is a proposition. You cannot answer that about a question or an order.

The connectives

TEACH

BUILDING BIGGER STATEMENTS

Single propositions are not very interesting. We join them with **connectives** to make **compound propositions**, and the whole of this part is the study of how the truth of the pieces fixes the truth of the whole.

Each connective is defined by a **truth table**: a table listing every possible combination of truth values for the pieces, with the resulting value for the whole. A truth table is not an explanation of a connective, it *is* the connective. There is nothing more to know about "and" in logic than the four rows of its table.

MUST MEMORISE

THE CONNECTIVES

NAME	SYMBOL	READ AS	TRUE EXACTLY WHEN
Negation	$\neg p$	not p	p is false
Conjunction	$p \wedge q$	p and q	both are true
Disjunction	$p \vee q$	p or q	at least one is true
Exclusive or	$p \oplus q$	p or q but not both	exactly one is true
Implication	$p \rightarrow q$	if p then q	p is false, or q is true
Biconditional	$p \leftrightarrow q$	p if and only if q	both have the same truth value

◆ **Memory hook:** $\wedge$ points up like the A in "and"; $\vee$ is the other way up, for "or". $\rightarrow$ points from the assumption to the conclusion. $\leftrightarrow$ points both ways because it works both ways.

The tables themselves

Here are all six side by side. Learn to reproduce this from memory, because almost every logic question on every paper starts by using it.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \oplus Q$	$P \rightarrow Q$	$P \leftrightarrow Q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

Look at row 3 of $p \rightarrow q$: p is false, q is true, and the implication is **true**. Now row 4: both false, and it is **true** again.

Students reject this because it feels wrong. "If I pass, I will celebrate" seems to say nothing about what happens when I fail. Exactly: it says nothing, and a promise that says nothing about a case has not been broken in that case. An implication is a promise, and a promise is false only when it is **broken**, which needs p true and q false.

$p \rightarrow q$ is false in exactly one case: p true, q false. Learn that as a single fact and the other three rows follow.

TEACH

WHY "OR" NEEDS CARE

English uses "or" in two ways. "Tea or coffee?" usually means one or the other but not both. "Applicants must have a degree or five years of experience" means at least one, and having both is fine.

Logic splits these. **Disjunction** ($\vee$) is the inclusive one: true when at least one part is true, including when both are. **Exclusive or** ($\oplus$) is the other: true when exactly one is true. Compare the rows where p and q are both true: $\vee$ gives T, $\oplus$ gives F. That single row is the whole difference.

Building a table for a compound proposition

TEACH

THE METHOD, ONCE, PROPERLY

Any compound proposition, however tangled, is handled the same way.

Step 1. Count the letters. With n distinct propositional variables the table has 2^n rows: 2 letters give 4 rows, 3 give 8, 4 give 16.

Step 2. Fill the letter columns mechanically. Under the first letter write T for the top half of the rows and F for the bottom half; under the next, alternate in blocks half that size; keep halving. Done this way you cannot miss a combination or write one twice.

Step 3. Work outward from the innermost bracket. Give every intermediate expression its own column. Do not jump to the final answer in one step: you cannot check a step you did not write down, and the marks are for the working.

Step 4. Read the final column. All T is a tautology, all F a contradiction, a mixture a contingency.

A compound proposition is a

tautology if it is true for every assignment of truth values, that is, the final column is all T;

contradiction if it is false for every assignment, all F;

contingency if it is neither, true for some assignments and false for others.

◆ **Memory hook:** a tautology tells you nothing because it is always true, a contradiction is always false, and a contingency depends, which is what "contingent" means in ordinary English.

Now the real questions

The 2020/2021 paper asks for the definitions and the tables together. Read what it actually says, including its own spelling.

AS PRINTED

2020/2021 · Q5(A)

5. (a) Define and draw the truth table for each of the following logic proposition terminologies:

- (i) Negation (ii) Conjunction (iii) Exclusive or (iv) Implication
(v) Tautology (vi) Contingency

(6marks)

NOTE ON THE PAPER

The paper prints "Contingency". The word is **CONTINGENCY**. Write the correct spelling in your answer; nobody has ever lost a mark for spelling the examiner's typo correctly. Notice too that (v) and (vi) are not connectives at all, so "draw the truth table for a tautology" does not quite make sense as written. What is wanted is an example whose table is all T, and one whose table is mixed.

BREAK IT DOWN

GIVEN Six terms: four connectives and two names for a shape of final column.

ASKED Two things for each. **Define** it in words, and **draw** its truth table. Half the marks are in the words, so a page of tables with no definitions scores about half.

MARKS SAY 6 marks over six items is one mark each: one sentence and one small table apiece. Do not write an essay on implication.

TEACHES This section, 1.1.

TRAP Giving the table for $\vee$ when asked for exclusive or. They differ in exactly one row, and that row is the answer.

(i) Negation. The negation of p , written $\neg p$, is the proposition "it is not the case that p ". It reverses the truth value.

P	$\neg P$
T	F
F	T

(ii) Conjunction. The conjunction of p and q , written $p \wedge q$, is " p and q ". It is true only when both are true.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

(iii) Exclusive or. The exclusive or of p and q , written $p \oplus q$, is true when exactly one of them is true and false when they agree.

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

(iv) Implication. The implication $p \rightarrow q$, read "if p then q ", is false only when p is true and q is false. Here p is the **hypothesis** and q the **conclusion**.

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

(v) Tautology. A compound proposition that is true for every assignment of truth values to its variables. The standard example is $p \vee \neg p$, "either it is raining or it is not".

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

(vi) Contingency. A compound proposition that is true for some assignments and false for others, so it is neither a tautology nor a contradiction. Any ordinary compound will

do, for instance $p \wedge q$, whose column above reads T, F, F, F.

For completeness, the third member of that family: a **contradiction** is false for every assignment, for example $p \wedge \neg p$.

P	$\neg P$	$P \wedge \neg P$
T	F	F
F	T	F

Now redo it: cover this answer and write out all six definitions and tables on blank paper. Time yourself; you should be inside six minutes.

A harder table, from the recent papers

The same skill on propositions with more structure. This one is worth quoting carefully, because part of it appears on two different papers.

AS PRINTED

2024/2025 · Q1(B)

b. Construct a truth table for each of these compound propositions.

- i. $p \leftrightarrow \neg p$ ii. $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$
 iii. $(p \oplus q) \vee (p \oplus \neg q)$ iv. $((p \rightarrow q) \rightarrow r) \rightarrow s$

4 marks

AS PRINTED

2021/2022 · Q5(D)

(d) Construct a truth table for each of the following compound propositions

- (i) $(p \leftrightarrow q) \ominus (p \leftrightarrow \neg q)$
 (ii) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$

(5 marks)

(5 marks)

NOTE ON THE 2021/2022 VERSION

Part (i) prints the connective as $\ominus$, a circled *minus*. There is no standard logical connective with that symbol. In context it can only mean **EXCLUSIVE OR**, normally written $\oplus$, and the question is unanswerable on any other reading. Quote the glyph as printed, say in one line what it has to mean, and answer on that reading.

Note also that this paper prints "(5 marks)" with a space here, while every other mark on it is closed up as "5marks". Part (ii) is word for word the same proposition as 2024/2025 Q1 b(ii) above.

NOTE ON THE 2024/2025 VERSION

Its part (ii) is word for word the 2021/2022 part (ii), so one table answers both papers and that one table is worth 5 marks on one paper and a share of 4 on the other. Note also that part (iv) has **FOUR** letters, so its table needs 16 rows: four times the work of the others for the same share of the marks. In the hall, do i, ii and iii first.

BREAK IT DOWN

- GIVEN** Four compound propositions of increasing size: one letter, two, two, then four.
- ASKED** **Construct** a truth table. Not "simplify", not "say whether it is a tautology". Build the table; any conclusion you then draw from it is a bonus that shows you understood it.
- MARKS SAY** 4 marks for four tables is one mark each, and one of them is 16 rows. So the marks are for method, not endurance: set the columns out cleanly and the mark is yours.
- TEACHES** This section, 1.1, plus 1.3 for the shortcut on (ii).
- TRAP** Rushing (iv) and losing a row. Fill the letter columns by halving, as in Step 2, and you cannot go wrong.

WORKED EXAMPLE**PQ 24/25 Q1(B) I AND II**

(i) $p \leftrightarrow \neg p$. One letter, so 2 rows. The biconditional is true when both sides agree, and p never agrees with $\neg p$.

P	$\neg P$	$P \leftrightarrow \neg P$
T	F	F
F	T	F

Every row is F, so **$p \leftrightarrow \neg p$ is a contradiction**. Say so: naming it shows you understood the table you just drew.

(ii) $(q \rightarrow \neg p) \leftrightarrow (p \leftrightarrow q)$. Two letters, so 4 rows. Build the pieces first: $\neg p$, then $q \rightarrow \neg p$, then $p \leftrightarrow q$, then join them.

P	Q	$\neg P$	$Q \rightarrow \neg P$	$P \leftrightarrow Q$	WHOLE
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	F	F
F	F	T	T	T	T

Taking the rows one at a time, remembering that $\rightarrow$ is false only for T then F, and $\leftrightarrow$ is true only when its two sides match:

Row 1: q is T and $\neg p$ is F, so $q \rightarrow \neg p$ is F. $p \leftrightarrow q$ with both T is T. F against T gives F.

Row 2: q is F, so $q \rightarrow \neg p$ is T whatever follows it. $p \leftrightarrow q$ with T and F is F. T against F gives F.

Row 3: q is T and $\neg p$ is T, so T. $p \leftrightarrow q$ with F and T is F. T against F gives F.

Row 4: q is F so the implication is T. $p \leftrightarrow q$ with both F is T. T against T gives T.

Three F rows and one T row, so this is a **contingency**. It is true exactly when p and q are both false.

Now redo it: parts (iii) and (iv) are yours. For (iii) expect a surprise in the final column, and say what it means. For (iv) remember 16 rows, and fill the p column as 8 T then 8 F.

RECALL AND CHECK

1. In which single case is $p \rightarrow q$ false?
2. How many rows does a truth table with 5 letters have?
3. In which row do $p \vee q$ and $p \oplus q$ disagree?
4. Is " $x > 3$ " a proposition?

ANSWERS

1. When p is true and q is false, and in no other.
2. $2^5 = 32$.
3. When both are true: $\vee$ gives T, $\oplus$ gives F.
4. No. Its truth depends on x , so it has no truth value until x is fixed. It is a predicate, and Section 1.4 shows how to turn it into a proposition.

FILL IN THE GAP

1. A declarative sentence that is either true or false but not both is called a _____.
2. The connective $\wedge$ is called _____ and is true only when _____ parts are true.
3. A compound proposition whose final column is entirely T is a _____, entirely F a _____, and mixed a _____.
4. With n distinct variables a truth table has _____ rows.

ANSWERS

1. proposition. 2. conjunction, both. 3. tautology, contradiction, contingency. 4. 2^n .

PART ONE • 1.2

Implication, Converse, Inverse, Contrapositive

One implication has three relatives. Exactly one of them says the same thing as the original, and every paper in our set asks you which.

By the end of this section you will be able to form the converse, inverse and contrapositive of any conditional statement, in English or in symbols, and prove which one preserves truth value.

The three relatives

TEACH

BUILT FROM ONE IMPLICATION

Start with a conditional statement $p \rightarrow q$. There are exactly three ways to rearrange it, and each has a name.

The **converse** swaps the two parts: $q \rightarrow p$.

The **inverse** negates both parts, keeping the order: $\neg p \rightarrow \neg q$.

The **contrapositive** does both, swap and negate: $\neg q \rightarrow \neg p$.

Watch what happens to a concrete case. Let p be "it is a dog" and q "it is an animal", so $p \rightarrow q$ reads "if it is a dog then it is an animal", which is true.

NAME	FORM	THE DOG EXAMPLE	TRUE?
Original	$p \rightarrow q$	If it is a dog, it is an animal.	yes
Converse	$q \rightarrow p$	If it is an animal, it is a dog.	no , a cat is a counterexample
Inverse	$\neg p \rightarrow \neg q$	If it is not a dog, it is not an animal.	no , the same cat
Contrapositive	$\neg q \rightarrow \neg p$	If it is not an animal, it is not a dog.	yes

● MUST MEMORISE

THE ONE THAT MATTERS

For any conditional statement $p \rightarrow q$:

converse = $q \rightarrow p$

inverse = $\neg p \rightarrow \neg q$

contrapositive = $\neg q \rightarrow \neg p$

Only the contrapositive is logically equivalent to the original. The converse and the inverse are equivalent to each other, and neither follows from the original.

◆ **Memory hook:** Converse **C**hanges order. Inverse **I**nverts (negates) both. Contrapositive does both, and doing both is the only safe operation. In the dog example: knowing it is not an animal really does tell you it is not a dog.

TEACH

PROVING IT RATHER THAN ASSERTING IT

The 2024/2025 paper does not ask you to state this, it asks you to **prove** it, for six marks. A truth table is the proof.

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$	$\neg Q \rightarrow \neg P$
T	T	F	F	T	T	T	T
T	F	F	T	F	T	T	F
F	T	T	F	T	F	F	T
F	F	T	T	T	T	T	T

Read the columns, not the rows. The column for $p \rightarrow q$ is T, F, T, T. The column for the contrapositive $\neg q \rightarrow \neg p$ is T, F, T, T: **identical**, so they are equivalent.

The converse column is T, T, F, T and the inverse column is T, T, F, T: identical to each other, and different from the original in rows 2 and 3. So the converse and inverse are equivalent to each other and neither is equivalent to $p \rightarrow q$. That single table proves every claim in this section at once.

The real question

AS PRINTED

2024/2025 · Q1(C)

c. Consider the conditional statement:

"The home team wins whenever it is raining."

Let p represent: "It is raining."

Let q represent: "The home team wins."

- i. Using this, find the following: the contrapositive, the converse, and the inverse of the statement. **3 marks**
- ii. Prove that only the contrapositive of a conditional statement $p \rightarrow q$ has the same truth value as the original conditional statement. Use the statement from c(i) in your explanation. **6 marks**

NOTE ON THE PAPER

The paper hands you the letters, which is a gift: p is the *rain*, not the win. Read that assignment twice before writing, because the sentence puts the win first and the rain second, while the letters run the other way.

BREAK IT DOWN

GIVEN	The sentence "The home team wins whenever it is raining.", with p = it is raining and q = the home team wins.
ASKED	Part i: find three rearrangements, in English. Part ii: prove a general claim, then illustrate it with this sentence. "Prove" means a truth table or a full argument, not an assertion.
MARKS SAY	3 marks for three statements is one each, so short English sentences suffice. 6 marks for the proof is the bulk of the question: that is a full table with the columns compared and a conclusion written out.
TEACHES	This section, 1.2, and 1.1 for the table method.
TRAP	"Whenever" marks the hypothesis, so the statement is $p \rightarrow q$ (rain implies win), not $q \rightarrow p$. Getting that backwards inverts all three answers in part i and costs the lot.

GIVEN p = "It is raining", q = "The home team wins", and the sentence "The home team wins whenever it is raining."

FIND The contrapositive, converse and inverse, then a proof that only the contrapositive preserves truth value.

RULE converse $q \rightarrow p$, inverse $\neg p \rightarrow \neg q$, contrapositive $\neg q \rightarrow \neg p$.

Step 1. Get the direction right. "The home team wins whenever it is raining" rewrites as "if it is raining, then the home team wins". So the statement is $p \rightarrow q$.

Step 2 (part i). Apply each definition.

NAME	SYMBOLS	IN ENGLISH
Contrapositive	$\neg q \rightarrow \neg p$	If the home team does not win, then it is not raining.
Converse	$q \rightarrow p$	If the home team wins, then it is raining.
Inverse	$\neg p \rightarrow \neg q$	If it is not raining, then the home team does not win.

Step 3 (part ii). Prove it with a truth table. Build every column and compare them against $p \rightarrow q$.

P	Q	$\neg P$	$\neg Q$	$P \rightarrow Q$	$\neg Q \rightarrow \neg P$	$Q \rightarrow P$	$\neg P \rightarrow \neg Q$
T	T	F	F	T	T	T	T
T	F	F	T	F	F	T	T
F	T	T	F	T	T	F	F
F	F	T	T	T	T	T	T

The $p \rightarrow q$ column reads T, F, T, T. The contrapositive column reads T, F, T, T. They agree in every row, so $p \rightarrow q \equiv \neg q \rightarrow \neg p$. The converse reads T, T, F, T and disagrees with the original in rows 2 and 3, and the inverse matches the converse rather than the original.

Step 4. Say it in words, using their sentence. Suppose it rains and the home team loses. That single situation makes the original false, because the promise was broken. It also makes the contrapositive false: the team did not win, yet it was raining. The two statements fail together and hold together, which is what equivalence means.

Now the converse. Suppose the home team wins on a dry day. The original is untouched, since it only ever promised something about rainy days, so it stays true. But the converse, "if the home team wins then it is raining", is false. One statement true, the other false, so they are not equivalent. The inverse fails on that same dry, winning day.

Conclusion. Of the three rearrangements only the contrapositive always carries the same truth value as the original.

Now redo it: cover this and do the whole question for "I go to the beach whenever it is a sunny summer day", which is 2024/2025 Q1(a)(ii).

The rest of that question

AS PRINTED

2024/2025 · Q1(A)

a. State the converse, contrapositive, and inverse of each of these conditional statements.

- i. If it snows tonight, then I will stay at home.
- ii. I go to the beach whenever it is a sunny summer day.
- iii. When I stay up late, it is necessary that I sleep until noon.

4.5 marks

BREAK IT DOWN

GIVEN Three English sentences, each hiding an implication in a different disguise: a plain "if ... then", a "whenever", and a "necessary that".

ASKED **State** three rearrangements of each, so nine sentences in all. English is fine; symbols are not required but help you avoid errors.

MARKS SAY 4.5 marks over nine statements is half a mark each. Write them as short sentences and move on; there are no marks for commentary here.

TEACHES This section for the three forms, F.2 for decoding "whenever" and "necessary".

TRAP All three sentences put the pieces in a different order from $p \rightarrow q$. Identify p and q explicitly, in the margin, before writing a single answer.

WORKED EXAMPLE

PQ 24/25 Q1(A)

(i) "If it snows tonight, then I will stay at home."

Already in standard form. p = it snows tonight, q = I will stay at home.

Converse: If I stay at home tonight, then it will snow.

Inverse: If it does not snow tonight, then I will not stay at home.

Contrapositive: If I do not stay at home tonight, then it will not snow.

(ii) "I go to the beach whenever it is a sunny summer day."

"Whenever" marks the hypothesis. p = it is a sunny summer day, q = I go to the beach.

Rewritten: if it is a sunny summer day, then I go to the beach.

Converse: If I go to the beach, then it is a sunny summer day.

Inverse: If it is not a sunny summer day, then I do not go to the beach.

Contrapositive: If I do not go to the beach, then it is not a sunny summer day.

(iii) "When I stay up late, it is necessary that I sleep until noon."

"When" marks the hypothesis and "it is necessary that" marks the conclusion. p = I stay up late, q = I sleep until noon. Rewritten: if I stay up late, then I sleep until noon.

Converse: If I sleep until noon, then I stayed up late.

Inverse: If I do not stay up late, then I do not sleep until noon.

Contrapositive: If I do not sleep until noon, then I did not stay up late.

Now redo it: nine sentences, blank paper, four minutes. If any of yours reverses the roles of p and q , go back to F.2 and re-read the phrase table.

The negation of "I will stay at home" is "I will not stay at home". It is **not** "I will go out", however natural that sounds. Staying at home and going out might not exhaust the possibilities, and logic only knows what you wrote down.

Similarly, the negation of " $x > 3$ " is " $x \leq 3$ ", not " $x < 3$ ". Dropping the equality case is the commonest numeric slip on this topic.

RECALL AND CHECK

1. Which one of the three relatives is equivalent to the original?
2. Write the contrapositive of "if n is even then n^2 is even".
3. The converse is equivalent to which other statement?
4. What is the negation of " $x \geq 5$ "?

ANSWERS

1. The contrapositive.
2. If n^2 is not even, then n is not even.
3. The inverse. Both are T, T, F, T.
4. $x < 5$.

FILL IN THE GAP

1. The converse of $p \rightarrow q$ is _____.
2. The inverse of $p \rightarrow q$ is _____.
3. The contrapositive of $p \rightarrow q$ is _____.
4. Only the _____ has the same truth value as the original.

ANSWERS

1. $q \rightarrow p$. 2. $\neg p \rightarrow \neg q$. 3. $\neg q \rightarrow \neg p$. 4. contrapositive.

PART ONE · 1.3

Logical Equivalence and the Laws

Two statements that always agree can be swapped freely. That is what lets you simplify a condition, redesign a circuit, or rewrite a database query without changing what it does.

By the end of this section you will be able to test two propositions for equivalence with a truth table, state De Morgan's laws properly as laws, and use the standard laws to transform one expression into another.

What equivalence means

● MUST MEMORISE

DEFINITION

Two compound propositions are **logically equivalent** when they have the same truth value for every possible assignment of truth values to their variables. We write $P \equiv Q$. Equivalently: $P \equiv Q$ exactly when the biconditional $P \leftrightarrow Q$ is a **tautology**.

◆ **Memory hook:** $\equiv$ is the equals sign with an extra bar, meaning "equal in every row, not just this one". It is a claim *about* two propositions, so it is not itself a connective.

TEACH

HOW TO SHOW TWO THINGS ARE EQUIVALENT

There are two accepted methods, and the question usually tells you which it wants.

Method 1: truth table. Build a column for each side and check they match row for row. Always works, never elegant, and cannot fail through a wrong step. Use it when the question says "show that" or "construct a truth table".

Method 2: a chain of known laws. Rewrite one side step by step until it becomes the other, naming the law at every step. Shorter when it works, and it is what "using logical equivalences" means when a paper says so, as 2024/2025 Q3(c) does.

The naming is not decoration. In a law-chain proof the marks are for the justifications, so a chain of correct lines with no names attached scores poorly.

De Morgan's laws

DE MORGAN'S LAWS

The claim. Negating a conjunction turns it into the disjunction of the negations, and negating a disjunction turns it into the conjunction of the negations. In words: *not (both) is the same as (not one) or (not the other)*, and *not (either) is the same as (not one) and (not the other)*.

In symbols.

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Conditions. None beyond p and q being propositions. The laws hold universally, which is what makes them safe to apply anywhere.

The limiting case, and why the wrong version survives. The common error is to write $\neg(p \wedge q)$ as $\neg p \wedge \neg q$, keeping the connective instead of flipping it. Those two agree in exactly the rows where p and q have the **same** truth value, and differ in the other two. So a student who checks only the TT and FF rows sees the wrong law confirmed. Check a row where p and q disagree, and the error appears at once.

The same laws for sets. Replacing "and" by intersection, "or" by union and "not" by complement gives the set forms, which Part Two uses:

the complement of $(A \cap B) = (\text{complement of } A) \cup (\text{complement of } B)$

the complement of $(A \cup B) = (\text{complement of } A) \cap (\text{complement of } B)$

Consequence. A negation can always be pushed inwards until it sits on single variables. That is why every Boolean expression can be simplified mechanically, and why a compiler can re-write your `if` conditions.

Say it about breakfast. "It is not the case that I had both tea and toast" means "I did not have tea, or I did not have toast" (at least one of them was missing). It does **not** mean I had neither.

"It is not the case that I had either tea or toast" is the stronger one: it means I had no tea **and** no toast.

So negation flips the connective as it moves inside. Forgetting to flip it is the mistake De Morgan exists to prevent.

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Columns four and seven are both F, T, T, T. The law is proved.

The full list of laws

● MUST MEMORISE

THE LAWS OF LOGIC

Here **T** means a tautology (always true) and **F** a contradiction (always false).

NAME	LAW
Identity	$p \wedge T \equiv p$ $p \vee F \equiv p$
Domination	$p \vee T \equiv T$ $p \wedge F \equiv F$
Idempotent	$p \vee p \equiv p$ $p \wedge p \equiv p$
Double negation	$\neg(\neg p) \equiv p$
Commutative	$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$
Associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
Distributive	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption	$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$
Negation	$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$
Implication	$p \rightarrow q \equiv \neg p \vee q$
Contrapositive	$p \rightarrow q \equiv \neg q \rightarrow \neg p$
Biconditional	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$

◆ **Memory hook:** the one that unlocks most problems is **implication**: $p \rightarrow q \equiv \neg p \vee q$. An arrow is hard to manipulate; an "or" is easy. Turn every arrow into an or as your first move, and most chains fall out.

A real question

AS PRINTED

2020/2021 · Q4(C)

(c) Show that the following are logically equivalent

(i) $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$.

(3marks)

(ii) $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$

(3marks)

BREAK IT DOWN

- GIVEN** Two pairs of compound propositions, the first on three letters and the second on two.
- ASKED** **Show that** each pair is equivalent. A truth table is a complete answer, and is safer than a law chain under time pressure.
- MARKS SAY** 3 marks each. Part (i) has three letters so its table is 8 rows; part (ii) has two letters and 4 rows. Both are quick if the columns are set out before you start filling.
- TEACHES** This section for equivalence, 1.1 for building the table.
- TRAP** Part (i) needs 8 rows, and the temptation is to do 4. Count the distinct letters, not the brackets.

WORKED EXAMPLE**PQ 20/21 Q4(C)(I)****GIVEN** $(p \rightarrow q) \wedge (p \rightarrow r)$ and $p \rightarrow (q \wedge r)$.**FIND** Whether the two columns agree in every row.**RULE** Three letters, so $2^3 = 8$ rows.

P	Q	R	$P \rightarrow Q$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (P \rightarrow R)$	$Q \wedge R$	$P \rightarrow (Q \wedge R)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	F	F	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Column 6 reads T, F, F, F, T, T, T, T. Column 8 reads T, F, F, F, T, T, T, T. Identical in all eight rows, so $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$.

Sanity check in words: promising q and promising r separately, on the same condition p , is the same as promising both together on that condition. It should feel obvious, and the table is why it is certain.

GIVEN $p \leftrightarrow q$ and $(p \wedge q) \vee (\neg p \wedge \neg q)$.

FIND Whether the two agree in every row.

RULE Two letters, 4 rows.

P	Q	$P \leftrightarrow Q$	$P \wedge Q$	$\neg P$	$\neg Q$	$\neg P \wedge \neg Q$	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$
T	T	T	T	F	F	F	T
T	F	F	F	F	T	F	F
F	T	F	F	T	F	F	F
F	F	T	F	T	T	T	T

Column 3 is T, F, F, T and column 8 is T, F, F, T. Equivalent.

In words: "p and q agree" is the same as "either both are true, or both are false". That is exactly what the biconditional means, so the right-hand side is really the definition written out.

Now redo it: on blank paper, show that $p \rightarrow q \equiv \neg p \vee q$ with a truth table. It is the law you will lean on most.

⚠ TRAP

EQUIVALENT IS NOT "SIMILAR"

Equivalence needs agreement in **every** row. Two propositions that match in three rows out of four are not "nearly equivalent", they are simply different, and the one row where they differ is the counterexample you should quote.

When a question asks you to show two things are **not** equivalent, do not build the whole table and say "they differ". Name the row: "take p true and q false; then the first is F and the second is T". One row, fully specified, is a complete disproof.

RECALL AND CHECK

1. State both of De Morgan's laws.
2. Rewrite $p \rightarrow q$ without using the arrow.
3. Simplify $p \vee (p \wedge q)$.
4. How many rows are needed to test an equivalence on four letters?

ANSWERS

1. $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.
2. $\neg p \vee q$.
3. p, by the absorption law.
4. $2^4 = 16$.

FILL IN THE GAP

1. $P \equiv Q$ means the two have the same truth value for _____ assignment.
2. $\neg(p \vee q) \equiv$ _____.
3. The implication law says $p \rightarrow q \equiv$ _____.
4. $p \wedge \neg p \equiv$ _____.

ANSWERS

1. every. 2. $\neg p \wedge \neg q$. 3. $\neg p \vee q$. 4. F (a contradiction).

PART ONE · 1.4

Predicate Logic and Quantifiers

Propositional logic cannot say "every student has a laptop" without inventing a letter for each student. Predicate logic can, and that is the whole reason it exists.

By the end of this section you will be able to say exactly how predicate logic differs from propositional logic, use both quantifiers correctly, translate an English sentence with "some", "all", "no" or "every" into symbols, negate a quantified statement, and decide the truth of a quantified statement over the real numbers.

The limit of propositional logic

TEACH

WHY WE NEED SOMETHING MORE

In Part One so far, p was an entire sentence with a fixed truth value. That is fine for "it is raining", but it cannot see inside a sentence.

Consider "every student in this class has a laptop". With propositional letters you would need one letter per student, and you would still have no way to say "every". Worse, " $x + 1 = 3$ " cannot be a proposition at all, because its truth depends on x .

Predicate logic fixes both problems by splitting a sentence into a **subject** and a **property**, and then adding words for "all" and "some".

A **predicate** is a statement containing one or more variables, which becomes a proposition once the variables are given values. We write $P(x)$, read "P of x".

If $P(x)$ is " $x > 3$ ", then $P(5)$ is " $5 > 3$ ", which is true, and $P(1)$ is " $1 > 3$ ", which is false. $P(x)$ on its own is neither.

AXIS	PROPOSITIONAL LOGIC	PREDICATE LOGIC
Smallest unit	a whole sentence, p	a predicate with variables, $P(x)$
Can see inside a sentence	no	yes, subject and property are separate
Quantifiers	none	$\forall$ and $\exists$
Can express "all" and "some"	no	yes
Example	$p = \text{"Ada has a laptop"}$	$L(x) = \text{"x has a laptop"}$, so $\forall x L(x)$

◆ **Memory hook:** propositional logic treats a sentence as a sealed box. Predicate logic opens the box and finds a subject and a property inside.

The two quantifiers

● MUST MEMORISE

UNIVERSAL AND EXISTENTIAL

The **universal quantifier** $\forall$, read "for all" or "for every". $\forall x P(x)$ is true exactly when $P(x)$ is true for *every* x in the domain.

The **existential quantifier** $\exists$, read "there exists" or "for some". $\exists x P(x)$ is true exactly when $P(x)$ is true for *at least one* x in the domain.

The **domain** (or universe of discourse) is the collection the variable ranges over. It must always be stated, because the same statement can flip truth value when the domain changes.

	$\forall$ UNIVERSAL	$\exists$ EXISTENTIAL
Read as	for all, for every, for each	there exists, for some, at least one
True when	every element satisfies P	at least one element satisfies P
False when	one counterexample exists	every element fails P
To prove it	a general argument covering all cases	produce one witness
To disprove it	produce one counterexample	a general argument that none works
Pairs with	$\rightarrow$ usually	$\wedge$ usually

◆ **Memory hook:** $\forall$ is an upside-down **A** for **All**. $\exists$ is a back-to-front **E** for **Exists**. The asymmetry in the last two rows is worth learning on its own: *one example kills a "for all" and one example makes a "there exists"*.

⚠ TRAP

$\forall$ TAKES AN ARROW, $\exists$ TAKES AN AND

This is the single most common error in translation, and it appears on every paper that asks for one.

"Every student in the class has a laptop" is $\forall x (S(x) \rightarrow L(x))$, with an arrow. Writing $\forall x (S(x) \wedge L(x))$ instead would claim that *everything in the universe* is a student and has a laptop, which is far stronger and almost certainly false.

"Some student has a laptop" is $\exists x (S(x) \wedge L(x))$, with an and. Writing $\exists x (S(x) \rightarrow L(x))$ instead is nearly useless: it is satisfied by any non-student at all, since a false hypothesis makes the implication true.

If the domain is already restricted to students, the qualifier disappears and you write $\forall x L(x)$ or $\exists x L(x)$. Say which domain you are using.

Negating a quantified statement

DE MORGAN'S LAWS FOR QUANTIFIERS

The claim. A negation moved across a quantifier flips it. Saying "not everything has the property" is the same as saying "something lacks it", and saying "nothing has the property" is the same as saying "everything lacks it".

In symbols.

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Conditions. None. They hold for any predicate over any domain.

The limiting case. Over a **finite** domain $\{a_1, \dots, a_n\}$, $\forall$ is a chain of conjunctions and $\exists$ a chain of disjunctions, so these two laws collapse into the propositional De Morgan laws of section 1.3 applied n times. They are not a second pair of laws but the same pair, stretched over a domain. Over an **empty** domain they still hold, and the check is worth doing once: $\forall x P(x)$ is true there (nothing fails) and $\exists x P(x)$ is false (nothing succeeds), so both sides of the first law come out false.

Consequence. Any negation can be pushed inwards until it rests on a bare predicate, flipping every quantifier it passes. This is how "no student has a cat, a dog and a ferret" gets written two different but equivalent ways, which 2024/2025 Q2(a)(iv) exploits.

TEACH

IN PLAIN ENGLISH

"Not all swans are white" means "some swan is not white". You have not claimed all swans are non-white, only that at least one is.

"No swan is green" means "every swan is not green". Here the negation does apply to all of them.

Notice that "no" is a negated **existential**: "no x has property P " is $\neg \exists x P(x)$, which becomes $\forall x \neg P(x)$. Whenever an exam sentence starts with "No", write it as a negated "there exists" first, then convert if you want.

The translation question

AS PRINTED

2024/2025 · Q2(A)

a. Let $C(x)$ be the statement "x has a cat," let $D(x)$ be the statement "x has a dog," and let $F(x)$ be the statement "x has a ferret." Express each of these statements in terms of $C(x)$, $D(x)$, $F(x)$, quantifiers, and logical connectives. Let the domain consist of all students in your class.

- i. A student in your class has a cat, a dog, and a ferret.
- ii. All students in your class have a cat, a dog, or a ferret.
- iii. Some student in your class has a cat and a ferret, but not a dog.
- iv. No student in your class has a cat, a dog, and a ferret.
- v. For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet.

7.5 marks

NOTE ON THE PAPER

The domain is given as "all students in your class", which is a real gift: it means you do **NOT** need a student predicate $S(x)$, and the $\forall/\rightarrow$ pairing collapses. Students who invent $S(x)$ anyway are not wrong, but they do more work for the same marks. Note also the paper's spacing, " $F(x)$ " with a space, which is just typesetting.

BREAK IT DOWN

GIVEN	Three predicates C, D, F , and a domain restricted to students in the class.
ASKED	Express five English sentences in symbols. Nothing is to be proved or evaluated; the marks are purely for correct translation.
MARKS SAY	7.5 over five items is 1.5 each, so each answer is one line of symbols. There is no credit for prose here.
TEACHES	This section, 1.4.
TRAP	Item (v) is not three separate statements joined by "and" about one student. It says each animal has <i>some</i> owner, and the owners may be three different people. That distinction is the whole mark.

Throughout, the domain is the students in the class, so a bare $\forall x$ or $\exists x$ already means "every student" or "some student".

(i) "A student in your class has a cat, a dog, and a ferret."

"A student" here means at least one, so it is existential, and the three pets belong to the same student, so they join with "and".

$$\exists x (C(x) \wedge D(x) \wedge F(x))$$

(ii) "All students in your class have a cat, a dog, or a ferret."

Universal, and "or" is inclusive: having two of them is fine.

$$\forall x (C(x) \vee D(x) \vee F(x))$$

(iii) "Some student in your class has a cat and a ferret, but not a dog."

Existential. "But" is simply "and" with a change of tone, and "not a dog" negates D.

$$\exists x (C(x) \wedge F(x) \wedge \neg D(x))$$

(iv) "No student in your class has a cat, a dog, and a ferret."

"No" is a negated existential, so start there and then convert.

$$\neg \exists x (C(x) \wedge D(x) \wedge F(x))$$

which by the quantifier law equals $\forall x \neg(C(x) \wedge D(x) \wedge F(x))$, and by De Morgan equals $\forall x (\neg C(x) \vee \neg D(x) \vee \neg F(x))$.

All three are correct and any one earns the mark. Quoting the first and then showing the conversion demonstrates more, and costs one line.

Notice this is exactly the negation of (i), which is worth saying in your answer.

(v) "For each of the three animals, cats, dogs, and ferrets, there is a student in your class who has this animal as a pet."

Three separate existence claims, one per animal, and they need not be the same student. So it is a conjunction of three existentials, each with its own variable.

$$\exists x C(x) \wedge \exists y D(y) \wedge \exists z F(z)$$

Using three different letters is not fussiness. Writing $\exists x (C(x) \wedge D(x) \wedge F(x))$ instead would say one student owns all three, which is statement (i), a different and much stronger claim. That contrast is exactly what the examiner set (i) and (v) to test.

Now redo it: cover the answers and translate all five again. Then translate a sixth of your own: "exactly one student has a ferret", which is harder than it looks.

Deciding the truth of a quantified statement

a. Determine the truth value of each of these statements if the domain consists of all real numbers.

i. $\exists x(x^3 = -1)$ ii. $\exists x(x^4 < x^2)$ iii. $\forall x((-x)^2 = x^2)$ iv. $\forall x(2x > x)$

4 marks

BREAK IT DOWN

GIVEN	Four quantified statements, domain the real numbers.
ASKED	Determine the truth value. That means a verdict plus the reason: a witness for a true existential, a counterexample for a false universal, an argument otherwise.
MARKS SAY	1 mark each, so one line each. But a bare "true" is worth little; name the number.
TEACHES	This section, and the asymmetry table above.
TRAP	Part (iv) looks obviously true and is false. The domain is <i>all</i> reals, which includes zero and the negatives.

WORKED EXAMPLE

PQ 24/25 Q4(A)

(i) $\exists x(x^3 = -1)$. An existential, so one witness settles it. Take $x = -1$: then $x^3 = -1$. **TRUE.**

(ii) $\exists x(x^4 < x^2)$. Another existential. Try a fraction: $x = \frac{1}{2}$ gives $x^2 = 0.25$ and $x^4 = 0.0625$, and $0.0625 < 0.25$. **TRUE.**

Worth understanding rather than guessing: $x^4 < x^2$ means $x^2(x^2 - 1) < 0$, which needs $x^2 < 1$ and $x \neq 0$. So it holds exactly for the non-zero numbers strictly between -1 and 1 .

(iii) $\forall x((-x)^2 = x^2)$. A universal, so it needs an argument covering everything, not an example. $(-x)^2 = (-1)^2 \times x^2 = 1 \times x^2 = x^2$, for every real x . **TRUE.**

(iv) $\forall x(2x > x)$. A universal, so one counterexample kills it. Take $x = 0$: then $2x = 0$ and $x = 0$, and $0 > 0$ is false. **FALSE.**

Take $x = -3$ for a second counterexample: $2x = -6$, and $-6 > -3$ is false. In fact $2x > x$ only for the positive reals, and the domain here is all of them.

Now redo it: decide $\forall x(x^2 \geq x)$ over the reals, then over the integers. The answers differ, which is the point of always stating the domain.

Translating a sentence with structure

AS PRINTED

2024/2025 · Q2(C)

- c. Express the statement "If a person is female and is a parent, then this person is someone's mother" as a logical expression involving predicates, and quantifiers with a domain consisting of all people, and logical connectives. **4 marks**

BREAK IT DOWN

GIVEN	One English sentence and a domain of all people.
ASKED	Express it in symbols. You must invent the predicates yourself, which is part of the marks: define them before you use them.
MARKS SAY	4 marks for one sentence is generous, so they are paying for the setup: naming predicates, choosing quantifiers, nesting correctly.
TEACHES	This section. The nesting of two quantifiers is the new element.
TRAP	"Someone's mother" hides a second quantifier. It means there exists a person of whom she is the mother, so the answer nests $\exists$ inside $\forall$.

Step 1. Name the predicates. Over the domain of all people, let

$F(x)$ = "x is female"

$P(x)$ = "x is a parent"

$M(x, y)$ = "x is the mother of y"

M takes two arguments because motherhood is a relation between two people, not a property of one. Part Three studies such relations properly.

Step 2. Read the shape of the sentence. It says: for every person, if (female and parent) then (mother of someone).

Step 3. Write it.

$\forall x ((F(x) \wedge P(x)) \rightarrow \exists y M(x, y))$

Step 4. Check each piece. The outer quantifier is universal because the claim is about every person, and it takes an arrow, as universals do. The hypothesis is a conjunction, because both conditions must hold. The conclusion is existential, because being someone's mother only requires one such someone. The y is bound inside, so it cannot be confused with the x .

An equally correct variant pulls the existential out to the front: $\forall x \exists y ((F(x) \wedge P(x)) \rightarrow M(x, y))$. Both are accepted.

Now redo it: express "every student has an adviser" and then "some adviser advises every student". They differ only in the order of the two quantifiers, and they mean very different things.

⚠ TRAP

ORDER OF NESTED QUANTIFIERS CHANGES THE MEANING

$\forall x \exists y L(x, y)$ says: for each x there is *some* y , and it may be a different y each time. "Everyone has a mother."

$\exists y \forall x L(x, y)$ says: there is *one* y that works for every x . "Someone is the mother of everyone."

Same symbols, opposite claims, and the first is true while the second is absurd. Two quantifiers of the same kind can be swapped freely; two of different kinds never can.

RECALL AND CHECK

1. What does $\forall$ pair with, an arrow or an and? And $\exists$?
2. Write "no student passed" using $\exists$, then using $\forall$.
3. How do you disprove $\forall x P(x)$?
4. Is $\forall x \exists y L(x, y)$ the same as $\exists y \forall x L(x, y)$?

ANSWERS

1. $\forall$ pairs with $\rightarrow$, $\exists$ pairs with $\wedge$.
2. $\neg \exists x P(x)$, equivalently $\forall x \neg P(x)$, where $P(x)$ is "x passed".
3. Produce a single counterexample: one x with $P(x)$ false.
4. No. The first lets y depend on x ; the second demands one y for all x .

FILL IN THE GAP

1. $\forall$ is the _____ quantifier and is read _____.
2. $\exists$ is the _____ quantifier and is read _____.
3. $\neg\forall x P(x)$ is equivalent to _____.
4. A statement with a variable that becomes a proposition once the variable is given a value is called a _____.

ANSWERS

1. universal, "for all". 2. existential, "there exists". 3. $\exists x \neg P(x)$. 4. predicate.

PART ONE • 1.5

Rules of Inference and Valid Arguments

An argument moves from things you accept to something new. The rules below are the moves that are always safe, and the two famous ones that are not.

By the end of this section you will be able to say what makes an argument valid, name and apply the standard rules of inference, spot the two classic fallacies, and construct acceptable logic statements leading to a conclusion, which the 2025/2026 paper asks for directly.

What "valid" means

● MUST MEMORISE

VALIDITY IS ABOUT SHAPE, NOT TRUTH

An **argument** is a list of propositions called **premises**, followed by one called the **conclusion**.

An argument is **valid** when the conclusion is true in every case where all the premises are true. Equivalently, the conjunction of the premises implies the conclusion, and that implication is a tautology.

Validity says nothing about whether the premises are actually true. A valid argument from false premises can reach a false conclusion. What validity guarantees is that the *step* is safe.

◆ **Memory hook:** valid means the wiring is sound. Sound means the wiring is good *and* the input is true. An exam asks about the wiring.

TEACH

A VALID ARGUMENT WITH A FALSE CONCLUSION

Premise 1: If it is a bird, it can fly.

Premise 2: A penguin is a bird.

Conclusion: A penguin can fly.

The conclusion is false, yet the argument is **valid**: the shape is the safest one there is. The fault lies in premise 1, which is false. That is the distinction the word "valid" exists to make.

The rules of inference

● MUST MEMORISE

THE STANDARD RULES

Each rule reads: given the premises above the line, you may write the conclusion below it.

NAME	PREMISES	CONCLUSION	IN WORDS
Modus ponens	$p \rightarrow q$, and p	q	The rule fired, so the result follows.
Modus tollens	$p \rightarrow q$, and $\neg q$	$\neg p$	The result did not happen, so the cause did not.
Hypothetical syllogism	$p \rightarrow q$, and $q \rightarrow r$	$p \rightarrow r$	Chain two implications.
Disjunctive syllogism	$p \vee q$, and $\neg p$	q	One of two, and not the first, so the second.
Addition	p	$p \vee q$	If p is true, " p or anything" is true.
Simplification	$p \wedge q$	p	If both, then certainly one.
Conjunction	p , and q	$p \wedge q$	Two separate facts combine.
Resolution	$p \vee q$, and $\neg p \vee r$	$q \vee r$	The rule automated theorem provers run on.

◆ **Memory hook:** *ponens* is Latin for affirming and *tollens* for denying. Modus ponens affirms the hypothesis; modus tollens denies the conclusion. Those two carry most of the marks.

TEACH

MODUS TOLLENS IS THE CONTRAPOSITIVE AT WORK

Modus tollens says: from $p \rightarrow q$ and $\neg q$, conclude $\neg p$.

That is nothing more than section 1.2 in action. Since $p \rightarrow q$ is equivalent to its contrapositive $\neg q \rightarrow \neg p$, and you have $\neg q$, modus ponens on the contrapositive gives $\neg p$.

So the two rules are the same rule, seen from either end of an implication. Saying that in an answer shows the marker you understand the structure rather than having memorised a table.

The two fallacies

⚠ TRAP

THE TWO INVALID MOVES THAT LOOK VALID

Affirming the consequent. From $p \rightarrow q$ and q , concluding p . This is invalid, and it is the converse mistake from 1.2 wearing a different hat.

If it is raining, the ground is wet. The ground is wet. Therefore it is raining.

No: a burst pipe would do it. The wet ground is consistent with rain but does not establish it.

Denying the antecedent. From $p \rightarrow q$ and $\neg p$, concluding $\neg q$. Also invalid, and it is the inverse mistake.

If it is raining, the ground is wet. It is not raining. Therefore the ground is not wet.

No: the pipe again.

Both fallacies come from treating $p \rightarrow q$ as if it ran both ways. Only the contrapositive runs the other way, which is precisely why 1.2 matters here.

The question the newest paper asks

AS PRINTED

2025/2026 · Q4(C)

(c) Develop three good examples of acceptable logic statements leading to a conclusion.
(3 marks)

NOTE ON THE PAPER

The wording is loose. "Acceptable logic statements leading to a conclusion" means **VALID ARGUMENTS**: premises followed by a conclusion that genuinely follows. It does not mean three true sentences. Answer it as "give three valid arguments, naming the rule of inference each uses", which is what earns the marks, and say in one line that this is how you are reading it.

BREAK IT DOWN

GIVEN Nothing but the instruction. You supply the content.

ASKED **Develop** three examples, each being premises plus a conclusion that follows. Since you choose them, choose ones you can name.

MARKS SAY 3 marks, one per example. Each needs premises, a conclusion, and the name of the rule; a bare pair of sentences is half an answer.

TEACHES This section, 1.5.

TRAP Offering an argument that is not actually valid, most often by affirming the consequent. Stick to modus ponens, modus tollens and hypothetical syllogism, which are impossible to get wrong.

Reading the question as asking for three valid arguments, each named.

Example 1, modus ponens.

Premise 1: If a program compiles without errors, then it produces an executable file.

Premise 2: This program compiles without errors.

Conclusion: Therefore it produces an executable file.

Form: $p \rightarrow q$, p , therefore q . Valid, because the only row where $p \rightarrow q$ and p are both true also has q true.

Example 2, modus tollens.

Premise 1: If a set has 5 elements, then it has 32 subsets.

Premise 2: This set does not have 32 subsets.

Conclusion: Therefore this set does not have 5 elements.

Form: $p \rightarrow q$, $\neg q$, therefore $\neg p$. Valid. (Section 2.1 established the premise: a set of n elements has 2^n subsets, and $2^5 = 32$.)

Example 3, hypothetical syllogism.

Premise 1: If a student passes the practical, then the student qualifies for the project.

Premise 2: If a student qualifies for the project, then the student graduates.

Conclusion: Therefore if a student passes the practical, that student graduates.

Form: $p \rightarrow q$, $q \rightarrow r$, therefore $p \rightarrow r$. Valid, by chaining.

Finish with one line that earns the last mark: in each case the conclusion is true in every situation where the premises are both true, which is the definition of a valid argument.

Now redo it: write three more of your own, one of them using disjunctive syllogism, and check each against the rules table.

Logic in computing

(b) Discuss the place of Logic in Computer Science highlighting practical areas of relevance?

(2 marks)

NOTE ON THE PAPER

Printed with a question mark on a command, which is a habit of this examiner and carries no meaning. Two marks means two or three sentences with named areas, not an essay.

Logic is the formal language in which computer science states precise conditions and reasons about them, and it appears at every level of the subject.

Hardware. Every digital circuit is a Boolean expression made physical: AND, OR and NOT gates implement the connectives directly, so simplifying an expression means using fewer components. Part Eight is this.

Programming. Every condition in an if or while is a proposition, and the compiler applies De Morgan's laws and the implication law to optimise them.

Databases. A query is a predicate; the engine uses quantifiers and logical equivalence to rewrite it into a faster but equivalent form.

Verification. Proving that a program meets its specification, and automated theorem proving, both run on the rules of inference, with resolution as the workhorse.

Artificial intelligence. Knowledge representation and rule-based reasoning are predicate logic with an inference engine attached.

For two marks, three of these named with a one-line reason each is a full answer.

RECALL AND CHECK

1. State modus ponens and modus tollens.
2. Name the fallacy: "if p then q ; q ; therefore p ".
3. Can a valid argument have a false conclusion?
4. From " $p \vee q$ " and " $\neg p$ ", what follows and by which rule?

ANSWERS

1. Modus ponens: from $p \rightarrow q$ and p , conclude q . Modus tollens: from $p \rightarrow q$ and $\neg q$, conclude $\neg p$.
2. Affirming the consequent. It is invalid.
3. Yes, if a premise is false. Validity constrains the step, not the inputs.
4. q , by disjunctive syllogism.

FILL IN THE GAP

1. An argument is _____ when the conclusion is true in every case where all premises are true.
2. From $p \rightarrow q$ and $\neg q$ we conclude _____, by _____.
3. Concluding $\neg q$ from $p \rightarrow q$ and $\neg p$ is the fallacy of _____.
4. Chaining $p \rightarrow q$ with $q \rightarrow r$ to get $p \rightarrow r$ is called _____.

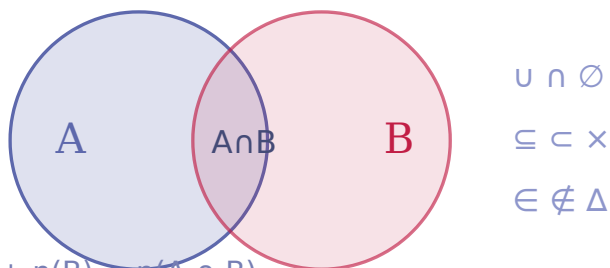
ANSWERS

1. valid. 2. $\neg p$, modus tollens. 3. denying the antecedent. 4. hypothetical syllogism.

PART TWO

Sets and Their Operations

A set is the simplest idea in this course and the one everything else is built from. Relations are sets, functions are sets, graphs are sets, and a database table is a set with a nice interface.



$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

Sets, Membership and the Nine Types

The 2025/2026 paper asks you to discuss nine types of set with an example of each. Here they are, with the vocabulary that surrounds them.

By the end of this section you will be able to define a set, state its cardinality, name and give an example of each of the nine standard types, and use the subset symbols correctly.

The definition and the vocabulary

● MUST MEMORISE

SET, ELEMENT, CARDINALITY

A **set** is a well-defined collection of distinct objects. The objects are its **elements** or **members**.

Well-defined is doing real work: for any object it must be unambiguous whether it belongs. "The set of tall students" is not a set until "tall" is pinned down. "The set of students over 1.8 m" is.

The number of elements in a set A is its **cardinality**, written $n(A)$ or $|A|$. So if $A = \{2, 4, 6\}$ then $n(A) = 3$.

◆ **Memory hook:** distinct and well-defined. Repeats do not count twice, and "maybe" is not an answer to whether something is in.

The nine types of set

● MUST MEMORISE

THE NINE TYPES, WITH EXAMPLES

TYPE	DEFINITION	EXAMPLE
Empty (null, void)	Contains no elements at all. Written $\emptyset$ or $\{ \}$.	$\{x \mid x \in \mathbb{R}, x^2 = -1\} = \emptyset$
Singleton (unit)	Contains exactly one element.	$\{5\}$, or the set of even prime numbers $\{2\}$
Finite	Its elements can be counted and the count ends.	$\{a, e, i, o, u\}$, $n = 5$
Infinite	Counting its elements never ends.	$\mathbb{N}$, the natural numbers
Equal	Two sets with exactly the same elements. Order and repeats are irrelevant.	$\{1, 2, 3\} = \{3, 2, 1, 1\}$
Equivalent	Two sets with the same cardinality, whatever their elements.	$\{a, b, c\}$ and $\{1, 2, 3\}$
Subset	Every element of A is also in B. Written $A \subseteq B$.	$\{1, 2\} \subseteq \{1, 2, 3\}$
Universal	The set containing everything under discussion. Written U.	$U = \{1, 2, \dots, 10\}$ in a question about single digits
Disjoint	Two sets with no element in common, so $A \cap B = \emptyset$.	$\{1, 3, 5\}$ and $\{2, 4, 6\}$

Two more you should be able to name if pressed, since some books count them among the nine: a **proper subset** ($A \subset B$, a subset that is not the whole of B) and the **power set** (the set of all subsets, covered in 2.3).

◆ **Memory hook:** group them in threes. By *size*: empty, singleton, finite/infinite. By *comparison*: equal, equivalent, subset. By *role*: universal, disjoint, power set.

⚠ TRAP

EQUAL VERSUS EQUIVALENT

These two words are close in English and precise in mathematics, and the examiner knows it.

Equal sets have the same *elements*: $\{1, 2, 3\} = \{3, 2, 1\}$.

Equivalent sets have the same *number* of elements: $\{a, b, c\}$ and $\{1, 2, 3\}$ are equivalent but very much not equal.

Every pair of equal sets is equivalent. The reverse fails constantly.

⚠ TRAP

$\emptyset$ IS NOT $\{\emptyset\}$

$\emptyset$ is the empty set: it has nothing in it, so $n(\emptyset) = 0$.

$\{\emptyset\}$ is a set containing one thing, and that thing happens to be the empty set. So $n(\{\emptyset\}) = 1$. It is a singleton, not an empty set.

Think of $\emptyset$ as an empty box and $\{\emptyset\}$ as a box with an empty box inside it. The second is not empty.

Also: never write $\{\emptyset\}$ meaning "the empty set", and never write " $\emptyset = \{ \}$ " with anything between the braces. Both cost marks in the power-set questions of 2.3.

Subsets

● MUST MEMORISE

$\subseteq$ AND $\subset$

$A \subseteq B$ ("A is a subset of B") means every element of A is also an element of B. A may equal B.

$A \subset B$ ("A is a proper subset of B") means $A \subseteq B$ and $A \neq B$, so B has at least one element A lacks.

Standard facts, all worth knowing on sight:

$\emptyset \subseteq A$ for every set A. The empty set is a subset of everything, because it has no element that could fail to be in A.

$A \subseteq A$ for every set A.

If $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$, so subethood carries through a chain.

A set with n elements has 2^n subsets.

◆ **Memory hook:** $\subseteq$ is to sets what $\leq$ is to numbers, and $\subset$ is what $<$ is. The extra line means "or equal".

⚠ TRAP

$\in$ AND $\subseteq$ ARE NOT INTERCHANGEABLE

Let $A = \{1, 2, 3\}$.

$1 \in A$ is true. The number 1 is an element.

$\{1\} \subseteq A$ is true. The *set* $\{1\}$ is a subset.

$1 \subseteq A$ is nonsense. The number 1 is not a set, so it cannot be a subset.

$\{1\} \in A$ is false. The elements of A are numbers, not sets, so the set $\{1\}$ is not among them.

The rule: **$\in$ relates an element to a set, $\subseteq$ relates a set to a set.** Check which side of the symbol each object belongs on before you write it.

Listing the members of a described set

AS PRINTED

2021/2022 · Q1(A)

1. (a) List the members of the following sets. (5marks)
- (i) $\{x \mid x \text{ is a real number such that } x^2 = 1\}$
 - (ii) $\{x \mid x \text{ is a positive prime number less than } 20\}$
 - (iii) $\{x \mid x \text{ is a perfect square and } x < 100\}$
 - (iv) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$
 - (v) $\{x \mid x \text{ is an integer such that } 2x = x\}$

NOTE ON THE PAPER

The marks are printed *before* the five items, on the instruction line, which is why the quote above carries them there.

ITEM (III) IS GENUINELY AMBIGUOUS AND YOU SHOULD SAY SO. It reads " $\{x \mid x \text{ is a perfect square and } x < 100\}$ " without saying that x is a positive integer. Read strictly, 0 qualifies, and so do non-integer squares such as 0.25, which would make the set infinite. The intended answer is the perfect squares of whole numbers below 100. State the reading you use and give the set; a stated assumption protects the mark, a silent one does not.

BREAK IT DOWN

- GIVEN** Five sets in set-builder notation.
- ASKED** **List the members**, that is, convert each from set-builder to roster form.
- MARKS SAY** 5 marks, one per item. Each is one line. Do not explain; just list, with braces.
- TEACHES** This section, 2.1, and section F.1 for the notation.
- TRAP** Two items have **fewer** members than they look. Item (iv) is empty, and item (v) has exactly one member. Writing "no solution" instead of the empty set loses the mark: the answer to "list the members" is always a set.

(i) $\{x \mid x \text{ is a real number such that } x^2 = 1\}$

$x^2 = 1$ gives $x = 1$ or $x = -1$, and both are real.

$\{1, -1\}$

(ii) $\{x \mid x \text{ is a positive prime number less than } 20\}$

A prime has exactly two positive divisors, itself and 1, so 1 is not prime.

$\{2, 3, 5, 7, 11, 13, 17, 19\}$, eight members.

(iii) $\{x \mid x \text{ is a perfect square and } x < 100\}$

Reading used: x is a whole number that is the square of a whole number. Then $0^2 = 0$ up to $9^2 = 81$ qualify, and $10^2 = 100$ does not, because the condition is strictly less than 100.

$\{0, 1, 4, 9, 16, 25, 36, 49, 64, 81\}$, ten members.

If your reading excludes 0, say so and give the same set without it, which is nine members. Either is defensible; leaving the reading unstated is not.

(iv) $\{x \mid x \text{ is an integer such that } x^2 = 2\}$

$x^2 = 2$ gives $x = \sqrt{2}$ or $x = -\sqrt{2}$, and neither is an integer. No integer qualifies, so the set is empty.

$\emptyset$, also written $\{ \}$.

Do not write $\{ \emptyset \}$. That is the set *containing* the empty set, which has one member and is not empty at all.

(v) $\{x \mid x \text{ is an integer such that } 2x = x\}$

$2x = x$ gives $x = 0$, which is an integer, so exactly one value qualifies.

$\{0\}$, a singleton set.

Note the contrast between (iv) and (v), which is almost certainly why the examiner set them next to each other: one is the empty set with no members, the other is a singleton whose single member happens to be 0. $\emptyset$ and $\{0\}$ are different sets, and confusing them is the standard slip here.

Now redo it: list $\{x \mid x \text{ is an integer such that } x^2 < 5\}$. The answer is $\{-2, -1, 0, 1, 2\}$, five members.

The real question

b) i) Discuss nine (9) types of sets, giving an example of each.

(3 marks)

ii) List two mathematical ways to describe a set.

(2 marks)

BREAK IT DOWN

GIVEN	Nothing. Both parts are pure recall.
ASKED	Part i: discuss nine types giving an example of each . The example is compulsory and carries marks. Part ii: list two ways to describe a set.
MARKS SAY	3 marks for nine types is a third of a mark each, so a table with one line and one example per row is exactly right. Do not write paragraphs. Part ii is 2 marks for two named methods, so name them <i>and</i> show one line of each.
TEACHES	This section for the types, F.1 for the two descriptions.
TRAP	Giving nine types but forgetting the examples throws away half the marks, because "giving an example of each" is part of the instruction.

WORKED EXAMPLE

PQ 25/26 Q1(B)

(i) Nine types of set. Answer as a table; it is faster to write and easier to mark.

TYPE	MEANING	EXAMPLE
Empty set	has no elements	$\emptyset = \{ \}$
Singleton set	has exactly one element	$\{7\}$
Finite set	has a countable, ending number of elements	$\{a, e, i, o, u\}$
Infinite set	counting never ends	$\mathbb{N} = \{0, 1, 2, \dots\}$
Equal sets	same elements	$\{1, 2, 3\}$ and $\{3, 1, 2\}$
Equivalent sets	same number of elements	$\{a, b\}$ and $\{9, 4\}$
Subset	every element of one lies in the other	$\{1, 2\} \subseteq \{1, 2, 3\}$
Universal set	contains everything under discussion	$U = \{1, 2, \dots, 10\}$
Disjoint sets	no element in common	$\{1, 3\}$ and $\{2, 4\}$

(ii) Two mathematical ways to describe a set.

1. The roster (listing or tabular) method: write the elements between braces.

Example: $A = \{2, 4, 6, 8\}$.

2. Set-builder (rule) notation: state the property membership requires. Example: $A = \{x \mid x \text{ is even and } 0 < x < 10\}$, read "the set of all x such that x is even and x lies between 0 and 10".

Now redo it: cover the table and write all nine types with fresh examples of your own. Three minutes.

RECALL AND CHECK

1. What is $n(\{\emptyset\})$?
2. Give two sets that are equivalent but not equal.
3. How many subsets does $\{a, b, c, d\}$ have?
4. True or false: $\{2\} \in \{1, 2, 3\}$.

ANSWERS

1. It is a singleton whose only element is the empty set.
2. $\{a, b, c\}$ and $\{1, 2, 3\}$, for instance. Same size, different elements.
3. $2^4 = 16$.
4. False. $\{2\}$ is a subset, not an element: $\{2\} \subseteq \{1, 2, 3\}$ is the true statement.

FILL IN THE GAP

1. The number of elements in a set is its _____.
2. Two sets with no element in common are _____.
3. A set with n elements has _____ subsets.
4. Sets with the same elements are _____; sets with the same number of elements are _____.

ANSWERS

1. cardinality. 2. disjoint. 3. 2^n . 4. equal, equivalent.

PART TWO · 2.2

Set Operations and Venn Diagrams

Five operations, each with a picture. Draw the picture and the answer is usually obvious; skip the picture and the same question becomes a guessing game.

By the end of this section you will be able to compute union, intersection, difference, symmetric difference and complement, draw and read a Venn diagram, and state De Morgan's laws for sets.

The five operations

● MUST MEMORISE

THE FIVE OPERATIONS

OPERATION	SYMBOL	DEFINITION	IN WORDS
Union	$A \cup B$	$\{x \mid x \in A \text{ or } x \in B\}$	everything in either, counted once
Intersection	$A \cap B$	$\{x \mid x \in A \text{ and } x \in B\}$	only what is in both
Difference	$A - B$	$\{x \mid x \in A \text{ and } x \notin B\}$	in A but not in B
Symmetric difference	$A \triangle B$	$(A - B) \cup (B - A)$	in one or the other, not in both
Complement	A' or $\bar{A}$	$\{x \mid x \in U \text{ and } x \notin A\}$	everything outside A

◆ **Memory hook:** $\cup$ is a **u**p-cup that holds everything, so it is **u**nion. $\cap$ points down and only what is caught in both bowls stays. The complement needs a universal set to be meaningful at all, so U must always be stated.

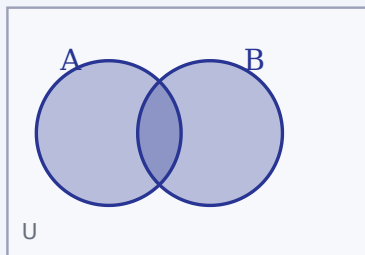
The pictures

TEACH

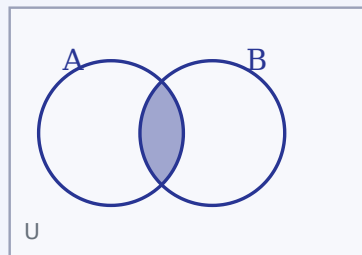
VENN DIAGRAMS

A **Venn diagram** draws each set as a circle inside a rectangle, where the rectangle is the universal set U . Overlapping circles create regions, and every element of U lives in exactly one region.

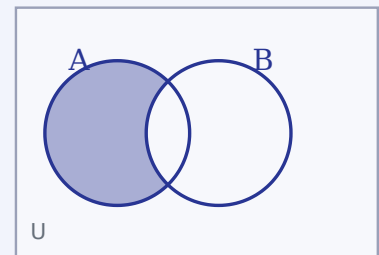
Shade the region the operation describes, and the answer becomes something you can read off rather than compute.



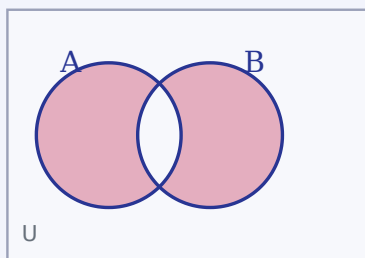
$A \cup B$ union



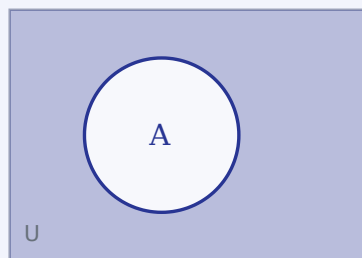
$A \cap B$ intersection



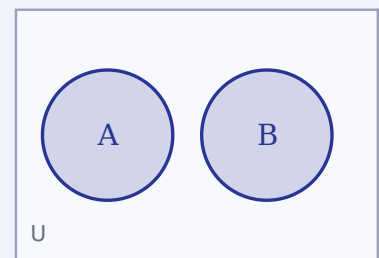
$A - B$ difference



$A \triangle B$ symmetric difference



A' complement



disjoint: $A \cap B = \emptyset$

WORKED EXAMPLE

METHOD

GIVEN $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$.

FIND All five operations.

$A \cup B$ = everything in either = $\{1, 2, 3, 4, 5, 6\}$. Note 3 and 4 appear once, not twice.

$A \cap B$ = what is in both = $\{3, 4\}$.

$A - B$ = in A, not in B = $\{1, 2\}$.

$B - A$ = in B, not in A = $\{5, 6\}$. Note this differs from $A - B$, so the order of the two sets matters.

$A \triangle B = (A - B) \cup (B - A) = \{1, 2\} \cup \{5, 6\} = \{1, 2, 5, 6\}$. It is the union minus the intersection.

A' = everything in U outside A = $\{5, 6, 7, 8\}$.

Check. $n(A) = 4$, $n(B) = 4$, $n(A \cap B) = 2$, and $n(A \cup B) = 4 + 4 - 2 = 6$, which matches the six elements listed. Always run that check; it catches a dropped element instantly.

The real question

AS PRINTED

2025/2026 · Q1(C)

(c) $A = \{11, 12, 13, 14\}$

$B = \{13, 14, 15, 16\}$

From the two sets above, calculate:

- | | | |
|------|-------------------------|----------|
| i) | Intersection set | (½ mark) |
| ii) | Difference ($A - B$) | (1 mark) |
| iii) | Difference ($B - A$). | (1 mark) |

BREAK IT DOWN

GIVEN $A = \{11, 12, 13, 14\}$ and $B = \{13, 14, 15, 16\}$, which overlap in two elements.

ASKED **Calculate** three sets. Answers are sets, so write them in braces.

MARKS SAY 2.5 marks total and one line each. This is a gift question: take it, check it, move on.

TEACHES This section, 2.2.

TRAP Parts ii and iii are asked separately *because* they differ. Writing the same answer twice, or writing the symmetric difference for both, throws away two of the 2.5 marks.

GIVEN $A = \{11, 12, 13, 14\}$, $B = \{13, 14, 15, 16\}$.

FIND $A \cap B$, $A - B$, $B - A$.

RULE Intersection keeps what is in both; $A - B$ keeps what is in A and not in B.

(i) Intersection. Go through A one element at a time and ask whether it is in B. 11: no. 12: no. 13: yes. 14: yes.

$$A \cap B = \{13, 14\}$$

(ii) $A - B$. Elements of A that are *not* in B, which is the two that just failed the test.

$$A - B = \{11, 12\}$$

(iii) $B - A$. Now start from B. 13: in A, drop it. 14: in A, drop it. 15: not in A, keep. 16: not in A, keep.

$$B - A = \{15, 16\}$$

Check. The three answers should partition the union: $\{11, 12\} \cup \{13, 14\} \cup \{15, 16\} = \{11, 12, 13, 14, 15, 16\}$, which is $A \cup B$, with 6 elements. And $4 + 4 - 2 = 6$. Consistent.

Now redo it: add $A \triangle B$ and A' taking $U = \{10, 11, \dots, 18\}$. Answers: $\{11, 12, 15, 16\}$ and $\{10, 15, 16, 17, 18\}$.

De Morgan's laws for sets

DE MORGAN'S LAWS (SET FORM)

The claim. The complement of an intersection is the union of the complements, and the complement of a union is the intersection of the complements. Complementing flips the operation.

In symbols.

$$(A \cap B)' = A' \cup B'$$

$$(A \cup B)' = A' \cap B'$$

Conditions. A and B must be subsets of a stated universal set U, since complement is meaningless without one.

The limiting case. Take $B = U$. Then B' is the empty set, and the first law reads $(A \cap U)' = A' \cup \emptyset$, which is $A' = A'$: true, and empty of content. The law only says something when both sets are proper parts of U, and it says most when they overlap. The same collapse happens at $A = B$, where it reduces to $A' = A'$ again. A worked answer that lands on one of these has almost certainly lost a term somewhere.

Why it is the same law as in logic. Membership in an intersection is a conjunction ("x is in A and x is in B") and membership in a union is a disjunction. Complement is negation. So the set law is the logic law of section 1.3, read through the definition of membership. That is exactly the proof the 2024/2025 paper asks for.

c. Use set builder notation and logical equivalences to establish the first De Morgan law $A \cap \bar{B} = \bar{A} \cup \bar{B}$

3 marks

NOTE ON THE PAPER

Two typesetting faults in one line. The overbar sits above **B** alone when the law needs it above the whole of $A \cap B$, and the relation symbol images as a short dash where an equals sign belongs. The law being asked for is unambiguous from its name: the complement of an intersection equals the union of the complements. Answer that, and note the misprint in one line so the marker knows you read it rather than copied it.

BREAK IT DOWN

- GIVEN** The name of the law and a garbled rendering of it.
- ASKED** **Establish** it, using **set builder notation and logical equivalences**. The method is prescribed: translate membership into logic, apply the logic law, translate back. A Venn diagram alone will not answer this question.
- MARKS SAY** 3 marks for about five lines of derivation, one mark per genuine step.
- TEACHES** This section, plus 1.3 for the logical form.
- TRAP** Drawing two Venn diagrams and asserting they match. That illustrates the law, it does not establish it, and the question named the method it wants.

WORKED EXAMPLE

PQ 24/25 Q3(C)

To prove: $(A \cap B)' = A' \cup B'$.

Two sets are equal when they have exactly the same elements, so take an arbitrary element x and show that belonging to the left is equivalent to belonging to the right, step by step.

LINE	STATEMENT	REASON
1	$(A \cap B)' = \{x \mid x \notin (A \cap B)\}$	definition of complement
2	$= \{x \mid \neg(x \in A \cap B)\}$	meaning of $\notin$
3	$= \{x \mid \neg(x \in A \wedge x \in B)\}$	definition of intersection
4	$= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$	De Morgan's law for logic
5	$= \{x \mid x \notin A \vee x \notin B\}$	meaning of $\notin$
6	$= \{x \mid x \in A' \vee x \in B'\}$	definition of complement
7	$= A' \cup B'$	definition of union

Line 4 is the only real step; everything else is unpacking a definition. That is worth saying out loud in your answer, because it shows you know where the content is.

The other law, for completeness: the same seven lines with $\cap$ and $\cup$ exchanged, and $\wedge$ and $\vee$ exchanged, give $(A \cup B)' = A' \cap B'$.

Now redo it: write the seven lines for the second law from memory, naming the reason on each.

For any two finite sets,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$n(A)$ the number of elements in A . $n(A \cup B)$ the number in the union. $n(A \cap B)$ the number in both, subtracted because adding $n(A)$ and $n(B)$ counts those elements twice.

This is the two-set case of the inclusion and exclusion principle, which Part Five develops for three sets and more. It is worth 7 marks on the 2025/2026 paper by itself.

◆ **Memory hook:** add both, then pay back the overlap you counted twice.

RECALL AND CHECK

1. With $A = \{1,2,3\}$, $B = \{3,4\}$, find $A \triangle B$.
2. Is $A - B$ the same as $B - A$?
3. State $(A \cup B)'$ in terms of A' and B' .
4. $n(A) = 20$, $n(B) = 15$, $n(A \cap B) = 6$. Find $n(A \cup B)$.

ANSWERS

1. $\{1, 2, 4\}$. The 3 is in both, so it is excluded.
2. No. $A - B = \{1,2\}$ but $B - A = \{4\}$.
3. $A' \cap B'$.
4. $20 + 15 - 6 = 29$.

FILL IN THE GAP

1. $A \cap B$ is the set of elements in _____ A and B .
2. $A \triangle B$ equals $(A - B) \cup (B - A)$.
3. $(A \cap B)' = \text{_____}$.
4. $n(A \cup B) = n(A) + n(B) - \text{_____}$.

ANSWERS

1. both. 2. $\cup$ (union). 3. $A' \cup B'$. 4. $- n(A \cap B)$.

PART TWO · 2.3

Bit Strings, Power Sets and Cartesian Products

Three ways of building something new out of a set. Each one is asked about directly, and the bit-string representation is how a computer actually stores a set.

By the end of this section you will be able to write any subset of a universal set as a bit string and compute unions, intersections and complements by bitwise operations, list a power set without missing a subset, and form a Cartesian product and count its size.

Bit strings

NOT IN THE COURSE TEXT

The official 161-page text never mentions bit strings, yet the 2023/2024 and 2024/2025 papers together spend more than 15 marks on them. Everything you need is here.

Fix a universal set U and write its elements in a definite order, usually increasing. A subset A of U can then be described by a string of 0s and 1s with one position per element of U :

- put **1** in position i if the i -th element of U is in A ,
- put **0** in position i if it is not.

That string is the **bit string** (or characteristic vector) of A . With $U = \{1, 2, \dots, 10\}$ there are ten positions, so every subset is ten bits, and a computer can hold it in a single machine word.

This is not a classroom curiosity. It is how set types are implemented, and it is why testing membership is one machine instruction rather than a search.

● MUST MEMORISE

SETS AS BIT STRINGS

With U ordered, the subset A has bit string $b_1b_2 \dots b_n$ where $b_i = 1$ exactly when the i -th element of U is in A .

WHERE

U is the universal set, **listed in a fixed order** that is agreed before anything is encoded. Change the order and every bit string changes. n is the number of elements in U , so every bit string for that U is exactly n bits long. b_i is the bit in position i , reading left to right from $i = 1$.

The operations then become **bitwise**:

SET OPERATION	BIT OPERATION	RULE PER POSITION
$A \cup B$	bitwise OR	1 if either bit is 1
$A \cap B$	bitwise AND	1 only if both bits are 1
A'	bitwise NOT	flip every bit
$A \triangle B$	bitwise XOR	1 if the bits differ

◆ **Memory hook:** union is or, intersection is and, complement is not. The set operations and the logic connectives are the same operations wearing different clothes, which is exactly what section 1.3 and 2.2 kept saying.

WORKED EXAMPLE

METHOD

GIVEN $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ in increasing order, $A = \{1, 3, 5, 7, 9\}$.

FIND The bit string of A.

RULE 1 where the position's element is in A, 0 otherwise.

Walk the positions in order. Position 1 holds the element 1, which is in A, so 1. Position 2 holds 2, not in A, so 0. Continuing:

POSITION	1	2	3	4	5	6	7	8	9	10
ELEMENT	1	2	3	4	5	6	7	8	9	10
IN A?	yes	no	yes	no	yes	no	yes	no	yes	no
BIT	1	0	1	0	1	0	1	0	1	0

So the bit string is **1010101010**, which the papers group in fours from the left as **10 1010 1010**. The grouping is only for legibility, like spaces in a phone number.

AS PRINTED

2023/2024 · Q3(A)

3. (a) Suppose that a universal set, $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express the following sets with bit strings where the i^{th} bit in the string is 1 if i is in the set and 0 otherwise.

(i) $\{3, 4, 5\}$ **(1 marks)**

(ii) $\{1, 3, 6, 10\}$ **(1 marks)**

(iii) $\{2, 3, 4, 7, 8, 9\}$ **(1 marks)**

NOTE ON THE PAPER

"(1 marks)" is printed three times: singular value, plural noun. Harmless. The question helpfully defines the convention itself, so use exactly the ordering it gives, with position i holding the element i .

BREAK IT DOWN

GIVEN $U = \{1, \dots, 10\}$, and the convention that bit i is 1 when i is in the set.

ASKED **Express** three sets as bit strings. Ten bits each, no working needed.

MARKS SAY 1 mark each, so speed matters more than presentation. Write the ten bits and move on.

TEACHES This section, 2.3.

TRAP Producing nine or eleven bits. Count them before moving on; the length must equal $n(U)$ every time.

WORKED EXAMPLE

PQ 23/24 Q3(A)

(i) {3, 4, 5}. Ones in positions 3, 4, 5, zeros elsewhere.

0011100000 (grouped: 00 1110 0000)

(ii) {1, 3, 6, 10}. Ones in positions 1, 3, 6, 10.

1010010001 (grouped: 10 1001 0001)

(iii) {2, 3, 4, 7, 8, 9}. Ones in positions 2, 3, 4, 7, 8, 9.

0111001110 (grouped: 01 1100 1110)

Check every time: count the 1s and compare with the size of the set. Three, four and six ones respectively, matching $n = 3, 4, 6$. Then count the total length: ten in each. Two checks, three seconds, and they catch every slip this question can produce.

AS PRINTED

2024/2025 · Q3(D)

d. The bit string for the set {1, 3, 5, 7, 9} with universal set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10} is 10 1010 1010.

i. What is the bit string for the complement of this set?

2.5 marks

NOTE ON THE PAPER

The universal set closes with two round brackets instead of a brace, and there is no space before the 10. Both are typing slips and change nothing: U is {1, 2, ..., 10}.

WORKED EXAMPLE

PQ 24/25 Q3(D)(I)

GIVEN The bit string 1010101010 for {1, 3, 5, 7, 9} over $U = \{1, \dots, 10\}$.

FIND The bit string of the complement.

RULE Complement is bitwise NOT: flip every bit.

Flipping 1010101010 gives **0101010101**, grouped as **01 0101 0101**.

Reading it back as a set, the 1s now sit in positions 2, 4, 6, 8, 10, so the complement is {2, 4, 6, 8, 10}, the even numbers up to 10. That is exactly right, since the original held the odd ones.

Always convert back and sanity-check the meaning. It takes five seconds and turns a mechanical answer into one you know is correct.

AS PRINTED

2024/2025 · Q3(D)(II)

ii. The bit strings for the sets {1, 2, 3, 4, 5} and {1, 3, 5, 7, 9} are 11 1110 0000 and 10 1010 1010, respectively. Use bit strings to find the union and intersection of these sets.

2.5 marks

BREAK IT DOWN

GIVEN	Two ten-bit strings, already computed for you.
ASKED	Find the union and intersection, and the method is prescribed: use bit strings . So do the bitwise work and show it.
MARKS SAY	2.5 marks for two operations. Show the two strings lined up and the result underneath; that layout is the working.
TEACHES	This section, 2.3.
TRAP	Converting to sets, doing the union there, and converting back. That gets the right answer by the wrong method and risks losing the method marks, since the question named the method.

WORKED EXAMPLE

PQ 24/25 Q3(D)(II)

Line the strings up in columns and apply the rule position by position.

POSITION	1	2	3	4	5	6	7	8	9	10
{1,2,3,4,5}	1	1	1	1	1	0	0	0	0	0
{1,3,5,7,9}	1	0	1	0	1	0	1	0	1	0
OR (UNION)	1	1	1	1	1	0	1	0	1	0
AND (MEET)	1	0	1	0	1	0	0	0	0	0

Union = 1111101010, grouped **11 1110 1010**. Reading the 1s back: positions 1, 2, 3, 4, 5, 7, 9, so the union is **{1, 2, 3, 4, 5, 7, 9}**.

Intersection = 1010100000, grouped **10 1010 0000**. The 1s sit at 1, 3, 5, so the intersection is **{1, 3, 5}**.

Check. $n(\text{union}) = 7$, $n(\text{first}) = 5$, $n(\text{second}) = 5$, $n(\text{intersection}) = 3$, and $5 + 5 - 3 = 7$. The counting formula from 2.2 agrees.

Now redo it: compute the symmetric difference of the same two sets by XOR. You should get 01 0100 1010, that is {2, 4, 7, 9}.

The power set

● MUST MEMORISE

POWER SET

The **power set** of A, written **P(A)**, is the set of *all* subsets of A, including the empty set and A itself.

If $n(A) = n$ then **$n(P(A)) = 2^n$** .

n the number of elements in A. **2^n** the number of subsets, because each element is independently either in or out, giving two choices repeated n times.

◆ **Memory hook:** the power set is one binary decision per element, in or out. That is why its size is 2^n , and why bit strings and power sets are the same idea: **every subset is a bit string**, so the number of subsets is the number of n-bit strings.

GIVEN $A = \{1, 2, 3\}$.

FIND $P(A)$.

RULE $2^3 = 8$ subsets. List by size so none is missed.

Listing by size is the safe method, because you can count each row against the binomial coefficients of Part Five.

SIZE	SUBSETS	HOW MANY
0	$\emptyset$	1
1	$\{1\}, \{2\}, \{3\}$	3
2	$\{1,2\}, \{1,3\}, \{2,3\}$	3
3	$\{1,2,3\}$	1

Total $1 + 3 + 3 + 1 = 8 = 2^3$. Correct.

Listing by bit pattern is the method a program uses, and it explains the order the 2024/2025 paper prints. Count from 000 to 111 in binary, reading each pattern as "which elements are in":

COUNT	BITS (FOR 3,2,1)	SUBSET
0	000	$\{ \}$
1	001	$\{1\}$
2	010	$\{2\}$
3	011	$\{1, 2\}$
4	100	$\{3\}$
5	101	$\{1, 3\}$
6	110	$\{2, 3\}$
7	111	$\{1, 2, 3\}$

That is exactly the order printed in 2024/2025 Q3(b), and knowing why makes the program in Part Eleven write itself.

⚠ TRAP

DO NOT FORGET THE TWO FREE SUBSETS

Every power set contains $\emptyset$ and the whole set A . They are the two most commonly forgotten, and forgetting either makes the count wrong, which is the first thing a marker checks.

Count first, then list. If you expect 2^n and you have written $2^n - 1$, you know exactly what is missing before you hand it in.

The Cartesian product

● MUST MEMORISE

CARTESIAN PRODUCT

The **Cartesian product** of A and B, written $A \times B$, is the set of all **ordered pairs** whose first entry comes from A and second from B:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

$$n(A \times B) = n(A) \times n(B).$$

(a, b) an ordered pair: round brackets, and the order matters, so $(1, 2) \neq (2, 1)$. $n(A) \times n(B)$ the count, by the product rule of Part Five: $n(A)$ choices for the first entry, and for each of them $n(B)$ choices for the second.

◆ **Memory hook:** braces for a set, where order does not matter; round brackets for a pair, where it does. $A \times B$ is every way of picking one from each, so it is a grid, and a grid's area is the product of its sides.

AS PRINTED

2025/2026 · Q2(B)

(b) A university has the set of students:

$S = \{\text{John, Mary, Peter}\}$ and the set of courses:

$C = \{\text{CSC701, CSC702}\}$

Find the Cartesian product $S \times C$

(2 marks)

BREAK IT DOWN

GIVEN S with 3 students, C with 2 courses.

ASKED **Find** $S \times C$. Note the order: students first, courses second, because it says $S \times C$ and not $C \times S$.

MARKS SAY 2 marks for six pairs. List them all; a formula with no list does not answer "find".

TEACHES This section, 2.3.

TRAP Writing the pairs with braces instead of round brackets, or writing them in the order (course, student). Both are marked wrong because the pair is ordered and the question fixed the order.

GIVEN $S = \{\text{John, Mary, Peter}\}$, $C = \{\text{CSC701, CSC702}\}$.

FIND $S \times C$.

RULE Every ordered pair with the student first. Expect $n(S) \times n(C) = 3 \times 2 = 6$ pairs.

Take each student in turn and pair with every course. Working systematically is what stops a pair going missing.

$S \times C = \{(\text{John, CSC701}), (\text{John, CSC702}), (\text{Mary, CSC701}), (\text{Mary, CSC702}), (\text{Peter, CSC701}), (\text{Peter, CSC702})\}$

Six pairs, matching 3×2 . It is a grid: three rows of students, two columns of courses.

What it means here: $S \times C$ is every possible student and course combination, that is, every enrolment that *could* exist. The enrolments that actually exist form a subset of it, and a subset of a Cartesian product is precisely what Part Three calls a **relation**. That single sentence is the bridge from sets to relations, and it is worth writing in your answer.

Now redo it: write $C \times S$ and confirm it is a different set with the same six students and courses. That difference is why the product is not commutative.

a. Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same.

3.5 marks

BREAK IT DOWN

GIVEN Two ways of bracketing a product of four sets.

ASKED **Explain why** they differ. This is an argument about the *shape* of the elements, not a calculation.

MARKS SAY 3.5 marks for a short explanation, so they want the structure spelled out, ideally with a small concrete example.

TEACHES This section, 2.3.

TRAP Answering "because the Cartesian product is not associative" and stopping. That names the fact without explaining it. The marks are in showing what a typical element of each side looks like.

The point: an ordered pair has exactly two slots, and what sits in each slot is decided by the brackets.

Left side, $(A \times B) \times (C \times D)$. This is a product of two things: the set $A \times B$, and the set $C \times D$. So a typical element is an ordered pair whose first entry is itself a pair from $A \times B$ and whose second entry is a pair from $C \times D$:

$((a, b), (c, d))$

Two slots at the top level, each holding a pair.

Right side, $A \times (B \times C) \times D$. This is a product of three things: A , then $B \times C$, then D . A typical element is an ordered *triple*:

$(a, (b, c), d)$

Three slots at the top level, with the middle one holding a pair.

Why that settles it. An element of the left is a pair; an element of the right is a triple. A pair is never equal to a triple, whatever is inside them, so the two sets share no elements at all and cannot be equal. The Cartesian product is **not associative**.

A concrete example seals the marks. Take $A = B = C = D = \{1\}$. Then

$$(A \times B) \times (C \times D) = \{((1,1), (1,1))\}$$

$$A \times (B \times C) \times D = \{(1, (1,1), 1)\}$$

Both sets have exactly one element, so they are *equivalent*, but those elements are differently shaped, so the sets are not *equal*. Section 2.1's distinction between equal and equivalent is doing real work here, and saying so is worth a mark.

Worth adding: because the counts multiply the same way either side, n of both is $n(A)n(B)n(C)n(D)$. Equal size, different structure. That is why people expect them to be the same and why they are not.

RECALL AND CHECK

1. Write the bit string of $\{2, 5\}$ over $U = \{1, 2, 3, 4, 5\}$.
2. How many elements are in $P(A)$ when $n(A) = 6$?
3. $n(A) = 4$ and $n(B) = 7$. Find $n(A \times B)$.
4. Which bitwise operation gives the intersection?

ANSWERS

1. 01001.
2. $2^6 = 64$.
3. $4 \times 7 = 28$.
4. AND.

FILL IN THE GAP

1. In a bit string, position i holds a 1 when the i -th element of _____ is in the set.
2. The power set of A is the set of all _____ of A , and it has _____ elements.
3. $A \times B$ is the set of all _____ (a, b) .
4. Union corresponds to the bitwise _____ operation.

ANSWERS

1. the universal set U . 2. subsets, 2^n . 3. ordered pairs. 4. OR.

PART THREE

Relations, Matrices and Ordered Sets

A relation is any set of ordered pairs. From that one idea come relational databases, dependency graphs, equivalence of states, and the ordering that makes a scheduler work.

$$R = \{(1,1), (1,2), (2,1), (2,2), (3,4)\}$$

1	1	0	0
1	1	0	0
0	0	0	1
0	0	0	0

reflexive

symmetric

transitive



antisymmetric



equivalence



partial order

Relations and Their Representations

A relation records which things are connected to which. There are four ways to write one down, and the exam uses all four.

By the end of this section you will be able to define a relation on a set, list the pairs of a relation given by a rule, and move freely between the set of pairs, the matrix, the arrow diagram and the directed graph.

NOT IN THE COURSE TEXT

The official 161-page text never defines a relation as a set of ordered pairs, never mentions reflexive, symmetric, antisymmetric or transitive, and never mentions partial or total orders. Every paper we hold examines them: 2024/2025 gives 7.5 marks to it, 2023/2024 gives 10, 2025/2026 gives 6. Everything you need is taught here from zero.

What a relation is

TEACH

BUILT STRAIGHT OUT OF PART TWO

Section 2.3 ended with the Cartesian product $A \times B$, the set of *every* ordered pair with a first entry from A and a second from B . That is the set of all connections that *could* exist.

A **relation** is simply a choice of which of them actually do. It is any subset of $A \times B$.

If $A = \{\text{John, Mary}\}$ and $B = \{\text{CSC701, CSC702}\}$, the product has four pairs. The relation "is enrolled in" might be $\{(\text{John, CSC701}), (\text{Mary, CSC702})\}$: a subset, recording the two enrolments that exist.

That is the whole definition. Everything in this part is a consequence.

● MUST MEMORISE

DEFINITION

A **relation** from A to B is a subset of $A \times B$.

A **relation on A** is a subset of $A \times A$, that is, a relation from a set to itself. Almost every exam question is of this kind.

We write **$a R b$** to mean $(a, b) \in R$, read "a is related to b".

A relation on a set of n elements is a subset of $A \times A$, which has n^2 pairs, so there are 2^{n^2} different relations on it.

◆ **Memory hook:** the Cartesian product lists every possible arrow; a relation is the set of arrows you actually drew.

⚠ TRAP

THE ORDER IN THE PAIR IS THE DIRECTION OF THE ARROW

$(a, b) \in R$ means a is related to b , **not** b to a . In the relation "is the parent of", (Ada, Ben) says Ada is Ben's parent, and (Ben, Ada) would say the reverse.

Every property in section 3.2 is a statement about which direction the pairs run, so getting the convention backwards inverts every answer. Fix it now: **first entry acts on second entry**.

The four representations

TEACH

ONE RELATION, FOUR PICTURES

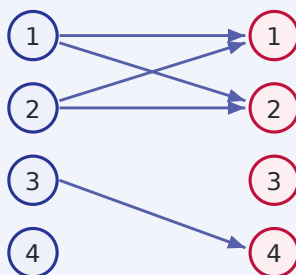
Take $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,2), (2,1), (2,2), (3,4)\}$. Here is the same relation written four ways. Learn to convert between them instantly, because different exam questions hand you different forms.

1. As a set of ordered pairs

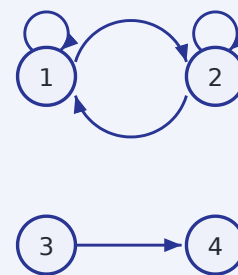
$$R = \{(1,1), (1,2), (2,1), (2,2), (3,4)\}$$

2. As a matrix (row = first entry, column = second)

	1	2	3	4
1	1	1	0	0
2	1	1	0	0
3	0	0	0	1
4	0	0	0	0



4. As a directed graph



loops at 1 and 2 = the pair

● MUST MEMORISE

THE RELATION MATRIX

For a relation R on $A = \{a_1, \dots, a_n\}$, the **relation matrix** (also called the logical matrix) M has

$$M_{ij} = 1 \text{ if } (a_i, a_j) \in R, \text{ and } 0 \text{ otherwise.}$$

i the row, indexing the **first** entry of the pair. **j** the column, indexing the **second**. M_{ij} the entry in row i , column j .

Every property of a relation becomes a visible pattern in this matrix, which is what makes 3.2 easy rather than fiddly.

◆ **Memory hook: row acts on column.** Row i , column j means "does a_i relate to a_j ". Same order as the pair.

Listing a relation given by a rule

AS PRINTED

2023/2024 · Q3(D)(I)

(i) Which ordered pairs are in the relation $R = \{(a, b) \mid a \text{ divides } b\}$?

(2 marks)

NOTE ON THE PAPER

The set is given earlier in the same question as $A = \{1, 2, 3, 4\}$, so the relation is on that set. "a divides b" means b is a whole multiple of a, so 2 divides 4 but 4 does not divide 2, so the pairs run one way only. That is the point of choosing divisibility, as section 3.2 will show.

BREAK IT DOWN**GIVEN** $A = \{1, 2, 3, 4\}$ and the rule "a divides b".**ASKED** Which ordered pairs are in the relation. So list them, in braces, as a set of pairs.**MARKS SAY** 2 marks for a list. Work through it systematically rather than by inspiration.**TEACHES** This section, 3.1.**TRAP** Forgetting that every number divides itself, so (1,1), (2,2), (3,3), (4,4) are all in. And forgetting that 1 divides everything.**WORKED EXAMPLE****PQ 23/24 Q3(D)(I)****GIVEN** $A = \{1, 2, 3, 4\}$, $R = \{(a, b) \mid a \text{ divides } b\}$.**FIND** Every pair in R.**RULE** a divides b when $b \div a$ leaves no remainder. Test all $4 \times 4 = 16$ pairs.

Take each a in turn and ask which b it divides.

A	DIVIDES WHICH OF 1, 2, 3, 4?	PAIRS
1	all of them, since 1 divides everything	(1,1), (1,2), (1,3), (1,4)
2	2 and 4	(2,2), (2,4)
3	3 only	(3,3)
4	4 only	(4,4)

 $R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

Eight pairs. As a matrix, with rows and columns in the order 1, 2, 3, 4:

	1	2	3	4
1	1	1	1	1
2	0	1	0	1
3	0	0	1	0
4	0	0	0	1

Two patterns in that matrix are worth noticing now, because section 3.2 gives each of them a name and tests you on it. The whole diagonal is 1, because every number divides itself. And every 1 sits on or above the diagonal, so no two different numbers divide each other.

Now redo it: list the same relation on $\{1, 2, 3, 4, 5, 6\}$. You should get 14 pairs.

(c) Consider these relations on the set of integers:

$$R1 = \{(a, b) \mid a \leq b\} \quad R2 = \{(a, b) \mid a > b\}$$

$$R3 = \{(a, b) \mid a = b \text{ or } a = -b\}$$

$$R4 = \{(a, b) \mid a = b\} \quad R5 = \{(a, b) \mid a = b + 1\}$$

$$R6 = \{(a, b) \mid a + b \leq 3\}$$

(i) List 5 ordered pairs in each of the relation.

(5marks)

(ii) Which of the relations are: (a) Reflexive (b) Symmetric (c) Transitive (6½marks)

NOTE ON THE PAPER

The domain here is **THE INTEGERS**, not a small finite set, so these relations are infinite. You cannot list them, only sample them, which is exactly why part (i) asks for five pairs each rather than all of them. Part (ii) is answered in section 3.2, once the properties are defined.

WORKED EXAMPLE

PQ 20/21 Q5(C)(I)

Five pairs from each. Choose a spread, including negatives and a repeated value, because that also prepares part (ii).

RELATION	RULE	FIVE PAIRS
R1	$a \leq b$	(1,1), (1,2), (0,5), (-3,2), (-7,-7)
R2	$a > b$	(2,1), (5,0), (0,-4), (-2,-9), (10,3)
R3	$a = b \text{ or } a = -b$	(3,3), (3,-3), (-5,5), (0,0), (7,7)
R4	$a = b$	(0,0), (1,1), (-2,-2), (9,9), (100,100)
R5	$a = b + 1$	(1,0), (2,1), (0,-1), (-4,-5), (8,7)
R6	$a + b \leq 3$	(1,1), (0,3), (-5,2), (2,1), (-1,-1)

Check each one before writing it. For R5, (1,0) works because $1 = 0 + 1$. For R6, (0,3) works because $0 + 3 = 3$, and the rule says " ≤ 3 ", so equality is allowed. That boundary case is the one people get wrong.

Now redo it: give five more pairs for each, all different from these.

RECALL AND CHECK

1. Complete the definition: a relation on A is a subset of ____.
2. In the matrix of R , what does a 1 in row 3, column 2 mean?
3. How many relations are there on a set with 3 elements?
4. Is $(2,4)$ in the relation "a divides b"? Is $(4,2)$?

ANSWERS

1. $A \times A$.
2. That $(a_3, a_2) \in R$, so the third element relates to the second.
3. $2^{3^2} = 2^9 = 512$.
4. $(2,4)$ yes, since $4 = 2 \times 2$. $(4,2)$ no, since 2 is not a whole multiple of 4.

FILL IN THE GAP

1. A relation from A to B is a _____ of $A \times B$.
2. In a relation matrix, the row indexes the _____ entry of the pair.
3. A loop in the directed graph of R corresponds to a pair of the form _____.
4. There are _____ relations on a set of n elements.

ANSWERS

1. subset. 2. first. 3. (a, a) . 4. 2^{n^2} .

PART THREE · 3.2

The Five Properties of a Relation

Reflexive, symmetric, antisymmetric, transitive and irreflexive. Three of the five papers we hold ask you to classify a list of relations against these, so this is the highest-value page in Part Three.

By the end of this section you will be able to state each property precisely, test any relation against all five quickly and reliably, spot each one in a matrix at a glance, and produce the counterexample that proves a property fails.

The five definitions

● MUST MEMORISE

THE FIVE PROPERTIES

Throughout, R is a relation on a set A .

PROPERTY	DEFINITION	IN PLAIN WORDS
Reflexive	$(a, a) \in R$ for every $a \in A$	everything relates to itself, with no exceptions
Irreflexive	$(a, a) \notin R$ for every $a \in A$	nothing relates to itself
Symmetric	whenever $(a, b) \in R$, also $(b, a) \in R$	every arrow has a matching return arrow
Antisymmetric	whenever $(a, b) \in R$ and $(b, a) \in R$, then $a = b$	no two <i>different</i> elements point both ways
Transitive	whenever $(a, b) \in R$ and $(b, c) \in R$, also $(a, c) \in R$	any two-step path has a one-step shortcut

◆ **Memory hook:** reflexive looks in a mirror, symmetric goes both ways, antisymmetric forbids both ways unless it is the same thing twice, transitive takes shortcuts.

⚠ TRAP

ANTISYMMETRIC IS NOT THE OPPOSITE OF SYMMETRIC

This is the mistake that costs the most marks in this whole part.

A relation can be **both** symmetric and antisymmetric. Take $R = \{(1,1), (2,2)\}$ on $\{1, 2, 3\}$. Every arrow has its return arrow, since each is its own return, so it is symmetric. And no two *different* elements point both ways, so it is antisymmetric too.

A relation can also be **neither**. Take $R = \{(1,2), (2,1), (2,3)\}$. It fails symmetry, because $(2,3)$ has no $(3,2)$. It fails antisymmetry, because $(1,2)$ and $(2,1)$ are both in with $1 \neq 2$.

The four combinations all occur. So test the two properties **separately** and never infer one from the other.

⚠ TRAP

REFLEXIVE NEEDS EVERY ELEMENT

$R = \{(1,1), (2,2), (3,3)\}$ on $A = \{1, 2, 3, 4\}$ is **not** reflexive, because $(4,4)$ is missing. Three out of four is a fail.

This is exactly the trap in 2025/2026 Q2(c), and it is why "determine whether R is reflexive" is worth two marks rather than one: the mark is for naming the missing pair, not for the verdict.

Also note that **not reflexive** and **irreflexive** are different. $R = \{(1,1), (2,3)\}$ on $\{1,2,3\}$ is not reflexive (2 and 3 lack loops) and not irreflexive either (1 has one). Most relations are neither.

Seeing the properties in the matrix

TEACH

THE FASTEST RELIABLE METHOD

Write the matrix and every property becomes something you can see. This is far quicker than checking pairs one at a time, and much harder to get wrong.

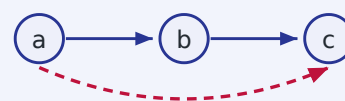
PROPERTY	WHAT TO LOOK AT	PASSES WHEN
Reflexive	the main diagonal	every diagonal entry is 1
Irreflexive	the main diagonal	every diagonal entry is 0
Symmetric	the matrix versus its mirror image in the diagonal	$M_{ij} = M_{ji}$ everywhere
Antisymmetric	each off-diagonal pair of positions	never both M_{ij} and M_{ji} equal to 1 when $i \neq j$
Transitive	every two-step path	$M_{ij} = M_{jk} = 1$ forces $M_{ik} = 1$

Reflexive: diagonal all 1 **Symmetric: mirror across diagonal** **Transitive: shortcut must exist**

$\boxed{1} \ 0 \ 1$
 $0 \ 1 \ \boxed{0}$
 $1 \ 1 \ 1 \ \boxed{}$

$\begin{matrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{matrix}$

$a \rightarrow b$ and $b \rightarrow c$
 forces $a \rightarrow c$



the dashed arrow must be present

● MUST MEMORISE

HOW TO DISPROVE EACH PROPERTY

Every one of these properties is a "for all" statement, so a **single counterexample** disproves it, and quoting that counterexample is what earns the mark. Never write "R is not transitive" alone.

TO SHOW R IS NOT ...	PRODUCE
reflexive	one element a with $(a, a) \notin R$
symmetric	one pair $(a, b) \in R$ with $(b, a) \notin R$
antisymmetric	a pair (a, b) and (b, a) both in R with $a \neq b$
transitive	(a, b) and (b, c) in R with $(a, c) \notin R$

◆ **Memory hook:** one bad case kills a "for all". Find it, write it, move on.

The repeated question

AS PRINTED

2024/2025 · Q4(B)

b. Consider the following relations on $\{1, 2, 3, 4\}$:

$R1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,

$R2 = \{(1, 1), (1, 2), (2, 1)\}$,

$R3 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,

$R4 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,

$R5 = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4)\}$,

$R6 = \{(3, 4)\}$.

i. Which of these relations are reflexive?

ii. Which of these relations are symmetric?

iii. Which of these relations are antisymmetric?

iv. Which of these relations are transitive?

v. Which of these relations are reflexive?

7.5 marks

NOTE ON THE PAPER

PARTS I AND V ARE THE SAME QUESTION. Both read "Which of these relations are reflexive?" Since ii, iii and iv cover symmetric, antisymmetric and transitive, part v was almost certainly meant to be **IRREFLEXIVE** or **ASYMMETRIC**. Answer what is printed: give the reflexive answer again, and add one line noting the apparent duplication and supplying the irreflexive answer as well. That covers you whichever was intended, and costs three lines. Never silently answer a question the paper did not ask.

BREAK IT DOWN

GIVEN Six relations on $\{1, 2, 3, 4\}$, listed as sets of pairs.

ASKED **Which of these relations are** [property]. A list of names is the answer. Reasons are not demanded, but one short reason per failure is cheap insurance and shows method.

MARKS SAY 7.5 marks over five parts is 1.5 each, so about six verdicts per part. That is a lot of checking for the time, which is why the matrix method matters.

TEACHES This section, 3.2.

TRAP Confusing symmetric with antisymmetric, and forgetting that a relation can be both or neither. Also: $R6 = \{(3,4)\}$ is a single pair, and single-pair relations are vacuously symmetric-free, antisymmetric and transitive. Vacuous cases are where marks go missing.

Method. Build the matrix of each relation first. Then read the diagonal for reflexive, compare with the mirror for symmetric and antisymmetric, and check two-step paths for transitive.

The six matrices (rows and columns in the order 1, 2, 3, 4):

R1	1	2	3	4	R2	1	2	3	4	R3	1	2	3	4
1	1	1	0	0	1	1	1	0	0	1	1	1	0	1
2	1	1	0	0	2	1	0	0	0	2	1	1	0	0
3	0	0	0	1	3	0	0	0	0	3	0	0	1	0
4	1	0	0	1	4	0	0	0	0	4	1	0	0	1

R4	1	2	3	4	R5	1	2	3	4	R6	1	2	3	4
1	0	0	0	0	1	1	1	1	1	1	0	0	0	0
2	1	0	0	0	2	0	1	1	1	2	0	0	0	0
3	1	1	0	0	3	0	0	1	1	3	0	0	0	1
4	1	1	1	0	4	0	0	0	1	4	0	0	0	0

(i) Which are reflexive?

Read each diagonal and require all four 1s.

R	DIAGONAL	REFLEXIVE?
R1	1, 1, 0, 1	no, (3,3) missing
R2	1, 0, 0, 0	no, (2,2), (3,3), (4,4) missing
R3	1, 1, 1, 1	yes
R4	0, 0, 0, 0	no, and in fact irreflexive
R5	1, 1, 1, 1	yes
R6	0, 0, 0, 0	no, and irreflexive

Answer: R3 and R5.

(ii) Which are symmetric?

Check $M_{ij} = M_{ji}$ for every off-diagonal position.

R	TEST	SYMMETRIC?
R1	(3,4) is in but (4,3) is not	no
R2	(1,2) and (2,1) both in; no other off-diagonal pairs	yes
R3	(1,4) and (4,1) both in, (1,2) and (2,1) both in	yes
R4	(2,1) is in but (1,2) is not	no
R5	(1,2) is in but (2,1) is not	no
R6	(3,4) is in but (4,3) is not	no

Answer: R2 and R3.

(iii) Which are antisymmetric?

Look for any off-diagonal position where both M_{ij} and M_{ji} are 1. One such pair kills it.

R	TEST	ANTISYMMETRIC?
R1	(1,2) and (2,1) both in, $1 \neq 2$	no
R2	(1,2) and (2,1) both in	no
R3	(1,2) and (2,1) both in	no
R4	no returning pair anywhere	yes
R5	no returning pair anywhere	yes
R6	only one pair, so nothing to return	yes, vacuously

Answer: R4, R5 and R6.

(iv) Which are transitive?

For each, hunt for a two-step path with no shortcut.

R	TEST	TRANSITIVE?
R1	(4,1) and (1,2) are in, but (4,2) is not	no
R2	(2,1) and (1,2) are in, but (2,2) is not	no
R3	(4,1) and (1,2) are in, but (4,2) is not	no
R4	(4,3) and (3,1) are in, and (4,1) is in; (4,3) and (3,2) in, and (4,2) in; (3,2) and (2,1) in, and (3,1) in. All paths close.	yes
R5	every path closes; it is the relation $a \leq b$	yes
R6	only (3,4); no two-step path exists	yes, vacuously

Answer: R4, R5 and R6.

(v) Which are reflexive? (as printed, repeating part i)

Answer: R3 and R5, as in part (i).

Note for the marker: parts (i) and (v) are identically worded on the paper. If (v) was intended as **irreflexive**, the answer is **R4 and R6**, since those are the two whose diag-

onals are entirely 0.

Summary table, which is what you should transcribe into the answer book:

	R1	R2	R3	R4	R5	R6
REFLEXIVE	no	no	yes	no	yes	no
SYMMETRIC	no	yes	yes	no	no	no
ANTISYMMETRIC	no	no	no	yes	yes	yes
TRANSITIVE	no	no	no	yes	yes	yes
IRREFLEXIVE	no	no	no	yes	no	yes

Now redo it: cover everything and rebuild the summary table from the six lists of pairs. Twelve minutes is a realistic target, which tells you how much of the 44 minutes this question would consume.

WORKED EXAMPLE

PQ 20/21 Q5(C)(II)

The same skill on the infinite relations from 3.1, where you cannot write a matrix and must argue instead.

R	REFLEXIVE?	SYMMETRIC?	TRANSITIVE?
R1: $a \leq b$	yes , $a \leq a$ always	no, $1 \leq 2$ but not $2 \leq 1$	yes , $a \leq b$ and $b \leq c$ give $a \leq c$
R2: $a > b$	no, $a > a$ is never true	no, $2 > 1$ but not $1 > 2$	yes , $a > b$ and $b > c$ give $a > c$
R3: $a = b$ or $a = -b$	yes , $a = a$ always	yes , the rule is unchanged by swapping	yes , see below
R4: $a = b$	yes	yes	yes
R5: $a = b + 1$	no, $a = a + 1$ is impossible	no, $2 = 1 + 1$ but $1 \neq 2 + 1$	no, $(3, 2)$ and $(2, 1)$ in but $3 \neq 1 + 1$
R6: $a + b \leq 3$	no, take $a = 2$: $2 + 2 = 4 > 3$	yes , $a + b = b + a$	no, $(3, 0)$ and $(0, 3)$ in but $3 + 3 = 6 > 3$

Why R3 is transitive. $a R3 b$ means $|a| = |b|$, since $a = b$ or $a = -b$ is exactly the statement that they have the same absolute value. So if $|a| = |b|$ and $|b| = |c|$ then $|a| = |c|$. Rewriting the rule in terms of absolute value turns a fiddly case check into one line, and R3 is then visibly reflexive, symmetric and transitive all at once, which is the combination section 3.3 takes up.

Now redo it: decide antisymmetry for all six. You should find R1, R2, R4 and R5 antisymmetric, and R3, R6 not.

RECALL AND CHECK

1. Give a relation that is both symmetric and antisymmetric.
2. What single pair would you name to show $\{(1,2),(2,3)\}$ is not transitive?
3. Is the empty relation on $\{1,2\}$ transitive?
4. Where do you look in a matrix to test reflexivity?

ANSWERS

1. $\{(1,1)\}$ on $\{1\}$, or any relation consisting only of loops.
2. $(1,3)$ is missing, although $(1,2)$ and $(2,3)$ are present.
3. Yes, vacuously: there is no two-step path to fail. It is also symmetric and antisymmetric, and not reflexive.
4. The main diagonal; it must be all 1s.

FILL IN THE GAP

1. R is reflexive when $(a, a) \in R$ for _____ a in A .
2. R is _____ when (a,b) and (b,a) in R forces $a = b$.
3. To disprove transitivity, produce (a,b) and (b,c) in R with _____ not in R .
4. In the matrix, symmetry means $M_{ij} =$ _____.

ANSWERS

1. every. 2. antisymmetric. 3. (a, c) . 4. M_{ji} .

PART THREE • 3.3

Equivalence Relations and Partitions

Three properties together make the most useful relation in mathematics: one that behaves like equality without being equality.

By the end of this section you will be able to define an equivalence relation, test one in the order that finds a failure fastest, and explain the partition it creates.

The definition

● MUST MEMORISE

EQUIVALENCE RELATION

A relation R on a set A is an **equivalence relation** when it is all three of:

reflexive: $(a, a) \in R$ for every a

symmetric: $(a, b) \in R$ forces $(b, a) \in R$

transitive: (a, b) and (b, c) in R force $(a, c) \in R$

All three, with no exceptions. Failing any one means it is not an equivalence relation, however close it looks.

◆ **Memory hook:** Reflexive, Symmetric, Transitive: **RST**. Say it out loud before every answer, and check them in that order.

An equivalence relation is "sameness in some respect". Ordinary equality is the strictest example, but most useful ones are looser.

"Has the same remainder when divided by 3" is an equivalence relation on the integers. So is "was born in the same year", "has the same absolute value", and "computes the same output". None of these is equality, yet each lets you treat different things as interchangeable for a purpose.

That is why it matters in computing: two states of a machine that behave identically can be merged, two expressions that always evaluate the same can be swapped, and a cache can return one answer for many requests. Every one of those is an equivalence relation being exploited.

Testing in the fastest order

Reflexivity is the cheapest to test and the most commonly broken. It needs one glance at the diagonal, and a single missing loop settles the whole question.

So the efficient order is:

- 1. Reflexive?** Scan the diagonal. If one loop is missing, stop: the answer is no, and you have your counterexample.
- 2. Symmetric?** Compare each pair with its reverse. One unmatched pair settles it.
- 3. Transitive?** The expensive one, so do it last, and only if the first two passed.

Under exam time this ordering routinely saves several minutes on a four-mark question.

The real questions

(c) A social networking platform records users who can view their own profiles. The relation is defined as:

$$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)\}$$

(2 marks)

Determine whether R is reflexive. Justify your answer.

NOTE ON THE PAPER

The mark text is printed *above* the instruction rather than after it, which is only a layout quirk. Note that the underlying set is not stated explicitly; it is implied to be $\{1, 2, 3, 4\}$ by the pairs present. Say that you are taking $A = \{1, 2, 3, 4\}$ in your answer, because the verdict depends on it: if the platform had a fifth user, R would not be reflexive.

BREAK IT DOWN

- GIVEN** Six pairs, four of them loops, on an implied set $\{1, 2, 3, 4\}$.
- ASKED** **Determine whether** R is reflexive, and **justify**. The justification is explicitly demanded, so a bare "yes" is half an answer at best.
- MARKS SAY** 2 marks: one for the verdict, one for the reason. Name the loops.
- TEACHES** 3.2 for the property, this section for the context.
- TRAP** Being distracted by $(1,2)$ and $(2,3)$. Reflexivity is only about loops; the other pairs are irrelevant to this question. Also, state the set: an unstated domain is where this question hides its difficulty.

WORKED EXAMPLE**PQ 25/26 Q2(C)**

- GIVEN** $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (2,3)\}$, on the set $A = \{1, 2, 3, 4\}$ implied by the elements appearing.
- FIND** Whether R is reflexive, with justification.
- RULE** R is reflexive when $(a, a) \in R$ for every a in A .

Step 1. Fix the set. The elements mentioned are 1, 2, 3 and 4, so $A = \{1, 2, 3, 4\}$.

Step 2. Check each element for its loop.

A	IS (A, A) IN R?
1	yes, (1,1) is listed
2	yes, (2,2) is listed
3	yes, (3,3) is listed
4	yes, (4,4) is listed

Step 3. Verdict and justification.

Yes, R is reflexive. Every element of $A = \{1, 2, 3, 4\}$ is related to itself, because all four pairs $(1,1)$, $(2,2)$, $(3,3)$ and $(4,4)$ are present in R . The remaining pairs $(1,2)$ and $(2,3)$ do not affect reflexivity, which depends only on the loops.

Worth one extra line, and it is free: R is *not* symmetric, since $(1,2)$ is in R but $(2,1)$ is not, and it is not transitive, since $(1,2)$ and $(2,3)$ are in R but $(1,3)$ is not. So R is reflexive but not an equivalence relation. In context that says every user can view their own profile, but viewing is not mutual and does not chain.

Now redo it: what single pair would you add to make R symmetric? And what would you then need for transitivity?

AS PRINTED**2025/2026 · Q2(D)**

- (d) Let $A = \{1,2,3,4\}$ and let $R = \{(1,1),(2,2),(3,3),(4,4),(1,3),(3,1),(2,4),(4,2)\}$. Determine whether R is an equivalence relation. Provide detailed explanations. **(4 marks)**

BREAK IT DOWN

GIVEN	$A = \{1, 2, 3, 4\}$ explicitly, and eight pairs: four loops and two mutual pairs.
ASKED	Determine whether R is an equivalence relation, with detailed explanations . So all three properties must be checked individually and the reasoning shown.
MARKS SAY	4 marks: roughly one per property plus one for the conclusion. Structure the answer under three headings and the marks fall out.
TEACHES	This section, 3.3.
TRAP	Transitivity looks fiddly with eight pairs. Work it systematically: list every two-step path and check its shortcut. Do not eyeball it and assert.

GIVEN $A = \{1, 2, 3, 4\}$, $R = \{(1, 1), (2, 2), (3, 3), (4, 4), (1, 3), (3, 1), (2, 4), (4, 2)\}$.

FIND Whether R is reflexive, symmetric and transitive, hence whether it is an equivalence relation.

RULE All three properties must hold. Test in the order reflexive, symmetric, transitive.

The matrix makes all three visible at once:

	1	2	3	4
1	1	0	1	0
2	0	1	0	1
3	1	0	1	0
4	0	1	0	1

1. Reflexive

The diagonal reads 1, 1, 1, 1. All four pairs $(1, 1)$, $(2, 2)$, $(3, 3)$, $(4, 4)$ are present, so every element relates to itself. **R is reflexive.**

2. Symmetric

Take each non-loop pair and look for its reverse.

PAIR IN R	REVERSE NEEDED	PRESENT?
$(1, 3)$	$(3, 1)$	yes
$(3, 1)$	$(1, 3)$	yes
$(2, 4)$	$(4, 2)$	yes
$(4, 2)$	$(2, 4)$	yes

The loops are their own reverses, so they hold automatically. The matrix is its own mirror image across the diagonal. **R is symmetric.**

3. Transitive

Every two-step path must have its one-step shortcut. Ignoring paths where one step is a loop, since those give back a pair already present, the genuine two-step paths are:

FIRST STEP	SECOND STEP	SHORTCUT NEEDED	PRESENT?
$(1, 3)$	$(3, 1)$	$(1, 1)$	yes
$(3, 1)$	$(1, 3)$	$(3, 3)$	yes
$(2, 4)$	$(4, 2)$	$(2, 2)$	yes
$(4, 2)$	$(2, 4)$	$(4, 4)$	yes

There is no path from the $\{1, 3\}$ side to the $\{2, 4\}$ side, because no pair joins them, so no further check is needed. **R is transitive.**

Conclusion

R is reflexive, symmetric and transitive, so R IS an equivalence relation on A .

The detail that earns the last mark. An equivalence relation cuts its set into groups, and this one is easy to see. Element 1 is related to 1 and 3; element 2 is related to 2 and 4. So A splits into

{1, 3} and {2, 4}

Those are the **equivalence classes**. Every element lies in exactly one, none is empty, and together they make up all of A. Here the relation is "has the same parity", since 1 and 3 are odd while 2 and 4 are even, which is why it behaves so neatly.

Now redo it: remove (2,4) and (4,2) from R and re-test. It stays an equivalence relation, but the classes become {1,3}, {2}, {4}.

Equivalence classes and partitions

● MUST MEMORISE

CLASSES AND PARTITIONS

Given an equivalence relation R on A, the **equivalence class** of a, written [a], is the set of everything related to a:

$$[a] = \{x \in A \mid (a, x) \in R\}$$

A **partition** of A is a collection of non-empty subsets that are pairwise disjoint and whose union is all of A. Every element is in exactly one piece.

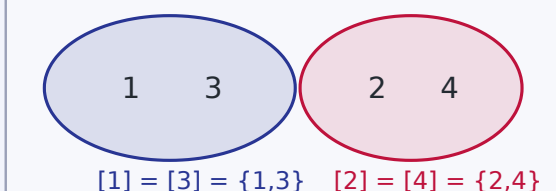
The fundamental fact: the equivalence classes of an equivalence relation always form a partition of A, and conversely every partition defines an equivalence relation ("lies in the same piece").

◆ **Memory hook:** an equivalence relation and a partition are the same information written two ways. Sorting things into boxes and declaring things equivalent are one act.

TEACH

THE PICTURE

A, partitioned by R



Every element sits in exactly one class.

No class is empty.

The classes together make up all of A.

That is precisely a partition.

RECALL AND CHECK

1. Name the three properties an equivalence relation must have.
2. Which property should you test first, and why?
3. Is "is less than or equal to" on the integers an equivalence relation?
4. What are the equivalence classes of "same remainder on division by 3" over the integers?

ANSWERS

1. Reflexive, symmetric, transitive.
2. Reflexive: it is the cheapest to check and the most often broken, and one missing loop ends the question.
3. No. It is reflexive and transitive but not symmetric, since $1 \leq 2$ while $2 \leq 1$ is false. It is a partial order instead, which is section 3.4.
4. Three classes: the multiples of 3, those leaving remainder 1, and those leaving remainder 2.

FILL IN THE GAP

1. An equivalence relation is reflexive, _____ and _____.
2. The set of everything related to a is called the _____ of a.
3. The equivalence classes of an equivalence relation form a _____ of the set.
4. In a partition, every element lies in _____ piece.

ANSWERS

1. symmetric, transitive. 2. equivalence class. 3. partition. 4. exactly one.

PART THREE · 3.4

Partial and Total Orders

Swap symmetry for antisymmetry and you stop describing sameness and start describing ranking. That is the difference between an equivalence relation and an order.

By the end of this section you will be able to define a partially ordered set and a totally ordered set, tell them apart by the one question that separates them, and prove which a given table is.

NOT IN THE COURSE TEXT

The official text never uses the words "partially ordered", "totally ordered" or "poset". The 2025/2026 paper gives 4.5 marks to exactly this, in Q2(e).

The definitions

● MUST MEMORISE

PARTIAL ORDER

A relation R on A is a **partial order** when it is

reflexive: every element relates to itself

antisymmetric: (a,b) and (b,a) both in R forces $a = b$

transitive: (a,b) and (b,c) in R force (a,c) in R

The set together with the order, written (A, R) , is a **partially ordered set**, or **poset**.

◆ **Memory hook: RAT**: Reflexive, Antisymmetric, Transitive. Compare with **RST** for an equivalence relation. The only difference is the middle letter, and that one letter is the difference between "the same as" and "no worse than".

● MUST MEMORISE

TOTAL ORDER

A partial order is a **total order** (also called a linear order or a chain) when it also satisfies **comparability**:

for every a and b in A , either $(a, b) \in R$ or $(b, a) \in R$.

That is, **no two elements are incomparable**. Every pair can be put in order.

A set with a total order is a **totally ordered set**, or a **chain**.

◆ **Memory hook**: every total order is a partial order. The extra demand is that nothing sits off to the side: you can line all the elements up in a single row, which is why it is also called linear.

Once you know a relation is a partial order, ask one thing:

"Is there a pair I cannot compare?"

If yes, it is partial only. If no, it is total.

Examples of partial but not total:

Divisibility on $\{1, 2, 3, 4\}$: 2 and 3 are incomparable, since neither divides the other.
Subset inclusion on the subsets of $\{1,2\}$: $\{1\}$ and $\{2\}$ are incomparable, since neither contains the other.

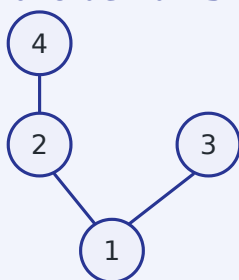
"Is a prerequisite of" on university courses: two unrelated electives are incomparable.

Examples of total:

" $\leq$ " on the integers, on the reals, or on any set of numbers: any two numbers can be compared.

Alphabetical order on words: any two words have a first one.

Partial order: divisibility on $\{1,2,3,4\}$



2 and 3 are incomparable:
no line joins them, so it is
partial and not total.

Total order: $\leq$ on $\{1,2,3,4\}$



A single chain. Every pair
is comparable, so it is total.

⚠ TRAP

ANTISYMMETRIC DOES NOT MEAN "NOT SYMMETRIC"

Section 3.2 made this point and it returns here with marks attached. Antisymmetry forbids two **different** elements pointing at each other. It says nothing about loops, which are always allowed and indeed required by reflexivity.

So a partial order is full of pairs (a, a) , and that is not a violation of antisymmetry. The definition already says "then $a = b$ ", which those pairs satisfy.

The real question

AS PRINTED

2025/2026 · Q2(E)

(e) Differentiate between a partially ordered set and a totally-ordered set. Hence or otherwise determine with proof the status of the table below in terms of partially-ordered or total ordered set.

Student	CGPA
A	4.80
B	4.50
C	4.10

(4½ marks)

NOTE ON THE PAPER

The relation is never stated. The table gives three students and three distinct CGPAs, and you are expected to supply the obvious order, "has a CGPA at least as high as". **SAY WHICH RELATION YOU ARE USING IN YOUR FIRST LINE.** Without it the question has no determinate answer, and stating it is part of what the 4.5 marks are for. Note also the inconsistent hyphenation ("totally-ordered" then "total ordered"), which changes nothing.

BREAK IT DOWN

GIVEN	Three students with three distinct CGPAs.
ASKED	Two tasks. First differentiate the two kinds of ordered set, which needs a stated point of contrast, not two definitions side by side. Then determine with proof which the table is, so all the order axioms must be checked.
MARKS SAY	4.5 marks: roughly 2 for the difference and 2.5 for the classification with proof. The word "proof" means check every property explicitly.
TEACHES	This section, 3.4, plus 3.2 for the properties.
TRAP	Answering "it is a total order" with no proof. The marks are in the four checks: reflexive, antisymmetric, transitive, comparable. The last one is what settles total versus partial.

Part 1: the difference

Both are sets carrying a relation that is reflexive, antisymmetric and transitive, so both rank their elements consistently and neither allows two different elements to outrank each other.

They differ on comparability. In a **partially ordered set** there may be pairs of elements that the relation does not connect in either direction; such elements are *incomparable*, and the ranking has gaps. In a **totally ordered set** every pair is comparable, so all the elements line up in one chain with no gaps.

Consequently every total order is a partial order, but not the reverse. Divisibility on $\{1, 2, 3, 4\}$ is partial only, because 2 and 3 are incomparable; the relation $\leq$ on numbers is total, because any two numbers can be compared.

Part 2: classifying the table

- GIVEN** $S = \{A, B, C\}$ with $\text{CGPA}(A) = 4.80$, $\text{CGPA}(B) = 4.50$, $\text{CGPA}(C) = 4.10$.
FIND Whether (S, R) is partially or totally ordered, with proof.
RELATION Taking the natural relation **$x R y$ when $\text{CGPA}(x) \geq \text{CGPA}(y)$** , that is, x ranks at least as high as y . This must be stated because the paper does not state it.

Writing the pairs out, since $\text{CGPA}(A) > \text{CGPA}(B) > \text{CGPA}(C)$:

$$R = \{(A,A), (B,B), (C,C), (A,B), (A,C), (B,C)\}$$

1. Reflexive. Each student's CGPA is at least as high as itself, since $4.80 \geq 4.80$, $4.50 \geq 4.50$ and $4.10 \geq 4.10$. So (A,A) , (B,B) and (C,C) are all in R . **Reflexive.**

2. Antisymmetric. Suppose $x R y$ and $y R x$. Then $\text{CGPA}(x) \geq \text{CGPA}(y)$ and $\text{CGPA}(y) \geq \text{CGPA}(x)$, so $\text{CGPA}(x) = \text{CGPA}(y)$. The three CGPAs listed are distinct, so this forces $x = y$. **Antisymmetric.**

3. Transitive. Suppose $x R y$ and $y R z$. Then $\text{CGPA}(x) \geq \text{CGPA}(y) \geq \text{CGPA}(z)$, so $\text{CGPA}(x) \geq \text{CGPA}(z)$ and $x R z$. Checking the only non-trivial case directly: (A,B) and (B,C) are in R , and (A,C) is in R . **Transitive.**

So (S, R) is a **partially ordered set**. Now the deciding test.

4. Comparability. Take the three pairs of distinct students:

PAIR	COMPARISON	COMPARABLE?
A and B	$4.80 \geq 4.50$, so $A R B$	yes
A and C	$4.80 \geq 4.10$, so $A R C$	yes
B and C	$4.50 \geq 4.10$, so $B R C$	yes

Every pair is comparable, because any two real numbers can be compared.

Conclusion. (S, R) is reflexive, antisymmetric and transitive, so it is a partially ordered set; and every pair of students is comparable, so it is in fact a **totally ordered set**. The students form a single chain

$$A > B > C \quad (4.80, 4.50, 4.10)$$

which is exactly what a class ranking is.

The remark that shows real understanding: totality here depends on the three CGPAs being *distinct* and on CGPA being a single number. If two students shared a CGPA, antisymmetry would fail and R would not be a partial order at all. And if students were ranked on two criteria at once, say CGPA *and* attendance, two students could each lead on one measure, making them incomparable and the order merely partial. That is precisely how a partial order arises in practice.

Now redo it: add student D with CGPA 4.50 and re-classify. Antisymmetry now fails, so it is no longer a partial order at all.

TEACH

WHERE ORDERS APPEAR IN COMPUTING

Worth a line if a question asks for relevance.

Task scheduling. "Must finish before" is a partial order on jobs; independent jobs are incomparable and can run in parallel. Sorting a partial order into a legal sequence is a topological sort.

Version control. "Is an ancestor of" is a partial order on commits; branches are incomparable commits.

Type systems. "Is a subtype of" is a partial order.

Sorting. Every sorting algorithm assumes a *total* order on the keys. That is why a comparison function must never report two different items as mutually less-than: it would break antisymmetry and the sort would misbehave.

RECALL AND CHECK

1. Name the three properties of a partial order.
2. What one extra condition makes it a total order?
3. Give two incomparable elements under divisibility on $\{1, \dots, 6\}$.
4. Which property differs between an equivalence relation and a partial order?

ANSWERS

1. Reflexive, antisymmetric, transitive.
2. Comparability: for every a and b , either $a R b$ or $b R a$.
3. 4 and 6, for instance: neither divides the other. Also 2 and 5, or 3 and 4.
4. The middle one. An equivalence relation is symmetric; a partial order is antisymmetric.

FILL IN THE GAP

1. A partial order is reflexive, _____ and transitive.
2. A set with a partial order is called a _____.
3. A partial order in which every pair is comparable is a _____ order.
4. Two elements neither of which relates to the other are _____.

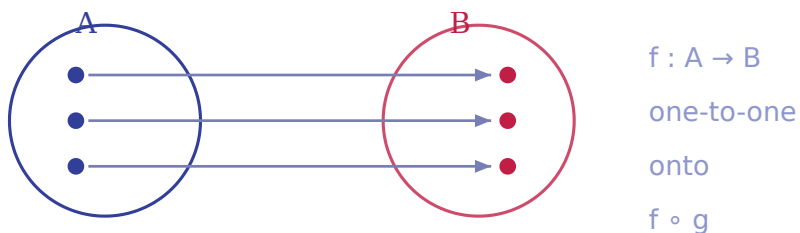
ANSWERS

1. antisymmetric. 2. partially ordered set (poset). 3. total (linear). 4. incomparable.

PART FOUR

Functions and Correspondence

A function is a relation that never gives two answers to the same question. Every subprogram you write is one, and the five kinds of correspondence in this part appear on paper after paper.



Functions, Domain and Range

A function assigns exactly one output to each input. That single word "exactly" is the whole definition, and it is what separates a function from an ordinary relation.

By the end of this section you will be able to say whether a given rule or diagram is a function, name the domain, codomain and range, and explain the difference between the last two.

From relations to functions

● MUST MEMORISE

DEFINITION

A **function** f from A to B , written $f : A \rightarrow B$, is a relation from A to B in which **every** element of A is paired with **exactly one** element of B .

If f pairs a with b we write $f(a) = b$, and call b the **image** of a , and a a **pre-image** of b .

Two conditions are hiding in "every ... exactly one":

Totality: nothing in A is left unassigned.

Single-valuedness: nothing in A gets two different outputs.

◆ **Memory hook:** a function is a machine with one slot and one answer. Put the same thing in twice and you must get the same thing out; put anything valid in and something must come out.

● MUST MEMORISE

DOMAIN, CODOMAIN, RANGE

For $f : A \rightarrow B$,

domain = A , the set of all allowed inputs.

codomain = B , the set the outputs are declared to live in.

range (or image) = the set of values actually produced, $\{f(a) \mid a \in A\}$.

The range is always a subset of the codomain, and it may be smaller.

◆ **Memory hook:** the codomain is what you *promised*, the range is what you *delivered*. When they are equal, the function is onto, which is section 4.2.

TEACH

WHY RANGE AND CODOMAIN DIFFER

Take $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(x) = x^2$.

The domain is all reals and the codomain is all reals, because that is how the function was declared. But no square is negative, so the range is only the non-negative reals. The function promised to return a real number and delivered only half of them.

In programming terms the codomain is the declared return type and the range is the set of values the routine can actually return. A routine declared to return an integer might in fact only ever return 0 or 1.

⚠ TRAP

TWO WAYS A RELATION FAILS TO BE A FUNCTION

An input with no output. On $A = \{1, 2, 3\}$, the relation $\{(1,a), (2,b)\}$ is not a function: 3 has no image. Every element of the domain must be used.

An input with two outputs. $\{(1,a), (1,b), (2,c), (3,d)\}$ is not a function either: 1 is sent to two places.

An output used twice is **fine**: $\{(1,a), (2,a), (3,a)\}$ is a perfectly good function. The restriction is on inputs, never on outputs. Getting that backwards is the commonest error in the diagram question of 4.2.

WORKED EXAMPLE

METHOD

Which of these relations on $A = \{1, 2, 3\}$ to $B = \{a, b, c\}$ are functions?

RELATION	FUNCTION?	WHY
$\{(1,a), (2,b), (3,c)\}$	yes	each input once, each with one output
$\{(1,a), (2,a), (3,a)\}$	yes	repeated outputs are allowed
$\{(1,a), (2,b)\}$	no	3 has no image, so it is not total
$\{(1,a), (1,b), (2,c), (3,c)\}$	no	1 has two images

For the second one, the domain is $\{1,2,3\}$, the codomain is $\{a,b,c\}$, and the range is just $\{a\}$.

Now redo it: is $\{(1,c), (2,c), (3,b)\}$ a function? What is its range? (Yes; range $\{b, c\}$.)

TEACH

THE TWO-LINE NOTATION

For a function on a small finite set, the tidiest way to write it is a two-line table with each input directly above its output:

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

meaning $f(1) = 2$, $f(2) = 3$, $f(3) = 1$. It is only notation, and it is not a matrix in the sense of section 3.1: you cannot multiply two of these or row-reduce them. The name is borrowed, the operations are not.

RECALL AND CHECK

1. What two conditions must a relation satisfy to be a function?
2. For $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$, state the codomain and the range.
3. May two different inputs share an output?
4. May one input have two outputs?

ANSWERS

1. Every element of the domain is used (totality), and none is sent to more than one place (single-valuedness).
2. Codomain $\mathbb{R}$; range the non-negative reals.
3. Yes, always allowed.
4. No. That is exactly what a function forbids.

FILL IN THE GAP

1. In $f : A \rightarrow B$, the set A is the _____ and B is the _____.
2. The set of values actually produced is the _____.
3. If $f(a) = b$ then b is the _____ of a .
4. The range is always a _____ of the codomain.

ANSWERS

1. domain, codomain. 2. range (image). 3. image. 4. subset.

PART FOUR • 4.2

The Five Correspondence Types

Ten marks on two separate papers, for one diagram you can learn in ten minutes. This is the best value in the whole course.

By the end of this section you will be able to define one-to-one and onto, draw all five correspondence types from memory, and say which is which from a diagram.

The two properties

● MUST MEMORISE

ONE-TO-ONE AND ONTO

Let $f : A \rightarrow B$.

One-to-one (also **injective**): different inputs always give different outputs. Formally, $f(a_1) = f(a_2)$ forces $a_1 = a_2$. *No output is hit twice.*

Onto (also **surjective**): every element of B is hit by something. Formally, for every $b \in B$ there is an $a \in A$ with $f(a) = b$. *Equivalently, the range equals the codomain.*

Bijective: both one-to-one and onto. A perfect pairing, with every element of A matched to exactly one of B and nothing left over on either side.

◆ **Memory hook**: one-to-one is about **collisions** in B (there are none). Onto is about **coverage** of B (it is complete). One looks for doubles, the other looks for gaps.

Compare the sizes of A and B and some cases become impossible, which is a fast sanity check on any answer.

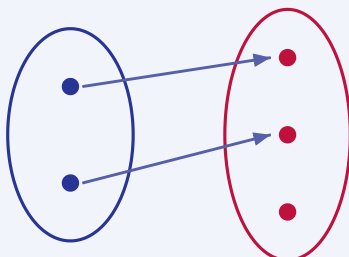
SIZES	ONE-TO-ONE POSSIBLE?	ONTO POSSIBLE?	BIJECTION POSSIBLE?
$n(A) < n(B)$	yes	no , too few inputs to cover B	no
$n(A) = n(B)$	yes	yes	yes
$n(A) > n(B)$	no , two inputs must collide	yes	no

The "no" in the last row is the pigeonhole principle of section 5.3 arriving early: with more inputs than outputs, some output is used twice.

The five diagrams

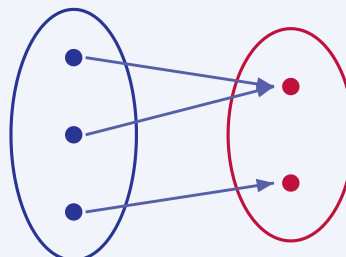
The exam asks for a diagram, so the diagram is the answer. Each one below uses a small A and B so the property is visible at a glance.

(i) One-to-one, not onto



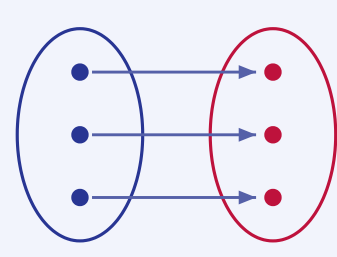
no collisions, but one target is never reached

(ii) Onto, not one-to-one



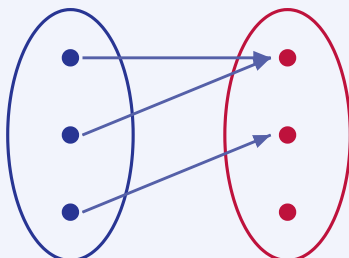
every target reached, but the top one is hit twice

(iii) One-to-one and onto

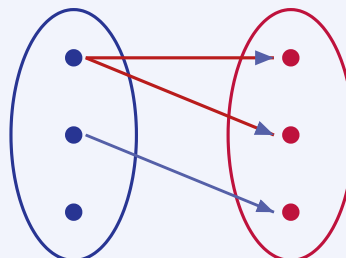


a perfect pairing: no collisions, no gaps

(iv) Neither one-to-one nor onto (v) Not a function



the top target is hit twice AND the bottom one is missed



the top input has TWO arrows, and the bottom input has none

Reading a diagram

Look at the RIGHT circle for both i and ii. Two arrows landing on one dot means not one-to-one.

A dot with no arrow landing on it means not onto.

Then check the LEFT circle: two arrows leaving one dot, or a dot with none, and it is not a function at all.

The real question

AS PRINTED

2021/2022 · Q2(C)

(c) With the aid of a diagram, explain the following types of function correspondences types:

- (i) One-to-One, not Onto (ii) Onto, not One-to-One (iii) One-to-One and Onto
(iv) Neither One-to-One nor Onto (v) Not a Function **(10 marks)**

AS PRINTED

2020/2021 · Q4(B)

(b) With the aid of a diagram, explain the following types of function correspondences types:

- (i) One-to-one, not onto (ii) Onto, not one-to-one (iii) One-to-one and onto
(iv) Neither one-to-one nor onto (v) Not a function **(10 marks)**

NOTE ON THE PAPER

"the following types of function correspondences types" says "types" twice, on both papers. Harmless. The two versions are the same question down to the five items and the 10 marks, and differ only in capitalisation: 2021/2022 writes "One-to-One, not Onto", 2020/2021 writes "One-to-one, not onto". One prepared answer serves both.

BREAK IT DOWN

GIVEN Five named correspondence types.

ASKED **With the aid of a diagram, explain** each. Two things are demanded per item: a drawing **and** a sentence. A page of diagrams with no words scores about half, and so does a page of words with no diagrams.

MARKS SAY 10 marks over five items is 2 each, so one diagram plus one or two sentences per item. This is the highest marks-per-minute question in the whole set: it is pure recall and needs no calculation.

TEACHES This section, 4.2, and 4.1 for what a function is.

TRAP Item (v) is not a correspondence type at all, it is the failure case. Its diagram must break the **function** rule, on the left circle, and not merely be an odd pattern of arrows on the right.

Open with the two definitions, because every item below refers to them and stating them once earns the understanding marks.

One-to-one (injective): different inputs give different outputs, so no element of B receives two arrows.

Onto (surjective): every element of B receives at least one arrow, so the range equals the codomain.

(i) One-to-one, not onto. Every element of B receives at most one arrow, but at least one receives none. Take $A = \{1, 2\}$, $B = \{a, b, c\}$ with $f(1) = a$, $f(2) = b$. Nothing maps to c . This needs $n(A) < n(B)$.

(ii) Onto, not one-to-one. Every element of B is reached, but at least one is reached twice. Take $A = \{1, 2, 3\}$, $B = \{a, b\}$ with $f(1) = a$, $f(2) = a$, $f(3) = b$. Both targets are covered, and a is hit twice. This needs $n(A) > n(B)$.

(iii) One-to-one and onto (bijective). Every element of B receives exactly one arrow. Take $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ with $f(1) = a$, $f(2) = b$, $f(3) = c$. This is a perfect pairing and requires $n(A) = n(B)$. Only a bijection has an inverse, which is section 4.3.

(iv) Neither one-to-one nor onto. Some element of B is hit twice and some other is missed. Take $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ with $f(1) = a$, $f(2) = a$, $f(3) = b$. Here a is hit twice and c is never reached.

(v) Not a function. The mapping breaks the definition in 4.1, so the fault is on the *input* side. Either some element of A has two arrows leaving it, or some element of A has none. Take $A = \{1, 2, 3\}$, $B = \{a, b, c\}$ with 1 sent to both a and b , and 3 sent nowhere. Both faults are present, and either alone is enough.

Draw the five pictures exactly as above, label the circles A and B , and write the one-line reason under each. Keep the reason on the right circle for (i) to (iv) and on the left circle for (v).

Now redo it: draw all five from memory on blank paper in four minutes, then check the arrow counts against the diagrams above.

The exam sometimes gives a formula instead of a picture. The test is the same, done algebraically.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = 2x + 1.$$

One-to-one? Suppose $f(a) = f(b)$. Then $2a + 1 = 2b + 1$, so $2a = 2b$ and $a = b$. **Yes.**

Onto? Given any y , solve $y = 2x + 1$ for $x = (y - 1)/2$, which is a real number. **Yes.** So f is a bijection.

$$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2.$$

One-to-one? $f(2) = 4 = f(-2)$ with $2 \neq -2$. **No.**

Onto? There is no real x with $x^2 = -1$. **No.** So it is type (iv), neither.

Notice how each "no" is settled by a single explicit counterexample, and each "yes" by a general argument. That asymmetry is section 1.4's rule about quantifiers, showing up again.

RECALL AND CHECK

1. Define one-to-one and onto in one sentence each.
2. If $n(A) = 5$ and $n(B) = 3$, can $f : A \rightarrow B$ be one-to-one?
3. What must be true of the range for f to be onto?
4. In a diagram, what tells you it is not a function?

ANSWERS

1. One-to-one: different inputs always give different outputs. Onto: every element of the codomain is an output of something.
2. No. With more inputs than outputs two inputs must share a target.
3. The range equals the codomain.
4. An element of A with two arrows leaving it, or with none.

FILL IN THE GAP

1. Another word for one-to-one is _____.
2. Another word for onto is _____.
3. A function that is both is called _____.
4. A function is onto exactly when its _____ equals its codomain.

ANSWERS

1. injective. 2. surjective. 3. bijective (a bijection). 4. range.

PART FOUR • 4.3

Composition and Inverse Functions

Feeding one function's output into another is composition, and the exam asks for the same worked pair on two different papers.

By the end of this section you will be able to compose two functions in either order, explain why order matters, state when an inverse exists, and answer the conditions question on commuting linear functions.

Composition

● MUST MEMORISE

COMPOSITION

For $f : B \rightarrow C$ and $g : A \rightarrow B$, the **composition** $f \circ g$ is the function from A to C given by

$$(f \circ g)(x) = f(g(x))$$

g runs first, on x . **f** runs second, on whatever g produced. $\circ$ is read "composed with", or "after".

◆ **Memory hook: read it right to left.** In $f \circ g$ the letter nearest the x is the one that goes first. Say it as "f after g" and the order stops being confusing.

⚠ TRAP**THE ORDER IS NOT DECORATION**

$f \circ g$ and $g \circ f$ are usually **different functions**, and the exam asks for both precisely to see whether you noticed.

Everyday version: putting on socks then shoes is not the same as shoes then socks. Composition, like getting dressed, is not commutative.

So compute each one separately, from its own definition. Never compute one and assume the other.

The question that appears twice

AS PRINTED**2023/2024 · Q3(C)**

(c) Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from $\mathbb{R}$ to $\mathbb{R}$. State if $f \circ g = g \circ f$.

(3½ marks)**AS PRINTED****2024/2025 · Q2 B(III)**

iii. Find $f \circ g$ and $g \circ f$, where $f(x) = x^2 + 1$ and $g(x) = x + 2$, are functions from $\mathbb{R}$ to $\mathbb{R}$

2 marks**NOTE ON THE PAPER**

The same question on two papers, one year apart, with the same functions. The only differences: 2023/2024 adds "State if $f \circ g = g \circ f$ " and pays 3½ marks, while 2024/2025 pays 2 and does not ask for the comparison. Both print "are functions from $\mathbb{R}$ to $\mathbb{R}$ ", which reads oddly because the subject of the sentence is f and g . Answer both with the same working, and add the comparison line either way: it costs one sentence and 2023/2024 pays for it.

BREAK IT DOWN

GIVEN $f(x) = x^2 + 1$ and $g(x) = x + 2$, both from the reals to the reals.

ASKED **Find** both compositions, then **state** whether they are equal. Two computations and one verdict.

MARKS SAY 3.5 marks: roughly 1.5 per composition and 0.5 for the comparison. Show the substitution step; that is where the method mark sits.

TEACHES This section, 4.3.

TRAP Expanding $(x + 2)^2$ as $x^2 + 4$. It is $x^2 + 4x + 4$. That single slip loses both the answer and the comparison.

GIVEN $f(x) = x^2 + 1$, $g(x) = x + 2$, both $\mathbb{R}$ to $\mathbb{R}$.

FIND $f \circ g$, $g \circ f$, and whether they are equal.

RULE $(f \circ g)(x) = f(g(x))$: g first, then f .

$f \circ g$

Apply g first, then feed the result into f .

$$\begin{aligned}(f \circ g)(x) &= f(g(x)) \\ &= f(x + 2) \quad [g \text{ has run}] \\ &= (x + 2)^2 + 1 \quad [\text{put } x + 2 \text{ wherever } f \text{ has } x] \\ &= x^2 + 4x + 4 + 1 \\ &= \mathbf{x^2 + 4x + 5}\end{aligned}$$

$g \circ f$

Now the other order: f first, then g .

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= g(x^2 + 1) \quad [f \text{ has run}] \\ &= (x^2 + 1) + 2 \quad [g \text{ adds 2 to whatever it is given}] \\ &= \mathbf{x^2 + 3}\end{aligned}$$

Are they equal?

No. $x^2 + 4x + 5$ and $x^2 + 3$ are different functions.

Saying so is enough for the mark, but naming a witness is better and costs one line. At $x = 0$:

$$(f \circ g)(0) = 0 + 0 + 5 = 5, \quad (g \circ f)(0) = 0 + 3 = 3.$$

Since $5 \neq 3$, the two compositions differ, so $\mathbf{f \circ g \neq g \circ f}$.

They agree only where $x^2 + 4x + 5 = x^2 + 3$, that is $4x = -2$, so at the single point $x = -\frac{1}{2}$. Agreeing at one point is not being the same function.

Now redo it: compose the same f with $h(x) = 3x$. You should get $(f \circ h)(x) = 9x^2 + 1$ and $(h \circ f)(x) = 3x^2 + 3$.

When do two functions commute?

bi. Let $f(x) = ax + b$ and $g(x) = cx + d$, where a, b, c and d are constants. Determine necessary and sufficient conditions on the constants a, b, c and d so that $f \circ g = g \circ f$.

2 marks

NOTE ON THE PAPER

The label is printed "bi.", welding part b to its first roman numeral. The phrase "necessary and sufficient" is section F.2's vocabulary: it means find the condition that holds **EXACTLY** when the two compositions agree, so the argument must run both ways, not just one.

GIVEN $f(x) = ax + b$, $g(x) = cx + d$, with a, b, c, d constant.

FIND The condition on a, b, c, d that holds exactly when $f \circ g = g \circ f$.

RULE Compose both ways, then compare coefficients.

Step 1. Compose both ways.

$$(f \circ g)(x) = f(cx + d) = a(cx + d) + b = acx + ad + b$$

$$(g \circ f)(x) = g(ax + b) = c(ax + b) + d = acx + cb + d$$

Step 2. Compare. Both have the same coefficient of x , namely ac , so the x terms always agree whatever the constants are. The two functions are equal exactly when the constant terms agree:

$$ad + b = cb + d$$

Step 3. Say why this is both necessary and sufficient. If $ad + b = cb + d$ then the two expressions are identical, so the compositions agree everywhere: the condition is **sufficient**. Conversely, if the compositions agree everywhere then they agree at $x = 0$, where they take the values $ad + b$ and $cb + d$, so those must be equal: the condition is **necessary**. Hence it is exactly the right condition.

A tidier form, worth a line: rearranging gives $ad - d = cb - b$, that is $d(a - 1) = b(c - 1)$. In that shape it is easy to read off the familiar special cases: if $a = c = 1$ both sides are 0, so any two translations commute; and if $b = d = 0$ both sides are 0, so any two functions through the origin commute.

Now redo it: check the earlier example. There $f(x) = x^2 + 1$ is not linear, so this condition does not apply; but for $f(x) = 2x + 3$ and $g(x) = 4x + 1$, is $d(a-1) = b(c-1)$? Here $1 \times 1 = 1$ and $3 \times 3 = 9$, so no, and the two do not commute.

ii. If f and $f \circ g$ are one-to-one, does it follow that g is one-to-one? Justify your answer.

2 marks

Answer: yes, g must be one-to-one.

Proof. Suppose $g(a) = g(b)$ for some a and b in the domain of g . Applying f to both sides gives $f(g(a)) = f(g(b))$, that is $(f \circ g)(a) = (f \circ g)(b)$. But $f \circ g$ is one-to-one, so $a = b$. Since $g(a) = g(b)$ forced $a = b$, g is one-to-one.

Notice what was not needed. The hypothesis that f is one-to-one was never used. Only the injectivity of the composition mattered. Saying that shows the marker you followed the logic rather than pattern-matching, and it is true: $f \circ g$ one-to-one alone forces g one-to-one.

The converse fails, which is worth one more line. If g is one-to-one, $f \circ g$ need not be: take $g(x) = x$ on $\{1, 2\}$ and f sending both 1 and 2 to the same value. Then g is one-to-one and $f \circ g$ is not.

Inverse functions

● MUST MEMORISE

INVERSE

A function $f : A \rightarrow B$ has an **inverse** $f^{-1} : B \rightarrow A$ exactly when f is a **bijection**, that is, one-to-one *and* onto.

The inverse undoes f : $f^{-1}(f(a)) = a$ for every a in A , and $f(f^{-1}(b)) = b$ for every b in B .

Why both conditions are needed. If f is not one-to-one, two inputs share an output and the inverse would not know which to return, so it would not be single-valued. If f is not onto, some b has no pre-image and the inverse would be undefined there, so it would not be total. Those are exactly the two conditions of section 4.1.

◆ **Memory hook:** only a perfect pairing can be run backwards. Collisions break single-valuedness, gaps break totality.

WORKED EXAMPLE

METHOD

Find the inverse of $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 1$.

Section 4.2 showed this f is a bijection, so an inverse exists.

Write $y = 2x + 1$ and solve for x :

$$y = 2x + 1 \Rightarrow y - 1 = 2x \Rightarrow x = (y - 1)/2$$

So $f^{-1}(y) = (y - 1)/2$, usually rewritten with x as the variable: $f^{-1}(x) = (x - 1)/2$.

Check it, always. $f^{-1}(f(x)) = ((2x + 1) - 1)/2 = 2x/2 = x$. Correct.

By contrast $f(x) = x^2$ on the reals has no inverse: it is neither one-to-one nor onto, and "the square root" cannot decide between 2 and -2 .

RECALL AND CHECK

1. In $f \circ g$, which function acts first?
2. With $f(x) = x + 1$ and $g(x) = 3x$, find both compositions.
3. What must be true of f for f^{-1} to exist?
4. Is composition commutative?

ANSWERS

1. g , the one written nearest x .
2. $(f \circ g)(x) = 3x + 1$ and $(g \circ f)(x) = 3(x + 1) = 3x + 3$.
3. f must be a bijection: one-to-one and onto.
4. No. $f \circ g$ and $g \circ f$ generally differ.

FILL IN THE GAP

1. $(f \circ g)(x)$ means _____.
2. Composition is read from _____ to _____.
3. An inverse exists exactly when the function is a _____.
4. For $f(x) = x^2 + 1$ and $g(x) = x + 2$, $(f \circ g)(x) =$ _____.

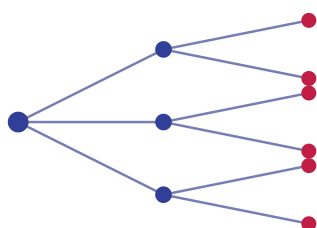
ANSWERS

1. $f(g(x))$. 2. right, left. 3. bijection. 4. $x^2 + 4x + 5$.

PART FIVE

Counting, Permutations and Combinations

Counting sounds like the easiest thing in this book and is the one that loses the most marks. Every question here is answered by deciding two things: are the choices stages or alternatives, and does order matter.



$$3 \times 2 = 6$$

$$P(n, r)$$

$$C(n, r)$$

$$n(A \cup B)$$

stages multiply

order matters

order does not

overlap subtracts

The Product and Sum Rules

Two rules answer most counting questions on these papers, and choosing between them is the whole skill. Stages multiply. Alternatives add.

By the end of this section you will be able to break a counting task into parts, decide whether those parts are compulsory or alternative, apply the product rule, the sum rule and the subtraction rule, and answer the auditorium chairs, ATM PIN, three-digit-number and password questions in full.

What a counting question is asking

TEACH

COUNTING WITHOUT LISTING

A counting question asks **how many** objects of some kind exist. The honest way to answer is to list them all and count the list. That works for six letters and fails for a million PIN codes, so we need rules that give the size of the list without writing it out.

Every such rule comes from the same move: **break the task into parts**, count each part, then combine. The only question is how to combine, and there are exactly two answers.

IF THE PARTS ARE	THEN	BECAUSE
compulsory: you do part 1 <i>and</i> part 2 <i>and</i> part 3	multiply	every way of doing part 1 pairs with every way of doing part 2
alternative: you do part 1 <i>or</i> part 2 <i>or</i> part 3, never two at once	add	the possibilities are separate piles and nothing sits in two piles

Those two words, **compulsory** and **alternative**, are the ones the 2025/2026 paper uses when it asks you to name the parts of a task. They are worth learning in exactly that language.

The product rule

● MUST MEMORISE

PRODUCT RULE

If a task is carried out in a sequence of stages, and
 stage 1 can be done in n_1 ways,
 stage 2 in n_2 ways *whatever* stage 1 did,
 ... and stage k in n_k ways,

then the whole task can be carried out in

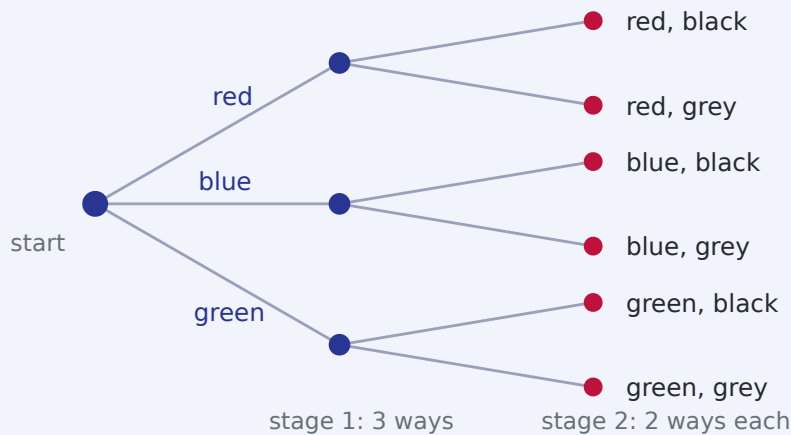
$n_1 \times n_2 \times \dots \times n_k$ ways.

WHERE

k is the number of stages, one for each decision you must make. n_i is the number of choices at stage i . $\times$ is ordinary multiplication.

◆ **Memory hook: and means times.** A shirt *and* a pair of trousers: 3 shirts and 2 trousers give 6 outfits, not 5. Say the task out loud; if the joining word is "and", multiply.

Draw the choices as a tree. Each stage adds a level, and each complete path from the root to a tip is one outcome. Three first choices, each splitting into two, gives $3 \times 2 = 6$ tips.



The product rule is a tree. Three shirts, then two pairs of trousers for each: six paths, so six outfits.

The condition that stage 2 has n_2 ways **whatever stage 1 did** is what makes every branch split the same number of times. If the second choice depended on the first, the branches would have different widths and the count would not be a single product.

The sum rule

● MUST MEMORISE

SUM RULE

If a task can be done in one of k alternative ways, and

alternative 1 can be done in n_1 ways, alternative 2 in n_2 ways, ... alternative k in n_k ways,

and **no outcome counts under two alternatives at once**, then the task can be done in

$n_1 + n_2 + \dots + n_k$ ways.

WHERE

k is the number of alternatives, one for each separate case. **no overlap** is a genuine condition, not decoration. If two alternatives can both produce the same outcome, adding counts that outcome twice.

◆ **Memory hook: or means plus.** Piles you can choose between are added; stages you must pass through are multiplied.

⚠ TRAP

THE WORD "OR" IS NOT PROOF OF A SUM

English "or" is a bad guide on its own. In the password question below, each character is "an uppercase letter *or* a digit", and that or is counted by **adding**: $26 + 10 = 36$ symbols. But choosing eight characters is eight compulsory stages, so those 36s are **multiplied**. One sentence, both rules, in that order.

The reliable test is not the word. It is this question: **am I describing one decision with several options, or several decisions I must all make?** Options add. Decisions multiply.

The subtraction rule

● MUST MEMORISE

SUBTRACTION RULE

To count the objects with some property, when the ones *without* it are easier to count:

$$(\text{number with the property}) = (\text{total}) - (\text{number without it})$$

WHERE

total is the count with the condition ignored entirely, which is nearly always the easy number. **number without it** is the count of the cases the condition rules out, and it must be counted over the same total, not over some smaller pool. **the two groups** must cover everything between them and share nothing, which is what makes one subtraction exact.

This is the counting form of the complement from section 2.2: the objects with the property and the objects without it fill the whole collection between them and share nothing.

◆ **Memory hook: "at least one" means "not none"**. Whenever a question says at least one, count everything, count the cases with none, and subtract. Counting "at least one" directly means splitting into exactly one, exactly two, exactly three and so on, which is longer and easier to get wrong.

The auditorium chairs

AS PRINTED

2021/2022 · Q2(A)

2. (a) The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently? **(6marks)**

NOTE ON THE PAPER

"labeled" is the American spelling; the paper uses it throughout. The mark text is closed up as "(6marks)", which is how this whole paper prints marks. Neither affects the answer. What does affect the answer is "not exceeding 100": that means 100 is allowed, so the integers run 1 to 100 inclusive and there are 100 of them, not 99.

BREAK IT DOWN

GIVEN	A label is an uppercase English letter followed by a positive integer that is at most 100.
ASKED	The largest number of chairs that can be labelled differently , which is the number of distinct labels available.
PARTS	Two, and both are compulsory: every label has a letter and a number. So this is the product rule.
MARKS SAY	6 marks for one multiplication is generous, so the marks are for the reasoning: name the two stages, count each, justify the counts, then multiply.
TEACHES	This section, 5.1.
TRAP	Two of them. "Uppercase English letter" is 26, not 52. "Positive integer not exceeding 100" is 100, not 99 and not 101: positive excludes 0, and not exceeding includes 100.

WORKED EXAMPLE

PQ 21/22 Q2(A)

GIVEN label = uppercase letter, then a positive integer at most 100.

FIND How many different labels exist.

RULE Product rule: the two parts are compulsory.

Stage 1: choose the letter. The uppercase English letters are A to Z, so there are **26** choices.

Stage 2: choose the integer. Positive integers not exceeding 100 are 1, 2, 3, ..., 100. Counting them, there are **100** choices.

Stage 2 offers the same 100 numbers whichever letter was picked, so the product rule applies directly:

$$\text{number of labels} = 26 \times 100 = \mathbf{2600}$$

Answer: 2600 chairs can be labelled differently.

Say the last sentence in the paper's own terms: since each chair needs its own label, 2600 is the largest number of chairs that can be labelled differently.

Now redo it: if the label were two uppercase letters followed by a positive integer not exceeding 100, how many? Three compulsory stages now: $26 \times 26 \times 100 = 67600$.

The ATM PIN codes

AS PRINTED

2025/2026 · Q6(B)

(b) ATM PIN code in Nigeria can contain the digits 0,1,2,...,9 and digits may be repeated. In the light of the above, how many 4 to 6 digit pin codes are there? **(5 marks)**

NOTE ON THE PAPER

Two sentences have been run together with no full stop: "digits may be repeated In the light of the above". Read the capital "In" as the start of the second sentence. The phrase "4 to 6 digit" means codes of length 4, length 5 or length 6, and those three lengths are the alternatives that make this a sum-rule question sitting on top of a product-rule one.

BREAK IT DOWN

GIVEN	Each position holds one of the ten digits 0 to 9, and repeats are allowed.
ASKED	How many codes of length 4, 5 or 6 there are, in total.
PARTS	Two levels. Within one length the positions are compulsory parts, so multiply. Across the three lengths the cases are alternative parts, so add.
MARKS SAY	5 marks for three powers and one addition. Show each length separately, then total. A bare final number risks losing the method marks.
TEACHES	This section, 5.1.
TRAP	Answering only for 6 digits, or multiplying the three lengths together. A code is 4 or 5 or 6 digits long, never two lengths at once, so the three counts are added.

WORKED EXAMPLE

PQ 25/26 Q6(B)

- GIVEN** Digits 0 to 9, so 10 choices per position. Repetition allowed. Lengths 4, 5 and 6.
FIND The total number of such codes.
RULE Product rule within a length, sum rule across lengths.

Step 1. Count the codes of each length. Because repeats are allowed, every position still has all 10 digits available, no matter what the earlier positions took. So each position is a stage with 10 ways.

LENGTH	PRODUCT RULE	COUNT
4 digits	$10 \times 10 \times 10 \times 10 = 10^4$	10000
5 digits	10^5	100000
6 digits	10^6	1000000

Step 2. Combine the three lengths. A code has exactly one length, so no code is counted twice and the sum rule applies:

$$\begin{aligned}\text{total} &= 10^4 + 10^5 + 10^6 \\ &= 10000 + 100000 + 1000000 \\ &= \mathbf{1110000}\end{aligned}$$

Answer: there are 1110000 PIN codes of 4 to 6 digits. Written with separators for readability, 1 110 000.

Why repetition matters. If digits could not repeat, each position would remove one digit from the pool and the length-4 count would be $10 \times 9 \times 8 \times 7$ instead of 10^4 . Section 5.2 counts that case. The words "digits may be repeated" are the paper telling you which of the two applies.

Now redo it: how many codes of 4 to 6 digits start with a nonzero digit? Each length loses its first-position zero, giving $9 \times 10^3 + 9 \times 10^4 + 9 \times 10^5 = 9000 + 90000 + 900000 = 999000$.

Three-digit numbers, and naming the rule

AS PRINTED

2025/2026 · Q6(C)

- (c) i. How many three digit natural numbers can be formed using digits 0, 1, $\dots$, 9? (1 mark)
ii. Identify the number of parts in the task and the type of the parts (compulsory or alternative). (4 marks)
iii. Which rule applies here? (1 mark)

NOTE ON THE PAPER

Parts ii and iii say "the task" and "here" without saying which task. They sit directly under (c) i, and (c) i is the nearest task, so answer them about the three-digit numbers. Part (b) immediately above is also a counting task with a different answer, so the safe examination move is to answer for (c) i and then add one line covering (b) as well. That costs a sentence and cannot be marked wrong. Both answers are given below.

BREAK IT DOWN

GIVEN The ten digits 0 to 9, and a three-digit natural number to build from them.

ASKED (i) **How many** such numbers. (ii) **Identify** the number of parts and their type. (iii) **Which rule** applies. Only the first is a calculation; the other two ask you to describe your own method.

MARKS SAY 1, 4 and 1. The 4 marks are the largest on this part and they are for the description, not the arithmetic. Name each part, say how many ways it has, and say why it is compulsory.

TEACHES This section, 5.1.

TRAP Answering $10^3 = 1000$. A three-digit number cannot begin with 0, because 012 is the two-digit number 12. The hundreds position has 9 choices, not 10.

GIVEN Digits 0 to 9, forming a three-digit natural number.

FIND The count, the parts, and the rule.

RULE Product rule, with one restricted position.

i. How many three-digit natural numbers

A three-digit natural number runs from 100 to 999. Build it position by position:

POSITION	ALLOWED DIGITS	WAYS
hundreds	1 to 9, since a leading 0 would make it a shorter number	9
tens	0 to 9	10
units	0 to 9	10

$$\text{count} = 9 \times 10 \times 10 = \mathbf{900}$$

Answer: 900. Check it another way, which is worth doing on a 1-mark part: the numbers 100 to 999 inclusive number $999 - 100 + 1 = 900$. The two methods agree.

ii. The number of parts and their type

Three parts, all compulsory. Set out in full, since 4 marks are on offer:

1. **Part 1, choose the hundreds digit:** 9 ways. Compulsory, because every three-digit number has a hundreds digit.
2. **Part 2, choose the tens digit:** 10 ways. Compulsory.
3. **Part 3, choose the units digit:** 10 ways. Compulsory.

They are compulsory rather than alternative because the number is not finished until all three have been done. You do not pick *one* of the three parts; you do part 1 **and** part 2 **and** part 3.

For part (b) above, the structure is different and worth one line: that task has **three alternative parts**, one per allowed length (4, 5 and 6 digits), and each of those parts is itself built from compulsory sub-parts, one per digit position.

iii. Which rule applies

The product rule, because the parts of (c) i are compulsory: the counts 9, 10 and 10 are multiplied.

And again for (b): **the sum rule** across the three lengths, sitting on top of the product rule within each length.

Now redo it: how many three-digit natural numbers have no repeated digit? Hundreds 9 ways, tens 9 (zero is back in, but the hundreds digit is gone), units 8. So $9 \times 9 \times 8 = 648$.

The password question

(b) Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there? **(5 marks)**

NOTE ON THE PAPER

This is the hardest counting question in the whole set, and it is worth 5 marks out of 22.5. It combines all three rules of this section: add across the three lengths, multiply across the character positions, and subtract to handle "at least one digit". Nothing about it is printed wrongly. The difficulty is real, not a defect.

BREAK IT DOWN

GIVEN	Password length 6, 7 or 8. Each character is an uppercase letter or a digit. At least one character must be a digit.
ASKED	How many possible passwords there are in total.
PARTS	Three alternative lengths, so add. Within a length, one compulsory part per character position, so multiply. The "at least one" condition is handled by subtracting.
MARKS SAY	5 marks. Roughly: 1 for the 36 symbols, 1 for seeing that "at least one digit" means total minus letters-only, 2 for the three lengths, 1 for the total. Show every line.
TEACHES	This section, 5.1: product, sum and subtraction rules together.
TRAP	Trying to count "at least one digit" directly by cases (exactly one digit, exactly two, and so on). It can be done, but it is seven cases per length and almost nobody finishes it correctly under exam pressure. Subtract instead.

GIVEN Length 6, 7 or 8; each character from 26 uppercase letters or 10 digits; at least one digit required.

FIND The number of possible passwords.

RULE Sum over lengths; product over positions; subtraction for "at least one".

Step 1. Count the symbols available per character. A character is an uppercase letter **or** a digit, and no symbol is both, so the sum rule gives

$$26 + 10 = \mathbf{36} \text{ symbols.}$$

Step 2. Turn "at least one digit" into a subtraction. Let P_n be the number of valid passwords of length n . Then

$$\begin{aligned} P_n &= (\text{all strings of length } n \text{ over the 36 symbols}) - (\text{those using no digit at all}) \\ &= 36^n - 26^n \end{aligned}$$

A string with no digit is made only of the 26 letters, which is where 26^n comes from. Every string either contains a digit or does not, and never both, so the subtraction is exact.

Step 3. Evaluate each length.

$$P_6 = 36^6 - 26^6 = 2176782336 - 308915776 = \mathbf{1867866560}$$

$$P_7 = 36^7 - 26^7 = 78364164096 - 8031810176 = \mathbf{70332353920}$$

$$P_8 = 36^8 - 26^8 = 2821109907456 - 208827064576 = \mathbf{2612282842880}$$

Step 4. Add the three lengths. A password has exactly one length, so the sum rule applies:

$$P = P_6 + P_7 + P_8 = \mathbf{2684483063360}$$

Answer: 2684483063360 possible passwords, or 2 684 483 063 360 written with separators.

If you run out of time, the formula $P = (36^6 - 26^6) + (36^7 - 26^7) + (36^8 - 26^8)$ with the three rules named earns most of the 5 marks on its own. Write the formula first, then evaluate. Never the other way round.

Now redo it: same question but passwords are exactly six characters and must contain at least one letter. Answer: $36^6 - 10^6 = 2176782336 - 1000000 = 2175782336$.

RECALL AND CHECK

1. When do you multiply the counts of the parts, and when do you add them?
2. What condition must alternatives satisfy before the sum rule is valid?
3. How many three-digit natural numbers are there, and why is it not 1000?
4. Turn "at least one vowel" into a subtraction.
5. Labels are one uppercase letter then an integer from 1 to 100. How many?

ANSWERS

1. Multiply when the parts are compulsory, that is, when all of them must be done. Add when they are alternatives and exactly one of them happens.
2. No outcome may count under two alternatives, otherwise it is counted twice.
3. 900. The hundreds digit cannot be 0, so it has 9 choices, not 10: $9 \times 10 \times 10$.
4. (total number of strings) – (number of strings with no vowel).
5. $26 \times 100 = 2600$.

FILL IN THE GAP

1. Parts that must all be done are _____, and their counts are _____.
2. Parts you choose between are _____, and their counts are _____.
3. The number of PIN codes of 4 to 6 digits, repeats allowed, is _____.
4. "At least one" is counted as total minus _____.

ANSWERS

1. compulsory, multiplied. 2. alternative, added. 3. 1110000. 4. none.

PART FIVE • 5.2

Permutations and Combinations

One question decides which formula you need: does the order of the chosen objects change the answer? If it does, you are permuting. If it does not, you are combining.

By the end of this section you will be able to compute factorials, use the permutation and combination formulas, count arrangements of a word with repeated letters, handle "at least one" selections, and answer the SCHOOL, the six boys and four girls, and the prayer partners questions in full.

Factorial notation

● MUST MEMORISE

FACTORIAL

Section F.1 introduced this notation. It is restated here because every formula in this part is built from it, and you should not have to turn back mid-question.

For a positive integer n , **n factorial**, written $n!$, is the product of every positive integer from n down to 1:

$$n! = n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$$

and by definition

$$0! = 1$$

WHERE

n is a whole number that is not negative. $!$ is read "factorial", so $5!$ is read "five factorial". It is not an exclamation mark and never punctuation.

◆ **Memory hook: count down and multiply.** And $0! = 1$ is not a trick: there is exactly one way to arrange nothing, namely do nothing, so the count is 1, not 0.

TEACH

THE VALUES YOU SHOULD KNOW BY SIGHT

N	0	1	2	3	4	5	6	7	8	10
N!	1	1	2	6	24	120	720	5040	40320	3628800

Factorials grow violently. That is the reason counting questions ask for an expression as often as a number: $20!$ has nineteen digits and no examiner wants it written out.

The cancelling trick, which you will use constantly. Because $n! = n \times (n - 1)!$, a ratio of factorials collapses:

$$10! / 7! = (10 \times 9 \times 8 \times 7!) / 7! = 10 \times 9 \times 8 = 720$$

Never compute $10!$ and $7!$ separately and divide. Cancel first, always.

Permutations: order matters

● MUST MEMORISE

PERMUTATION

A **permutation** is an arrangement of objects **in a definite order**. An **r-permutation** of n distinct objects is an ordered selection of r of them, and the number of these is

$$P(n, r) = n! / (n - r)!$$

WHERE

n is how many distinct objects are available. r is how many are being chosen and arranged, with $0 \leq r \leq n$. $P(n, r)$ is also written nPr or nP_r , which is the label printed on most calculators.

The special case $r = n$: arranging all n objects gives $P(n, n) = n! / 0! = n!$. Every permutation question about "all of them" is just a factorial.

◆ **Memory hook: P for position.** A permutation cares where each object lands, so AB and BA are two different answers.

TEACH

WHERE THE FORMULA COMES FROM

You already have this from the product rule. To fill r ordered positions from n objects with no repeats:

position 1: n ways

position 2: $n - 1$ ways, since one object is used up

position 3: $n - 2$ ways

...

position r : $n - r + 1$ ways

Multiplying gives $n \times (n - 1) \times \dots \times (n - r + 1)$, which is r factors counting down from n . Writing that as a ratio of factorials:

$$n \times (n - 1) \times \dots \times (n - r + 1) = n! / (n - r)!$$

because dividing by $(n - r)!$ cancels exactly the tail of the product that we stopped short of. So $P(n, r)$ is not a new idea; it is the product rule packaged.

Worked: $P(6, 4) = 6! / 2! = 720 / 2 = 360$, and by the counting-down form $6 \times 5 \times 4 \times 3 = 360$. The two agree, as they must.

Combinations: order does not matter

● MUST MEMORISE

COMBINATION

A **combination** is a selection of objects **without regard to order**. The number of ways of choosing r objects from n distinct objects is

$$C(n, r) = n! / (r! (n - r)!)$$

WHERE

n is how many objects are available. r is how many are chosen, with $0 \leq r \leq n$. $C(n, r)$ is also written nCr or nC_r .

Why the extra $r!$ Each unordered selection of r objects can be arranged in $r!$ different orders, and $P(n, r)$ counts every one of those orders separately. Dividing by $r!$ collapses them back into one:

$$C(n, r) = P(n, r) / r!$$

◆ **Memory hook: C for committee.** A committee of three is the same committee however you read out the names, so order is thrown away and you divide by $r!$.

PERMUTATION: order matters

Choose 2 of A, B, C and line them up

A B B A

A C C A

B C C B

$$P(3, 2) = 3! / 1! = 6$$

six ordered pairs

A B and B A are different

COMBINATION: order does not

Choose 2 of A, B, C as a set

—— {A, B}

—— {A, C}

—— {B, C}

$$C(3, 2) = 3! / (2! 1!) = 3$$

three sets

each set collapses 2! orders into one

The same three objects, the same $r = 2$, and two different answers. Six divided by $2!$ is three.

Ask the question in this exact form: **if I swap two of the chosen objects, is that a different answer?**

TASK	SWAP CHANGES IT?	USE
Arranging letters into a word	yes, TAB is not BAT	permutation
Filling the posts of chair and secretary	yes, the two people swap jobs	permutation
Picking 4 students for a team	no, the same four people	combination
Choosing a subset of a set	no, a set has no order	combination

The wording that gives it away: **arrange, order, line up, rank, first and second** point to a permutation; **select, choose, pick, committee, team, subset, group** point to a combination.

One warning about that list. It is a guide, not a law. Read the task, not the verb: a question that says "select" and then gives the chosen people different jobs is a permutation despite the word.

Arranging a word with repeated letters

The number of distinct arrangements of n objects, of which one kind repeats n_1 times, another n_2 times, and so on, is

$$n! / (n_1! \times n_2! \times \dots \times n_k!)$$

WHERE

n is the total number of objects, letters counted as often as they appear. n_i is how many times the i -th repeated object occurs. Objects appearing once contribute $1! = 1$ and can be left out.

Why divide. Pretend the repeats are distinguishable for a moment, say O_1 and O_2 . Then all n objects are distinct and there are $n!$ arrangements. But swapping O_1 and O_2 produces a word that looks identical, so every genuine word has been counted $2!$ times. Divide by $2!$ to correct it.

◆ **Memory hook: divide out what you cannot tell apart.** One factorial on the bottom for each repeated letter, using how many times it repeats.

(d) Find out the number of ways that the letters of the word "SCHOOL" can be arranged?
(2½ marks)

NOTE ON THE PAPER

A command punctuated as a question, which this paper does in several places. Nothing else is wrong with it, and the quotation marks around the word are the paper's own. The word is the whole difficulty: SCHOOL has six letters and the letter O appears twice.

BREAK IT DOWN

- GIVEN** The six letters of the word SCHOOL: S, C, H, O, O, L.
- ASKED** **The number of ways** they can be arranged. Arranged means ordered, so this is a permutation.
- MARKS SAY** 2.5 marks. That is enough for the formula, the substitution and the answer, and not enough to list 360 words. Do not list them.
- TEACHES** This section, 5.2.
- TRAP** Answering $6! = 720$. The two O's are identical, so half of those 720 orderings spell the same word twice over. The answer is 360.

WORKED EXAMPLE

PQ 25/26 Q4(D)

- GIVEN** The letters S, C, H, O, O, L. Total $n = 6$, with the letter O repeated twice.
- FIND** The number of distinct arrangements.
- FORMULA** $n!$ divided by the factorial of each repeat count.

Step 1. Count the letters and the repeats.

LETTER	S	C	H	O	L
TIMES IT OCCURS	1	1	1	2	1

Total letters $n = 1 + 1 + 1 + 2 + 1 = 6$. One letter repeats, namely O, and it repeats twice.

Step 2. Apply the formula.

$$\begin{aligned}\text{arrangements} &= 6! / 2! \\ &= 720 / 2 \\ &= 360\end{aligned}$$

Answer: the letters of SCHOOL can be arranged in 360 ways.

State the reasoning in one line for the marker: if the six letters were all different there would be $6! = 720$ arrangements, but swapping the two O's changes nothing visible, so every distinct word has been counted $2!$ times and 720 must be halved.

Now redo it: how many arrangements has the word LEVEL? Five letters with L twice and E twice, so $5! / (2! 2!) = 120 / 4 = 30$.

Selecting with a condition

AS PRINTED

2025/2026 · Q3(B)

(b) In a group of students, there are 6 boys and 4 girls. Out of 10 students, 4 students have to be selected. Find out how many different ways the students can be selected such that at least one boy should be selected. **(5 marks)**

NOTE ON THE PAPER

"Out of 10 students" restates what the first sentence already said, since $6 + 4 = 10$. That is a helpful redundancy rather than a defect: it confirms the ten students are the whole pool. The phrase to act on is "at least one boy", which is the subtraction rule of section 5.1 in a combination setting.

BREAK IT DOWN

GIVEN	6 boys and 4 girls, 10 students in all. Four are to be selected.
ASKED	How many different ways the selection can be made with at least one boy among the four.
ORDER	A group of four is selected, with no jobs or ranking attached, so order does not matter. This is a combination.
MARKS SAY	5 marks. Enough for the total, the excluded case, the subtraction and a check. Showing the case-by-case method as a verification is worth doing if time allows.
TEACHES	This section, 5.2, using the subtraction rule from 5.1.
TRAP	Reading "at least one boy" as "exactly one boy" and answering $C(6, 1) \times C(4, 3) = 24$. At least one means one, two, three or four.

WORKED EXAMPLE**PQ 25/26 Q3(B)**

GIVEN	6 boys, 4 girls, 10 students; choose 4.
FIND	The number of selections containing at least one boy.
FORMULA	$C(n, r) = n! / (r! (n - r)!)$, with the subtraction rule for "at least one".

Step 1. Count all selections of four, ignoring the condition.

$$\begin{aligned}
 C(10, 4) &= 10! / (4! \times 6!) \\
 &= (10 \times 9 \times 8 \times 7) / (4 \times 3 \times 2 \times 1) \\
 &= 5040 / 24 \\
 &= \mathbf{210}
 \end{aligned}$$

Step 2. Count the selections that break the condition. A selection fails only when it contains **no boy**, that is, when all four chosen are girls. There are exactly 4 girls, so all four must be taken:

$$C(4, 4) = 4! / (4! \times 0!) = 1$$

Note $0! = 1$ here; that definition is doing real work.

Step 3. Subtract.

$$\text{at least one boy} = 210 - 1 = \mathbf{209}$$

Answer: 209 different selections contain at least one boy.

Verification by cases, which is the long way and a good check:

BOYS	GIRLS	WAYS	VALUE
1	3	$C(6, 1) \times C(4, 3)$	$6 \times 4 = 24$
2	2	$C(6, 2) \times C(4, 2)$	$15 \times 6 = 90$
3	1	$C(6, 3) \times C(4, 1)$	$20 \times 4 = 80$
4	0	$C(6, 4) \times C(4, 0)$	$15 \times 1 = 15$

$$24 + 90 + 80 + 15 = \mathbf{209}.$$
 The two methods agree.

The case table also shows why subtracting is the better exam move: four combinations and three additions, against one subtraction.

Now redo it: how many selections of four contain at least one girl? Total 210, minus the all-boy selections $C(6, 4) = 15$, gives 195.

The prayer partners

AS PRINTED

2025/2026 · Q3(C)

(c) Suppose 10 male and 10 female students needs to pray with one another in a one male-one female order:

- (i) How many Prayer sessions will there be? **(1 mark)**
- (ii) How many prayer units are generated here? **(1½ marks)**
- (iii) Calculate the choice of prayer partners. **(3 marks)**

NOTE ON THE PAPER

Two things to flag. First, "students needs" does not agree in number, and "Prayer" is capitalised in (i) but not in (ii). Those are harmless. Second, and this one matters: the paper never says what a **SESSION** is or what a **UNIT** is, so parts (i) and (ii) can be read more than one way. The answer below states the reading it uses before computing anything, gives the numbers that reading produces, and then gives the other reading's numbers too. Part (iii) is unambiguous and carries the most marks.

BREAK IT DOWN

GIVEN	10 male and 10 female students. They pray in pairs, each pair being one male and one female.
ASKED	(i) the number of sessions , (ii) the number of units , (iii) calculate the choice of prayer partners.
MARKS SAY	1, 1.5 and 3. The weight is on (iii), which is a full permutation, so budget your time there. Parts (i) and (ii) are one line each.
TEACHES	This section, 5.2, with the product rule from 5.1.
TRAP	Answering (iii) with $10 \times 10 = 100$. That counts the possible pairs, which is part (ii). The choice of partners means assigning every male a different female at once, and that is $10!$.

GIVEN 10 males, 10 females, praying one male with one female.

FIND Sessions, units, and the number of ways to choose partners.

RULE Product rule for the pairs, permutation for the assignment.

Step 0. State the reading. Write this on the script before answering, because the paper does not supply it:

A prayer unit is one male paired with one female. A session is one round in which all twenty students are paired at the same time, giving ten units per session. Every male is to pray with every female.

i. How many prayer sessions

In one session each of the 10 males prays with one female, so a session uses up 10 of the possible male-female pairings. Every male must eventually meet all 10 females, so each male needs 10 turns:

$$\text{number of sessions} = \mathbf{10}$$

ii. How many prayer units

A unit is one male with one female. Choosing the male is 10 ways and choosing the female is 10 ways, and both choices must be made, so the product rule gives

$$\text{units} = 10 \times 10 = \mathbf{100}$$

Consistency check: 10 sessions with 10 units in each is 100 units, which matches. The two answers agree, which is exactly why stating the reading first was worth a line.

iii. Calculate the choice of prayer partners

Now the real work. In a single session, every male is given a different female. Line the males up in a fixed order and hand out the females:

male 1: 10 choices of partner

male 2: 9 choices, since one female is taken

male 3: 8 choices

...

male 10: 1 choice

All ten choices must be made, so the product rule applies:

$$\text{choices} = 10 \times 9 \times 8 \times \dots \times 2 \times 1 = 10! = \mathbf{3628800}$$

In the language of this section, the ten females are being arranged among the ten males, so the count is $P(10, 10) = 10! / 0! = 10!$.

Answer: 3628800 ways, or 3 628 800 with separators.

The other reading, for safety. If instead a "session" means a single male-female meeting, then there are 100 sessions and 10 units in each round. Whichever reading a marker has in mind, the two numbers on the table are 10 and 100, and part (iii) is $10! = 3628800$ either way. Writing your reading down protects the marks.

Now redo it: with 5 males and 5 females, the units are $5 \times 5 = 25$ and the choice of partners is $5! = 120$.

RECALL AND CHECK

1. State the formulas for $P(n, r)$ and $C(n, r)$.
2. What is $0!$, and why?
3. Which is larger, $P(8, 3)$ or $C(8, 3)$, and by what factor?
4. How many arrangements has the word SCHOOL, and why is it not 720?
5. Four students are picked from ten for a team. Permutation or combination?

ANSWERS

1. $P(n, r) = n! / (n - r)!$ and $C(n, r) = n! / (r! (n - r)!)$.
2. $0! = 1$, because there is exactly one way to arrange nothing.
3. $P(8, 3)$ is larger, by a factor of $3! = 6$: $P(8, 3) = 336$ and $C(8, 3) = 56$.
4. 360. The two O's are identical, so $6!$ counts every word twice and must be divided by $2!$.
5. A combination: the team is the same team however the names are ordered.

FILL IN THE GAP

1. An arrangement in a definite order is a _____.
2. A selection with no regard to order is a _____.
3. $C(n, r)$ is $P(n, r)$ divided by _____.
4. Choosing 4 from 10 can be done in _____ ways.
5. Ten males paired with ten females can be partnered in _____ ways.

ANSWERS

1. permutation. 2. combination. 3. $r!$. 4. 210. 5. 3628800.

PART FIVE • 5.3

Inclusion, Exclusion and the Pigeonhole Principle

When two groups overlap, adding their sizes counts the overlap twice. Subtracting it once is the whole of inclusion and exclusion, and it is worth seven marks on the newest paper.

By the end of this section you will be able to state and apply the inclusion and exclusion principle for two and three sets, read the overlap off a Venn diagram, state the pigeonhole principle in both its forms, and answer the Ibibio and Efik question and the class-size question in full.

Why adding is wrong

TEACH

THE DOUBLE COUNT

The sum rule of section 5.1 came with a condition: **no outcome may count under two alternatives**. Sets of people almost always break that condition, because a person can be in both.

Suppose 30 students take Maths and 25 take Physics. Adding gives 55. But if 10 take both, those 10 were counted once in the 30 and again in the 25, so 55 overstates the class by exactly 10. The true total is 45.

That is the entire idea. Add the parts, then take the overlap back out once, because it went in twice.

● MUST MEMORISE

INCLUSION AND EXCLUSION, TWO SETS

Section 2.2 introduced this as the counting formula, where it belonged with the set operations. Here it is again, this time with the rearrangements the exam actually asks for.

For any two finite sets A and B,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

WHERE

$n(A)$ is the cardinality of A, the number of things in it, as defined in section 2.1. $A \cup B$ is the union: everything in A or in B or in both. $A \cap B$ is the intersection: the things in both, which is the overlap that was counted twice.

The rearrangements you will actually use. Exam questions give you three of the four quantities and ask for the fourth, so learn all three forms, not just the first:

$$n(A \cap B) = n(A) + n(B) - n(A \cup B) \quad [\text{find the overlap}]$$

$$n(A \text{ only}) = n(A) - n(A \cap B) \quad [\text{strip the overlap out of A}]$$

$$n(B \text{ only}) = n(B) - n(A \cap B)$$

◆ **Memory hook: in, in, out.** Put A in, put B in, take the middle out. And a completed answer always satisfies $A \text{ only} + \text{both} + B \text{ only} = n(A \cup B)$.

THE PRINCIPLE OF INCLUSION AND EXCLUSION

The claim. The size of a union is found by adding the sizes of the sets and then removing every overlap that the addition counted more than once. For two sets that means subtracting the single overlap; for more sets the corrections alternate in sign.

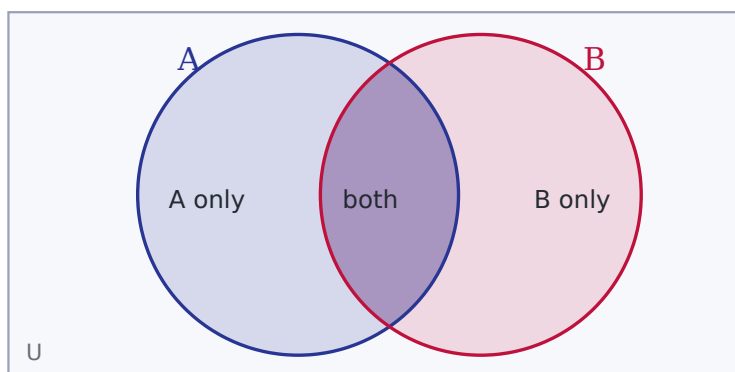
In symbols.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

Conditions. The sets must be **finite**, or the counts do not exist to be added. Nothing else is required: the sets may overlap in any way at all, which is precisely the point of the principle.

The limiting case. When the sets are disjoint, every overlap term is 0 and the principle collapses to the sum rule of section 5.1: $n(A \cup B) = n(A) + n(B)$. So the sum rule is not a separate law but the special case of this one where nothing is shared.



$n(A)$ counts
A only + both

$n(B)$ counts
B only + both

so "both" is in
the sum twice

Three regions, not two. Adding $n(A)$ and $n(B)$ covers the middle region twice, so it is subtracted once.

● MUST MEMORISE

INCLUSION AND EXCLUSION, THREE SETS

For three finite sets A, B and C,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

WHERE

$n(A)$, $n(B)$, $n(C)$ are the sizes of the three sets on their own, each counted with no regard to the others. $n(A \cap B)$ and the other two pair terms are the sizes of each two-way overlap, subtracted because each was counted in two of the singles. $n(A \cap B \cap C)$ is the size of the three-way overlap, added back because the singles put it in three times and the pairs took it out three times.

Read the pattern rather than memorising the letters: add the singles, subtract the pairs, add the triple. Signs alternate, and the number of sets in each term goes up by one.

The final term is added back because the elements in all three sets were added three times by the singles and then subtracted three times by the pairs, leaving them counted zero times. One addition restores them.

◆ **Memory hook: plus singles, minus pairs, plus triple.** Odd-sized groups are added, even-sized groups are subtracted.

The Ibibio and Efik question

AS PRINTED

2025/2026 · Q3(A)

(a) In a group of 100 students, 72 students can speak Ibibio and 43 students can speak Efik. Based on these data, answer the following questions:

- (i) Find the number of students who can speak both Ibibio and Efik. **(3 marks)**
- (ii) Find the number of students who can speak Ibibio only. **(2 marks)**
- (iii) Find the number of students who can speak Efik only. **(2 marks)**

NOTE ON THE PAPER

THIS QUESTION CONTRADICTS ITSELF, AND IT IS THE MOST IMPORTANT DEFECT ON THE PAPER. The opening sentence names the same language twice: "72 students can speak Ibibio and 43 students can speak Ibibio". Parts (i) to (iii) then ask about Ibibio **AND EFİK**, so the

second figure must have been meant as Efik. Read literally, the data mention Efik nowhere and 7 marks are unanswerable.

WHAT TO DO IN THE HALL. Do not stall and do not silently rewrite the question. Write one line: *"The second figure is taken as 43 students who can speak Efik, since parts (i) to (iii) ask about Efik and the question is otherwise unanswerable."* Then answer on that reading. A stated assumption that is obviously the intended one is normal mathematical practice and protects every mark; a silent one does not.

BREAK IT DOWN

GIVEN	100 students in the group. 72 speak Ibibio. 43 speak Efik, on the stated reading.
ASKED	(i) both languages, (ii) Ibibio only , (iii) Efik only . Three different regions of the same diagram.
HIDDEN ASSUMPTION	That every one of the 100 speaks at least one of the two, so $n(A \cup B) = 100$. The question does not say so, but without it the three parts have no unique answer. State this too.
MARKS SAY	3, 2 and 2. Part (i) is worth more because the other two follow from it in one subtraction each. Get (i) right and the rest are free.
TEACHES	This section, 5.3.
TRAP	Answering (ii) with 72. That is everyone who speaks Ibibio, including those who also speak Efik. "Only" means the overlap must come out.

GIVEN $n(U) = 100$, $n(I) = 72$ speak Ibibio, $n(E) = 43$ speak Efik.

FIND $n(I \cap E)$, then $n(I \text{ only})$ and $n(E \text{ only})$.

FORMULA $n(I \cup E) = n(I) + n(E) - n(I \cap E)$.

Step 0. State the two readings you are working under.

1. The second figure of 43 refers to Efik, not Ibibio.
2. Every student speaks at least one of the two languages, so $n(I \cup E) = 100$.

Assumption 2 is forced by the arithmetic as well as by common sense: $72 + 43 = 115$, which is 15 more than the group, so at least 15 students must be in both. If some students spoke neither, the overlap would be even larger, and the question gives no way to know how many.

i. Students who speak both

Rearrange the two-set formula to make the overlap the subject:

$$\begin{aligned} n(I \cap E) &= n(I) + n(E) - n(I \cup E) \\ &= 72 + 43 - 100 \\ &= 115 - 100 \\ &= \mathbf{15} \end{aligned}$$

Answer: 15 students speak both Ibibio and Efik.

ii. Students who speak Ibibio only

The 72 Ibibio speakers include the 15 who also speak Efik. Strip them out:

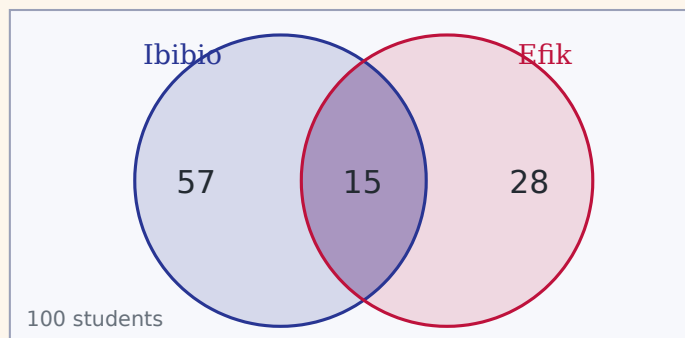
$$n(I \text{ only}) = n(I) - n(I \cap E) = 72 - 15 = \mathbf{57}$$

iii. Students who speak Efik only

$$n(E \text{ only}) = n(E) - n(I \cap E) = 43 - 15 = \mathbf{28}$$

Check, and write the check down. The three regions must fill the group exactly:

$$57 + 15 + 28 = 100 \quad \checkmark$$



$$57 + 15 = 72$$

$$28 + 15 = 43$$

$$\begin{aligned} 57 + 15 + 28 \\ = 100 \end{aligned}$$

The finished diagram. Every given number can be read back off it, which is what makes it a check and not a decoration.

Drawing this diagram is worth doing even when the question does not ask for it. It takes twenty seconds and it catches the "only" trap automatically.

Now redo it: in a group of 80, 50 speak Yoruba and 45 speak Hausa, everyone speaking at least one. Both: $50 + 45 - 80 = 15$. Yoruba only: 35. Hausa only: 30. Check: $35 + 15 + 30 = 80$.

The class-size question

AS PRINTED

2021/2022 · Q5(B)

(b) In a discrete structures' class, every student is a major in either Computer Science or Maths or both. There of 25 students who major in Computer, while 13 in Maths. If 8 students major in both Computing Science & Maths, how many students are in the entire class? **(4marks)**

NOTE ON THE PAPER

"There of 25 students" is a typing slip for "There are 25 students", and the subject is called "Computer Science" in one clause, "Computer" in the next and "Computing Science" in the third. All three name the same set. The sentence that saves the question is the first one: **EVERY STUDENT MAJORS IN COMPUTER SCIENCE OR MATHS OR BOTH**, which states outright the assumption that the Ibibio question left you to supply. So here $n(A \cup B)$ is the class, and it is what you are asked for.

BREAK IT DOWN

- GIVEN** $n(C) = 25$ major in Computer Science, $n(M) = 13$ major in Maths, $n(C \cap M) = 8$ major in both, and nobody majors in neither.
- ASKED** **How many students are in the entire class**, which is $n(C \cup M)$.
- MARKS SAY** 4 marks for one formula. Write the principle, then substitute, then evaluate, then answer in a sentence. A bare 30 earns less.
- TEACHES** This section, 5.3.
- TRAP** Answering $25 + 13 = 38$, forgetting the 8 who were counted in both figures.

WORKED EXAMPLE

PQ 21/22 Q5(B)

- GIVEN** $n(C) = 25$, $n(M) = 13$, $n(C \cap M) = 8$, every student in at least one.
- FIND** $n(C \cup M)$, the size of the class.
- FORMULA** $n(C \cup M) = n(C) + n(M) - n(C \cap M)$.

Because every student majors in one subject or the other or both, the class *is* the union, so the union is what to compute:

$$\begin{aligned}n(C \cup M) &= 25 + 13 - 8 \\&= 38 - 8 \\&= 30\end{aligned}$$

Answer: there are 30 students in the entire class.

Break it into regions as a check: Computer Science only is $25 - 8 = 17$, Maths only is $13 - 8 = 5$, and both is 8. Then $17 + 5 + 8 = 30$, which agrees.

One ambiguity worth a line on the script. The sentence "There of 25 students who major in Computer" could be misread as saying the class has 25 students in total. It cannot mean that: 8 of them major in both, and 13 major in Maths, so Maths-only alone would need 5 more students than a 25-strong class has room for once the 25 Computer Science majors are placed. The reading above is the only consistent one.

Now redo it: if instead 30 major in Computer Science, 20 in Maths and 12 in both, the class has $30 + 20 - 12 = 38$ students.

The pigeonhole principle

THE PIGEONHOLE PRINCIPLE

The claim. If more objects are placed into boxes than there are boxes, then some box must hold more than one object. Nothing about the objects, the boxes or the way they are placed can prevent it.

In symbols.

If $k + 1$ or more objects go into k boxes, some box holds at least 2.

If N objects go into k boxes, some box holds at least $\lceil N / k \rceil$.

Conditions. The number of boxes must be finite, and every object must go into some box. It says nothing about **which** box is crowded, and it cannot: that is a feature, since it lets you prove something exists without being able to find it.

The limiting case. When $N = k$ the principle promises only $\lceil k / k \rceil = 1$, which is no information at all, and correctly so: k objects can sit one to a box with nothing shared. The principle bites from $N = k + 1$ onward, and that single extra object is the whole of it.

● MUST MEMORISE

PIGEONHOLE PRINCIPLE

If $k + 1$ or more objects are placed into k boxes, then at least one box contains two or more objects.

WHERE

k is the number of boxes, sometimes called pigeonholes. **objects** are whatever is being sorted into them: people, numbers, letters, socks.

The generalised form, which is the one that answers real questions: if N objects are placed into k boxes, then at least one box contains at least

$\lceil N / k \rceil$ objects,

where $\lceil x \rceil$ is the **ceiling** of x : the smallest whole number that is not less than x . So $\lceil 7 / 3 \rceil = \lceil 2.33 \rceil = 3$.

◆ **Memory hook: more pigeons than holes means sharing.** The principle never tells you *which* box is crowded, only that one of them is. That is exactly why it is useful: it proves something exists without finding it.

Every pigeonhole argument has the same three steps, and the marks are in step 2.

1. **Name the objects.** What is being placed?
2. **Name the boxes.** This is the creative step. The boxes are usually the possible *values* of something, not physical containers.
3. **Compare the counts** and apply the principle.

Example. Among any 13 people, at least two were born in the same month. Objects: the 13 people. Boxes: the 12 months. Since $13 > 12$, two people share a box, that is, a birth month.

Example, generalised form. Among 100 people, at least $\lceil 100 / 12 \rceil = \lceil 8.33 \rceil = 9$ share a birth month.

Example with an unobvious box. In any group of 8 people, two have the same number of friends within the group. Objects: 8 people. Boxes: the possible friend counts, 0 to 7, which is 8 boxes, so the counting alone is not enough. The extra step: nobody can have 0 friends and somebody else have 7 at the same time, because a person with 7 friends is friends with everyone, including the supposed loner. So only 7 of the 8 boxes can be occupied, and 8 people into 7 boxes forces a repeat.

Where this stands on the papers: none of the five papers we hold has set a pigeonhole question outright. It is in the official course text, it takes two minutes to learn, and the 2023/2024 paper does ask a closely related impossibility question about five cities that is settled by a counting argument in Part Six. Learn the statement and the three steps; that is the whole of the return on it.

RECALL AND CHECK

1. State the inclusion and exclusion principle for two sets.
2. 40 students take Maths, 30 take Physics, 12 take both and everyone takes at least one. How many students are there?
3. In the same group, how many take Maths only?
4. Write the three-set formula from the pattern, without looking.
5. State the pigeonhole principle, and say what $\lceil 17 / 5 \rceil$ is.

ANSWERS

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.
2. $40 + 30 - 12 = 58$.
3. $40 - 12 = 28$.
4. Add the three singles, subtract the three pairs, add the triple.
5. If $k + 1$ or more objects go into k boxes, some box holds two or more. $\lceil 17 / 5 \rceil = \lceil 3.4 \rceil = 4$.

FILL IN THE GAP

1. $n(A \cup B) = n(A) + n(B) - \underline{\hspace{2cm}}$.
2. The number who speak Ibibio only is 72 minus $\underline{\hspace{2cm}}$, giving $\underline{\hspace{2cm}}$.
3. In the three-set formula the pairs are $\underline{\hspace{2cm}}$ and the triple is $\underline{\hspace{2cm}}$.
4. Placing 10 objects into 3 boxes forces some box to hold at least $\underline{\hspace{2cm}}$.

ANSWERS

1. $n(A \cap B)$. 2. 15, 57. 3. subtracted, added. 4. $\lceil 10 / 3 \rceil = 4$.

PART FIVE · 5.4

The Binomial Theorem

Expanding a bracket raised to a power is a counting problem in disguise, and the numbers that come out are the combinations you met in section 5.2.

By the end of this section you will be able to read the binomial coefficient notation, state and apply the binomial theorem, build and use Pascal's triangle, pick out a single term without expanding the whole thing, and explain why the coefficients of a row add to a power of two.

TEACH

WHERE THIS STANDS ON THE PAPERS

Say this plainly before spending time here. **None of the five papers we hold sets a binomial theorem question.** It is a full module of the official course text, so it is examinable and cheap to learn, but it should be the last thing you revise, not the first.

It earns its place for a second reason: it is the clearest proof that combinations are not just a formula for committees. The coefficients of an expansion *are* counts of selections, and seeing that makes section 5.2 harder to forget.

The binomial coefficient

● MUST MEMORISE

BINOMIAL COEFFICIENT

The **binomial coefficient** is the combination count of section 5.2 under a second name and a second notation. For whole numbers n and r with $0 \leq r \leq n$,

$C(n, r) = n! / (r! (n - r)!)$, also written **$\binom{n}{r}$** and read " **$n$ choose r** ".

WHERE

n is the power the bracket is raised to. **r** selects which term of the expansion you are looking at, counting from 0. " **n choose r** " is the spoken form. When a question says "choose", it means this.

Two identities worth knowing on sight.

$C(n, r) = C(n, n - r)$ [choosing r to keep is choosing $n - r$ to leave]

$C(n, 0) = C(n, n) = 1$ [one way to take none, one way to take all]

◆ **Memory hook: the coefficient counts the ways.** The number in front of a term is the number of ways of picking that many copies of one letter from the brackets.

The theorem

● MUST MEMORISE

BINOMIAL THEOREM

For any numbers x and y and any whole number n that is not negative,

$$(x + y)^n = C(n, 0) x^n + C(n, 1) x^{n-1} y + C(n, 2) x^{n-2} y^2 + \dots + C(n, n) y^n$$

The general term, the one to quote when a question asks for a single term rather than the whole expansion, is

$$\text{term } r = C(n, r) x^{n-r} y^r, \text{ for } r = 0, 1, 2, \dots, n.$$

WHERE

r counts from 0, so the expansion has $n + 1$ terms, not n . **the exponents** always add to n : as the power of x falls, the power of y rises by the same step.

◆ **Memory hook: the powers see-saw and the exponents always sum to n .** If a line of your working breaks that rule, it is wrong, and you can see it without checking anything else.

TEACH

WHY THE COEFFICIENTS ARE COMBINATIONS

Write $(x + y)^3$ as three brackets side by side and do not collect anything yet:

$$(x + y)(x + y)(x + y)$$

Multiplying out means going through the three brackets and taking either x or y from each. Every choice gives one product of three letters, and every such product is a term.

How many of those products come out as x^2y ? Exactly as many as there are ways of picking **which one** of the three brackets supplied the y , and that is $C(3, 1) = 3$. So the x^2y term is $3x^2y$, and it is a combination count, not a coincidence.

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

This also explains the second identity above. Choosing which brackets give y is the same job as choosing which give x , so $C(3, 1) = C(3, 2)$ and the coefficients read the same forwards and backwards.

Pascal's triangle

TEACH

GETTING THE COEFFICIENTS WITHOUT FACTORIALS

Write $C(n, r)$ in rows, n going down and r across. Each row starts and ends with 1, and every other entry is the sum of the two directly above it. That triangle is the fastest way to get the coefficients for small n , which is all any exam will ask for.

sum 1			1				$n = 0$
sum 2			1		1		$n = 1$
sum 4			1		2		$n = 2$
sum 8			1		3		$n = 3$
sum 16		1		4		6	$n = 4$
sum 32	1		5		10		$n = 5$

$4 + 6 = 10$

Row n gives the coefficients of $(x + y)$ to the power n . Each entry is the sum of the two above it, and each row adds to a power of two.

The rule behind the triangle, called Pascal's identity, is

$$C(n, r) = C(n - 1, r - 1) + C(n - 1, r)$$

and it says something sensible about choosing. Fix one particular object. Any selection of r either takes that object, leaving $r - 1$ to pick from the other $n - 1$, or it does not, leaving all r to pick from the other $n - 1$. Those two cases are alternatives with no overlap, so the sum rule of section 5.1 adds them.

● MUST MEMORISE

THE ROW SUM

The coefficients of row n add to 2^n :

$$C(n, 0) + C(n, 1) + \dots + C(n, n) = 2^n$$

WHERE

n is the row number, counting from 0 at the apex of the triangle. $C(n, r)$ is the entry r places along that row, also counting from 0. 2^n is 2 multiplied by itself n times, so row 4 adds to 16 and row 5 to 32.

Why, in one line. Put $x = 1$ and $y = 1$ in the binomial theorem. The left side becomes $(1 + 1)^n = 2^n$ and every power on the right becomes 1, leaving just the coefficients.

And what it means. $C(n, r)$ counts the subsets of size r of a set with n elements, so adding over every r counts *all* the subsets. That is 2^n , which is exactly the size of the power set from section 2.3. Two sections of this book, arrived at from opposite directions, meeting on the same number.

◆ **Memory hook: every row doubles.** 1, 2, 4, 8, 16, 32. If your coefficients do not add to the next power of two, one of them is wrong.

Using it

WORKED EXAMPLE

FULL EXPANSION

Expand $(x + 2)^4$.

GIVEN The bracket $x + 2$ raised to the power 4, so $n = 4$ and the second term is $y = 2$.

FIND The expansion in full.

FORMULA term $r = C(4, r) x^{4-r} 2^r$.

Step 1. Read the coefficients off row 4: 1, 4, 6, 4, 1.

Step 2. Build each term. The power of x falls from 4 to 0 and the power of 2 rises from 0 to 4.

R	C(4, R)	x^{4-R}	2^R	TERM
0	1	x^4	1	x^4
1	4	x^3	2	$8x^3$
2	6	x^2	4	$24x^2$
3	4	x	8	$32x$
4	1	1	16	16

$$(x + 2)^4 = x^4 + 8x^3 + 24x^2 + 32x + 16$$

Check it, and this check costs one line. Put $x = 1$ on both sides. The left is $(1 + 2)^4 = 3^4 = 81$, and the right is $1 + 8 + 24 + 32 + 16 = 81$. They agree, so no coefficient was dropped.

Now redo it: expand $(x + 1)^5$. Row 5 is 1, 5, 10, 10, 5, 1 and every power of 1 is 1, so it is $x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1$. Check at $x = 1$: the sum is $32 = 2^5$.

Find the coefficient of x^3 in $(2x + 1)^5$.

Expanding all six terms would work and would waste four minutes. Use the general term instead and solve for r .

GIVEN $n = 5$, first term $2x$, second term 1 .

FIND The number multiplying x^3 .

FORMULA term $r = C(5, r) (2x)^{5-r} 1^r$.

Step 1. Find which r gives x^3 . The power of x in term r is $5 - r$, so

$$5 - r = 3, \text{ so } r = 2$$

Step 2. Evaluate that one term.

$$\begin{aligned} \text{term } 2 &= C(5, 2) (2x)^3 1^2 \\ &= 10 \times 8x^3 \times 1 \\ &= 80x^3 \end{aligned}$$

Answer: the coefficient of x^3 is 80.

The mistake to avoid is taking the coefficient to be $C(5, 2) = 10$ and forgetting that the 2 in $2x$ is also raised to the power 3. Whenever the bracket has a number attached to a letter, that number carries its own power.

Now redo it: the coefficient of x^2 in $(x + 3)^4$. Here $4 - r = 2$ so $r = 2$, and the term is $C(4, 2) x^2 3^2 = 6 \times 9 x^2 = 54x^2$.

RECALL AND CHECK

1. State the binomial theorem.
2. How many terms has the expansion of $(x + y)^7$?
3. Write row 4 of Pascal's triangle and say what its entries add to.
4. Expand $(x + 2)^4$.
5. Why is $C(n, r) = C(n, n - r)$?

ANSWERS

1. $(x + y)^n$ is the sum over r of $C(n, r) x^{n-r} y^r$, for r from 0 to n .
2. Eight. The terms run $r = 0$ to $r = 7$, so there are $n + 1$ of them.
3. 1, 4, 6, 4, 1, adding to $16 = 2^4$.
4. $x^4 + 8x^3 + 24x^2 + 32x + 16$.
5. Choosing which r objects to take is the same act as choosing which $n - r$ to leave behind, so the two counts are equal.

FILL IN THE GAP

1. In every term of $(x + y)^n$ the two exponents add to _____.
2. The expansion of $(x + y)^n$ has _____ terms.
3. Row n of Pascal's triangle adds to _____.
4. The coefficient of x^3 in $(2x + 1)^5$ is _____.

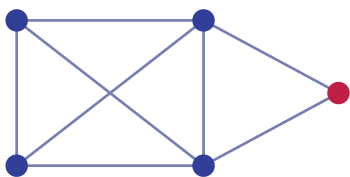
ANSWERS

1. n . 2. $n + 1$. 3. 2^n . 4. 80.

PART SIX

Graph Theory: Vertices and Edges

The heaviest part of the exam and the emptiest part of the course text. Every paper we hold gives graphs at least a third of its marks, and the official text does not define a vertex. Everything here is taught from zero.



$\deg(v)$ degree of a vertex

$\sum \deg = 2e$ handshaking

$v - e + f = 2$ planarity

$K_5, K_{3,3}$ the non-planar pair

Graphs, Vertices, Edges and Degree

A graph is dots and lines. Almost every mark in this part comes from naming the dots and lines precisely, so the vocabulary is the subject, not the preamble.

By the end of this section you will be able to define a graph, a vertex and an edge, count the order and the size of a graph, compute the degree of a vertex, say what incidence and adjacency mean, name loops, parallel edges, pendant and isolated vertices, and answer the incidence and short-notes questions in full.

NOT IN THE COURSE TEXT

The official 161-page text contains no graph theory at all. It never uses the word vertex, never defines an edge, and never states the Handshaking theorem. Meanwhile 2025/2026 gives graphs roughly 30 of its marks, 2024/2025 gives a whole 17.5-mark question to one graph, and 2020/2021, 2021/2022 and 2023/2024 each build two questions on them. This part is your only source, so nothing here assumes you have seen a graph before.

What a graph is

● MUST MEMORISE

GRAPH, VERTEX, EDGE

A **graph** $G = (V, E)$ consists of a set V of **vertices** and a set E of **edges**.

A **vertex** (plural **vertices**) is a point, drawn as a dot or a small circle, standing for an object: a city, a computer, a person, a state of a program. A vertex is also called a **node**.

An **edge** is a line that joins two vertices, standing for a relationship between the objects: a road, a cable, a friendship, a transition.

WHERE

V is the vertex set, and it must not be empty: a graph has at least one vertex. E is the edge set, and it may be empty. A graph with no edges is still a graph. $G = (V, E)$ is the standard way of naming a graph by its two parts, and it is worth writing out in an answer.

◆ **Memory hook: dots and lines, objects and relationships.** The picture is not the graph. Two very different-looking drawings can be the same graph, because only which dots are joined matters, never where they sit on the page.

● MUST MEMORISE

ORDER AND SIZE

The **order** of a graph is the number of vertices, written $|V|$ or n .

The **size** of a graph is the number of edges, written $|E|$ or e .

WHERE

$|V|$ and $|E|$ are cardinalities, the same bars used for set size in section 2.1. **order** counts dots. **size** counts lines. The two words are not interchangeable, and 2025/2026 Q4(a) pays 2 marks for knowing which is which.

◆ **Memory hook: order counts the objects, size counts the connections.** Everyday English suggests "size" should mean how big the graph is; in graph theory it means how many edges it has. Learn it against your instinct.

Degree, and the words that go with it

● MUST MEMORISE

DEGREE AND VALENCY

The **degree of a vertex** v , written **$\deg(v)$** , is the number of edges meeting v , with a loop counted twice.

Valency is another word for exactly the same thing. Some papers use one word, some the other, and 2025/2026 asks you to tell them apart. They are the same quantity under two names.

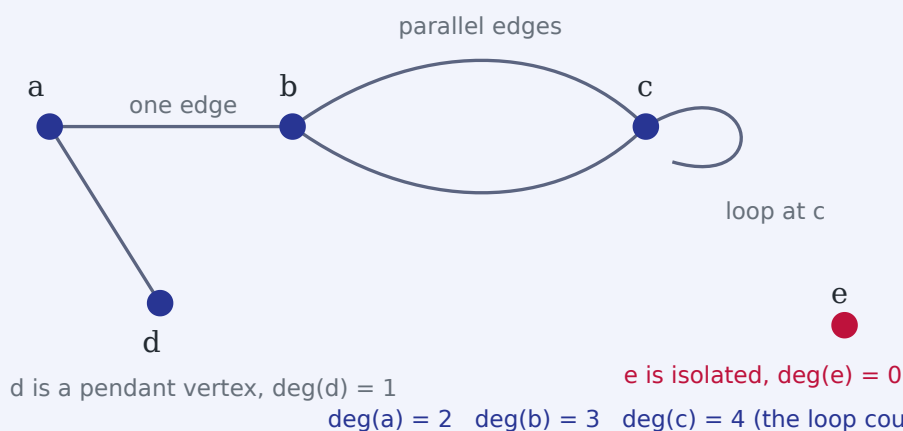
WHERE

$\deg(v)$ is a whole number, 0 or more. **a loop** at v counts **2**, because both of its ends meet v . **parallel edges** each count separately: two edges between the same pair contribute 2 to each endpoint.

◆ **Memory hook: count the line-ends touching the dot, not the neighbours.** A vertex joined to one neighbour by three parallel edges has degree 3, not 1.

TERM	MEANING	TEST FOR IT
incident	A vertex is incident to an edge when the edge has that vertex as one of its two ends.	Does the line touch the dot?
adjacent	Two vertices are adjacent when an edge joins them.	Is there a line between the two dots?
loop	An edge whose two ends are the same vertex.	Does the line start and finish at one dot?
parallel edges	Two or more edges joining the same pair of vertices. Also called multiple edges.	More than one line between the same two dots?
pendant vertex	A vertex of degree 1. Also called a leaf or an end vertex.	Exactly one line touches it?
isolated vertex	A vertex of degree 0.	No line touches it at all?

Incidence and adjacency are not the same relation, and the exam asks about incidence specifically. **Incidence relates a vertex to an edge; adjacency relates a vertex to another vertex.** Saying "the vertices are incident" is wrong, and so is "the vertex is adjacent to the edge".



Every term at once. Note that c has degree 4: two parallel edges to b, plus a loop worth two.

Degree and valency, told apart

a) With the aid of two examples each, differentiate between the following:

- | | |
|--|-----------|
| i) Degree and Valency of a node in a graph | (2 marks) |
| ii) Predicate Logic and Propositional logic | (2 marks) |
| iii) Universal Quantifier and Existential Quantifier | (2 marks) |
| iv) Adjacency and Path Matrice | (2 marks) |
| v) A logical matrix and relation matrix | (2marks) |

NOTE ON THE PAPER

Three defects, all harmless once seen. Item ii) prints "Prepositional logic"; the word is **PROPOSITIONAL**, since a preposition is a part of speech and has nothing to do with logic. Item iv) prints "Path Matrice", a slip for "Path Matrix" or "Path Matrices". Item v) prints "(2marks)" closed up, unlike the four above it. Quote them as printed and answer the intended questions.

This section answers i). Item ii) is section 1.4, item iii) is section 1.4, and items iv) and v) are section 6.5. The 10 marks are spread across three parts of this book, which is worth knowing before you start writing.

BREAK IT DOWN

GIVEN Five pairs of terms, 2 marks each.

ASKED **Differentiate, with the aid of two examples each.** The examples are not optional decoration: they are named in the instruction, so an answer without them cannot score full marks.

MARKS SAY 2 marks per pair. Read as roughly 1 for the distinction and 1 for the two examples. Keep each answer to four or five lines and move on; there are ten items on this question.

TEACHES Item i) here in 6.1. See 1.4 for ii) and iii), and 6.5 for iv) and v).

TRAP Item i) is a trick in the sense that there is **no** numerical difference between degree and valency. Saying so confidently, and explaining where each word comes from, is the answer. Inventing a difference loses the marks.

Degree and valency of a node in a graph.

The distinction, stated plainly. They are two names for one quantity: the number of edges meeting the node, a loop counting twice. There is no graph and no node for which the degree and the valency differ. The difference is one of usage, not of value:

	DEGREE	VALENCY
Where the word comes from	graph theory and mathematics	chemistry, where an atom's valency is the number of bonds it forms
How it is written	$\deg(v)$, or $d(v)$	usually written out in words
Where you will meet it	the Handshaking theorem, degree sequences, most textbooks	older texts, chemical graph theory, and this course's own papers

Two examples, as the question demands.

1. In the figure above, vertex b is met by one edge to a and two parallel edges to c . So **$\deg(b) = 3$** , and equally the **valency of b is 3**.
2. Vertex c carries two parallel edges to b and one loop. The loop contributes 2, so **$\deg(c) = 2 + 2 = 4$** , and the **valency of c is 4**. Both words give the same 4, and both would give the same 4 for any vertex you choose.

The sentence that earns the mark: "Degree and valency are synonyms for the number of edge-ends at a vertex; the terms differ only in origin, degree being the graph-theoretic word and valency the term borrowed from chemistry."

Now redo it: what are the degree and the valency of the pendant vertex d ? Both are 1, and both would be 0 for the isolated vertex e .

Short notes: size, valency, and the rest

(a) With relevant examples, write short notes on the following

- | | |
|--|-----------|
| (i) Size of a graph | (2 marks) |
| (ii) The valency of a vertex in a graph | (2 marks) |
| (iii) In-degree and Out-degree of a vertex | (2 marks) |
| (iv) Graph isomorphism | (2 marks) |
| (v) Eulerian graph | (2 marks) |

NOTE ON THE PAPER

Ten marks in five two-mark notes, and they run right across this part: (i) and (ii) belong here, (iii) is section 6.4, and (iv) and (v) are section 6.6. Each answer wants a definition plus a relevant example, and two marks buys about four lines. Items (i) and (ii) are answered below; the other three are answered where they are taught, and cross-referenced back here.

(i) Size of a graph

Definition. The size of a graph $G = (V, E)$ is the number of edges it has, written $|E|$. It is not the number of vertices; that is the **order**, $|V|$.

Example 1. The graph in the figure above has edges ab , bc , bc again, the loop at c , and ad . That is five edges, so its size is 5. Its order is 5 as well, since it has vertices a , b , c , d and e , and the coincidence is worth pointing out precisely because it shows the two counts are different questions.

Example 2. A graph on 4 vertices with every pair joined has $C(4, 2) = 6$ edges, so its size is 6 and its order is 4.

(ii) The valency of a vertex in a graph

Definition. The valency of a vertex is the number of edges incident to it, a loop counting twice. It is the same quantity as the degree, $\deg(v)$.

Example 1. In a triangle, every vertex is joined to the other two, so every vertex has valency 2.

Example 2. In a star with one centre joined to four outer vertices, the centre has valency 4 and each outer vertex has valency 1, so each outer vertex is a pendant vertex.

Incidence

AS PRINTED

2020/2021 · Q1(C)

(c) When is a Vertex said to be Incident to an Edge? Identify all the edges each of the vertices is incident to in the diagram below. (5marks)

AS PRINTED

2021/2022 · Q5(C)

(c) When is a Vertex said to be Incident to an Edge? Show an example (3½marks)

NOTE ON THE PAPER, AND ON THE FIGURE

The same question on two papers, one wanting a worked figure for 5 marks and the other an example for 3.5. The definition is identical; only the second half changes.

THE 2020/2021 FIGURE CANNOT BE FULLY READ. It shows four vertices numbered 1 to 4 and six edges labelled e_1 to e_6 . Vertices 1, 2 and 3 and the edges e_1 to e_5 are clear, including that e_1 and e_2 are parallel, as are e_4 and e_5 . The sixth edge, e_6 , is drawn as a long curve on the right passing around vertex 4, and at the resolution we hold it cannot be settled whether it is a loop at vertex 2 or a second path from 2 to 3. Vertex 4 is joined to nothing either way, which is plainly the point the question is testing. The answer below states the reading it uses, tabulates on that reading, and shows what changes under the other. **READ THE ARROWLESS CURVE ON YOUR OWN COPY BEFORE COMMITTING.**

BREAK IT DOWN

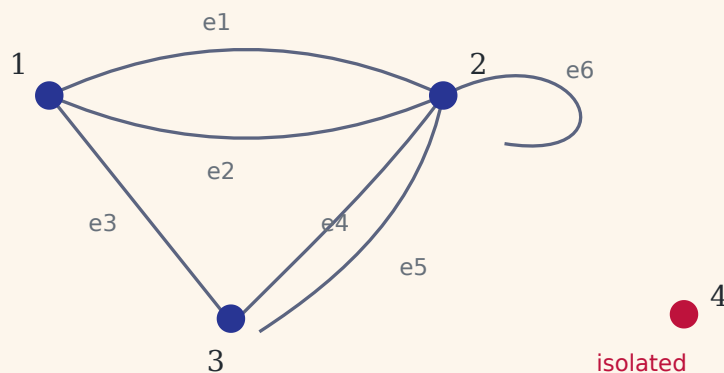
GIVEN	A graph with vertices 1, 2, 3, 4 and edges e_1 to e_6 .
ASKED	Two things. When a vertex is said to be incident to an edge, which is a definition; and identify all the edges each vertex is incident to , which is a table with one row per vertex.
MARKS SAY	5 marks: about 1.5 for the definition and 3.5 for the four rows. Do not answer only the first half; it is worth under a third of the marks.
TEACHES	This section, 6.1.
TRAP	Skipping vertex 4 because nothing touches it. Its row is the one the examiner is watching for: it is incident to no edges, so it is an isolated vertex, and saying so is worth a mark on its own.

- GIVEN** Vertices 1, 2, 3, 4. Edges e1 and e2 both joining 1 and 2; e3 joining 1 and 3; e4 and e5 both joining 2 and 3; e6 a loop at 2 on the reading used here.
- FIND** The definition of incidence, then every edge at every vertex.
- RULE** A vertex is incident to an edge exactly when it is one of that edge's two endpoints.

Part 1: the definition. A vertex v is said to be **incident** to an edge e when v is one of the two endpoints of e ; equivalently, when e touches v in the drawing. The relation runs both ways in speech: we say that e is incident to v , and that v is incident to e . For a loop, the single vertex is incident to the edge, and the edge contributes 2 to that vertex's degree.

Part 2: the table. State the reading first, then tabulate.

Reading used: e6 is a loop at vertex 2.



The figure as read, with e6 taken as a loop at vertex 2 and vertex 4 joined to nothing.

VERTEX	EDGES IT IS INCIDENT TO	DEGREE
1	e1, e2, e3	3
2	e1, e2, e4, e5, e6 (the loop, twice)	6
3	e3, e4, e5	3
4	none; vertex 4 is an isolated vertex	0

Check the total. The degrees add to $3 + 6 + 3 + 0 = 12$, which is twice the six edges. That is the Handshaking theorem of section 6.3, and it is the fastest way to catch a miscounted table.

Under the other reading, where e6 joins 2 to 3 rather than looping, vertex 2 is incident to e1, e2, e4, e5, e6 with degree 5 and vertex 3 to e3, e4, e5, e6 with degree 4. The degrees then add to $3 + 5 + 4 + 0 = 12$, still twice six, so the check passes either way. Only the two middle rows change.

For the 2021/2022 version, which asks only to "Show an example", one edge and its two vertices is enough: draw an edge e joining u and v , and write " u is incident to e and v is incident to e ; u and v are adjacent". Add a loop as a second example to show you know it counts twice.

Now redo it: in a triangle with vertices x, y, z and edges $f_1 = xy, f_2 = yz, f_3 = zx$, which edges is y incident to? f_1 and f_2 , so $\deg(y) = 2$.

Where graphs are used

AS PRINTED

2020/2021 · Q2(B)

(b) Highlight, with two relevant examples each, four real-life scenarios where graph models are applicable. **(6marks)**

AS PRINTED

2020/2021 · Q3(C)

(c) Highlight, with a relevant example each, four application areas in computing where graph models are applicable. **(8marks)**

AS PRINTED

2021/2022 · Q3(B)

(b) Highlight, with two relevant examples each, four real-life scenarios where graph models are applicable. **(6marks)**

AS PRINTED

2021/2022 · Q5(A)

(a) Highlight, with a relevant example each, four computing domains where graph models are applicable. **(6marks)**

NOTE ON THE PAPER

One question in three costumes, worth 6, 8 and 6 marks, and 2021/2022 Q3(b) repeats the first of them word for word. Two things change between them and both change your answer. **REAL-LIFE** versus **COMPUTING**: the first wants roads and social networks, the second wants compilers and routing tables. And **TWO EXAMPLES EACH** versus **A RELEVANT EXAMPLE EACH**: 2020/2021 Q2(b) asks for eight examples in total, the others for four. Count them before you start writing.

Answer every version from one table, taking the top half for the real-life question and the bottom half for the computing one. In each case say what the vertices are and what the edges are; that is the modelling step, and it is what separates a full answer from a list of words.

Four real-life scenarios, with two examples each

SCENARIO	VERTICES	EDGES	TWO EXAMPLES
Transport networks	towns, junctions, stations	roads, rail lines, flight routes	a road map used to find the shortest drive; an airline route map used to plan connections
Social networks	people	friendships, follows, calls	a friend graph used to suggest people you may know; a call graph used to spot a fraud ring
Utility distribution	substations, reservoirs, junctions	power lines, pipes	an electricity grid checked for a route that survives one cable failing; a water network sized to reach every district
Scheduling and allocation	tasks, exams, rooms	a conflict between two of them	an examination timetable where joined exams may not share a slot; radio frequencies assigned so neighbouring masts differ

Four computing application areas, with an example each

AREA	VERTICES	EDGES	EXAMPLE
Computer networks	routers and hosts	links between them	a routing table computed as shortest paths across the link graph
Compilers	basic blocks of a program	a jump or fall-through from one block to another	the control-flow graph a compiler walks to decide which code is unreachable
Databases and the web	pages, records, entities	hyperlinks, foreign keys	a search engine ranking pages by how the link graph points at them
Artificial intelligence and search	states of a problem	a legal move from one state to the next	a route planner or a game solver searching the state graph for a goal

One closing line worth writing, because it shows why the topic exists rather than just listing where it appears: in every one of these, the objects and the connections are all that matter, and the geometry of the drawing is irrelevant. That is precisely what a graph abstracts.

RECALL AND CHECK

1. Define a graph, a vertex and an edge.
2. What is the difference between the order and the size of a graph?
3. How much does a loop add to a vertex's degree, and why?
4. When is a vertex incident to an edge?
5. What is a pendant vertex, and what is an isolated vertex?
6. Is there any graph in which a vertex's degree and its valency differ?

ANSWERS

1. A graph $G = (V, E)$ is a set V of vertices, which are points standing for objects, together with a set E of edges, each joining two vertices and standing for a relationship.
2. Order is the number of vertices, $|V|$. Size is the number of edges, $|E|$.
3. Two, because both of its ends meet that vertex.
4. When the vertex is one of the edge's two endpoints.
5. A pendant vertex has degree 1; an isolated vertex has degree 0.
6. No. They are two names for the same number.

FILL IN THE GAP

1. A graph is written $G = (\text{_____}, \text{_____})$.
2. The number of edges is the _____ of the graph.
3. A vertex of degree 1 is called a _____ vertex.
4. Two vertices joined by an edge are _____; a vertex at the end of an edge is _____ to it.
5. Valency is another word for _____.

ANSWERS

1. V, E . 2. size. 3. pendant. 4. adjacent, incident. 5. degree.

PART SIX · 6.2

The Types of Undirected Graph

This is the single most repeated question in the whole set. It has appeared on three of the five papers we hold, worth 12, 12 and 9 marks, and it is always phrased the same way.

By the end of this section you will be able to name and draw every standard type of undirected graph, count the edges of a complete graph, decide which type a given drawing is, and answer the types question and the communication-links questions in full.

Three papers, one question:

PAPER	AS ASKED	MARKS
2020/2021 Q1(a)	with the aid of relevant examples	12
2021/2022 Q1(c)	with diagrammatical illustrations	12
2023/2024 Q4(a)	in details, and with diagrammatic illustration	9

That is 33 marks across three papers for one prepared answer. Every version demands **drawings**, so an answer in words alone throws away most of the marks however correct it is. Learn the list, and practise sketching all of it in under ten minutes.

The types

● MUST MEMORISE

THE FAMILIES BY WHAT EDGES ARE ALLOWED

The first three types are told apart by one question: **what kinds of edge does the graph permit?**

TYPE	LOOPS?	PARALLEL EDGES?	IN ONE LINE
Simple graph	no	no	at most one edge between any two vertices, and none from a vertex to itself
Multigraph	no	yes	a simple graph that also allows repeated edges between the same pair
Pseudograph	yes	yes	anything goes: the most general undirected graph

WHERE

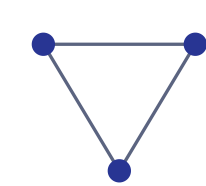
loop is an edge from a vertex to itself, defined in section 6.1. **parallel edges** are two or more edges joining the same pair, also from 6.1. **the nesting** matters: every simple graph is a multigraph, and every multigraph is a pseudograph. The families contain one another, they do not exclude one another.

◆ **Memory hook: simple, multi, pseudo.** Simple forbids both, multi allows repeats, pseudo allows repeats and self-loops. Say them in that order and the permissions only ever grow.

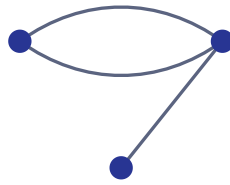
The remaining types are told apart by **shape** rather than by permission.

TYPE	DEFINITION	SYMBOL
Null (empty) graph	Has vertices but no edges at all. Every vertex is isolated.	N_n
Trivial graph	Has exactly one vertex and no edges. The smallest graph there is.	K_1
Complete graph	Simple, and <i>every</i> pair of distinct vertices is joined by an edge.	K_n
Regular graph	Every vertex has the same degree. If that degree is k it is k-regular .	
Cycle graph	n vertices joined in a single closed ring; 2-regular.	C_n
Path graph	n vertices in a line, not closed up.	P_n
Wheel graph	A cycle C_n plus one extra vertex joined to every vertex of the cycle.	W_n
Bipartite graph	The vertices split into two sets so that every edge runs between the sets and none inside either.	
Complete bipartite	Bipartite, and every vertex of one set is joined to every vertex of the other.	$K_{m,n}$
Connected graph	There is a route between every pair of vertices. Otherwise disconnected , and it falls into components .	
Weighted graph	Every edge carries a number: a distance, a cost, a capacity.	
Subgraph	A graph made from some of another graph's vertices and some of its edges.	$H \subseteq G$

◆ **Memory hook: K for complete, C for cycle, P for path, W for wheel, N for null.** The subscript is always the number of vertices, except in $K_{m,n}$ where it is the two side sizes.



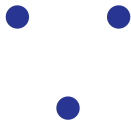
Simple



Multigraph



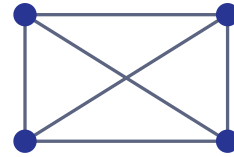
Pseudograph



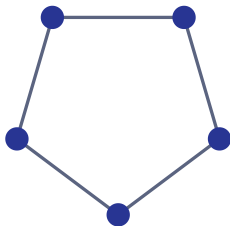
Null N_3



Trivial K_1



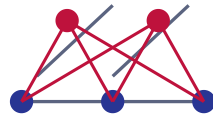
Complete K_4



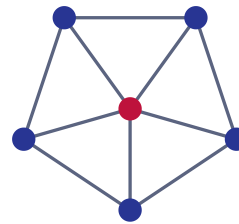
Cycle C_5



Path P_4



Complete bipartite $K_{2,3}$



Wheel W_5

Nine of the types at a glance. Crimson marks the vertex or the side that makes each shape what it is: the hub of the wheel, the two sides of the bipartite graph.

● MUST MEMORISE

EDGES OF A COMPLETE GRAPH

The complete graph on n vertices, K_n , has

$$C(n, 2) = n(n - 1) / 2 \text{ edges,}$$

and every vertex has degree $n - 1$.

WHERE

n is the number of vertices. $C(n, 2)$ is the combination count of section 5.2: an edge is an unordered pair of distinct vertices, and order does not matter because an edge from a to b is the same edge as one from b to a . $n - 1$ is the degree because each vertex is joined to every other vertex, that is, to all but itself.

The values you will need:

N	2	3	4	5	6	7
EDGES OF K_N	1	3	6	10	15	21

◆ **Memory hook: every pair, counted once.** The "counted once" is the divide by 2, and forgetting it is the standard error: $n(n - 1)$ counts each edge from both ends.

The question that appears three times

AS PRINTED

2020/2021 · Q1(A)

1. (a) Explain, with the aid of relevant examples, the types of undirected graph.

(12marks)

AS PRINTED

2021/2022 · Q1(C)

(c) Explain the types of undirected graph, with diagrammatical illustrations.

(12marks)

AS PRINTED

2023/2024 · Q4(A)

(a) Explain in details (and with diagrammatic illustration) the types of undirected graph.

(9 marks)

NOTE ON THE PAPER

The same question three times, differing only in wording and marks. The 2023/2024 version has no question number printed above it: the scan runs straight from Q3(d) into "(a) Explain in details ...", which is plainly Question 4 but is nowhere labelled as such. The phrase "Explain in details" is not standard English but its meaning is not in doubt. All three demand pictures.

BREAK IT DOWN

GIVEN Nothing. This is a recall question with no data.

ASKED **Explain** the types, **with examples** or **with diagrams**. Two things per type: what it is, and a drawing of one.

MARKS SAY 12, 12 and 9. At roughly 1 to 1.5 marks per type, aim for **eight to ten types** on the 12-mark versions and six or seven on the 9-mark version. More types briefly beats three types at length.

TEACHES This section, 6.2.

TRAP Two. Writing prose with no drawings, when the instruction names them. And drifting into **directed** graphs: the question says undirected, so digraphs and the two directional degrees earn nothing here. They are section 6.4 and a different question.

How to lay the answer out. One short paragraph per type, each with a small sketch beside it. Open with a single framing sentence, then work down the list. The framing sentence is worth writing because it shows the list is organised rather than remembered:

"An undirected graph is one whose edges have no direction. The types divide first by which edges are permitted, and then by the shape the edges make."

Group 1: told apart by what edges are allowed

1. **Simple graph.** No loops and no parallel edges, so at most one edge between any two vertices. *Example:* a triangle on a, b, c.
2. **Multigraph.** Parallel edges are allowed, loops are not. *Example:* two towns joined by two separate roads.
3. **Pseudograph.** Both loops and parallel edges are allowed; the most general undirected graph. *Example:* a network where a router also has a cable looping back to itself for testing.

Group 2: told apart by shape

1. **Null or empty graph N_n .** Vertices but no edges; every vertex is isolated. *Example:* three computers with no cables between them.
2. **Trivial graph K_1 .** One vertex, no edges. The smallest graph there is, and a special case of the null graph.
3. **Complete graph K_n .** Simple, with every pair of distinct vertices joined. It has $n(n-1)/2$ edges and every vertex has degree $n-1$. *Example:* K_4 , four vertices and six edges.
4. **Regular graph.** Every vertex has the same degree; k -regular if that degree is k . *Example:* a cycle is 2-regular, and K_n is $(n-1)$ -regular.
5. **Cycle graph C_n .** The vertices form one closed ring. *Example:* C_5 , a pentagon.
6. **Path graph P_n .** The vertices in a line, the ring left open. Its two end vertices are pendant.
7. **Wheel graph W_n .** A cycle plus a hub joined to every vertex on it. *Example:* a bicycle wheel, rim and spokes.
8. **Bipartite graph.** The vertices split into two sets with every edge running between the sets and none inside either. *Example:* students on one side, courses on the other, an edge for each enrolment. If every cross pair is joined it is **complete bipartite**, $K_{m,n}$.
9. **Connected and disconnected graphs.** Connected means a route exists between every pair of vertices. A disconnected graph falls into separate **components**. *Example:* the figure in section 6.1 is disconnected, because vertex 4 is joined to nothing.
10. **Weighted graph.** Each edge carries a number such as a distance or a cost. *Example:* the five-city map of section 6.8, whose edges are labelled with mileages.
11. **Subgraph.** A graph formed from some of the vertices and some of the edges of another. *Example:* a triangle inside K_4 .

Close with one sentence on the nesting, which few candidates write and which shows real understanding: *every simple graph is a multigraph, and every multigraph is*

a pseudograph, so the three families contain one another rather than exclude one another. The shape-based types cut across all three.

Now redo it: which type is a single square with its two diagonals drawn in? Four vertices, every pair joined, no loops or repeats: it is the complete graph K_4 , and it is also 3-regular.

Counting the possible links

AS PRINTED

2020/2021 · Q1(B)

(b) Assuming you are required to set up a simple communication link between CSIS, CST Lab, CLR, CUCRID and Senate Building, how many of such simple communication links are possible? Support your answer with a diagrammatical illustration. **(6½marks)**

AS PRINTED

2021/2022 · Q2(B)

(b) Assuming CU has contracted you to help set up simple communication links between the SENATE Building, CUCRID, CLR & CSIS NOC. How many of such simple communication links are possible? Support your answer with a diagrammatical illustration. **(7½marks)**

NOTE ON THE PAPER

These two look identical and are not. **COUNT THE BUILDINGS.** The 2020/2021 version names **FIVE** (CSIS, CST Lab, CLR, CUCRID, Senate Building) and the 2021/2022 version names **FOUR** (SENATE Building, CUCRID, CLR, CSIS NOC). Five gives 10 links and four gives 6. Copying the answer across from one paper to the other is the fastest way to lose 6.5 marks, and it is exactly what a candidate who has memorised "the answer is 10" will do.

BREAK IT DOWN

GIVEN	A list of named buildings, and the word simple , which is doing real work: no building links to itself, and no pair is linked twice.
ASKED	How many such links are possible , plus a diagrammatical illustration .
MARKS SAY	6.5 and 7.5, which is far more than one combination is worth. The marks are in the modelling and the drawing: say that buildings are vertices and links are edges, name the complete graph, give the formula, then draw it with every edge shown.
TEACHES	This section, 6.2, using $C(n, 2)$ from section 5.2.
TRAP	Answering $n(n - 1)$ instead of $n(n - 1)/2$. A link between CLR and CUCRID is the same link as one between CUCRID and CLR, so each pair must be counted once.

- GIVEN** Five buildings in the 2020/2021 version, four in the 2021/2022 version. Links are simple: no self-links, no repeats.
- FIND** The number of possible links, and a diagram.
- FORMULA** Links = $C(n, 2) = n(n - 1)/2$, the edge count of the complete graph K_n .

Step 1. Model it. Let each building be a **vertex** and each possible communication link an **edge**. "How many links are possible" asks for the case where every pair is joined, which is the **complete graph** K_n .

Step 2. Count for five buildings (2020/2021).

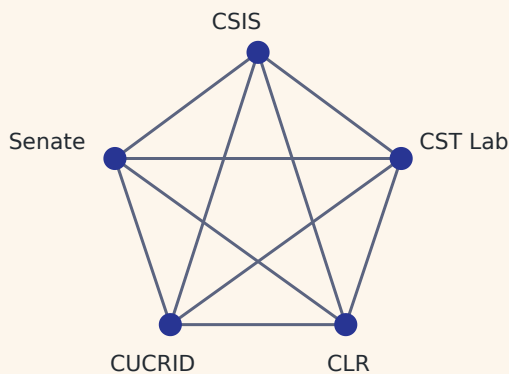
$$\text{links} = C(5, 2) = 5 \times 4 / 2 = 20 / 2 = 10$$

List them once to show the count is right: CSIS to each of the other four is 4 links; CST Lab to the three not yet used is 3; CLR to the two remaining is 2; CUCRID to Senate is 1. Then $4 + 3 + 2 + 1 = 10$, and no pair is repeated because each building only reaches forward down the list.

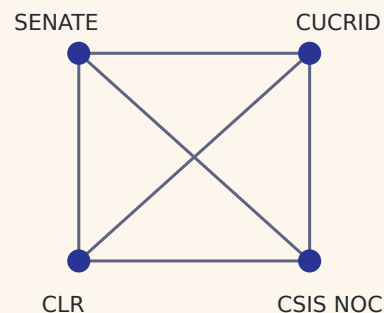
Step 3. Count for four buildings (2021/2022).

$$\text{links} = C(4, 2) = 4 \times 3 / 2 = 12 / 2 = 6$$

By the same listing, $3 + 2 + 1 = 6$.



$$K_5: 5 \times 4 / 2 = 10 \text{ links}$$



$$K_4: 4 \times 3 / 2 = 6 \text{ links}$$

Both answers drawn. Five buildings give ten links; four give six. Draw every edge, and label the vertices with the names the question used.

The sentence that ties it together: "Since every pair of buildings may be linked and a link has no direction, the possible links are the edges of the complete graph on the buildings, so there are $n(n - 1)/2$ of them."

Now redo it: with six buildings, how many links? $C(6, 2) = 6 \times 5 / 2 = 15$.

RECALL AND CHECK

1. What separates a simple graph, a multigraph and a pseudograph?
2. How many edges has K_7 , and what is the degree of each vertex?
3. What is a regular graph? Give two examples.
4. Define a bipartite graph.
5. Name the type: five vertices in a closed ring, nothing else.
6. A graph on 8 vertices has no edges. Name it.

ANSWERS

1. What edges are allowed: simple forbids loops and parallel edges; multigraph allows parallel edges but no loops; pseudograph allows both.
2. $7 \times 6 / 2 = 21$ edges, each vertex of degree 6.
3. Every vertex has the same degree. A cycle is 2-regular; K_4 is 3-regular.
4. Its vertices split into two sets with every edge running between the sets and none inside either.
5. The cycle graph C_5 .
6. The null or empty graph N_8 .

FILL IN THE GAP

1. A graph that allows loops and parallel edges is a _____.
2. K_n has _____ edges and every vertex has degree _____.
3. Five buildings can be joined by _____ simple links; four buildings by _____.
4. A graph with vertices but no edges is the _____ graph.

ANSWERS

1. pseudograph. 2. $n(n - 1)/2$, $n - 1$. 3. 10, 6. 4. null or empty.

PART SIX · 6.3

The Handshaking Theorem

One line of arithmetic that every graph must obey. It is asked for by name on three papers, and it silently decides a fourth: the whole of 2024/2025 Question Five turns on it.

By the end of this section you will be able to state and prove the Handshaking theorem, use it to find a missing degree count, use its corollary to prove that a described graph cannot exist, and answer the 17.5-mark Question Five of 2024/2025 in full.

THE HANDSHAKING THEOREM

The claim. In any undirected graph, the sum of the degrees of all the vertices equals twice the number of edges. Counting degrees counts every edge exactly twice, once from each end.

In symbols.

$\sum \deg(v) = 2|E|$, the sum running over every vertex v in V .

The proof, which is one sentence. Every edge has two endpoints, so each edge contributes exactly 1 to the degree of each of its two ends, that is, 2 to the total. Adding over all $|E|$ edges gives $2|E|$. A loop is no exception: both of its ends are at the same vertex, which is precisely why a loop is defined to add 2 to that vertex's degree.

Conditions. The graph must be **finite** and **undirected**. Loops and parallel edges are allowed, provided a loop counts 2. For a directed graph the corresponding statement is different and is given in section 6.4.

The corollary that does the real work. The number of vertices of **odd** degree is always **even**. Since the total $2|E|$ is even, and the even-degree vertices contribute an even amount, the odd-degree vertices must also contribute an even amount between them, which is only possible if there is an even number of them.

The limiting case. With no edges at all the theorem reads $0 = 0$, true and empty. It is at its most useful in the opposite direction: given a degree sum that is **odd**, no graph with those degrees exists at all, and that impossibility is what 2023/2024 Q5(b) is testing.

● MUST MEMORISE

HANDSHAKING, AND WHAT TO DO WITH IT

$$\sum \deg(v) = 2|E|$$

WHERE

Σ is summation, introduced in section F.1: add the quantity that follows over everything listed under it. $\deg(v)$ is the degree of vertex v from section 6.1, a loop counting twice. $|E|$ is the number of edges, the size of the graph. $2|E|$ is even **always**, whatever the graph, and that single fact is the source of every impossibility proof in this section.

The three ways it is used, and the exam uses all three:

1. **Find the edges from the degrees.** $|E| = (\sum \deg(v)) / 2$.
2. **Find a missing degree.** If all degrees but one are known and $|E|$ is known, the last one is forced.
3. **Prove something cannot exist.** If the degrees a question describes add to an odd number, no such graph exists, and saying so *is* the answer.

◆ **Memory hook: every handshake is counted twice, once by each hand.** That is where the name comes from: at a party, the total number of hands shaken is twice the number of handshakes.

Stating it, as two papers ask

AS PRINTED

2021/2022 · Q4(A)

4. (a) Briefly explain the Handshaking Theorem?

(5½marks)

5. (a) Define the Handshaking Theorem.

(3 marks)

NOTE ON THE PAPER

2021/2022 punctuates a command as a question ("Briefly explain ...?"), which it also does in Q1(a) and Q2(c). The two versions want the same thing at different lengths: 3 marks buys the statement and the reason, 5.5 buys the statement, the proof, the corollary and an example. Write to the marks.

WORKED EXAMPLE

PQ 21/22 Q4(A) · PQ 23/24 Q5(A)

For 3 marks (2023/2024)

Statement. In any undirected graph, the sum of the degrees of all the vertices is equal to twice the number of edges: $\sum \deg(v) = 2|E|$.

Reason. Each edge has two endpoints and so contributes 1 to the degree of each, that is, 2 to the total. Summing over all edges gives $2|E|$.

Example. A triangle has 3 vertices each of degree 2 and 3 edges: $2 + 2 + 2 = 6 = 2 \times 3$.

For 5.5 marks (2021/2022), add these

The corollary. The number of vertices of odd degree is even.

Why the corollary follows. Split the sum into the even-degree vertices and the odd-degree ones. The even-degree part is even. The whole sum is $2|E|$, also even. So the odd-degree part is even too; but a sum of odd numbers is even only when there is an even count of them.

A use. No graph can have exactly three vertices of degree 5 and nothing else, because $5 + 5 + 5 = 15$ is odd.

Where the name comes from, worth one line: if each vertex is a person and each edge a handshake, the degree of a person is the number of hands they shook, and the total counts every handshake from both sides.

Now redo it: a graph has 6 vertices each of degree 3. How many edges? $\sum \deg = 18$, so $|E| = 18 / 2 = 9$.

Proving a graph cannot exist

(b) An Airplane pilot has been scheduled for a trip to five (5) different cities. He can travel directly between two cities through the same route.

(i) Can he visit exactly three (3) cities from each of the cities on the flight plan?

Compute your answer.

(3 marks)

(ii) Support your answer with a diagrammatical illustration.

(2 marks)

NOTE ON THE PAPER

The second sentence is doing the work of "every pair of cities is joined by at most one route, and a route may be flown in either direction". That is the paper's way of saying the graph is **SIMPLE** and **UNDIRECTED**. Read part (i) as: can a simple undirected graph on 5 vertices have every vertex of degree exactly 3? The word "Compute" is the clue that an arithmetic argument, not a drawing, is what earns the 3 marks.

BREAK IT DOWN

GIVEN	5 cities. Routes are undirected and there is at most one between any pair.
ASKED	(i) Can every city connect to exactly 3 others, and compute the answer. (ii) Support it with a diagram.
MARKS SAY	3 and 2. The 3 marks are for the degree-sum computation. Answering only "no" earns almost nothing; the arithmetic is the answer.
TEACHES	This section, 6.3.
TRAP	Trying to draw it, failing, and concluding it is impossible. Failing to draw something proves nothing about whether it exists. The degree sum proves it.

- GIVEN** $n = 5$ vertices (cities), each required to have degree exactly 3, graph simple and undirected.
- FIND** Whether such a graph exists.
- FORMULA** $\sum \deg(v) = 2|E|$, so the degree sum must be even.

Step 1. Compute the degree sum. If each of the 5 cities connects to exactly 3 others, then every vertex has degree 3, so

$$\sum \deg(v) = 5 \times 3 = 15$$

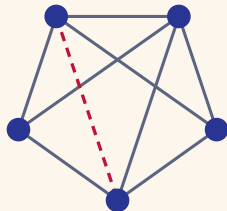
Step 2. Apply the theorem. By the Handshaking theorem this sum must equal $2|E|$, so

$$2|E| = 15, \text{ giving } |E| = 7.5$$

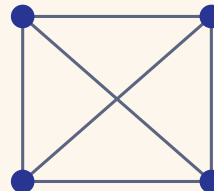
Step 3. Conclude. The number of edges must be a whole number, and 7.5 is not. Equivalently, 15 is odd and $2|E|$ is always even. **No such flight plan exists: the pilot cannot visit exactly three cities from each of five cities.**

The corollary says the same thing faster, and is worth adding: every one of the 5 vertices would have odd degree, giving 5 odd-degree vertices, but the number of odd-degree vertices is always even.

Part (ii): the illustration. A drawing cannot show something impossible, so show **why** it fails and **what does** work instead.



5 cities, trying for degree 3 each:
whichever edge is added last,
one city is left at degree 2 or 4



4 cities, degree 3 each: this works.
Sum = $4 \times 3 = 12 = 2 \times 6$ edges.
An odd degree needs an even number of cities

Left: the attempt on five cities always strands one vertex at the wrong degree. Right: on four cities it works, which shows the obstruction is the parity, not the degree 3.

Now redo it: can six cities each connect to exactly three others? $\sum \deg = 18$, even, so $|E| = 9$ and there is no obstruction. It can be done, for instance as two triangles joined by a perfect matching.

The 2024/2025 Question Five

AS PRINTED

2024/2025 · QUESTION FIVE

A graph designed by a 400-level Graphics student consists of 24 vertices and 30 edges. The graph has five vertices of degree 4, seven pendant vertices, and seven vertices of degree 2. All other vertices have a degree of either 3 or 4. After completing the design, the student seeks your assistance as a computer science student to help solve the following questions for further review:

- a. How many vertices of degree 4 are there in the given graph? **4 marks**
- b. What is the total number of vertices of degree 4 in the corresponding complete graph? **1 mark**
- c. Provide a proof to determine whether the Handshaking Theorem holds for this scenario. **3 marks**
- d. Write a C++ code snippet to compute the remaining number of vertices that have either degree 3 or 4. **3 marks**
- e. Demonstrate how the Handshaking Theorem can be applied in Data Communication. **2 marks**
- f. Prove that if G is a graph with n vertices and n edges, and no vertex has a degree of 0 or 1, then the degree of every vertex in G is 2. **3 marks**
- g. Explain what it means for two graphs G and H to be isomorphic. **1.5 marks**

NOTE ON THE PAPER

THIS QUESTION LOOKS SELF-CONTRADICTIONARY AND IS NOT. It tells you there are "five vertices of degree 4" and then asks in part (a) how many vertices of degree 4 there are. The answer is **NOT** five, and a candidate who writes five has fallen into the best trap on any of these papers.

The sentence that resolves it is "All other vertices have a degree of either 3 or 4". More degree-4 vertices are hiding in that remainder, and part (a) is asking for the **TOTAL**. Read the preamble as a partial inventory, not a complete one.

Part (d) is a C++ question and is answered in section 11.2, where the code can be compiled and run. Part (g) is answered in section 6.6, where isomorphism is taught. Everything else is answered below.

BREAK IT DOWN

- GIVEN** $|V| = 24$, $|E| = 30$. Five vertices of degree 4, seven pendant vertices (degree 1 by definition, section 6.1), seven of degree 2. All remaining vertices have degree 3 or 4.
- ASKED** Seven parts. The whole 17.5-mark question rests on one degree sum, so compute that first and almost everything follows.
- MARKS SAY** 4 for part (a) alone, which tells you part (a) is not a one-line read-off. If your answer to a 4-mark question is a number copied from the question, re-read the question.
- TEACHES** This section, 6.3, with (d) in 11.2 and (g) in 6.6.
- TRAP** Answering (a) with 5. Also forgetting that **pendant means degree 1**: the question never says "degree 1" anywhere, and if you do not know the word you cannot start.

- GIVEN** $|V| = 24$, $|E| = 30$; 5 vertices of degree 4, 7 pendant (degree 1), 7 of degree 2, the rest of degree 3 or 4.
- FIND** (a) the total number of degree-4 vertices; (b) the same count in the corresponding complete graph; (c) a proof that Handshaking holds; (e) an application; (f) a proof about n vertices and n edges.
- FORMULA** $\sum \deg(v) = 2|E|$.

a. How many vertices of degree 4 are there

Step 1. The total degree. By the Handshaking theorem,

$$\sum \deg(v) = 2|E| = 2 \times 30 = \mathbf{60}$$

Step 2. Account for the vertices the question names. A pendant vertex has degree 1.

GROUP	HOW MANY	DEGREE EACH	DEGREE TOTAL
stated degree 4	5	4	20
pendant	7	1	7
degree 2	7	2	14
accounted for	19		41

Step 3. What is left.

$$\text{vertices remaining} = 24 - 19 = 5$$

$$\text{degree remaining} = 60 - 41 = \mathbf{19}$$

Step 4. Split the remainder. Those 5 vertices have degree 3 or 4. Let x have degree 4 and y have degree 3. Then

$$x + y = 5 \quad \text{and} \quad 4x + 3y = 19$$

Substituting $y = 5 - x$ into the second equation:

$$4x + 3(5 - x) = 19$$

$$4x + 15 - 3x = 19$$

$$x + 15 = 19$$

$$x = \mathbf{4}, \text{ so } y = \mathbf{1}$$

So the remainder is four vertices of degree 4 and one of degree 3, and this is the only fit.

Step 5. Total.

$$\text{vertices of degree 4} = 5 \text{ (stated)} + 4 \text{ (found)} = \mathbf{9}$$

Answer: the graph has 9 vertices of degree 4.

Check the whole inventory, and write the check down: $9 \times 4 + 1 \times 3 + 7 \times 2 + 7 \times 1 = 36 + 3 + 14 + 7 = 60$, which is 2×30 . The vertices count too: $9 + 1 + 7 + 7 = 24$. Both agree.

b. Degree-4 vertices in the corresponding complete graph

State the reading, because "corresponding" is not defined. The complete graph corresponding to this one is the complete graph on the same 24 vertices, K_{24} .

In K_n every vertex is joined to every other, so every vertex has degree $n - 1$. Here degree of every vertex of $K_{24} = 24 - 1 = 23$

Since $23 \neq 4$, **no vertex of K_{24} has degree 4, so the answer is 0.**

Add the alternative reading in one line, since it costs nothing and the phrase is genuinely ambiguous: the only complete graph in which every vertex has degree 4 is K_5 , which has 5 such vertices. If "corresponding" was meant to point at the complete graph of degree 4, the answer is 5. State whichever you use.

c. Proof that the Handshaking theorem holds here

"Provide a proof to determine whether" means: check it, and show the check.

Left side, the degree sum. Using the full inventory from part (a):

$$\begin{aligned}\Sigma \deg(v) &= (9 \times 4) + (1 \times 3) + (7 \times 2) + (7 \times 1) \\ &= 36 + 3 + 14 + 7 \\ &= \mathbf{60}\end{aligned}$$

Right side, twice the edges.

$$2|E| = 2 \times 30 = \mathbf{60}$$

Conclusion. The two sides agree, so $\Sigma \deg(v) = 2|E|$ and the Handshaking theorem holds for this graph. Add the reason it had to: the theorem holds for *every* finite undirected graph, so agreement is not a coincidence but a confirmation that the degree inventory in part (a) is consistent. Had the two sides differed, the fault would have been in the inventory, not in the theorem.

e. The Handshaking theorem in Data Communication

Model a network as a graph: each **node** (router, switch, host) is a vertex, and each physical **link** between two nodes is an edge. The degree of a node is the number of its ports that are in use.

Application 1: counting cables from port counts. The theorem says the total number of used ports across the network is twice the number of links. So a network of 40 nodes with 6 used ports each needs $(40 \times 6) / 2 = 120$ cables, and a bill of materials that says anything else is wrong.

Application 2: validating a proposed topology. If a design specifies an odd total of used ports, that design cannot be built: no graph has an odd degree sum. This catches a specification error before any cable is bought, which is the practical point.

Application 3: the odd-degree corollary in redundancy planning. In a topology where every node must have an odd number of links, the number of nodes must be even. That constrains what a resilient ring or mesh design can look like.

f. If G has n vertices and n edges and no degree is 0 or 1

To prove: every vertex of G has degree exactly 2.

Proof. By the Handshaking theorem,

$$\Sigma \deg(v) = 2|E| = 2n$$

No vertex has degree 0 or 1, so every vertex has degree at least 2. There are n vertices, so

$$\Sigma \deg(v) \geq 2 + 2 + \dots + 2 \text{ (n times)} = 2n$$

But the sum *equals* $2n$. So the inequality is an equality.

Now suppose some vertex u had degree 3 or more. Then the other $n - 1$ vertices still contribute at least 2 each, so

$$\sum \deg(v) \geq 3 + 2(n - 1) = 2n + 1 > 2n,$$

contradicting the theorem. Hence no vertex has degree 3 or more, and since none has degree 0 or 1, **every vertex has degree exactly 2**. $\square$

What such a graph looks like, worth a closing line: a graph in which every vertex has degree 2 is a disjoint union of cycles. With n vertices and n edges and no isolated or pendant vertices, G is a cycle, or several cycles side by side.

Now redo it: a graph has 10 vertices and 12 edges, with four pendant vertices and four of degree 2. What degrees do the remaining two vertices share? Degree sum 24, accounted $4 \times 1 + 4 \times 2 = 12$ over 8 vertices, leaving 2 vertices sharing 12, so both have degree 6.

RECALL AND CHECK

1. State the Handshaking theorem and prove it in one sentence.
2. A graph has 7 vertices each of degree 4. How many edges?
3. Can a graph have exactly five vertices of degree 3 and no others? Why?
4. What is the degree of every vertex of K_{24} ?
5. Why is the number of odd-degree vertices always even?

ANSWERS

1. $\sum \deg(v) = 2|E|$, because each edge contributes 1 to the degree of each of its two endpoints.
2. $\sum \deg = 28$, so $|E| = 14$.
3. No. The degree sum would be 15, which is odd, and $2|E|$ is always even.
4. 23, since every vertex is joined to all the other 23.
5. The total degree is even and the even-degree vertices contribute an even amount, so the odd ones must too, which needs an even number of them.

FILL IN THE GAP

1. $\sum \deg(v) =$ _____.
2. A pendant vertex has degree _____.
3. In 2024/2025 Question Five the degree sum is _____ and the number of degree-4 vertices is _____.
4. The number of odd-degree vertices in any graph is always _____.

ANSWERS

1. $2|E|$. 2. 1. 3. 60, 9. 4. even.

PART SIX · 6.4

Directed Graphs, In-degree and Out-degree

Put an arrowhead on every edge and the degree splits in two: arrows coming in, and arrows going out. Two papers set the same tabulation question on a hand-drawn figure.

By the end of this section you will be able to define a directed graph, compute in-degree and out-degree, state the directed form of the Handshaking theorem, tabulate a digraph vertex by vertex,

and answer both tabulation questions and the short note on in-degree and out-degree.

Directed graphs

● MUST MEMORISE

DIGRAPH, ARC, IN-DEGREE, OUT-DEGREE

A **directed graph**, or **digraph**, is a graph in which every edge has a direction. A directed edge is called an **arc**, and it is written as an **ordered** pair (u, v) , drawn as an arrow from u to v .

For a vertex v :

the **in-degree** $\deg^-(v)$ is the number of arcs pointing **into** v ;

the **out-degree** $\deg^+(v)$ is the number of arcs pointing **out of** v .

WHERE

(u, v) is an ordered pair, so (u, v) and (v, u) are **different arcs**. This is the whole difference from section 6.1, where an edge joining u and v was one thing however you named it.

$\deg^-(v)$ counts arrowheads landing on v . $\deg^+(v)$ counts arrow tails leaving v . a **loop** at v adds 1 to each, since it both leaves v and arrives at v .

◆ **Memory hook: minus is in, plus is out.** The superscript minus looks like an arrow flattened as it arrives; the plus is the vertex sending traffic outward. And a mnemonic that never fails: the **in-degree** counts the arrow**heads**.

● MUST MEMORISE

HANDSHAKING FOR DIGRAPHS

In any directed graph,

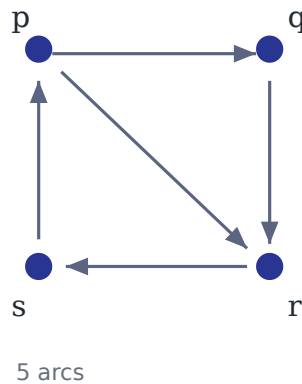
$$\sum \deg^-(v) = \sum \deg^+(v) = |E|$$

WHERE

$|E|$ is the number of arcs. **the two sums** are each equal to $|E|$, **not** to $2|E|$: every arc has exactly one tail and exactly one head, so it is counted once in each sum rather than twice in one.

How this differs from section 6.3, which is the point most often lost: in an undirected graph the degrees sum to $2|E|$; in a digraph the in-degrees sum to $|E|$ and the out-degrees sum to $|E|$, and adding the two sums together recovers the $2|E|$.

◆ **Memory hook: one tail and one head per arc.** Both column totals of your table must equal the arc count. That is a free check, and it catches a misread arrowhead immediately.



vertex	in	out
p	1	2
q	1	1
r	2	1
s	1	1
total	5	5

both totals equal the arc count

A digraph and its table. The arcs are p to q , q to r , r to s , s to p , and p to r . Both column totals come to 5, which is the check.

The short note

WORKED EXAMPLE

PQ 25/26 Q4(A)(III)

The question is quoted in full in section 6.1: "(iii) In-degree and Out-degree of a vertex", 2 marks, wanting short notes with relevant examples.

Definition. In a directed graph, the in-degree of a vertex v , written $\deg^-(v)$, is the number of arcs whose arrowhead lands on v . The out-degree, $\deg^+(v)$, is the number of arcs whose tail leaves v . A loop at v adds 1 to each.

Example 1. In the figure above, vertex p has one arc arriving (from s) and two leaving (to q and to r), so $\deg^-(p) = 1$ and $\deg^+(p) = 2$.

Example 2. On a web link graph, a page's in-degree is the number of pages linking *to* it and its out-degree the number it links *out* to. Search engines rank on the first and not the second, which is a one-line way of showing you understand the difference matters.

The line worth adding: $\sum \deg^-(v) = \sum \deg^+(v) = |E|$, because every arc has exactly one head and one tail.

Tabulating a digraph

AS PRINTED

2021/2022 · Q1(B)

(b) In the Graph diagram below, present in tabular form the following:

(6½marks)

- (i) The number of Vertices
- (ii) The number of edges
- (iii) The total number of In-degree & Out-degree from all the vertices.

(c) In the Graph diagram below, present in tabular form the following:

- (i) Identify each vertex and state the total number of vertices. **(1 marks)**
- (ii) Identify each edge per vertex and state total number of edges. **(1 marks)**
- (iii) Identify each vertex's in-degree and state the total number of the in-degrees. **(2 marks)**
- (iv) Identify each vertex's out-degree and state the total number of the out-degrees. **(2 marks)**

NOTE ON THE PAPER, AND ON BOTH FIGURES

The same question twice, expanded from three parts to four. 2023/2024 prints "(1 marks)" twice, a singular value with a plural noun.

NEITHER FIGURE CAN BE FULLY READ FROM THE SCANS WE HOLD, AND THIS MANUAL WILL NOT PRETEND OTHERWISE. Both are small directed graphs on four labelled vertices with a fifth vertex sitting alone and joined to nothing, drawn with several curved arcs between the same pairs and with arrowheads a couple of pixels across.

For 2021/2022 Q1(b) the arc count and the arc directions are **NOT** recoverable at the resolution available, so no reading of it is given here at all. For 2023/2024 Q5(c) the multi-arc bundles are clear but the two crossing diagonals in the middle are not; that reading is given below, flagged, with the arithmetic that lets you re-do it in seconds once you have read the arrowheads off your own copy.

THE MARKS ARE FOR THE METHOD, NOT FOR THE FIGURE. Both questions pay for a table laid out correctly with totals that reconcile. Learn the layout, and fill it from whatever your own copy shows.

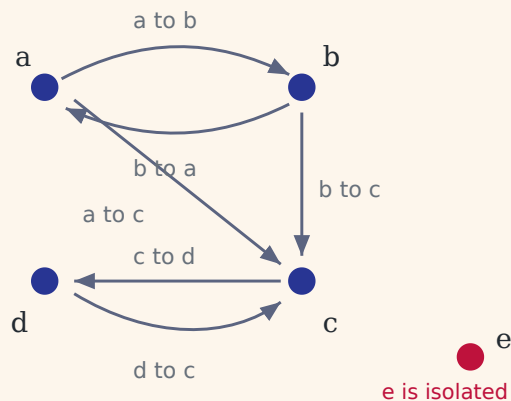
BREAK IT DOWN

GIVEN	A directed graph drawn on the paper, with labelled vertices, several arcs, and at least one vertex joined to nothing.
ASKED	Present in tabular form: the vertices and their count, the edges and their count, then each vertex's in-degree and out-degree with totals. The instruction "in tabular form" is explicit, so a paragraph loses marks even if every number in it is right.
MARKS SAY	6.5 and 6 respectively, split roughly evenly across the parts. In-degree and out-degree carry 2 marks each on 2023/2024, which is where the care belongs.
TEACHES	This section, 6.4.
TRAP	Three. Forgetting the isolated vertex, which must still appear in the table with 0 and 0. Counting a two-arc bundle as one edge. And giving a single "degree" column instead of two.

The method, in four steps. It works on any figure, including the one on your own paper.

1. **List the vertices first**, reading the labels off the drawing and including any vertex with no arcs. Write the count.
2. **Walk the arcs one bundle at a time**, writing each as an ordered pair (tail, head). Where several curves join the same pair, count the curves, not the pairs. Write the count.
3. **Build a row per vertex** with three columns: arcs out, arcs in, and the two degrees. Fill it from the list in step 2, not from the drawing, so each arc is used exactly once.
4. **Total both degree columns.** Each total must equal the arc count from step 2. If they do not, an arc was missed or double counted.

Worked on a clearly drawn example, so the method is visible without an unreadable figure in the way. *This figure is mine, not the paper's.*



Six arcs on five vertices, one of them isolated. Every arrowhead is unambiguous, which is the point of drawing it again.

Step 1. Vertices: a, b, c, d, e. **Total vertices = 5.** Note e, joined to nothing.

Step 2. Arcs as ordered pairs: (a, b), (b, a), (b, c), (c, d), (d, c), (a, c). **Total arcs = 6.**

Step 3 and 4. The table.

VERTEX	ARCS OUT	OUT-DEGREE	ARCS IN	IN-DEGREE
a	(a, b), (a, c)	2	(b, a)	1
b	(b, a), (b, c)	2	(a, b)	1
c	(c, d)	1	(b, c), (a, c), (d, c)	3
d	(d, c)	1	(c, d)	1
e	none	0	none	0
Total		6		6

The check. Both totals are 6, which is the arc count from step 2. That is $\sum \deg^+(v) = \sum \deg^-(v) = |E|$, and it is the only verification available in the exam hall. Do it every time.

Now redo it: add an arc (e, a) to the figure. Then $|E| = 7$, $\deg^+(e) = 1$, $\deg^-(a) = 2$, and both totals become 7.

- GIVEN** A digraph on vertices a, b, c, d and an isolated vertex e. Bundles read from the scan: one arc b to a; two arcs a to b; two arcs d to a; two arcs b to c; three arcs d to c. The two crossing diagonals are read as two arcs a to c and one arc b to d.
- FIND** The vertices and their count, the arcs and their count, each in-degree with its total, each out-degree with its total.
- RULE** $\sum \deg^-(v) = \sum \deg^+(v) = |E|$.

Flag first, then answer. Write this line before the table: "*The two crossing diagonals are taken as two arcs from a to c and one from b to d; the remaining bundles are unambiguous.*" An examiner marking a hand-drawn figure will accept a stated reading; a wrong reading presented as certain earns nothing.

(i) Vertices. a, b, c, d, e. **Total = 5.** Vertex e carries no arcs at all and must still be listed.

(ii) Arcs, per vertex of origin.

BUNDLE	ARCS
b to a	1
a to b	2
d to a	2
b to c	2
d to c	3
a to c	2 (stated reading)
b to d	1 (stated reading)
Total edges	13

(iii) and (iv) In-degree and out-degree.

VERTEX	OUT-DEGREE	FROM	IN-DEGREE	FROM
a	4	2 to b, 2 to c	3	1 from b, 2 from d
b	4	1 to a, 2 to c, 1 to d	2	2 from a
c	0	none	7	2 from b, 3 from d, 2 from a
d	5	2 to a, 3 to c	1	1 from b
e	0	none	0	none
Total	13		13	

The check passes: both totals are 13, which equals the arc count. Note also that c is a **sink**, with arcs arriving and none leaving, and that e is isolated in both directions.

If your copy reads the diagonals the other way, say one arc each way, then $|E| = 13$ still holds only if the count of diagonal arcs is unchanged; with two arcs a to c replaced by one a to c and one c to a, the arc total stays 13, $\deg^+(a)$ falls to 3, $\deg^+(c)$ rises to 1

and $\deg^-(c)$ falls to 6, while $\deg^-(a)$ rises to 4. Redo the two affected rows and check the totals again. Nothing else changes.

For the 2021/2022 version, whose figure is not readable at all from the scan, use the four-step method above on your own copy. Its part (iii) asks only for "the total number of In-degree & Out-degree from all the vertices", which is the pair of column totals, and by the digraph Handshaking rule both of them equal the number of arcs. If you can count the arcs you can answer part (iii) even if you misread an arrowhead.

⚠ TRAP

NEVER PRESENT AN UNREADABLE FIGURE AS READ

Both of these questions are drawn by hand and scanned. If an arrowhead is genuinely ambiguous on your copy, the professional move is the same one this manual has used: **state the reading you are working under, then answer under it**. One sentence protects the whole table.

What loses marks is silence. A table with no stated reading and a misread arc looks like a candidate who cannot count, rather than one working from a poor photocopy.

RECALL AND CHECK

1. What makes a digraph different from an undirected graph?
2. Define in-degree and out-degree.
3. What do the in-degrees sum to, and what do the degrees of an undirected graph sum to?
4. How much does a loop add to each of the two degrees in a digraph?
5. A digraph has 13 arcs. What must each of your two column totals be?

ANSWERS

1. Every edge has a direction, so an arc is an ordered pair and (u, v) differs from (v, u) .
2. In-degree counts arcs arriving at the vertex; out-degree counts arcs leaving it.
3. In-degrees sum to $|E|$. Undirected degrees sum to $2|E|$.
4. One to each, since it both leaves and arrives at that vertex.
5. 13 each. If either differs, an arc has been missed or counted twice.

FILL IN THE GAP

1. A directed edge is called an _____ and is written as an _____ pair.
2. $\deg^-(v)$ is the _____-degree.
3. $\sum \deg^+(v) =$ _____.
4. A vertex with arcs arriving and none leaving is a _____.

ANSWERS

1. arc, ordered. 2. in. 3. $|E|$. 4. sink.

Adjacency, Path and Relation Matrices

A graph can be stored as a grid of ones and zeros. Once it is, questions about routes become questions about multiplying the grid by itself, and the newest paper gives 6.5 marks to how a computer holds it.

By the end of this section you will be able to write the adjacency matrix of a graph or digraph, read degrees off it, build the path matrix from powers of the adjacency matrix, tell a logical matrix from a relation matrix, and answer the four matrix questions on the 2025/2026 paper.

The adjacency matrix

● MUST MEMORISE

ADJACENCY MATRIX

Number the vertices 1 to n in a fixed order. The **adjacency matrix** A of the graph is the n by n grid whose entry in row i , column j is

a_{ij} = the number of edges from vertex i to vertex j , which for a simple graph is **1** if they are joined and **0** if they are not.

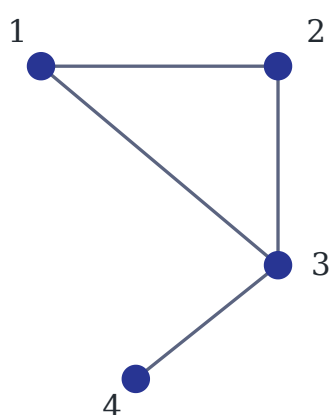
WHERE

n is the order of the graph, so A is always square and always n by n . **the vertex order** must be fixed and stated first. Re-number the vertices and you get a different-looking matrix for the same graph. a_{ii} , on the leading diagonal, records loops at vertex i .

Three properties to state whenever you are asked to explain one.

1. For an **undirected** graph, A is **symmetric**: $a_{ij} = a_{ji}$, because an edge joining i and j joins j and i . For a **digraph** it usually is not, since an arc has a direction.
2. **The row sum is the degree**. Adding across row i gives $\deg(i)$ for an undirected simple graph, and $\deg^+(i)$ for a digraph; the column sum gives $\deg^-(i)$.
3. A **simple** graph has zeros all along the leading diagonal, because a simple graph has no loops.

◆ **Memory hook: row to column, tail to head.** Read a_{ij} as "from i to j ". For an undirected graph the direction does not matter and the matrix comes out symmetric, which is itself the proof that the graph is undirected.



to:	1	2	3	4	
from 1	0	1	1	0	deg 2
from 2	1	0	1	0	deg 2
from 3	1	1	0	1	deg 3
from 4	0	0	1	0	deg 1

symmetric, zero diagonal, row sums are the degrees
degree sum $8 = 2 \times 4$ edges

A graph and its adjacency matrix. Everything you can say about the graph can be read off the grid, which is the point of building it.

The path matrix

● MUST MEMORISE

WALKS, AND THE PATH MATRIX

The key fact. If A is the adjacency matrix, then the entry in row i , column j of A^k is the number of **walks of length exactly k** from vertex i to vertex j .

The **path matrix** (also called the reachability matrix) P records only whether a route exists at all:

$p_{ij} = 1$ if there is a path from i to j , and 0 if there is none.

WHERE

a **walk of length k** is a sequence of k edges followed end to end; vertices and edges may repeat. A^k is A multiplied by itself k times, using ordinary arithmetic. P is built from $A + A^2 + \dots + A^{n-1}$ by replacing every non-zero entry with 1. Nothing longer than $n - 1$ edges is needed: a path that visits no vertex twice cannot use more.

The difference the exam asks about, in one line: the **adjacency** matrix answers "can I get there in **one** step?"; the **path** matrix answers "can I get there **at all**?"

◆ **Memory hook: adjacency is one hop, path is any number of hops.** Powers of A count the routes of each exact length; flattening the sum to ones and zeros forgets how many and remembers only whether.

GIVEN The graph in the figure above: vertices 1, 2, 3, 4 with edges 1-2, 1-3, 2-3 and 3-4.

FIND A, then A^2 , then the path matrix P.

RULE The (i, j) entry of A^k counts walks of length k ; P flattens the sum of the powers to ones and zeros.

Step 1. Write A, stating the vertex order 1, 2, 3, 4.

	1	2	3	4
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1
4	0	0	1	0

Step 2. Square it. Entry (i, j) of A^2 is row i of A dotted with column j of A. For instance entry $(1, 1)$:

$$(0 \times 0) + (1 \times 1) + (1 \times 1) + (0 \times 0) = 2$$

and there are indeed two walks of length 2 from vertex 1 back to itself, namely 1 to 2 to 1 and 1 to 3 to 1.

A^2	1	2	3	4
1	2	1	1	1
2	1	2	1	1
3	1	1	3	0
4	1	1	0	1

Read one entry aloud to prove you understand it. Entry $(3, 3)$ is 3, so there are three walks of length 2 from vertex 3 to itself: out and back along each of its three edges, to 1, to 2 and to 4. Entry $(3, 4)$ is 0, because there is no walk of exactly two edges from 3 to 4, only the direct one of length 1.

Step 3. Build the path matrix. Add A, A^2 and A^3 , then replace every non-zero by 1. Every entry comes out non-zero, so

P	1	2	3	4
1	1	1	1	1
2	1	1	1	1
3	1	1	1	1
4	1	1	1	1

What P tells you at a glance: every vertex reaches every other, so **the graph is connected**. A path matrix with a block of zeros in one corner is the signature of a disconnected graph, and that is the usual reason for computing it.

Now redo it: delete the edge 3-4. Vertex 4 then reaches nothing, so row 4 and column 4 of P become all zero apart from the diagonal, and the graph is disconnected.

Adjacency against path matrix

WORKED EXAMPLE

PQ 25/26 Q1 A) IV)

The question is quoted in full in section 6.1: "iv) Adjacency and Path Matrice", 2 marks, wanting the difference **with two examples each**. The paper's "Matrice" is a slip for "Matrix" or "Matrices".

	ADJACENCY MATRIX	PATH MATRIX
Records	direct connection, one edge	reachability, any number of edges
Entry a_{ij} or p_{ij}	1 if i and j are joined by an edge	1 if some path runs from i to j
Obtained from	the drawing directly	$A, A^2, \dots$ summed and flattened to 1s and 0s
Answers	"is there a road between these two towns?"	"can I drive from this town to that one?"

Two examples for the adjacency matrix. (1) In the graph above, $a_{12} = 1$ because 1 and 2 are joined; $a_{14} = 0$ because they are not. (2) In K_3 , every off-diagonal entry is 1 and every diagonal entry is 0.

Two examples for the path matrix. (1) In the same graph $p_{14} = 1$, because 1 to 3 to 4 is a path, even though $a_{14} = 0$. That single pair is the clearest possible illustration of the difference and is worth writing down. (2) In a graph of two separate triangles, P has ones inside each triangle's block and zeros between the blocks, showing at once that the graph is disconnected.

Logical matrix against relation matrix

● MUST MEMORISE

LOGICAL AND RELATION MATRICES

A **logical matrix** (also called a Boolean or binary matrix) is any matrix whose entries are all 0 or 1, where the entries are read as **false** and **true** rather than as numbers.

A **relation matrix** is the particular logical matrix that records a relation R from a set A to a set B , as defined in section 3.1:

$m_{ij} = 1$ exactly when (a_i, b_j) is in R , and **0** otherwise.

WHERE

a_i is the i -th element of A in a fixed listed order, giving the rows. b_j is the j -th element of B , giving the columns. For a relation on one set, A and B are the same and the matrix is square. m_{ij} is 1 or 0, and it is a **truth value**, not a count.

The relationship between the three terms, which is what a "differentiate between" question is really asking for:

- ▶ Every relation matrix **is** a logical matrix. The reverse does not hold: a logical matrix is only a relation matrix once you say which sets index its rows and columns.
- ▶ An adjacency matrix of a **simple** graph is also a logical matrix, and it is exactly the relation matrix of the relation "is joined to". A multigraph's adjacency matrix is **not** logical, because its entries count parallel edges and may exceed 1.

◆ **Memory hook: logical says what the entries are, relation says what they mean.** The first is a description of the contents; the second is a description of the job.

The question is quoted in full in section 6.1: "v) A logical matrix and relation matrix", printed "(2marks)" with no space.

The distinction. A logical matrix is any matrix all of whose entries are 0 or 1, read as false and true. A relation matrix is a logical matrix whose rows are labelled by the elements of one set and whose columns are labelled by the elements of another, with a 1 marking exactly the pairs that belong to the relation. So every relation matrix is a logical matrix given a job to do; not every logical matrix has been given that job.

Two examples of a logical matrix. (1) The 2 by 2 grid with rows (1, 0) and (0, 1) is logical, and until its rows and columns are labelled it records nothing in particular. (2) The truth table of a two-variable connective, written as a grid of Ts and Fs, is a logical matrix in disguise.

Two examples of a relation matrix. (1) On $A = \{1, 2, 3\}$ the relation "is less than" has 1 exactly above the diagonal:

	1	2	3
1	0	1	1
2	0	0	1
3	0	0	0

(2) The adjacency matrix of the simple graph drawn above is the relation matrix of "is joined to" on $\{1, 2, 3, 4\}$.

The line that earns the second mark: section 3.2 showed that the properties of a relation can be read straight off its matrix, the diagonal for reflexive and the mirror image for symmetric. That is only possible because the rows and columns are labelled, which is precisely what makes it a relation matrix rather than a bare logical one.

Explaining the adjacency matrix, and its memory cost

(a) i. Clearly explain Adjacency matrix

(2½ marks)

ii. Discuss memory management in Adjacency matrix

(4 marks)

NOTE ON THE PAPER

Part (ii) is the most unusual question on any of the five papers: it is a **SYSTEMS** question wearing a discrete maths hat, and at 4 marks it is worth more than part (i). "Memory management" here means how the matrix is stored, what it costs, and when a different structure is the better choice. A candidate who only defines the matrix again cannot score it.

BREAK IT DOWN

GIVEN	Nothing. Both parts are explanation.
ASKED	(i) Clearly explain the adjacency matrix. (ii) Discuss memory management in it. "Discuss" invites several points, not one.
MARKS SAY	2.5 and 4. Part (ii) wants roughly four distinct points, one per mark: the cost, why it is that cost, when it is wasteful, and what to do instead.
TEACHES	This section, 6.5.
TRAP	Answering (ii) by repeating (i). They are different questions and the marker is looking for storage, not definition.

i. Clearly explain the adjacency matrix

Definition. Fix an order on the n vertices of a graph. The adjacency matrix A is the n by n matrix whose entry a_{ij} is the number of edges from vertex i to vertex j , which is 1 or 0 for a simple graph.

What it is for. It turns a drawing into data. A picture cannot be given to a program; a grid of numbers can, and every graph algorithm in computing starts from a representation of this kind.

Its properties, which is where the half marks live: it is square; it is symmetric exactly when the graph is undirected; its diagonal is zero exactly when there are no loops; its row sums are the degrees; and the entries of A^k count the walks of length k .

An example. Give the four-vertex matrix drawn above, with its vertex order stated, and point at one entry: $a_{34} = 1$ because vertices 3 and 4 are joined, while $a_{14} = 0$ because they are not.

ii. Memory management in the adjacency matrix

Four points, one per mark.

1. **The cost is n^2 , whatever the graph.** An adjacency matrix stores an entry for every ordered pair of vertices, so a graph of n vertices needs n^2 cells regardless of how many edges it actually has. A graph on 1000 vertices needs 1000000 cells even if it has only 5 edges. The storage depends on the **order**, never on the **size**.
2. **It is wasteful on sparse graphs, and most real graphs are sparse.** A graph is **sparse** when its edge count is far below the $n(n - 1)/2$ maximum, and **dense** when it is near it. Road networks, social graphs and computer networks are all sparse: a city has millions of junctions and each meets three or four roads. For these the matrix is almost entirely zeros, and storing zeros costs exactly as much as storing ones.
3. **Two savings are available without changing structure.** First, an undirected graph's matrix is symmetric, so only the upper triangle need be held, roughly halving the space to $n(n - 1)/2$ cells. Second, a simple graph's entries are single bits, so 64 of them pack into one machine word instead of one integer each; that is a factor of 32 or 64 on the same data.
4. **The alternative is an adjacency list.** Store, for each vertex, only the list of its neighbours. That costs about $n + 2|E|$ for an undirected graph rather than n^2 , which is dramatically less when $|E|$ is small. The trade-off is the point to state: the matrix answers "are i and j adjacent?" in constant time by one lookup, while the list must scan a neighbour list; but the list walks all the neighbours of a vertex in time proportional to their number, while the matrix must scan a whole row of n cells to find them.

The closing sentence that makes it a discussion rather than a list: choose the adjacency matrix for small or dense graphs and for algorithms that ask "is this pair joined?" repeatedly; choose the adjacency list for large sparse graphs and for algorithms that walk neighbours, which is most traversal and shortest-path work.

	ADJACENCY MATRIX	ADJACENCY LIST
Space	n^2	about $n + 2 E $
"Are i and j joined?"	one lookup	scan i 's list
"List the neighbours of i "	scan n cells	read the list, $\deg(i)$ items
Best for	small or dense graphs	large sparse graphs

Now redo it: a graph has 500 vertices and 800 edges. The matrix needs $500^2 = 250000$ cells; the list needs about $500 + 1600 = 2100$. The list is the right choice by a factor of over a hundred.

RECALL AND CHECK

1. What does a_{ij} record, and what must be fixed before you can write A at all?
2. Why is the adjacency matrix of an undirected graph symmetric?
3. What do the entries of A^2 count?
4. Give the one-line difference between an adjacency matrix and a path matrix.
5. How much space does an adjacency matrix need, and how much does an adjacency list need?

ANSWERS

1. The number of edges from vertex i to vertex j . The order of the vertices must be fixed and stated.
2. An edge joining i and j also joins j and i , so $a_{ij} = a_{ji}$.
3. The number of walks of length exactly 2 between each pair.
4. Adjacency records one-step connection; the path matrix records whether any route exists at all.
5. n^2 cells for the matrix, about $n + 2|E|$ for the list.

FILL IN THE GAP

1. The row sums of an adjacency matrix are the _____.
2. The entries of A^k count the _____ of length _____.
3. Every relation matrix is a _____ matrix, but not conversely.
4. An adjacency matrix costs _____ cells whatever the number of edges.

ANSWERS

1. degrees. 2. walks, k . 3. logical. 4. n^2 .

PART SIX • 6.6

Isomorphism, Euler and Hamilton

Three questions about the same drawing: is this the same graph as that one, can I walk every edge once, and can I visit every vertex once. Only the middle one has an easy test, and knowing that is half the marks.

By the end of this section you will be able to define graph isomorphism and use invariants to disprove it, define Euler paths and circuits and apply Euler's test, define Hamilton paths and circuits and say why no simple test exists, and answer both short notes and the isomorphism part of 2024/2025 Question Five.

Isomorphism

● MUST MEMORISE

GRAPH ISOMORPHISM

Two graphs G and H are **isomorphic**, written $G \cong H$, when there is a **bijection** f from the vertices of G to the vertices of H such that

u and v are adjacent in G exactly when $f(u)$ and $f(v)$ are adjacent in H .

WHERE

f is a bijection, that is, one-to-one and onto, as defined in section 4.2. So G and H must have the same number of vertices before anything else can be checked. "**exactly when**" is two conditions in one: f must carry edges to edges **and** non-edges to non-edges. $\cong$ is read "is isomorphic to".

In plain words: the two graphs are the same graph with the vertices renamed and the picture redrawn. Nothing about position, length or crossing matters; only which vertices are joined.

◆ **Memory hook: same structure, different labels.** An isomorphism is a relabelling that preserves every join and every non-join.

● MUST MEMORISE

INVARIANTS: HOW TO DISPROVE ISOMORPHISM

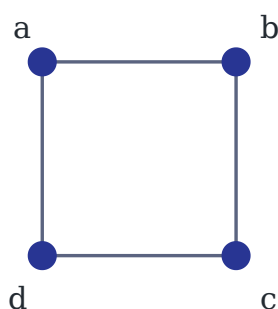
An **invariant** is a property that isomorphic graphs must share. If two graphs differ on any invariant, they are **not** isomorphic, and naming the invariant is a complete proof.

- ▶ the number of vertices;
- ▶ the number of edges;
- ▶ the **degree sequence**, that is, the list of all degrees in order;
- ▶ the number of components, and whether the graph is connected;
- ▶ the length of the shortest cycle, and whether any cycle exists at all;
- ▶ whether the graph is bipartite.

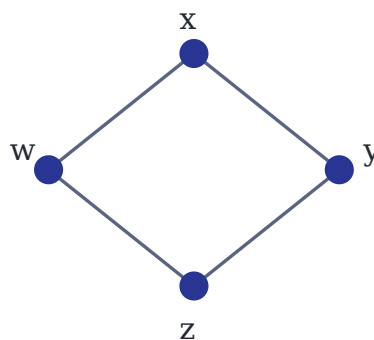
Section 6.7 adds one more to this list once it has the vocabulary for it.

The asymmetry that the exam rewards. Invariants can only ever **disprove**. Two graphs agreeing on every invariant in the list above may still fail to be isomorphic, so to **prove** isomorphism you must exhibit the bijection itself and check it. Say that sentence in any answer about isomorphism; it is the difference between knowing the definition and understanding it.

◆ **Memory hook: invariants say no, only a mapping says yes.** Count vertices, count edges, sort the degrees. If any of the three differ, stop and write the answer.



G: a square



H: a diamond

$$G \cong H$$

a to w

b to x

c to y

d to z

same graph, redrawn

Two drawings, one graph. Both are C_4 : four vertices, four edges, every degree 2, and the named bijection carries every edge to an edge.

WORKED EXAMPLE

PQ 24/25 Q5(G) · PQ 25/26 Q4(A)(IV)

The two questions are quoted in full in sections 6.3 and 6.1: 2024/2025 Q5(g) asks you to "Explain what it means for two graphs G and H to be isomorphic" for 1.5 marks, and 2025/2026 Q4(a)(iv) asks for a short note on "Graph isomorphism" with a relevant example for 2 marks. One answer serves both; give the example either way.

The explanation. Two graphs G and H are isomorphic when there is a one-to-one and onto function f from the vertices of G to the vertices of H such that two vertices u and v are adjacent in G if and only if $f(u)$ and $f(v)$ are adjacent in H . Informally, H is G with its vertices renamed and its picture redrawn: the two are the same graph in every respect except the labels and the geometry.

What must match, and what need not. The number of vertices, the number of edges and the degree of every corresponding vertex must match. The positions of the vertices on the page, the lengths of the edges and whether the edges cross need not: none of those is part of what a graph is.

The example. The square a, b, c, d and the diamond w, x, y, z above are isomorphic under a to w , b to x , c to y , d to z . Check it edge by edge: ab goes to wx , bc to xy , cd to yz , da to zw , and all four are edges of H . The two drawings look different and are the same graph.

The extra line for the full 1.5 marks: to disprove isomorphism it is enough to find one invariant on which the graphs differ, such as the degree sequence; but to prove it, the bijection must actually be exhibited, because agreeing on every standard invariant is not sufficient.

Now redo it: are a 4-cycle and a path on 4 vertices isomorphic? No. Both have 4 vertices, but the cycle has 4 edges and degree sequence 2, 2, 2, 2, while the path has 3 edges and degree sequence 1, 2, 2, 1. Either invariant settles it.

Euler paths and circuits

● MUST MEMORISE

EULER PATH AND EULER CIRCUIT

An **Euler path** is a walk that uses **every edge of the graph exactly once**. An **Euler circuit** is an Euler path that finishes where it started.

A graph that has an Euler circuit is called an **Eulerian graph**.

WHERE

every edge exactly once is the condition. Vertices may be revisited as often as you like; it is the **edges** that are rationed. **circuit** means the walk returns to its starting vertex; a path need not.

◆ **Memory hook: Euler is about edges, Hamilton is about vertices.** Both names begin with a letter that helps: **E**uler for **e**edges. Get that one association right and neither definition can be confused again.

EULER'S THEOREM ON CIRCUITS AND PATHS

The claim. Whether a graph can be traced without lifting the pen or repeating an edge depends only on how many of its vertices have odd degree, and on nothing else.

In symbols.

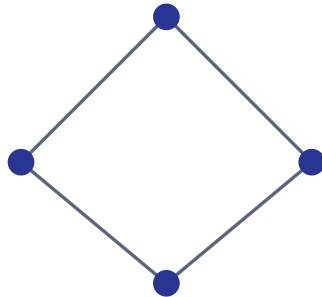
A connected graph has an **Euler circuit** exactly when **every** vertex has even degree. A connected graph has an **Euler path but no circuit** exactly when it has **exactly two** vertices of odd degree, and any such path must start at one of them and finish at the other.

Conditions. The graph must be **connected**, apart from isolated vertices, which no walk needs to reach. Loops and parallel edges are allowed. Note that "exactly one odd vertex" never arises: by the Handshaking corollary of section 6.3, the number of odd-degree vertices is always even.

Why the degree condition is the right one. Every time a walk passes through a vertex it uses one edge to arrive and one to leave, so it consumes edges there in pairs. A vertex of odd degree therefore cannot be passed through cleanly: the walk must either start or finish there. With no odd vertices the walk can close up into a circuit; with two it runs between them; with four or more it cannot be a single walk at all.

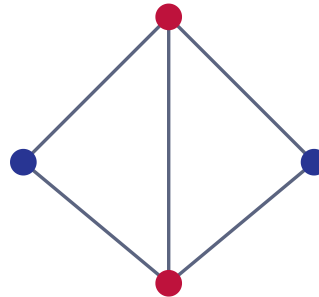
The limiting case. A graph with no edges satisfies the condition vacuously and has an empty Euler circuit, which is true and of no use. The theorem earns its keep in the negative direction: the seven bridges of Königsberg form a graph with **four** odd vertices, so no walk crosses every bridge once, and that impossibility is the result the whole subject grew from.

all degrees 2, all even



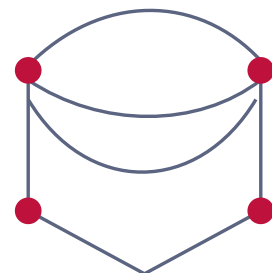
Euler circuit exists

two odd vertices, in crimson



Euler path only, between them

four odd vertices



neither path nor circuit

The whole theorem in three pictures. Count the odd-degree vertices: none, two, or more than two.

WORKED EXAMPLE

PQ 25/26 Q4(A)(V)

The question is quoted in full in section 6.1: "(v) Eulerian graph", 2 marks, wanting a short note with a relevant example.

Definition. A connected graph is **Eulerian** when it contains an Euler circuit, that is, a closed walk using every edge of the graph exactly once. By Euler's theorem a connected graph is Eulerian exactly when every one of its vertices has even degree.

Example 1, Eulerian. The 4-cycle drawn above: every vertex has degree 2, which is even, so the circuit a to b to c to d to a uses all four edges once and returns to the start.

Example 2, not Eulerian. The seven bridges of Königsberg, the problem Euler solved in 1736 and the origin of the whole subject. Four land masses joined by seven bridges give a graph whose four vertices all have odd degree (3, 3, 3 and 5), so no walk crosses every bridge exactly once. The townspeople had been trying; Euler proved it could not be done.

The distinction worth adding for the second mark: a graph with exactly two odd vertices has an Euler **path** but no circuit, and is therefore **not** Eulerian. Only the all-even case earns the name.

Hamilton paths and circuits

● MUST MEMORISE

HAMILTON PATH AND CIRCUIT

A **Hamilton path** is a walk that visits **every vertex of the graph exactly once**. A **Hamilton circuit** is a Hamilton path that returns to its starting vertex.

WHERE

every vertex exactly once is the condition, so here it is the **vertices** that are rationed and edges may be left unused entirely. **the starting vertex** of a circuit is visited twice in the sense that the walk returns to it, which is not counted as a repeat visit.

There is no easy test, and saying so is part of the answer. Euler circuits are settled by counting degrees. For Hamilton circuits **no such simple criterion is known**, and deciding whether one exists is one of the classic computationally hard problems. Only sufficient conditions exist, such as Dirac's: a simple graph on $n \geq 3$ vertices in which every vertex has degree at least $n/2$ has a Hamilton circuit. That is a one-way test, so a graph failing it may still have one.

◆ **Memory hook: Euler counts degrees, Hamilton counts on luck.** If an exam question asks you to find a Hamilton circuit, you are expected to find it by inspection; if it asks whether one exists in general, the answer involves saying there is no efficient test.

⚠ TRAP

EULER AND HAMILTON ARE EASY TO SWAP

	EULER	HAMILTON
Uses each	edge exactly once	vertex exactly once
May repeat	vertices	nothing, and may skip edges
Test	count odd-degree vertices	none known; only sufficient conditions
Everyday version	a postal round covering every street	a delivery visiting every house

The travelling salesman question of section 6.8 is a Hamilton circuit question with weights attached: visit every city once, return home, and do it as cheaply as possible. That is why it has no quick method either.

GIVEN The four-vertex graph of section 6.5: vertices 1, 2, 3, 4 with edges 1-2, 1-3, 2-3 and 3-4.

FIND Whether it has an Euler circuit, an Euler path, and a Hamilton path or circuit.

RULE Euler by counting odd degrees; Hamilton by inspection.

Step 1. The degrees, read off the adjacency matrix row sums in section 6.5: $\deg(1) = 2$, $\deg(2) = 2$, $\deg(3) = 3$, $\deg(4) = 1$.

Step 2. Count the odd ones. Vertices 3 and 4 have odd degree, and the other two are even. That is **exactly two** odd vertices.

Step 3. Apply Euler's theorem. Exactly two odd vertices means **no Euler circuit, but an Euler path exists**, and it must run between the two odd vertices, that is, between 3 and 4. One such path:

4 to 3 to 1 to 2 to 3

Check it: the edges used are 34, 31, 12, 23. That is all four edges, each once. Vertex 3 is visited twice, which an Euler path is allowed to do.

Step 4. Hamilton, by inspection. A Hamilton path must touch each of 1, 2, 3, 4 exactly once. Try 4 to 3 to 1 to 2: every vertex once, so **a Hamilton path exists**.

A Hamilton **circuit** would have to return from 2 to 4, and there is no edge 2-4. In fact vertex 4 has degree 1, and a circuit must both enter and leave every vertex, needing degree at least 2. So **no Hamilton circuit exists**, and the pendant vertex is the whole reason.

That last observation generalises and is worth remembering: a graph with any vertex of degree 1 can never have a Hamilton circuit.

Now redo it: add the edge 2-4. Degrees become 2, 3, 3, 2, so still exactly two odd vertices and still an Euler path but no circuit. Now 1 to 2 to 4 to 3 to 1 is a Hamilton circuit, so the answers to the two questions moved independently.

RECALL AND CHECK

1. Define graph isomorphism.
2. Name three invariants, and say what invariants can and cannot prove.
3. State Euler's condition for a circuit and for a path.
4. What is an Eulerian graph?
5. What is the difference between an Euler circuit and a Hamilton circuit?
6. Why can a graph with a pendant vertex have no Hamilton circuit?

ANSWERS

1. A bijection between the vertex sets under which two vertices are adjacent in one graph exactly when their images are adjacent in the other.
2. Vertex count, edge count, degree sequence (also connectedness, shortest cycle, bipartiteness). They can disprove isomorphism but never prove it; only exhibiting the bijection proves it.
3. A connected graph has an Euler circuit exactly when every degree is even, and an Euler path but no circuit exactly when exactly two degrees are odd.
4. A connected graph that has an Euler circuit, equivalently one whose every vertex has even degree.
5. Euler uses every edge once and may repeat vertices; Hamilton visits every vertex once and may skip edges.
6. A circuit must enter and leave every vertex it visits, which needs degree at least 2.

FILL IN THE GAP

1. Isomorphic graphs must have the same _____ sequence.
2. An Euler circuit exists exactly when every vertex has _____ degree.
3. Euler is about _____; Hamilton is about _____.
4. The Königsberg graph has _____ odd vertices, so no such walk exists.

ANSWERS

1. degree. 2. even. 3. edges, vertices. 4. four.

PART SIX · 6.7

Planarity, Euler's Formula and Kuratowski

Can this graph be drawn on paper with no edges crossing? Two graphs cannot, every other obstruction is built from those two, and the newest paper gives 10 marks to proving it.

By the end of this section you will be able to define a planar graph, state and use Euler's formula for planar graphs, derive and apply the two edge bounds, state Kuratowski's theorem, say why K_5 and $K_{3,3}$ matter, and prove both non-planar in full.

Planar graphs

● MUST MEMORISE

PLANAR GRAPH AND FACES

A graph is **planar** when it can be drawn in the plane with **no two edges crossing** except at their shared endpoints. Such a drawing is called a **plane drawing** of the graph.

A plane drawing cuts the plane into **faces**, or regions. The unbounded region outside the whole drawing is a face too, and forgetting it is the standard error.

WHERE

planar is a property of the **graph**: it asks whether *some* crossing-free drawing exists.

plane is a property of a particular **drawing**, one that happens to have no crossings. **a**

face is a region enclosed by edges, counting the infinite outer region as one.

Planarity is an invariant, in the sense of section 6.6: isomorphic graphs are either both planar or both non-planar, since a relabelling cannot create or remove a crossing. So it joins the list of properties that can disprove isomorphism.

Why a drawing with crossings proves nothing. K_4 is usually drawn as a square with both diagonals, which cross; but it can be redrawn as a triangle with one vertex inside, which does not. So a crossing you happen to have drawn never shows a graph is non-planar. Only a counting argument does, which is exactly what this section supplies.

◆ **Memory hook: planar means it can be drawn flat, not that you drew it flat.** And never forget the outside face.

EULER'S FORMULA FOR PLANAR GRAPHS

The claim. Every plane drawing of the same connected planar graph cuts the plane into the same number of regions, and that number is fixed by the vertex and edge counts alone. Vertices minus edges plus faces is always two, whatever the graph and however it is drawn.

In symbols.

$$v - e + f = 2$$

Conditions. The graph must be **connected** and **planar**, and f must be counted from a **plane drawing**, that is, one with no crossings, including the unbounded outer face. For a graph with k components the formula becomes $v - e + f = 1 + k$.

The limiting case. For a tree, which section 7.1 defines as a connected graph with no cycles, there are no enclosed regions at all, so $f = 1$, and the formula gives $v - e + 1 = 2$, that is, $e = v - 1$. The edge count of a tree is not a separate fact to memorise but this formula at its smallest possible f .

For a **simple connected planar** graph with $v \geq 3$ vertices,

$$e \leq 3v - 6$$

and if in addition the graph has **no triangles**, for example if it is bipartite,

$$e \leq 2v - 4$$

WHERE

v is the number of vertices and e the number of edges. $3v - 6$ is the general bound, coming from the fact that every face of a simple graph is bounded by at least 3 edges. $2v - 4$ is the triangle-free bound, coming from the fact that without triangles every face needs at least 4 edges.

Where they come from, which is worth knowing because the derivation is short and it is what makes the bounds usable rather than magic. Count the pairs (face, edge on its boundary) in two ways. Every edge borders exactly two faces, so that count is $2e$. Every face is bounded by at least 3 edges in a simple graph, so that count is at least $3f$. Hence

$$2e \geq 3f, \text{ so } f \leq 2e/3.$$

Substituting into Euler's formula $f = 2 - v + e$ gives

$$2 - v + e \leq 2e/3$$

$$6 - 3v + 3e \leq 2e$$

$$e \leq 3v - 6$$

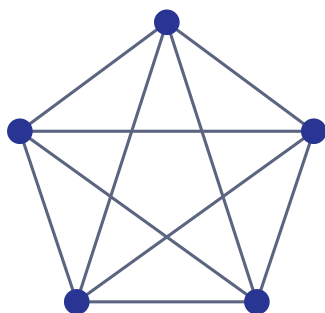
With no triangles every face needs at least 4 edges, so $2e \geq 4f$ and the same substitution gives $e \leq 2v - 4$.

How to use them. These are **one-way** tests. If a graph *breaks* the bound it is definitely not planar. If it satisfies the bound it may or may not be planar, and the bound has told you nothing.

◆ **Memory hook: too many edges for too few vertices means crossings are forced.**

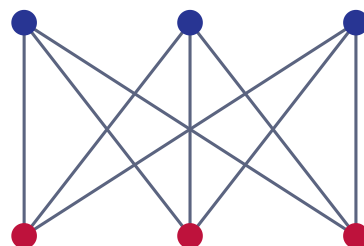
Three v minus six in general; two v minus four when there are no triangles to help.

The two graphs that matter



K_5 $v = 5, e = 10$

every crossing drawn here is unavoidable



$K_{3,3}$ $v = 6, e = 9$, bipartite

The two obstructions. K_5 joins five vertices to each other; $K_{3,3}$ joins each of three vertices to each of three others.

KURATOWSKI'S THEOREM

The claim. There are only two ways a graph can fail to be planar. A graph is planar if and only if it contains no subgraph that is a subdivision of K_5 or of $K_{3,3}$. Every non-planar graph, however large, hides one of those two inside it.

In symbols.

G is planar $\Leftrightarrow G$ contains no subdivision of K_5 and no subdivision of $K_{3,3}$.

Conditions. A **subdivision** of a graph is what you get by adding extra vertices in the middle of its edges, which never changes whether the graph can be drawn flat. The theorem holds for all finite simple graphs. Note the "if and only if": unlike the edge bounds, this is a complete characterisation, not a one-way test.

The limiting case, and why it is remarkable. Any graph on 4 or fewer vertices is planar, because K_5 needs 5 vertices and $K_{3,3}$ needs 6, so neither can be hidden inside anything smaller. K_4 is therefore the largest complete graph that can be drawn flat, and K_5 is the first that cannot. The force of the theorem is that the list stops at two: an infinite family of obstructions could have existed and does not.

The 10-mark planarity question

AS PRINTED

2025/2026 · Q5(B)

- (b) i. Briefly state and explain Kuratowski's Theorem. (2 marks)
- ii. Why are K_5 and $K_{3,3}$ important in graph theory? (2 marks)
- iii. Show that K_5 is non-planar using Euler's formula, Recall that Euler's formula for a K_5 is $e \leq 3v - 6$ and for a $K_{3,3}$ is $e \leq 2v - 4$. Also show that $K_{3,3}$ is non-planar. (6 marks)

NOTE ON THE PAPER

Part iii has a comma where a full stop belongs ("using Euler's formula, Recall that"), and the subscripts are printed inline as " K_5 " and " $K_{3,3}$ " rather than as K_5 and $K_{3,3}$.

ONE THING IN IT IS MATHEMATICALLY LOOSE AND YOU SHOULD HANDLE IT CAREFULLY.

The paper calls $e \leq 3v - 6$ and $e \leq 2v - 4$ "Euler's formula". They are not. Euler's formula is $v - e + f = 2$; these two are **COROLLARIES** of it, derived by combining it with a count of the edges bounding each face. The safe answer states Euler's formula properly, notes that the two inequalities follow from it, and then uses them exactly as the question directs. That earns the marks the question offers and shows you know the difference, without arguing with the examiner.

Note also that the bound $e \leq 2v - 4$ is not a general planar bound: it needs the graph to have no triangles. $K_{3,3}$ is bipartite so it has none, which is exactly why the paper offers that bound for it.

BREAK IT DOWN

GIVEN	The two inequalities, handed to you in the question. Part iii is a "show that", so the tools are supplied and the marks are for using them correctly.
ASKED	(i) State and explain Kuratowski. (ii) Why the two graphs matter. (iii) Show both are non-planar, which means a computation for each with a stated conclusion.
MARKS SAY	2, 2 and 6. Part iii carries three marks per graph: roughly one for the counts, one for the substitution, one for the conclusion. Do both graphs; a candidate who does only K_5 loses half of the largest part.
TEACHES	This section, 6.7, with the complete and complete bipartite graphs from 6.2.
TRAP	Miscounting the edges. K_5 has $C(5, 2) = 10$ edges, not 5; $K_{3,3}$ has $3 \times 3 = 9$, not 6. Get either count wrong and the inequality comes out satisfied, and you will confidently prove a non-planar graph planar.

- GIVEN** K_5 , the complete graph on 5 vertices, and $K_{3,3}$, the complete bipartite graph with 3 vertices on each side. The bounds $e \leq 3v - 6$ and $e \leq 2v - 4$.
- FIND** A statement of Kuratowski's theorem, the reason these two graphs matter, and a proof that each is non-planar.
- FORMULA** Euler's formula $v - e + f = 2$, and the two bounds derived from it.

i. State and explain Kuratowski's theorem

Statement. A graph is planar if and only if it contains no subgraph that is a subdivision of K_5 or of $K_{3,3}$.

Explanation. A subdivision of a graph is obtained by inserting extra vertices along its edges, which cannot affect whether the graph can be drawn without crossings, since the inserted vertices simply sit on the curves already drawn. The theorem therefore says that these two graphs are the **only** obstructions to planarity: any graph that cannot be drawn flat contains a copy of one of them, possibly with extra vertices along its edges, and any graph containing neither can be drawn flat.

The line that shows you understand it: this is an "if and only if", so unlike the edge bounds it decides the question in both directions. The bounds can only ever say "not planar"; Kuratowski also says "planar".

ii. Why K_5 and $K_{3,3}$ are important

Three reasons, one per half mark and a spare:

1. **They are the two smallest non-planar graphs**, and by Kuratowski's theorem they are the **only** obstructions to planarity. Every non-planar graph in existence contains a subdivision of one of them, so understanding two small graphs settles a question about infinitely many large ones.
2. **They mark the exact boundary.** K_4 is planar and K_5 is not, so K_4 is the largest complete graph that can be drawn flat. $K_{2,3}$ is planar and $K_{3,3}$ is not. The two graphs are where planarity stops.
3. **They matter in practice.** $K_{3,3}$ is the classic "three utilities to three houses" puzzle, which has no solution for exactly this reason. Planarity decides whether a circuit can be laid out on a single-layer board without wires crossing, so these two graphs are the test for when a second layer becomes unavoidable.

iii. Show that K_5 is non-planar

State the tool first. Euler's formula for a connected planar graph is $v - e + f = 2$. Combining it with the fact that every face of a simple graph is bounded by at least three edges gives the bound the question supplies: for any simple connected planar graph with $v \geq 3$,

$$e \leq 3v - 6$$

Step 1. Count v and e for K_5 . It has 5 vertices, and being complete it joins every pair, so by section 6.2

$$v = 5, \quad e = C(5, 2) = 5 \times 4 / 2 = 10$$

Step 2. Evaluate the bound.

$$3v - 6 = 3 \times 5 - 6 = 15 - 6 = 9$$

Step 3. Compare and conclude.

$$e = 10 > 9 = 3v - 6$$

Every simple connected planar graph on at least 3 vertices satisfies $e \leq 3v - 6$. K_5 does not. **Therefore K_5 is non-planar.** $\square$

iii, continued. Show that $K_{3,3}$ is non-planar

Step 1. Count v and e for $K_{3,3}$. Three vertices on each side gives $3 + 3 = 6$ vertices, and every vertex of one side joins every vertex of the other, so by the product rule of section 5.1

$$v = 6, \quad e = 3 \times 3 = 9$$

Step 2. Check that the tighter bound applies. This is the step most answers skip and it is worth a mark. $K_{3,3}$ is bipartite, and a bipartite graph has no odd cycles, so in particular it has **no triangles**. Every face of a plane drawing of it would therefore be bounded by at least four edges, which is what gives the bound

$$e \leq 2v - 4$$

Step 3. Evaluate and compare.

$$2v - 4 = 2 \times 6 - 4 = 12 - 4 = 8$$

$$e = 9 > 8 = 2v - 4$$

Therefore $K_{3,3}$ is non-planar. $\square$

Why the general bound would have failed here, worth one line because it shows why the question supplied two different inequalities: for $K_{3,3}$, $3v - 6 = 12$, and $e = 9$ is comfortably under 12. The general bound is satisfied, so it proves nothing. Only the triangle-free bound is sharp enough, and it applies only because the graph is bipartite.

GRAPH	V	E	$3V - 6$	$2V - 4$	VERDICT
K_4	4	6	6	not applicable, has triangles	bound satisfied; K_4 is planar
K_5	5	10	9	not applicable, has triangles	$10 > 9$, non-planar
$K_{3,3}$	6	9	12, no help	8	$9 > 8$, non-planar

Now redo it: is K_6 planar? $v = 6$ and $e = C(6, 2) = 15$, while $3v - 6 = 12$. Since $15 > 12$ it is non-planar, which also follows from Kuratowski because K_6 contains K_5 .

- GIVEN** K_4 drawn as a triangle with one vertex inside, joined to all three corners. No edges cross.
- FIND** The number of faces, and a check of Euler's formula.
- FORMULA** $v - e + f = 2$.

Count. $v = 4$ and $e = 6$.

Solve for f.

$$4 - 6 + f = 2, \text{ so } f = 4$$

Verify by looking at the drawing. The inner vertex splits the triangle into three small regions, and there is the unbounded region outside: $3 + 1 = 4$. The formula and the picture agree, and the outer face is the one that is easy to miss.

The bound as a sanity check: $3v - 6 = 6$ and $e = 6$, so K_4 sits exactly on the bound. A graph on the bound is called **maximal planar**: it is planar, and adding any further edge would make it non-planar.

RECALL AND CHECK

1. Define a planar graph, and say why a drawing with crossings proves nothing.
2. State Euler's formula and say what f counts.
3. State the two edge bounds and the condition on the second one.
4. State Kuratowski's theorem.
5. Show K_5 is non-planar in two lines.
6. Why does the bound $3v - 6$ fail to settle $K_{3,3}$?

ANSWERS

1. A graph is planar when some drawing of it has no crossing edges. A particular drawing with crossings may just be a bad drawing, as with K_4 .
2. $v - e + f = 2$, where f counts the faces of a plane drawing, including the unbounded outer face.
3. $e \leq 3v - 6$ for a simple connected planar graph with $v \geq 3$, and $e \leq 2v - 4$ when it also has no triangles.
4. A graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$.
5. $v = 5$ and $e = 10$, but $3v - 6 = 9$, and $10 > 9$.
6. Because $3v - 6 = 12$ and $e = 9$ satisfies it. The bound is not sharp enough; the triangle-free bound of 8 is.

FILL IN THE GAP

1. Euler's formula is _____, where f includes the _____ face.
2. K_5 has _____ edges and the bound allows _____.
3. $K_{3,3}$ has _____ edges and the triangle-free bound allows _____.
4. Kuratowski's theorem says the only obstructions to planarity are _____ and _____.

ANSWERS

1. $v - e + f = 2$, outer. 2. 10, 9. 3. 9, 8. 4. K_5 , $K_{3,3}$.

PART SIX · 6.8

Shortest Routes and the Travelling Salesman

The same five-city map has appeared on three papers, worth 10 marks every time, always with the same distances and always asking the same thing. It is the most reliable 10 marks in this book.

By the end of this section you will be able to state the travelling salesman problem, count how many tours a graph has, apply the nearest-neighbour method and say honestly what it does and does not guarantee, and produce the optimal Detroit tour with a proof that it is optimal.

● MUST MEMORISE

WEIGHTED GRAPHS AND THE TRAVELLING SALESMAN PROBLEM

A **weighted graph** is a graph in which every edge carries a number $w(u, v)$, its **weight**: a distance, a cost, a time. The **weight of a route** is the sum of the weights of the edges it uses.

The **travelling salesman problem** asks: starting from one vertex, visit **every other vertex exactly once** and return to the start, with the **smallest possible total weight**.

WHERE

$w(u, v)$ is the weight on the edge joining u and v , read straight off the labels on the drawing. a **tour** is a **Hamilton circuit**, as defined in section 6.6: every vertex once, back to the start. **smallest total weight** is what makes it an optimisation problem rather than an existence one. On a complete graph a tour always exists; the work is finding the cheapest.

◆ **Memory hook: every city once, home at the end, cheapest possible.** Drop any one of those three and it is a different problem with a different answer.

On a complete graph with n vertices, the number of **distinct** tours starting and ending at a fixed vertex is

$$(n - 1)! / 2$$

WHERE

$(n - 1)!$ counts the orders in which the other $n - 1$ cities can be visited, by the permutation rule of section 5.2. **the division by 2** is because a tour and the same tour driven backwards use the same edges and cost the same, so they are one route, not two.

For the five-city map: $(5 - 1)! / 2 = 24 / 2 = 12$ distinct tours. That is few enough to list them all, which is why this question is answerable exactly in an exam.

Why the problem is famous, in one line worth writing: the count grows factorially, so 10 cities give 181440 tours and 20 cities give more than 10^{16} . No method is known that finds the exact optimum quickly for large n , which is why heuristics like nearest neighbour exist at all.

◆ **Memory hook: fix the start, permute the rest, halve for the reverse.**

The question that appears three times

AS PRINTED

2020/2021 · Q2(A)

2. (a) A Christian missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance? **(10marks)**

AS PRINTED

2021/2022 · Q4(C)

(c) Given the graph below, a travelling missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance? **(10marks)**

AS PRINTED

2023/2024 · Q4(B)

(b) A Christian missionary is to embark on a trip from the city of Detroit through all other cities and returns to Detroit, in which order should he visit these cities to travel the minimum distance?

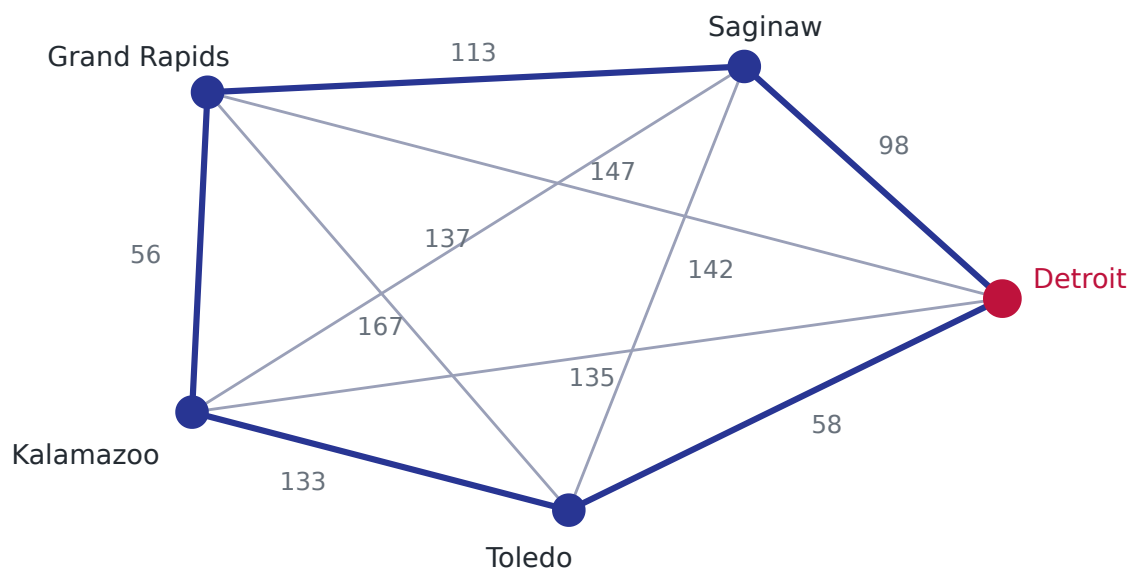
NOTE ON THE PAPER, AND ON THE FIGURE

Three papers, one question, 30 marks in total, and the same figure each time. The wording differs only in "A Christian missionary" against "a travelling missionary" and in whether the marks are printed closed up.

WHY THE 2023/2024 QUOTE ABOVE CARRIES NO MARK TEXT. On that paper the diagram sits between the question and its marks, and "(10 marks)" is printed on its own line *below* the figure. Splicing it onto the end of the sentence would join words the paper does not join, so the quote stops where the paper's sentence stops and the marks are reported here instead. They are 10, the same as the other two versions.

THE FIGURE IS THE SAME ON ALL THREE AND IT IS READABLE. It is a complete graph on five cities, so all $C(5, 2) = 10$ edges are drawn and labelled. The labels were read at 500 to 700 dpi and are listed in the worked answer below. Unlike the two digraphs of section 6.4, nothing here is in doubt.

WHAT "IN WHICH ORDER" MEANS FOR YOUR ANSWER. The question asks for an order, so the answer is a sequence of city names, not just a number. Give the route, give its total, and give the argument that no route is shorter. All three are needed for 10 marks.



The optimal tour, in indigo: Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit. Total

The five-city map as printed on all three papers, with the optimal tour drawn over it. All ten edges exist; only five are used.

BREAK IT DOWN

GIVEN	Five cities, all pairs joined, with the ten distances labelled on the figure. The tour must start and finish at Detroit.
ASKED	In which order the cities should be visited to travel the minimum distance. An order, a total, and a reason it is minimal.
MARKS SAY	10, the largest single part on any of these papers. Budget: 1 for tabulating the distances, 1 for the tour count, 4 for the enumeration or the method, 2 for the route, 2 for showing it is optimal.
TEACHES	This section, 6.8, with Hamilton circuits from 6.6 and permutations from 5.2.
TRAP	Giving the nearest-neighbour route and calling it the minimum without checking. Nearest neighbour is a heuristic: it is fast, it is often good, and it is not guaranteed to be optimal. With only 12 tours you can check them all, so there is no excuse for guessing here.

GIVEN Five cities forming a complete weighted graph, start and finish at Detroit. The ten distances are tabulated in step 1.

FIND The order of visits giving the minimum total distance, and that minimum.

RULE A tour is a Hamilton circuit; there are $(n - 1)!/2 = 12$ of them; the minimum is found by evaluating all 12.

Step 1. Tabulate the distances, reading them off the figure. Do this first, always: it takes thirty seconds and every later line depends on it.

	GRAND RAPIDS	SAGINAW	KALAMAZOO	TOLEDO	DETROIT
GRAND RAPIDS	0	113	56	167	147
SAGINAW	113	0	137	142	98
KALAMAZOO	56	137	0	133	135
TOLEDO	167	142	133	0	58
DETROIT	147	98	135	58	0

Step 2. Count the tours. Five cities with Detroit fixed as the start gives

$$(5 - 1)! / 2 = 4! / 2 = 24 / 2 = \mathbf{12} \text{ distinct tours.}$$

Twelve is small enough to evaluate exhaustively, and doing so is what turns "here is a short route" into "here is **the** shortest route".

Step 3. Evaluate all twelve. Abbreviate Grand Rapids as GR, Saginaw as S, Kalamazoo as K, Toledo as T and Detroit as D. Each row starts and ends at D.

TOUR	DISTANCES ADDED	TOTAL
D S GR K T D	$98 + 113 + 56 + 133 + 58$	458
D K GR S T D	$135 + 56 + 113 + 142 + 58$	504
D S K GR T D	$98 + 137 + 56 + 167 + 58$	516
D GR K S T D	$147 + 56 + 137 + 142 + 58$	540
D GR K T S D	$147 + 56 + 133 + 142 + 98$	576
D GR S K T D	$147 + 113 + 137 + 133 + 58$	588
D K GR T S D	$135 + 56 + 167 + 142 + 98$	598
D K S GR T D	$135 + 137 + 113 + 167 + 58$	610
D K T GR S D	$135 + 133 + 167 + 113 + 98$	646
D GR S T K D	$147 + 113 + 142 + 133 + 135$	670
D GR T K S D	$147 + 167 + 133 + 137 + 98$	682
D GR T S K D	$147 + 167 + 142 + 137 + 135$	728

Step 4. Read off the answer.

Order: Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit.

Minimum distance: 458.

Why it is optimal, which is the part worth two marks: all twelve possible tours have been evaluated and none is shorter than 458. This is not an estimate. The worst tour costs 728, so the choice of order matters by over 270, which is a good closing observation.

Note the route may be written either way round. Detroit, Toledo, Kalamazoo, Grand Rapids, Saginaw, Detroit is the same tour driven backwards and costs the same 458. Either is a correct answer; that is exactly what the division by 2 in step 2 was accounting for.

The nearest-neighbour method, and its honest status. If time is short, the heuristic is: from the current city, always go to the nearest unvisited one, and return home at the end.

AT	UNVISITED, WITH DISTANCES	NEAREST
Detroit	GR 147, S 98, K 135, T 58	Toledo, 58
Toledo	GR 167, S 142, K 133	Kalamazoo, 133
Kalamazoo	GR 56, S 137	Grand Rapids, 56
Grand Rapids	S 113	Saginaw, 113
Saginaw	return home	Detroit, 98

$$\text{total} = 58 + 133 + 56 + 113 + 98 = \mathbf{458}$$

On this map the heuristic happens to find the optimum, because the route it builds is the optimal tour driven backwards. Say "**happens to**". Nearest neighbour carries no guarantee: it commits to a cheap first step that can force an expensive last one, and on other maps it does exactly that. If you use it, use it to *find* a candidate and then justify the candidate by comparison, as above. A 10-mark answer that says "nearest neighbour gives 458, therefore 458 is the minimum" has asserted the thing it was asked to establish.

Now redo it: how many tours would six cities have? $(6 - 1)!/2 = 120/2 = 60$, five times as many for one extra city. That growth is the whole difficulty of the problem.

⚠ TRAP

A HEURISTIC IS NOT A PROOF

The three papers all say "the minimum distance". A minimum is a claim about **every** alternative, so it can only be established by ruling the alternatives out. With 12 tours, list them. With 60 you would explain the method and evaluate a well-chosen few, stating plainly that the result is a good route rather than a proven optimum.

Writing that distinction down costs one sentence and is the difference between an answer that is correct and one that is merely right.

RECALL AND CHECK

1. State the travelling salesman problem.
2. How many distinct tours has a complete graph on n vertices, and why the division by 2?
3. How many tours has the five-city map?
4. Give the optimal Detroit route and its total.
5. What does the nearest-neighbour method guarantee?

ANSWERS

1. Starting from one vertex of a weighted graph, visit every other vertex exactly once and return to the start, at minimum total weight.
2. $(n - 1)!/2$. Fixing the start leaves $(n - 1)!$ orders for the rest, and each tour is counted twice because driving it backwards costs the same.
3. $4!/2 = 12$.
4. Detroit, Saginaw, Grand Rapids, Kalamazoo, Toledo, Detroit, totalling 458.
5. Nothing. It is a heuristic that often gives a good route and is not guaranteed to give the shortest.

FILL IN THE GAP

1. A tour is a _____ circuit with weights attached.
2. A complete graph on n vertices has _____ distinct tours.
3. The five-city map has _____ tours and its minimum is _____.
4. Nearest neighbour is a _____, so it is not a _____.

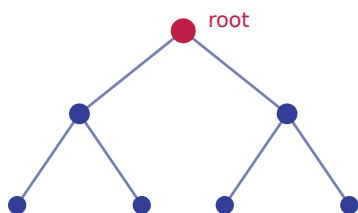
ANSWERS

1. Hamilton. 2. $(n - 1)!/2$. 3. 12, 458. 4. heuristic, proof.

PART SEVEN

Trees and Their Traversals

A tree is a graph with no way back. That single restriction makes it the structure computing actually uses: file systems, expressions, search, and the chain-letter question three of these papers keep asking.



$$e = v - 1$$

always

$$l = (m - 1)i + 1$$

full m-ary

at most m to the h

leaves at height h

pre, in, post

three traversals

Trees and Rooted-Tree Terminology

Two papers ask for the key terminology of a rooted tree, worth 7.5 marks each, and a third asks where trees are used. The vocabulary is the question.

By the end of this section you will be able to define a tree and a forest, prove that a tree has one fewer edge than it has vertices, name every part of a rooted tree with a labelled diagram, and answer the terminology and application questions in full.

NOT IN THE COURSE TEXT

The official text has no trees in it. Its twelve occurrences of the word "tree" are all in one Venn diagram about trees growing on a street. It never uses root, leaf or traversal in the technical sense. Yet 2024/2025 gives a whole 17.5-mark question to trees, and 2020/2021, 2021/2022 and 2023/2024 give them two parts each. Everything here is taught from zero.

What a tree is

● MUST MEMORISE

TREE AND FOREST

A **tree** is a connected undirected graph with **no cycles**.

A **forest** is an undirected graph with no cycles that need not be connected: each of its components is a tree.

WHERE

connected means there is a route between every pair of vertices, as defined in section 6.2.

no cycles means no route leaves a vertex and returns to it without repeating an edge.

This is the restriction that makes a tree a tree. **acyclic** is the one-word term for "no cycles", and a forest is exactly an acyclic graph.

Three equivalent definitions, any of which may be quoted. A graph on v vertices is a tree exactly when

1. it is connected and has no cycles; or
2. there is **exactly one** path between every pair of vertices; or
3. it is connected and has exactly $v - 1$ edges.

◆ **Memory hook: connected, and no way back.** Adding any edge to a tree creates a cycle; removing any edge disconnects it. A tree is as connected as a graph can be without being redundant.

A tree with v vertices has exactly

$$e = v - 1 \text{ edges.}$$

WHERE

v is the number of vertices and e the number of edges. **the $- 1$** is the root, or any one vertex you like: every other vertex is reached by exactly one edge.

It is not a separate fact. Section 6.7 gave Euler's formula $v - e + f = 2$ for a connected planar graph. A tree has no cycles, so it encloses no region and its only face is the outer one: $f = 1$. Then

$$v - e + 1 = 2, \text{ so } e = v - 1.$$

By the Handshaking theorem of section 6.3 the degrees therefore sum to $2(v - 1)$, which is a useful check on any tree you draw.

◆ **Memory hook: one edge per vertex, except the first.** A tree of 17 vertices has 16 edges, always.

Rooted trees

A **rooted tree** is a tree in which one vertex has been singled out as the **root**, and every edge is then understood as pointing away from the root.

WHERE

the root is chosen, not discovered: any vertex of a tree may be made the root, and different choices give different rooted trees from the same underlying tree. **away from the root** is what gives every other vertex a unique parent, and it is the whole reason the family vocabulary below makes sense.

◆ **Memory hook: pick a vertex and hang the tree from it.** Everything else in this section is the vocabulary of what happens once you have.

Let T be a rooted tree and v one of its vertices.

TERM	MEANING
Root	The single distinguished vertex. It has no parent, and it is the only vertex that does not.
Parent	The vertex immediately above v on the unique path from v to the root. Every vertex but the root has exactly one.
Child	Any vertex whose parent is v . A vertex may have any number of children, including none.
Siblings	Vertices that share a parent.
Ancestors	Every vertex on the path from v up to the root, not counting v itself.
Descendants	Every vertex having v as an ancestor, that is, everything hanging below v .
Internal vertex	A vertex with at least one child. Also called a branch vertex.
Leaf	A vertex with no children. Also called a terminal or external vertex.
Subtree at v	v together with all its descendants and the edges between them; itself a rooted tree with root v .
Level (depth) of v	The number of edges on the path from the root to v . The root is at level 0.
Height of T	The largest level of any vertex, that is, the length of the longest root-to-leaf path.
Ordered rooted tree	A rooted tree in which the children of each vertex are given a left-to-right order. The traversals of section 7.3 require this.

◆ **Memory hook: it is a family tree drawn upside down.** Parents above, children below, siblings alongside, ancestors all the way up, descendants all the way down. Every term means what it means in a family, except that the root is drawn at the top.

The terminology question

AS PRINTED

2020/2021 · Q2(C)

(c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. **(7½marks)**

AS PRINTED

2021/2022 · Q3(C)

(c) Highlight the various key terminologies of a Rooted Tree. Support your answer with a diagrammatical illustration. **(7½marks)**

NOTE ON THE PAPER

Word for word the same question on two papers, the same marks, the same capitalisation of "Rooted Tree". One prepared answer covers 15 marks across the set. "Highlight" here means list and define; "diagrammatical illustration" is not optional, and an answer without a drawing cannot reach full marks.

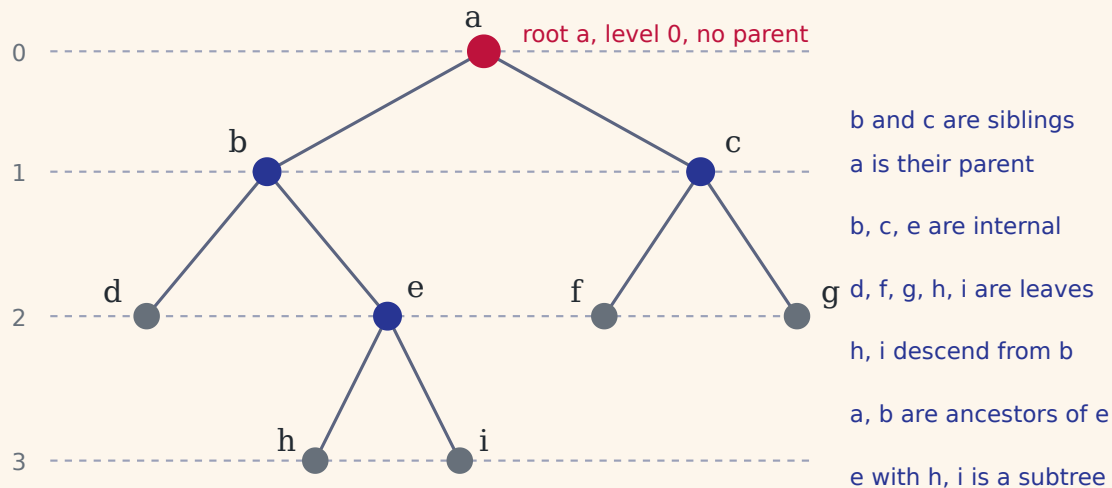
BREAK IT DOWN

GIVEN	Nothing. Pure recall.
ASKED	Highlight the various key terminologies , and support with a diagram . Two deliverables: a labelled list and one labelled tree.
MARKS SAY	7.5. At roughly half a mark per term, aim for ten to twelve terms plus the diagram. Define each in one line; do not write a paragraph on any of them.
TEACHES	This section, 7.1.
TRAP	Drawing a tree and labelling only the root and the leaves. The marks are spread across the terms, so the diagram should carry as many labels as it can hold: an internal vertex, a pair of siblings, a subtree, a level and the height.

Open with the definition, because every term below depends on it: a rooted tree is a connected acyclic graph in which one vertex has been designated the root, with all edges understood as directed away from it.

Then the terms. Use the table above, giving each in one line. Twelve terms at half a mark each covers the paper.

Then the diagram, fully labelled. This is the half of the answer that is easiest to earn and easiest to skip.



Levels down the left. The height is 3: the deepest level any vertex reaches.

9 vertices and 8 edges, as $e = v - 1$ requires.

One diagram carrying every term the question asks for. Draw the level lines: they earn the marks for level and height at no extra cost.

Close with the two counting facts, which show the terms are not just names: this tree has 9 vertices and 8 edges, satisfying $e = v - 1$, and its degrees sum to 16, which is twice the 8 edges as the Handshaking theorem requires.

Now redo it: in the diagram above, name the parent of h, the siblings of f, and the level of i.
Answers: e; g; level 3.

Where trees are used

Highlight, with a relevant example each, three application domains of Tree models in computing. **(4½ marks)**

NOTE ON THE PAPER

THIS PART HAS NO LETTER. After Q4(a) and Q4(b) the paper runs straight into this sentence with its marks but no "(c)", just as Question 4 itself carries no number. It is plainly the third part of Question 4, and it is quoted here exactly as printed, with no letter supplied.

Note the count: **THREE** domains with **ONE** example each, not the four with two examples that the graph version of this question asks for in section 6.1. Read the numbers before writing.

Three domains, each with what the vertices are, what the edges are, and one concrete example. Naming the modelling is what separates this from a list of words.

1. **File systems and hierarchical storage.** Vertices are directories and files; edges join a directory to what it contains. The root vertex is the root directory, every file has exactly one parent directory, and there are no cycles, which is precisely why the route from the root to any file is unique. *Example:* resolving a path name is a walk from the root down to a leaf.
2. **Compilers and expression evaluation.** Vertices are operators and operands; the children of an operator are the things it acts on. *Example:* a compiler parses source code into a syntax tree and then walks it to emit machine code, which is the expression-tree work of section 7.3.
3. **Searching and decision procedures.** Vertices are stored keys or states; edges are the comparisons or moves leading from one to the next. *Example:* a binary search tree finds a record in a number of comparisons proportional to the tree's height rather than to the number of records, which is why the height bound of section 7.2 matters.

A fourth, if the question asks for more: data compression, where a Huffman coding tree gives short codes to frequent symbols by placing them near the root.

RECALL AND CHECK

1. Define a tree, and give two other equivalent definitions.
2. How many edges has a tree with 17 vertices?
3. What is the difference between a tree and a rooted tree?
4. Define leaf, internal vertex, level and height.
5. Why does a tree have exactly one face?

ANSWERS

1. A connected acyclic undirected graph. Equivalently: exactly one path between every pair of vertices; or connected with exactly $v - 1$ edges.
2. 16, since $e = v - 1$.
3. A rooted tree has one vertex singled out as the root, which gives every other vertex a unique parent.
4. A leaf has no children; an internal vertex has at least one; the level of a vertex is its distance from the root; the height is the largest level.
5. It has no cycles, so it encloses no region; only the unbounded outer face remains, and Euler's formula then gives $e = v - 1$.

FILL IN THE GAP

1. A tree is a _____ graph with no _____.
2. A tree on v vertices has _____ edges.
3. The only vertex with no parent is the _____.
4. A vertex with no children is a _____.
5. The root sits at level _____.

ANSWERS

1. connected, cycles. 2. $v - 1$. 3. root. 4. leaf. 5. 0.

PART SEVEN • 7.2

m-ary and Balanced Trees

Two formulas answer almost every tree counting question on these papers, and the chain-letter problem they solve has appeared on three of the five in three different costumes.

By the end of this section you will be able to define m -ary, full m -ary, binary and balanced trees, use the two counting formulas to find the vertices, leaves or internal vertices of a full m -ary tree, prove the height bound on leaves, and answer the balanced-tree, chain-letter, quinary and playoff questions in full.

The families

● MUST MEMORISE

M-ARY, FULL AND BINARY

A rooted tree is **m -ary** when every internal vertex has **at most** m children.

It is a **full m -ary tree** when every internal vertex has **exactly** m children.

An m -ary tree with $m = 2$ is a **binary tree**; with $m = 3$ a **ternary tree**; with $m = 5$ a **quinary tree**, which is the word the 2024/2025 paper uses.

WHERE

m is a positive whole number, the branching factor. **at most m** is the m -ary condition; **exactly m** is the **full** condition, and the difference between them decides whether the counting formulas below apply at all. **a leaf** has no children, so the condition is only ever about internal vertices.

◆ **Memory hook: m -ary caps the branching, full fixes it.** Every full m -ary tree is m -ary; the reverse is not true, and the formulas need *full*.

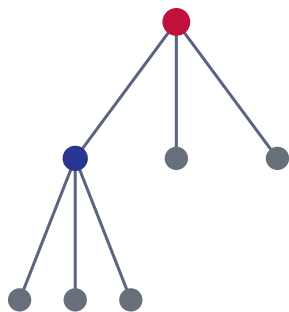
An m -ary tree of height h is **balanced** when every leaf lies at level h or at level $h - 1$.

WHERE

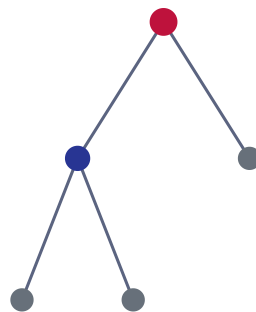
h is the height, the deepest level any vertex reaches, from section 7.1. **h or $h - 1$** is the whole condition: the leaves sit on the bottom two levels and nowhere higher. No leaf may dangle near the top while another branch runs deep.

Why balance is worth having. A balanced tree of n vertices has height about $\log_m n$, which is as small as it can be. Since searching costs one comparison per level, a balanced tree is what makes a search structure fast: an unbalanced tree can degenerate into a single long chain, at which point searching it is no better than reading a list.

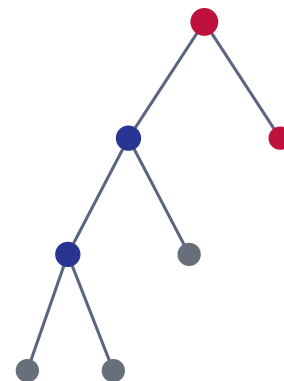
◆ **Memory hook: all the leaves on the bottom two shelves.** If any leaf is higher than that, the tree is not balanced.



Full ternary, balanced
every internal vertex has exactly 3
leaves at levels 1 and 2
leaves on levels 1 and 2 only



Full binary, balanced



Full binary, NOT balanced
a leaf at level 1, height 3

All three are full: every internal vertex has exactly m children. Only the first two are balanced. The crimson leaf on the right sits far above the bottom two levels, which is what breaks it.

The two counting formulas

● MUST MEMORISE

COUNTING A FULL M-ARY TREE

In a **full** m-ary tree with **i** internal vertices:

$$n = mi + 1 \quad (\text{total vertices})$$

$$l = n - i = (m - 1)i + 1 \quad (\text{leaves})$$

WHERE

m is the branching factor: every internal vertex has exactly m children. **i** is the number of **internal** vertices, the ones that do have children. **n** is the total number of vertices and **l** the number of leaves, so $n = i + l$ always.

Where $n = mi + 1$ comes from. Every vertex except the root is the child of exactly one internal vertex. Each of the i internal vertices supplies exactly m children, so there are mi non-root vertices, and adding the root gives $mi + 1$.

Then the leaves follow with no new idea: $l = n - i = (mi + 1) - i = (m - 1)i + 1$.

Rearranged for the exam, because questions usually give you the leaves and want the rest:

$$i = (l - 1) / (m - 1) \quad \text{and then} \quad n = mi + 1$$

◆ **Memory hook: every vertex but the root is somebody's child.** That one sentence gives $n = mi + 1$, and everything else in this box is algebra on it.

● MUST MEMORISE

THE HEIGHT BOUND

An m-ary tree of height h has **at most m^h leaves**.

WHERE

h is the height, from section 7.1. **m^h** is m multiplied by itself h times. **at most** is exact: equality holds precisely when the tree is full **and** every leaf is at level h.

Why. Level 0 holds the root alone. Each vertex has at most m children, so level 1 holds at most m vertices, level 2 at most $m \times m = m^2$, and level k at most m^k . Every leaf lies at some level no deeper than h, and the deepest level can hold at most m^h of them.

◆ **Memory hook: each level multiplies by m.** The product rule of section 5.1 applied h times: h independent choices of which child to descend to, m ways each.

The balanced m-ary tree question

AS PRINTED

2020/2021 · Q3(A)

3. (a) Explain, with examples, what a Balanced M-ary Tree means?

(7½marks)

AS PRINTED

2021/2022 · Q4(B)

(b) Explain, with examples, what a Balanced M-ary Tree means.

(8½marks)

NOTE ON THE PAPER

The same question twice, worth 7.5 and 8.5 marks, differing only in the final punctuation: 2020/2021 ends a command with a question mark, 2021/2022 with a full stop. Sixteen marks across the set for one prepared answer. "With examples" is in the instruction, so drawings are required, and the plural means more than one.

BREAK IT DOWN

GIVEN	Nothing. Recall with illustration.
ASKED	Explain what a balanced m -ary tree means, with examples . Two ideas are bundled in the phrase, and both need defining: <i>m</i> -ary and <i>balanced</i> .
MARKS SAY	7.5 and 8.5, which is a lot for one definition. The marks are for unpacking it: define m -ary, define full, define height and level, define balanced, then give a balanced example and an unbalanced one, and say why balance matters.
TEACHES	This section, 7.2, with the vocabulary of 7.1.
TRAP	Defining "balanced" and stopping. The term has two words in it, and an answer that never says what m -ary means has answered half the question. The second trap is giving only a balanced example: the counter example is what shows you know where the boundary is.

Build the answer in four layers, each earning its own marks.

1. m-ary

A rooted tree is m-ary when every internal vertex has at most m children. It is **full** m-ary when every internal vertex has exactly m children. For $m = 2$ it is called binary, for $m = 3$ ternary, for $m = 5$ quinary.

2. Level and height

The level of a vertex is the number of edges from the root to it, the root being at level 0. The height of the tree is the largest level any vertex reaches.

3. Balanced

An m-ary tree of height h is **balanced** when every leaf is at level h or at level $h - 1$: all the leaves lie on the bottom two levels. Putting the two words together, a **balanced m-ary tree** is a rooted tree in which no internal vertex has more than m children and every leaf sits on one of the two deepest levels.

4. The examples, drawn

Refer to the three trees drawn above. The first is a full ternary tree of height 2 whose leaves lie at levels 1 and 2, so it is balanced. The second is a full binary tree of height 2 with leaves at levels 1 and 2, also balanced. The third is a full binary tree of height 3 with one leaf at level 1, which is two levels above the bottom, so it is **not** balanced even though it is full.

That third example is the one that earns marks, because it shows that full and balanced are independent: a tree can be one without the other.

5. Why it matters, for the last mark

A balanced m-ary tree with n vertices has height roughly $\log_m n$, the smallest a tree of that size can have. Search cost is proportional to height, so balance is what keeps a search structure fast; an unbalanced tree can stretch into a chain and lose the advantage entirely.

Now redo it: is a full binary tree with leaves at levels 2, 3 and 3 balanced? Its height is 3, so leaves are allowed at levels 3 and 2. Yes, it is balanced.

The chain letter, in three costumes

(b) Suppose that someone starts a chain letter. Each person who receives the letter is asked to send it on to four other people. Some people do this, but others do not send any letters. How many people have seen the letter, including the first person, if no one receives more than one letter and if the chain letter ends after there have been 100 people who read it but did not send it out? How many people sent out the letter? **(8marks)**

3. (a) Suppose CU sends out convocation invitation cards to graduating students, each student who receives the cards is asked to send them on to four other students. While some students did send out the cards, others did not. If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out.

(i) How many people have seen the card, including the first person? **(5marks)**

(ii) How many people sent out the card? **(5marks)**

(d) Suppose CU sends out convocation invitation cards to graduating students, each student who receives the cards is asked to send them on to four (4) other students. While some students did send out the cards, others did not. If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out.

(i) How many people have seen the card, including the first person? **(6 marks)**

(ii) How many people sent out the card? **(3½ marks)**

NOTE ON THE PAPER

ONE PROBLEM, THREE PAPERS, 22.5 MARKS. 2020/2021 tells it as a chain letter and asks both questions in one run-on sentence for 8 marks; the two later papers tell it as convocation invitation cards and split it into (i) and (ii) for 10 and 9.5 marks. The numbers are identical every time: four recipients each, 100 people who did not pass it on.

Both card versions end with "If no one receives more than one card and if the card circulation ends after there have been 100 people who read it but did not send it out." That is a subordinate clause with no main clause, so the sentence never finishes. The numbered parts supply the question. Nothing is missing from the data.

The sentence doing the mathematical work is "**NO ONE RECEIVES MORE THAN ONE CARD**". That is what makes the picture a **TREE**: one parent per person, and no cycles.

BREAK IT DOWN

GIVEN	Each sender sends to exactly 4 others. Nobody receives twice. 100 people received it and did not send it on.
ASKED	(i) how many have seen it, including the first person; (ii) how many sent it out .
MODEL	A full 4-ary tree . Vertices are people. The root is the first person. A person's children are the four they sent to. Senders are the internal vertices; the 100 who did not send are the leaves .
MARKS SAY	8, 10 and 9.5 across the papers, far more than two formulas are worth. The marks are for the modelling: say what the vertices are, why it is a tree, why it is full, and which quantity is which.
TEACHES	This section, 7.2.
TRAP	Answering (i) with 100, or with 400. And getting the two questions the wrong way round: "seen it" is all the vertices n , "sent it out" is the internal vertices i .

GIVEN $m = 4$ recipients per sender; $l = 100$ people who read it but did not send it on; nobody receives it twice.

FIND n , the number who have seen it, and i , the number who sent it out.

FORMULA For a full m -ary tree, $l = (m - 1)i + 1$ and $n = mi + 1$.

Step 1. Model it, and say so. Draw each person as a vertex and each sending as an edge from sender to recipient. Because **no one receives more than one letter**, every person other than the first has exactly one parent, so the picture is a **rooted tree** with the first person at the root. Because everyone who sends, sends to **exactly four**, every internal vertex has exactly four children, so it is a **full 4-ary tree**.

senders = internal vertices = i

people who read but did not send = leaves = $l = 100$

people who have seen it = all vertices = n

Step 2. Find the internal vertices from the leaves.

$$l = (m - 1)i + 1$$

$$100 = (4 - 1)i + 1$$

$$100 = 3i + 1$$

$$99 = 3i$$

$$i = 33$$

Step 3. Find the total.

$$n = mi + 1 = 4 \times 33 + 1 = 132 + 1 = 133$$

Answers.

(i) 133 people have seen the letter, including the first person.

(ii) 33 people sent it out.

Check, and write the check down. The vertices split into senders and non-senders with nothing left over:

$$i + l = 33 + 100 = 133 = n \quad \checkmark$$

And count the edges: 33 senders each sending 4 gives 132 letters sent, which is exactly the 132 people other than the first. That is $e = v - 1$ from section 7.1, so the tree arithmetic closes twice over.

For the 2020/2021 version, which asks both questions in one sentence and pays 8 marks, give the same working and end with both numbers named: 133 have seen it, 33 sent it out.

Now redo it: same problem with each person sending to three others. Then $100 = 2i + 1$ gives $i = 49.5$, which is not a whole number, so **no such chain exists**: with $m = 3$ the leaf count is always odd. Saying that is the answer, and it is the same style of impossibility argument as section 6.3.

The quinary tree

AS PRINTED

2024/2025 · Q6(B)

- b. Consider the expression: $(3 - 4 \cdot 2)[\text{UNRESOLVED}] - (5 + (5 - 2)[\text{UNRESOLVED}])$
- i. Draw the rooted tree representing this computation **2 marks**
 - iv. Write this expression in prefix notation **2 marks**
 - v. Write this expression in postfix notation. **2 marks**
 - vi. A quinary tree has the height of 3. Find the total number of leaves present on it. **2.5 marks**
 - vii. Prove that a quinary tree is an m-ary tree of height 3 has at most m^h leaves. **3 marks**

NOTE ON THE PAPER, AND ON THE TWO MARKS IN GREY

THE NUMBERING SKIPS. The parts run i, then iv, v, vi, vii. There is no ii and no iii. The marks still total 17.5 for the question, so nothing is missing from the paper; only the labels are wrong.

[UNRESOLVED] IS NOT PRINTED ON THE PAPER. It is this manual's marker, shown in the quote as plain text, for a character that could not be read. The expression carries a single superscript after each of the two closing brackets, roughly two pixels tall in the source photograph. Rendered at 900 dpi they are still unreadable, and the PDF's own text layer dropped them too, returning "(3-4.2)- (5+ (5-2))". **THEY HAVE NOT BEEN GUESSED.** Section 7.3 solves the plain reading and says what changes if they turn out to be exponents.

PART VII IS NOT GRAMMATICAL: "Prove that a quinary tree is an m-ary tree of height 3 has at most m^h leaves" has two verbs where one belongs. The claim being asked for is the general one: an m-ary tree of height h has at most m^h leaves. Note also that the paper writes the exponent as a small raised h while the height in the sentence is fixed at 3, which is part of the same confusion. Prove the general statement and then apply it at $m = 5$, $h = 3$; that answers both readings.

Parts i, iv and v are answered in section 7.3, where expression trees are taught. Parts vi and vii are answered below.

BREAK IT DOWN

- GIVEN** A quinary tree, that is $m = 5$, of height $h = 3$.
- ASKED** (vi) **Find** the total number of leaves. (vii) **Prove** the bound. One computation and one proof.
- MARKS SAY** 2.5 and 3. Part vii is a proof, so induction or a level-by-level argument is expected, not an example.
- TEACHES** This section, 7.2.
- TRAP** Part vi says "the total number of leaves" as though it were determined, but a quinary tree of height 3 could have as few as 3 leaves. What is determined is the **maximum**, and that is what part vii is about. Say so, then give $5^3 = 125$.

GIVEN $m = 5$ (quinary), $h = 3$.

FIND The number of leaves, and a proof of the general bound.

FORMULA An m -ary tree of height h has at most m^h leaves.

vi. The number of leaves

Say what is determined. A quinary tree of height 3 need not have any particular number of leaves; the quantity fixed by m and h is the **maximum**, reached when the tree is full and every leaf sits at level 3. So:

$$\text{leaves} = m^h = 5^3 = 5 \times 5 \times 5 = 125$$

Answer: at most 125 leaves, and exactly 125 for the full quinary tree of height 3.

Check it against the other formula, which is worth a line. The full quinary tree of height 3 has internal vertices at levels 0, 1 and 2, that is

$$i = 1 + 5 + 25 = 31$$

and then $l = (m - 1)i + 1 = 4 \times 31 + 1 = 125$. The two formulas agree, as they must, and $n = mi + 1 = 5 \times 31 + 1 = 156$ total vertices.

vii. Prove the bound

To prove: an m -ary tree of height h has at most m^h leaves.

The method, in two sentences, before it is used. This is a proof by **mathematical induction**: show the claim holds at the smallest value of h , then show that whenever it holds at one height it must also hold at the next, and those two facts together carry it to every height. Section 9.2 sets the method out in full, with the reasons it works and several more examples; everything needed for this proof is in the sentence you have just read.

Proof, by induction on the height h .

Base case, $h = 0$. A tree of height 0 is a single vertex, which is the root and also a leaf. It has 1 leaf, and $m^0 = 1$. So the claim holds.

Inductive step. Suppose the claim holds for every m -ary tree of height $h - 1$, for some $h \geq 1$. Let T be an m -ary tree of height h . Remove the root. What remains is at most m subtrees, one for each child of the root, since T is m -ary. Each subtree has height at most $h - 1$, so by the inductive hypothesis each has at most m^{h-1} leaves.

Every leaf of T lies in one of those subtrees, so

$$\text{leaves of } T \leq m \times m^{h-1} = m^h$$

By induction the claim holds for every $h \geq 0$. $\square$

When is the bound attained? Exactly when the root has all m children, every subtree attains its own bound, and no leaf sits above level h ; that is, when the tree is **full** and all leaves are at level h . Adding that sentence shows the bound is sharp rather than merely true.

Applying it at $m = 5$, $h = 3$ gives $5^3 = 125$, which is part vi.

Now redo it: what is the greatest number of leaves a binary tree of height 4 can have? $m^h = 2^4 = 16$.

The playoff tree

AS PRINTED

2020/2021 · Q4(A)

4. (a) Playoff between two teams consists of at most five games. The first team that wins three games wins the playoff.

(i) Construct a tree diagram for five games.

(4marks)

(ii) In how many different ways can the playoff occur?

(3½marks)

NOTE ON THE PAPER

"Playoff between two teams" is missing its article; it should read "A playoff". More importantly, part (i) says "Construct a tree diagram for five games", but the playoff can finish in three or four games as well: the tree must show **EVERY WAY THE SERIES CAN END**, and its branches are of different lengths. Drawing exactly five levels everywhere would be wrong, because once a team has three wins no further game is played.

BREAK IT DOWN

GIVEN Two teams. At most five games. The first to three wins takes the playoff.

ASKED (i) **Construct** the tree diagram. (ii) **In how many different ways** the playoff can occur.

MODEL A **binary** tree. Each vertex is a state of the series; the two children are "team A wins the next game" and "team B wins it". A branch stops as soon as either team reaches three wins, so the leaves are exactly the possible outcomes.

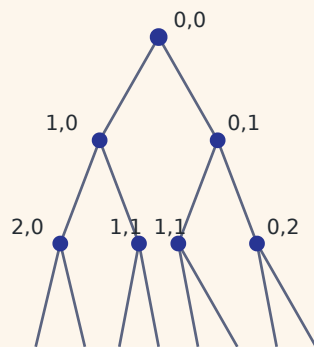
MARKS SAY 4 for the diagram, 3.5 for the count. The count is the number of **leaves**, so drawing the tree carefully answers both parts at once.

TEACHES This section, 7.2, with the combinations of 5.2.

TRAP Drawing all $2^5 = 32$ sequences of five games. Most of them are impossible, because the series stops the moment a team reaches three wins. The answer is 20, not 32.

- GIVEN** Two teams, first to 3 wins, at most 5 games.
- FIND** The tree of possible series, and the number of ways the playoff can occur.
- RULE** Each game branches two ways; a branch stops at 3 wins. The outcomes are the leaves.

Part (i): the tree. Label each vertex with the score so far as (wins for A, wins for B). The root is (0, 0). From any vertex, the left child adds a win for A and the right child a win for B. Stop as soon as either count reaches 3.



Series ending in	A wins	B wins
3 games	1	1
4 games	3	3
5 games	6	6
total	10	10

20 leaves in all

the first three levels: each game branches two ways
 a branch stops the moment a team reaches 3 wins
 so branches have different lengths

A winning in 4 games: the last game must be A's third win, so choose which 2 of the first 3 A also won: $C(3, 2) = 3$

The top of the tree, and the leaf count worked out by length. Every leaf is one possible series, so counting leaves answers part (ii).

Part (ii): how many ways. Count the leaves by how long the series runs. Take team A winning first.

- ▶ **A wins in 3 games.** Only AAA, so 1 way.
- ▶ **A wins in 4 games.** The fourth game must be A's third win, so A won exactly 2 of the first 3. Choosing which 2: $C(3, 2) = 3$.
- ▶ **A wins in 5 games.** The fifth game is A's third win, so A won exactly 2 of the first 4: $C(4, 2) = 6$.

A wins in $1 + 3 + 6 = 10$ ways.

By symmetry B also wins in 10 ways, and the two cases cannot both happen, so the sum rule of section 5.1 gives

$$\text{total} = 10 + 10 = 20$$

Answer: the playoff can occur in 20 different ways.

Why the last game must be the decider, which is the step most answers skip: if A had already won three games earlier, the series would have stopped there and the later games would never be played. So in any series of exactly k games, the k -th is the winner's third win.

Now redo it: best of three, first to two wins. A wins in 2 games: 1 way; in 3 games: $C(2, 1) = 2$ ways. So A wins in 3 ways, B in 3, total 6.

RECALL AND CHECK

1. What is the difference between an m-ary and a full m-ary tree?
2. When is an m-ary tree balanced?
3. State the two counting formulas for a full m-ary tree.
4. A full 4-ary tree has 100 leaves. How many vertices in total, and how many are internal?
5. What is the greatest number of leaves a quinary tree of height 3 can have?
6. In the playoff question, why is the answer not $2^5 = 32$?

ANSWERS

1. m-ary means every internal vertex has at most m children; full m-ary means exactly m.
2. When every leaf is at level h or h - 1, where h is the height.
3. $n = mi + 1$ and $l = (m - 1)i + 1$.
4. $i = (100 - 1)/3 = 33$ internal, and $n = 4 \times 33 + 1 = 133$ in total.
5. $5^3 = 125$.
6. The series stops as soon as a team reaches three wins, so most five-game sequences never occur. The answer is 20.

FILL IN THE GAP

1. In a full m-ary tree with i internal vertices, $n = \underline{\hspace{2cm}}$ and $l = \underline{\hspace{2cm}}$.
2. An m-ary tree of height h has at most $\underline{\hspace{2cm}}$ leaves.
3. The chain letter has $\underline{\hspace{2cm}}$ senders and $\underline{\hspace{2cm}}$ people who have seen it.
4. A tree is balanced when every leaf is at level $\underline{\hspace{2cm}}$ or $\underline{\hspace{2cm}}$.

ANSWERS

1. $mi + 1$, $(m - 1)i + 1$. 2. m^h . 3. 33, 133. 4. h, h - 1.

PART SEVEN • 7.3

Traversals and Expression Trees

Three ways to walk a tree, and each one writes an arithmetic expression in a different notation. The 2024/2025 paper gives 6 marks to the walks and 6 more to the notations.

By the end of this section you will be able to traverse an ordered rooted tree in preorder, inorder and postorder, build the expression tree of an arithmetic expression, read prefix and postfix notation off that tree, and answer both traversal questions on the 2024/2025 paper in full.

The three traversals

● MUST MEMORISE

PREORDER, INORDER, POSTORDER

Let T be an ordered rooted tree with root r and subtrees $T_1, T_2, \dots, T_k$ hanging from r , taken left to right. Each traversal is defined by where the root goes.

Preorder: visit **r first**, then traverse $T_1, \dots, T_k$ in preorder.

Inorder: traverse T_1 in inorder, then visit **r** , then traverse $T_2, \dots, T_k$ in inorder.

Postorder: traverse $T_1, \dots, T_k$ in postorder, then visit **r last**.

WHERE

ordered means the children of every vertex have a fixed left-to-right order. Without it none of the three is defined, which is why section 7.1 named ordered rooted trees separately. T_1 is the **leftmost** subtree, and inorder treats it differently from all the others. **a single vertex** is its own traversal in all three orders: the recursion stops there.

The names say it. "Pre", "in" and "post" describe where the **root** is visited relative to its subtrees: before them, in among them, or after them. Nothing else differs.

◆ **Memory hook: the prefix tells you when the root is visited.** Pre-order: root first. Post-order: root last. In-order: root in the middle, after the leftmost subtree only.

TEACH

HOW TO DO IT WITHOUT LOSING YOUR PLACE

The reliable method by hand is the **tracing** method. Draw a curve that starts at the root, hugs the outside of the whole tree anticlockwise, and returns to the root. Every vertex is passed **three** times: on its left, underneath it, and on its right.

- ▶ **Preorder:** list each vertex the **first** time the curve passes it, on the left.
- ▶ **Inorder:** list each vertex the **second** time, from below.
- ▶ **Postorder:** list each vertex the **third** time, on the right.

One curve gives all three lists, which matters when a question asks for all three of the same tree, as 2024/2025 Q6(a) does.

The check that catches nearly every slip: each list must contain **every vertex exactly once**, so its length equals the number of vertices. In preorder the **root comes first**; in postorder the **root comes last**. If either of those fails, stop and redo it.

The traversal question

AS PRINTED

2024/2025 · Q6(A)

- List the vertices in the order that you would visit them if you traversed the tree in preorder. **2 marks**
- List the vertices in the order that you would visit them if you traversed the tree in inorder. **2 marks**
- List the vertices in the order that you would visit them if you traversed the tree in postorder. **2 marks**

NOTE ON THE PAPER, AND ON THE FIGURE

The part opens with the line "a. Consider the following tree" and then prints the figure, with the three numbered parts below it. The quote above starts at part i for that reason: splicing the lead line onto part i would join text the paper separates with a whole diagram.

The tree is a drawing, so its structure is read off the picture. It has 17 vertices, and it is **NOT** binary: the root has 2 children but two other vertices have 3 each, which matters because the in-order rule treats the leftmost subtree differently from the rest. The structure read from the scan is:

a is the root with children b and c; b has children d, e, f; e has children j and k; k has children n and o; c has children g, h, i; h has children l and m; m has children p and q. The leaves are d, f, j, n, o, g, l, p, q and i.

One label is worth flagging: the vertex between d and f is blurred and could be misread as "c", but c is already the root's other child and the run of labels reads d, e, f, g, so it is **E**. Everything else on the figure is unambiguous.

BREAK IT DOWN

GIVEN An ordered rooted tree of 17 vertices, drawn on the paper.

ASKED **List the vertices** in each of the three orders. Three lists, nothing else. No explanation is asked for and none is needed.

MARKS SAY 2 each. That is one mark for the method and one for getting all seventeen right, so accuracy is the whole game. Draw the tracing curve once and read all three off it.

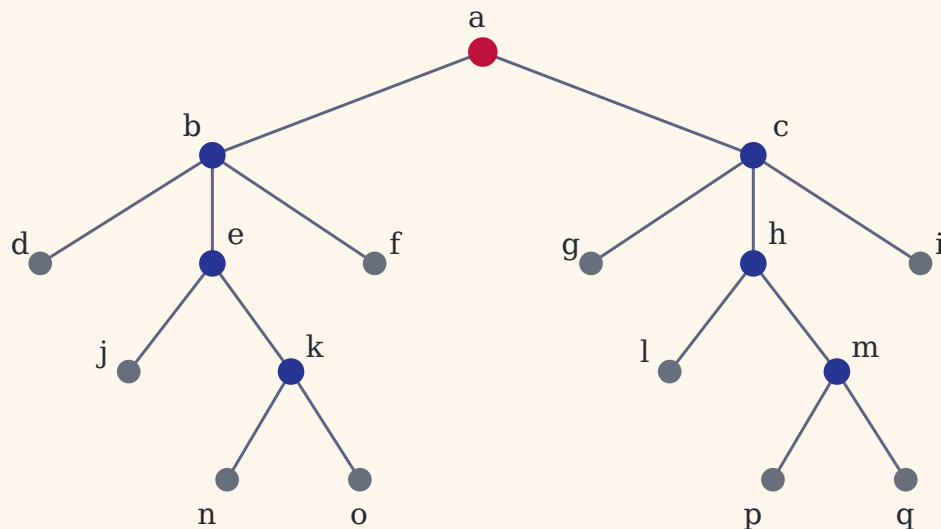
TEACHES This section, 7.3.

TRAP Applying the **binary** inorder rule, "left subtree, root, right subtree", to vertices that have three children. The general rule is: leftmost subtree, then the root, then **all** the remaining subtrees in order. For b, with children d, e, f, that gives d, then b, then e's subtree, then f.

GIVEN The 17-vertex ordered rooted tree drawn below, read off the paper's figure.

FIND The preorder, inorder and postorder lists.

RULE Preorder visits the root first, inorder after the leftmost subtree, postorder last.



17 vertices, 16 edges. Crimson is the root; indigo the internal vertices; grey the ten leaves.

The tree as read from the paper. Note that b and c each have three children, so this is not a binary tree.

i. Preorder

Root first, then each subtree in turn. Take a, then everything under b, then everything under c.

under b: b, then d, then e's subtree, then f

e's subtree: e, then j, then k's subtree, which is k, n, o

so under b: b, d, e, j, k, n, o, f

under c: c, g, then h's subtree, then i

h's subtree: h, l, then m's subtree, which is m, p, q

so under c: c, g, h, l, m, p, q, i

a, b, d, e, j, k, n, o, f, c, g, h, l, m, p, q, i

ii. Inorder

Leftmost subtree, then the root, then the rest. At a: everything under b, then a, then everything under c.

at b, children d, e, f: inorder(d), b, inorder(e), inorder(f)

at e, children j, k: j, e, inorder(k); at k, children n, o: n, k, o

so inorder(e) = j, e, n, k, o, and under b: d, b, j, e, n, k, o, f

at c, children g, h, i: g, c, inorder(h), i

at h, children l, m: l, h, inorder(m); at m: p, m, q

so inorder(h) = l, h, p, m, q, and under c: g, c, l, h, p, m, q, i

d, b, j, e, n, k, o, f, a, g, c, l, h, p, m, q, i

iii. Postorder

All the subtrees first, root last.

under b: d, then postorder(e), then f, then b
 postorder(e) = j, then postorder(k) = n, o, k, then e, giving j, n, o, k, e
 so under b: d, j, n, o, k, e, f, b
 under c: g, then postorder(h), then i, then c
 postorder(h) = l, then p, q, m, then h, giving l, p, q, m, h
 so under c: g, l, p, q, m, h, i, c

d, j, n, o, k, e, f, b, g, l, p, q, m, h, i, c, a

Check all three. Each list has 17 entries and each contains every vertex exactly once. Preorder starts with a; postorder ends with a; inorder has a in the middle, right after the whole of b's subtree. All four conditions hold.

Now redo it: traverse just the subtree at k, whose children are n and o. Preorder k, n, o. Inorder n, k, o. Postorder n, o, k.

Expression trees

● MUST MEMORISE

EXPRESSION TREE, PREFIX AND POSTFIX

An **expression tree** represents an arithmetic expression: each **leaf** is a number, and each **internal vertex** is an operator whose children are the things it acts on, in order.

Traversing that tree gives the three standard notations:

TRAVERSAL	NOTATION	ALSO CALLED	OPERATOR SITS
preorder	prefix notation	Polish notation	before its operands
inorder	infix notation	ordinary algebra	between its operands
postorder	postfix notation	reverse Polish	after its operands

WHERE

the tree records the grouping, so the brackets of the original expression are built into its shape. **prefix** and **postfix** need **no brackets at all** and are never ambiguous, which is exactly why compilers and calculators use them. **infix** is the notation people write, and it is the only one of the three that needs brackets and precedence rules.

Building the tree: find the operator that is applied **last**, which is the one *least* tightly bound. That becomes the root. Its two sides become its subtrees, and the same rule is applied to each of them.

◆ **Memory hook: the root is the last operation performed.** Work out what you would compute last, put it at the top, and recurse.

The expression question

THE QUESTION

2024/2025 Q6(b) is quoted in full in section 7.2, where its parts vi and vii are answered. Its parts i, iv and v are answered here: draw the rooted tree of the expression, then write it in prefix and in postfix notation, 2 marks each.

THE EXPRESSION CARRIES TWO CHARACTERS THAT COULD NOT BE READ. As recorded in 7.2, a single superscript follows each of the two closing brackets and neither is legible at any resolution available, including in the PDF's own text layer. They are not guessed. The answer below solves the **PLAIN READING**, in which those characters are absent, and then says exactly what changes if they turn out to be exponents.

BREAK IT DOWN

GIVEN	The expression $(3 - 4 \cdot 2) - (5 + (5 - 2))$ on the plain reading, where $\cdot$ is multiplication.
ASKED	(i) Draw the rooted tree. (iv) Write it in prefix notation. (v) Write it in postfix notation.
MARKS SAY	2 each. Parts iv and v are read straight off the tree from part i, so the tree is worth getting right first: one drawing earns all 6 marks.
TEACHES	This section, 7.3.
TRAP	Putting the wrong operator at the root. Inside the first bracket, multiplication binds tighter than subtraction, so the subtraction is done last and is the root of that subtree, not the multiplication. Getting the root wrong makes all three answers wrong together.

GIVEN $(3 - 4 \cdot 2) - (5 + (5 - 2))$, the plain reading of the printed expression.

FIND The expression tree, then the prefix and postfix forms.

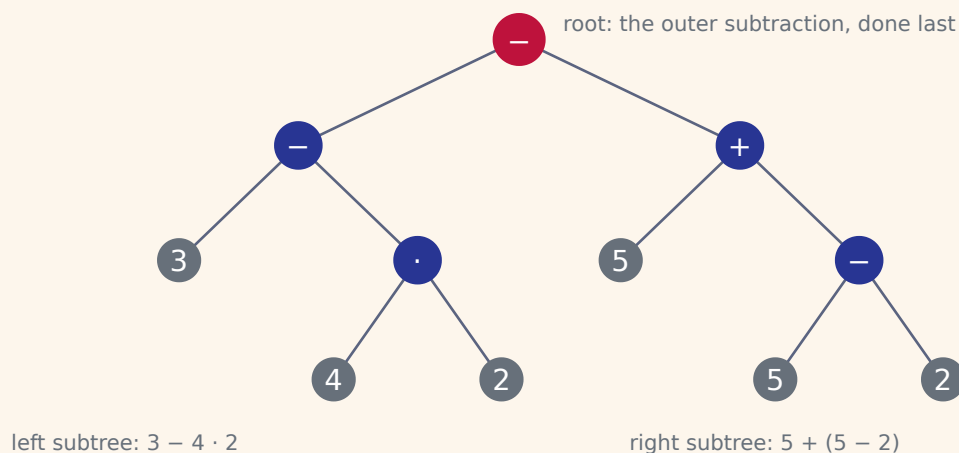
RULE The last operation performed is the root. Preorder gives prefix, postorder gives postfix.

i. The rooted tree

Find the root. The whole expression is (something) $-$ (something), and that outer subtraction is what you would do last. So the root is $-$.

Left subtree, $3 - 4 \cdot 2$. Multiplication binds tighter than subtraction, so $4 \cdot 2$ is computed first and the subtraction last. Its root is $-$, with children 3 and the subtree $4 \cdot 2$.

Right subtree, $5 + (5 - 2)$. The bracket is computed first, so the addition is last. Its root is $+$, with children 5 and the subtree $5 - 2$.



The expression tree. Operators at the internal vertices, numbers at the leaves, and no brackets anywhere: the shape holds the grouping.

iv. Prefix notation

Read the tree in **preorder**: root, then left subtree, then right.

root $-$, then the left subtree $(- 3 \cdot 4 2)$, then the right subtree $(+ 5 - 5 2)$

$- - 3 \cdot 4 2 + 5 - 5 2$

v. Postfix notation

Read the tree in **postorder**: left subtree, right subtree, then root.

left subtree $(3 4 2 \cdot -)$, then the right subtree $(5 5 2 - +)$, then the root $-$

$3 4 2 \cdot - 5 5 2 - + -$

Check by evaluating the postfix form, which costs four lines and proves the tree is right. Read left to right, pushing numbers and applying each operator to the two most recent values:

READ	STACK AFTERWARDS
3, 4, 2	3, 4, 2
.	3, 8 (4 × 2)
−	−5 (3 − 8)
5, 5, 2	−5, 5, 5, 2
−	−5, 5, 3 (5 − 2)
+	−5, 8 (5 + 3)
−	−13 (−5 − 8)

Directly: $(3 - 8) - (5 + 3) = -5 - 8 = -13$. The two agree, so the tree, the postfix form and the arithmetic are all consistent.

If the two unreadable characters are exponents. Say this in one line on the script rather than leaving it unaddressed. Each would become an extra internal vertex, an exponentiation operator, sitting above the bracket it follows: the left subtree would become $(3 - 4 \cdot 2)$ raised to that power, and the inner $(5 - 2)$ likewise. The tree gains two vertices, the root and the overall shape are unchanged, and both notations gain the two operators in the corresponding places. Nothing about the method changes.

Now redo it: give the prefix and postfix forms of $(2 + 3) \cdot 4$. The root is $\cdot$, so prefix is $\cdot + 2 3 4$ and postfix is $2 3 + 4 \cdot$.

⚠ TRAP

PREFIX IS NOT "THE EXPRESSION BACKWARDS"

Postfix is not the reverse of prefix, and neither is produced by moving the operators about by eye. Both are read off the **tree**, and the tree is where the grouping lives. Compare:

INFIX	PREFIX	POSTFIX
$3 - 4 \cdot 2$	$- 3 \cdot 4 2$	$3 4 2 \cdot -$
$(3 - 4) \cdot 2$	$\cdot - 3 4 2$	$3 4 - 2 \cdot$

Those two infix expressions differ only by a pair of brackets and they have different values, -5 and -2 . Their prefix and postfix forms use the same five symbols in different orders, with **no brackets at all**. That is the property that makes these notations worth having, and it only works if the tree was built correctly.

RECALL AND CHECK

1. Define preorder, inorder and postorder in one line each.
2. Which traversal gives prefix notation, and which gives postfix?
3. In the 17-vertex tree, which vertex comes first in preorder and which comes last in postorder?
4. How do you find the root of an expression tree?
5. Why do prefix and postfix need no brackets?
6. Give the postfix form of $(2 + 3) \cdot 4$.

ANSWERS

1. Preorder visits the root before its subtrees; postorder after them; inorder after the leftmost subtree and before the rest.
2. Preorder gives prefix; postorder gives postfix.
3. a in both cases: the root is first in preorder and last in postorder.
4. It is the operator applied last, the most loosely bound one.
5. The order of the symbols already records the grouping, so no punctuation is needed to disambiguate it.
6. $2\ 3\ +\ 4\ \cdot$

FILL IN THE GAP

1. Preorder visits the root _____; postorder visits it _____.
2. Prefix notation is also called _____ notation.
3. The root of an expression tree is the operator applied _____.
4. The postfix form of $(3 - 4 \cdot 2) - (5 + (5 - 2))$ is _____.

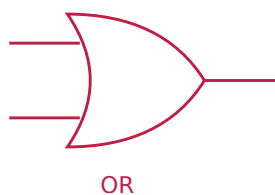
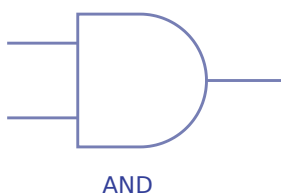
ANSWERS

1. first, last. 2. Polish. 3. last. 4. $3\ 4\ 2\ \cdot\ -\ 5\ 5\ 2\ -\ +\ -$

PART EIGHT

Boolean Algebra and Logic Circuits

The same laws as Part One and Part Two, written a third way. Once you see that AND, intersection and conjunction are one operation under three names, this part is a translation exercise.



$$A \cdot B$$

$$F + S$$

$$(A \cdot B)'$$

Boolean Algebra and Its Laws

Two values, three operations, and a set of laws you have already met twice. The newest paper spends 7.5 marks on turning two sentences about a door and an alarm into Boolean expressions.

By the end of this section you will be able to write Boolean expressions for AND, OR and NOT, build a Boolean table, name the laws of Boolean algebra and their duals, translate between Boolean algebra, logic and sets, draw the corresponding gates, and answer the door and alarm questions in full.

NOT IN THE COURSE TEXT

The official text has no Boolean algebra in it: no Boolean expression, no logic gate, no truth table written in ones and zeros. The 2025/2026 paper gives it 7.5 marks. Everything here is taught from zero, but almost none of it is new: it is section 1.3 in different clothing, and the translation table below is the whole trick.

The two values and three operations

● MUST MEMORISE

BOOLEAN ALGEBRA

A **Boolean variable** takes one of exactly two values, written **1** and **0**, read as **true** and **false**, or as **on** and **off**. The three operations are

OPERATION	WRITTEN	READ	VALUE IS 1 EXACTLY WHEN
Boolean product	$A \cdot B$, or AB	A AND B	both A and B are 1
Boolean sum	$A + B$	A OR B	at least one of A, B is 1
Complement	A' , or $\bar{A}$	NOT A	A is 0

WHERE

· is **not** ordinary multiplication and + is **not** ordinary addition, although · behaves exactly like multiplication on 0 and 1. The one place they part company is $1 + 1 = 1$, not 2: there is no value above 1 for it to be. A' is also written with an overbar. Both appear on these papers.

The complete arithmetic, which is worth writing out once:

$$\begin{aligned} 0 \cdot 0 &= 0, & 0 \cdot 1 &= 0, & 1 \cdot 0 &= 0, & 1 \cdot 1 &= 1 \\ 0 + 0 &= 0, & 0 + 1 &= 1, & 1 + 0 &= 1, & 1 + 1 &= 1 \\ 0' &= 1, & 1' &= 0 \end{aligned}$$

◆ **Memory hook: product needs both, sum needs one.** And the only surprise in the whole table is $1 + 1 = 1$, which is just "true or true is true".

Every law in this part has already been proved twice in this book, once for propositions in section 1.3 and once for sets in section 2.2. They are the same laws.

BOOLEAN ALGEBRA	LOGIC (PART ONE)	SETS (PART TWO)
1	T, a tautology	U, the universal set
0	F, a contradiction	$\emptyset$, the empty set
$A \cdot B$	$p \wedge q$	$A \cap B$
$A + B$	$p \vee q$	$A \cup B$
A'	$\neg p$	A' , the complement
$A = B$	$p \equiv q$	$A = B$

◆ **Memory hook: AND, intersection and conjunction are one operation with three names.** If you can do section 1.3, you can do this part: replace $\wedge$ by $\cdot$, $\vee$ by $+$, and $\neg$ by a dash.

For all Boolean variables A, B, C:

NAME	IN TERMS OF $\cdot$	IN TERMS OF $+$
Identity	$A \cdot 1 = A$	$A + 0 = A$
Domination	$A \cdot 0 = 0$	$A + 1 = 1$
Idempotent	$A \cdot A = A$	$A + A = A$
Complement	$A \cdot A' = 0$	$A + A' = 1$
Double complement	$(A')' = A$	
Commutative	$A \cdot B = B \cdot A$	$A + B = B + A$
Associative	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$	$(A + B) + C = A + (B + C)$
Distributive	$A \cdot (B + C) = A \cdot B + A \cdot C$	$A + B \cdot C = (A + B) \cdot (A + C)$
De Morgan	$(A \cdot B)' = A' + B'$	$(A + B)' = A' \cdot B'$
Absorption	$A \cdot (A + B) = A$	$A + A \cdot B = A$

The duality principle. Every law above comes in a pair, and each half is obtained from the other by swapping $\cdot$ with $+$ and **0** with **1**. That is not a coincidence: it is a theorem, and it means you only ever have to memorise half the table.

Two that are not in Part One's list and are worth noting. **Distributing $+$ over $\cdot$** , that is $A + B \cdot C = (A + B) \cdot (A + C)$, has no counterpart in ordinary arithmetic, where $a + bc$ is certainly not $(a + b)(a + c)$. Boolean algebra is not number algebra, and this is where the difference bites hardest. **Absorption** is the law that does most of the work in simplifying circuits.

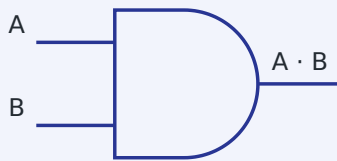
◆ **Memory hook: swap the dots for plusses and the zeros for ones, and the law still holds.** Learn one column and derive the other.

Gates

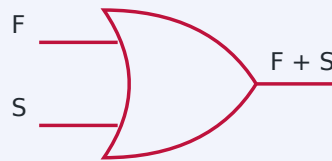
TEACH

THE THREE BASIC GATES

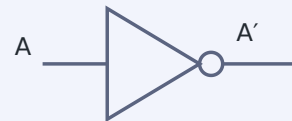
A **logic gate** is a physical component computing one Boolean operation. Circuits are drawn from them, and a Boolean expression and a circuit diagram are two views of the same thing.



AND gate
output 1 only if both inputs are 1



OR gate
output 1 if either input is 1



NOT gate
output is the opposite of t

The three gates the papers use. The small circle on the NOT gate is the standard mark for inversion.

Two more names appear often enough to be worth knowing: a **NAND** gate computes $(A \cdot B)'$ and a **NOR** gate computes $(A + B)'$, that is, an AND or an OR followed by an inversion.

The door and the alarm

AS PRINTED

2025/2026 · Q5(A)

- (a) i. A secure laboratory door opens only when a valid ID card (A) and a correct finger-print (B) are presented. Express this condition using Boolean Algebra and determine when the door opens. Draw the Boolean table. **(4½ marks)**
- ii. An alarm should sound if either a fire sensor (F) or a smoke sensor (S) is activated. Formulate the Boolean expression and determine the output when only the smoke sensor detects danger. **(3 marks)**

NOTE ON THE PAPER

Nothing is printed wrongly here, but two words are doing precise work and are easy to read past.

Part i says the door opens "**ONLY WHEN**" both are presented. Section F.2 covered that phrase: "only when" states a **NECESSARY** condition. Read strictly, it says the door does not open without both, which does not by itself promise that presenting both *does* open it. The intended reading, and the one to answer, is that both together are necessary and sufficient, so $\text{Door} = A \cdot B$. Say which reading you are using in one line; it costs nothing and shows you noticed.

Part ii says "**EITHER ... OR**", which in this context means the **INCLUSIVE** or: the alarm should certainly sound if *both* sensors fire. So it is $A + B$, not exclusive or. A candidate who reads "either ... or" as exclusive gets an alarm that stays silent in a real fire, which is worth saying in the answer.

BREAK IT DOWN

GIVEN	Part i: two conditions A and B, both required. Part ii: two sensors F and S, either sufficient.
ASKED	Part i: express the condition, determine when the door opens, draw the Boolean table. Three deliverables for 4.5 marks. Part ii: formulate the expression and determine the output for one specific input.
MARKS SAY	4.5 and 3. In part i the table is explicitly asked for and is worth roughly 2 of the 4.5, so an answer without it cannot pass. In part ii "determine the output when only the smoke sensor detects danger" is a single substitution: name $F = 0$, $S = 1$ and evaluate.
TEACHES	This section, 8.1.
TRAP	Answering part ii with the expression only. It asks for a specific output as well, and that is where the marks split.

GIVEN Part i: A = valid ID card presented, B = correct fingerprint presented, each 1 if so and 0 if not. Part ii: F = fire sensor activated, S = smoke sensor activated.

FIND The Boolean expression for each, the Boolean table for the door, and the alarm output when only S is 1.

RULE "and" is the Boolean product; "either or" is the inclusive Boolean sum.

i. The laboratory door

Assign the variables. Let A = 1 when a valid ID card is presented and 0 otherwise; B = 1 when a correct fingerprint is presented and 0 otherwise. Let D = 1 when the door opens.

The expression. The door opens only when **both** are presented, so the two conditions are combined with AND:

$$D = A \cdot B$$

Determine when the door opens. By the arithmetic of the Boolean product, $D = 1$ exactly when $A = 1$ and $B = 1$. In words: **the door opens only in the single case where a valid ID card and a correct fingerprint are both presented, and stays shut in the other three cases.**

The Boolean table. Two variables give $2^2 = 4$ rows, by the counting of section 5.1.

A	B	$D = A \cdot B$	DOOR
0	0	0	shut
0	1	0	shut
1	0	0	shut
1	1	1	opens

The circuit, if a diagram is wanted: a single AND gate with inputs A and B and output D, as drawn above.

A line worth adding. Rows 2 and 3 are the security property doing its job: a correct fingerprint with no card, or a card with the wrong fingerprint, both leave the door shut. That is what "only when" bought.

ii. The alarm

Assign the variables. Let F = 1 when the fire sensor is activated, S = 1 when the smoke sensor is activated, and L = 1 when the alarm sounds.

The expression. The alarm sounds if either sensor is activated, and certainly if both are, so this is the **inclusive** or, which is the Boolean sum:

$$L = F + S$$

Determine the output when only the smoke sensor detects danger. That is $F = 0$ and $S = 1$. Substituting:

$$L = F + S = 0 + 1 = 1$$

So the alarm sounds. Which is exactly what it should do: a smoke sensor firing on its own is the ordinary case of a fire being detected early.

The full table, which costs three more lines and secures the mark:

F	S	$L = F + S$	ALARM
0	0	0	silent
0	1	1	sounds (the case asked for)
1	0	1	sounds
1	1	1	sounds

Why it must be inclusive or. If exclusive or had been used, $L = F \oplus S$, then the last row would give $L = 0$: the alarm would fall silent precisely when both sensors detect danger. Saying that sentence shows you chose the operation rather than guessed it, and it is the difference between the two marks and the three.

Now redo it: the door should also open for a security override switch V , regardless of card and fingerprint. Then $D = A \cdot B + V$, and with $A = 0$, $B = 0$, $V = 1$ the output is $0 \cdot 0 + 1 = 0 + 1 = 1$, so the door opens.

Simplifying with the laws

WORKED EXAMPLE

METHOD · SIMPLIFICATION

Simplify $A \cdot B + A \cdot B'$. Name every law you use; that is where the marks are in any simplification question.

$$\begin{aligned}
 &A \cdot B + A \cdot B' \\
 &= A \cdot (B + B') \quad [\text{distributive}] \\
 &= A \cdot 1 \quad [\text{complement: } B + B' = 1] \\
 &= A \quad [\text{identity}]
 \end{aligned}$$

What it means: a circuit that fires when A is on and B is on, or when A is on and B is off, is just a circuit that fires when A is on. The B input can be removed entirely, which is a real saving in components.

A second one, using De Morgan. Simplify $(A + B)' + A' \cdot B$.

$$\begin{aligned}
 &(A + B)' + A' \cdot B \\
 &= A' \cdot B' + A' \cdot B \quad [\text{De Morgan}] \\
 &= A' \cdot (B' + B) \quad [\text{distributive}] \\
 &= A' \cdot 1 \quad [\text{complement}] \\
 &= A' \quad [\text{identity}]
 \end{aligned}$$

Check it on a table, which is always available and always convincing: $A = 0$ gives $(0 + B)' + 1 \cdot B = B' + B = 1$, and $A' = 1$. $A = 1$ gives $(1 + B)' + 0 \cdot B = 0 + 0 = 0$, and $A' = 0$. The two agree in both cases.

RECALL AND CHECK

1. What are the two Boolean values and the three operations?
2. What is $1 + 1$, and why is that not a mistake?
3. State De Morgan's laws in Boolean notation.
4. What is the duality principle?
5. Write the expression for the laboratory door and give its table in one line.
6. Why must the alarm use inclusive rather than exclusive or?

ANSWERS

1. 1 and 0; the Boolean product $\cdot$ (AND), the Boolean sum $+$ (OR), and the complement $'$ (NOT).
2. 1. The sum is 1 when at least one operand is 1, and there is no value above 1.
3. $(A \cdot B)' = A' + B'$ and $(A + B)' = A' \cdot B'$.
4. Swapping $\cdot$ with $+$ and 0 with 1 turns any law into another valid law.
5. $D = A \cdot B$, which is 0, 0, 0, 1 as A and B run 00, 01, 10, 11.
6. Because with exclusive or the alarm would fall silent when both sensors detect danger.

FILL IN THE GAP

1. The Boolean product $A \cdot B$ is 1 only when _____.
2. $1 + 1 =$ _____.
3. The door expression is _____ and the alarm expression is _____.
4. Duality swaps _____ with _____, and 0 with 1.
5. $A \cdot B + A \cdot B'$ simplifies to _____.

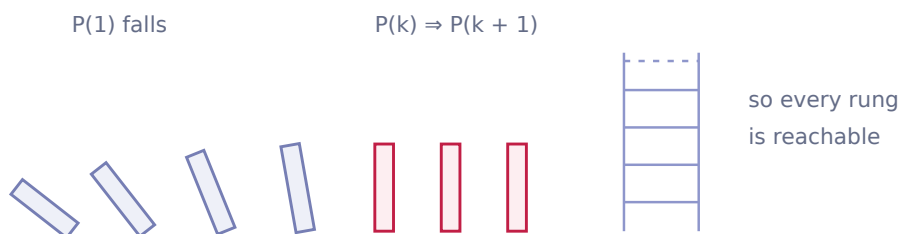
ANSWERS

1. both are 1. 2. 1. 3. $D = A \cdot B$, $L = F + S$. 4. $\cdot$, $+$. 5. A.

PART NINE

Proof, Induction and Recursion

Everywhere else in this book you were asked what something is. Here you are asked to show that it must be so. Six strategies cover every proof this course sets, and one of them, induction, is the ladder that climbs forever from a single rung.



Proof Techniques

A proof is not a feeling that something is true. It is a chain of steps from things already agreed to the thing claimed, where every link can be checked by someone who does not trust you. This section gives you the six chains this course uses, and the vocabulary the examiner uses to ask for them.

By the end of this section you will be able to say what a theorem, lemma, corollary, proposition, conjecture and axiom each are; write a trivial, vacuous, direct, contrapositive, contradiction or cases proof and choose between them; use the definitions of even, odd, rational and divides fluently; and lay a proof out so a marker can follow it line by line.

TEACH

WHAT IS BEING ASKED FOR

When a question says **prove**, **show that**, **establish** or **demonstrate**, it wants an argument that holds for **every** case covered by the claim, not one example. One example proves nothing except an existence claim.

The mirror image matters just as much. To **disprove** a claim about all cases you need exactly **one** counterexample, and no more. Section 3.2 already used this: a single pair (1, 2) with (2, 1) missing kills symmetry outright.

THE CLAIM SAYS	TO PROVE IT	TO DISPROVE IT
For all x , $P(x)$	an argument about an arbitrary x	one counterexample
There exists x with $P(x)$	one example (a witness)	an argument about an arbitrary x

That table is just the quantifier negation of section 1.4 read as advice: the negation of "for all" is "there exists ... not", so disproving a universal claim is proving an existential one, and finding a witness is the whole job.

The vocabulary of provable things

● MUST MEMORISE

THEOREM, LEMMA, COROLLARY AND THE REST

These words are not interchangeable, and a question that asks you to "differentiate" between them wants the distinction, not a definition of each in isolation.

WORD	WHAT IT IS	THE DISTINGUISHING FEATURE
Axiom	a statement taken as true without proof	the starting point; nothing is behind it
Theorem	a statement shown to be true	important in its own right
Proposition	a statement shown to be true	true, proved, but of lesser importance
Lemma	a statement proved in order to prove something else	a helper; it exists to serve a later proof
Corollary	a statement that follows at once from a theorem	comes after, and costs almost no extra work
Conjecture	a statement believed true but not yet proved	no proof exists yet; it may still be false

◆ **Memory hook: a lemma comes before, a corollary comes after, and a conjecture has not arrived at all.** The theorem is the thing in the middle that everyone remembers.

Worth one line in an answer. A conjecture that is proved becomes a theorem; a conjecture that is disproved becomes nothing at all. The word records the state of our knowledge, not a property of the statement.

The definitions every proof in this course leans on

● MUST MEMORISE

EVEN, ODD, RATIONAL, DIVIDES

Almost every proof set on this course is unlocked by writing one of these four definitions down. If you are stuck, the move is nearly always "replace the word by its definition".

WORD	DEFINITION
m is even	$m = 2k$ for some integer k
n is odd	$n = 2k + 1$ for some integer k
n is rational	$n = p / q$ for some integers p and q with $q \neq 0$
a divides b, written $a \mid b$	$b = ka$ for some integer k

WHERE

k, p, q are **integers**, and the phrase "for some integer k " is doing real work: it says such a k exists, and the proof's next move is always to name it and use it. $a \mid b$ is read "a divides b", and it points the opposite way to a fraction: $a \mid b$ means b is the multiple, a is the divisor. The bar is not a division sign and the statement is not a number, it is either true or false. $\neq$ means "is not equal to"; $q \neq 0$ because division by zero has no value.

An integer is either even or odd, never both and never neither. That sentence is what makes a proof by cases on parity exhaustive, and it is worth stating explicitly when you use it.

◆ **Memory hook: write the definition, then do algebra.** Nine proofs in ten on this course are that and nothing more.

The six strategies

● MUST MEMORISE

SIX WAYS TO PROVE $P \rightarrow Q$

Every strategy below proves the same shape of statement, the conditional $p \rightarrow q$ of section 1.2. They differ in what you assume at the start.

STRATEGY	YOU ASSUME	YOU DERIVE	USE IT WHEN
Trivial	nothing	q is true on its own	the conclusion is already known true
Vacuous	nothing	p is false	the hypothesis cannot hold
Direct	p	q	the default; always try this first
Contrapositive	$\neg q$	$\neg p$	the negations are easier to work with
Contradiction	p and $\neg q$	$r \wedge \neg r$	nothing to build from; the claim is a denial
Cases	p as $p_1 \vee \dots \vee p_n$	q from each p_i separately	the hypothesis splits naturally, such as odd or even

WHERE

p is the **hypothesis**, also called the premise or antecedent: what you are given. q is the **conclusion**, also called the consequent: what you must reach. $\neg q$ is the negation of q , from section 1.1. $r \wedge \neg r$ is a **contradiction**: some statement r and its own negation, both derived. Section 1.3 showed that is false in every row of its table, which is exactly why deriving one destroys the assumption that produced it.

◆ **Memory hook: direct goes forwards, contrapositive goes backwards, contradiction goes nowhere and shows the alternative is impossible.**

Neither is a trick. Both rest on a logical equivalence proved back in section 1.3, and quoting that equivalence by name is worth a mark.

Contrapositive. $p \rightarrow q \equiv \neg q \rightarrow \neg p$. The two have identical truth tables, so proving either one proves the other. Section 1.2 built that table and 2024/2025 Q1(c)(ii) asked you to prove it; the proof lives there, and this section is where you spend it.

Contradiction. If assuming $p \wedge \neg q$ leads to something false, then $p \wedge \neg q$ is itself false, so its negation $\neg p \vee q$ is true, and that is exactly $p \rightarrow q$ written the other way (section 1.3, the implication law).

Contrapositive is a special case of contradiction, which is why the two are so often confused. If you assume $\neg q$ and reach $\neg p$ while p was also assumed, you have $p \wedge \neg p$, a contradiction. The practical difference is in the writing: a contrapositive proof is short and clean, a contradiction proof carries an extra assumption all the way through.

Prefer the contrapositive when it is available.

Direct $p \longrightarrow q$ assume the given, reach the goal

Contrapositive $\neg q \longrightarrow \neg p$ flip both, then go forwards

Contradiction $p \wedge \neg q \longrightarrow r \wedge \neg r$ an impossibility
so the assumption was wrong

The three arrows worth drawing in the margin of a proof question before you start writing.

Direct proofs, worked

WORKED EXAMPLE

METHOD · THREE DIRECT PROOFS

1. The sum of two odd integers is even.

Proof. Let m and n be arbitrary odd integers. By the definition of odd there are integers a and b with

$$m = 2a + 1 \quad \text{and} \quad n = 2b + 1$$

Then

$$\begin{aligned} m + n &= (2a + 1) + (2b + 1) \\ &= 2a + 2b + 2 \\ &= 2(a + b + 1) \end{aligned}$$

Since $a + b + 1$ is an integer, $m + n$ is twice an integer, so $m + n$ is even. $\square$

2. If n is even then n^2 is even.

Proof. Let n be an arbitrary integer and suppose n is even. Then $n = 2k$ for some integer k , so

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

and $2k^2$ is an integer, so n^2 is even. $\square$

3. Divisibility is transitive: if $a \mid b$ and $b \mid c$ then $a \mid c$.

Proof. Let a, b, c be integers and assume $a \mid b$ and $b \mid c$. By the definition of divides there are integers k and j with

$$b = ka \quad \text{and} \quad c = jb$$

Substituting the first into the second, $c = j(ka) = (jk)a$. Since jk is an integer, $a \mid c$. $\square$

Notice the shape all three share. Assume, replace each word by its definition, do ordinary algebra, then read the definition backwards to name what you have reached. That is the entire method, and the last sentence, "since ... is an integer", is the one candidates leave out.

⚠ TRAP

TWO INTEGERS, TWO LETTERS

Writing $m = 2a + 1$ and $n = 2a + 1$ with the **same** letter says m and n are the same number. The proof then only covers the case $m = n$ and is worth almost nothing. Two independent objects get two independent letters, every time.

The same slip in reverse: proving something for "an arbitrary even integer $n = 2k$ " and then later writing $k = 3$ to check. Checking is fine, but say you are checking. A proof that fixes its arbitrary letter to a number has stopped being a proof.

Contrapositive, worked

WORKED EXAMPLE

METHOD · CONTRAPOSITIVE

1. If mn is even, then m is even or n is even.

Why not direct? Assuming $mn = 2k$ tells you almost nothing about m and n separately. But the negation of the conclusion is clean: " m is not even and n is not even" means both are odd, and odd numbers have a formula. That is the signal to flip.

The contrapositive. If m is odd and n is odd, then mn is odd. Note that negating " m is even or n is even" gives " m is odd **and** n is odd", by De Morgan's law of section 1.3. Getting that "and" is where the mark is.

Proof. Suppose m and n are both odd, so $m = 2a + 1$ and $n = 2b + 1$ for some integers a and b . Then

$$\begin{aligned} mn &= (2a + 1)(2b + 1) \\ &= 4ab + 2a + 2b + 1 \\ &= 2(2ab + a + b) + 1 \end{aligned}$$

and $2ab + a + b$ is an integer, so mn is odd. The contrapositive holds, therefore the original statement holds. $\square$

2. If $x^3 - 7x^2 + x - 7 = 0$, then $x \neq 10$.

The contrapositive. If $x = 10$, then $x^3 - 7x^2 + x - 7 \neq 0$.

Proof. Substitute $x = 10$:

$$10^3 - 7 \times 10^2 + 10 - 7 = 1000 - 700 + 10 - 7 = \mathbf{303}$$

and $303 \neq 0$. $\square$

Worth an extra line. The expression factorises as $x^2(x - 7) + (x - 7) = (x^2 + 1)(x - 7)$, and $x^2 + 1$ is never zero for real x , so the only real solution is $x = 7$. Saying so shows you understood the question rather than only answered it.

3. If $5a + 25b = 156$, then a or b is not an integer.

The contrapositive. If a and b are both integers, then $5a + 25b \neq 156$.

Proof. Suppose a and b are integers. Then

$$5a + 25b = 5(a + 5b)$$

and $a + 5b$ is an integer, so $5a + 25b$ is a multiple of 5. But $156 = 5 \times 31 + 1$ leaves remainder 1, so 156 is not a multiple of 5. Hence $5a + 25b \neq 156$. $\square$

Now redo it: show that if $6a + 9b = 100$ then a or b is not an integer. Same move: $6a + 9b = 3(2a + 3b)$ is a multiple of 3, and $100 = 3 \times 33 + 1$ is not.

Contradiction, worked

WORKED EXAMPLE

METHOD · CONTRADICTION

1. There is no largest integer.

Why contradiction? The claim is a denial: it says a thing does not exist. There is nothing to build forwards from, so assume the thing does exist and watch it break.

Proof. Suppose, for contradiction, that there is a largest integer, and call it N . Then $N + 1$ is also an integer, and $N + 1 > N$, so N is not the largest after all. That contradicts the assumption, which must therefore be false. There is no largest integer. $\square$

2. $\sqrt{2}$ is irrational. This is the classic, and it is worth knowing by heart because every part of it is a technique you have already met.

Proof. Suppose, for contradiction, that $\sqrt{2}$ is rational. Then by the definition of rational there are integers p and q with $q \neq 0$ and

$$\sqrt{2} = p / q$$

and we may take the fraction in **lowest terms**, so that p and q have no common factor other than 1. Squaring and multiplying by q^2 :

$$2 = p^2 / q^2, \text{ so } p^2 = 2q^2$$

So p^2 is even. By the contrapositive of worked example 2 above, if p were odd then p^2 would be odd, so **p is even**, say $p = 2k$. Substituting:

$$(2k)^2 = 2q^2, \text{ so } 4k^2 = 2q^2, \text{ so } q^2 = 2k^2$$

By the same argument **q is even**. But then p and q are both even and share the factor 2, contradicting "lowest terms". The assumption was false, so $\sqrt{2}$ is irrational. $\square$

Where each technique appeared. The definition of rational opened it, a contrapositive supplied the step from p^2 even to p even, and the contradiction closed it. Proofs at this level are almost always built from the smaller ones.

Cases, worked

WORKED EXAMPLE

METHOD · PROOF BY CASES

For every integer n , the number $n^3 - n$ is even.

Why cases? Nothing is given about n at all, so there is no definition to unfold. But every integer is even or odd, and each of those does come with a formula. Two cases, and they are exhaustive.

The rule that makes it legal, which is worth quoting:

$$(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q \equiv (p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)$$

which is another equivalence from section 1.3: proving each case separately is exactly proving the whole.

Case 1: n is even, so $n = 2k$ for some integer k .

$$n^3 - n = (2k)^3 - 2k = 8k^3 - 2k = 2(4k^3 - k)$$

and $4k^3 - k$ is an integer, so $n^3 - n$ is even.

Case 2: n is odd, so $n = 2k + 1$ for some integer k . Expanding the cube by the binomial theorem of section 5.4:

$$(2k + 1)^3 = 8k^3 + 12k^2 + 6k + 1$$

so

$$\begin{aligned} n^3 - n &= 8k^3 + 12k^2 + 6k + 1 - (2k + 1) \\ &= 8k^3 + 12k^2 + 4k \\ &= 2(4k^3 + 6k^2 + 2k) \end{aligned}$$

and $4k^3 + 6k^2 + 2k$ is an integer, so $n^3 - n$ is even.

Both cases are covered and both give the same conclusion, so $n^3 - n$ is even for every integer n . $\square$

A one-line check before you trust the algebra: $n = 3$ gives $27 - 3 = 24$, and $n = 4$ gives $64 - 4 = 60$. Both even. A check is not a proof, but a failed check saves you from writing out a proof of something false.

The slicker argument, worth adding after the marks are safe: $n^3 - n = n(n - 1)(n + 1)$ is a product of three consecutive integers, and among any two consecutive integers one is even. One line, no cases.

The official course text works this same example and drops two errors into the algebra. They are worth knowing because a reader who copies them into an exam loses the marks, and because spotting them is exactly the skill a proof question is testing.

First, in the even case it writes $8k^3 - 2k = 2(4k^2 - k)$. The bracket should be $4k^3 - k$: dividing $8k^3$ by 2 gives $4k^3$, and the cube does not become a square.

Second, in the odd case it expands $(2k + 1)^3$ as $8k^3 + 6k^2 + 6k + 1$. The correct expansion is $8k^3 + 12k^2 + 6k + 1$; the squared term is $3 \times (2k)^2 \times 1 = 12k^2$, not $6k^2$. The same line also prints the cube as a square, writing $(2k + 1)^2$ where it means $(2k + 1)^3$.

Why the conclusion survives anyway, which is the interesting part: both wrong brackets still come out as $2 \times (\text{an integer})$, so the final claim " $n^3 - n$ is even" is still true. A proof can reach a true conclusion through false steps, and that is precisely why every line has to be checked rather than the answer alone.

Trivial and vacuous proofs

Both come straight off the conditional's truth table in section 1.2, which is false in one row only, when p is true and q is false.

Trivial proof: q is already true, so no row can make $p \rightarrow q$ false, whatever p says.

If it is raining, then $1 = 1$.

Proof: $1 = 1$ is true, so the conditional is true regardless of the weather. $\square$

Vacuous proof: p is false, so the one false row is never reached.

If the door is both open and closed, then $2 + 2 = 5$.

Proof: a door cannot be both open and closed, so the hypothesis is a contradiction and is false. The conditional is therefore true. $\square$

These are not jokes, and they matter in computing. A loop body that never runs satisfies every claim about what it does; a specification that says "for all x in the empty list, ..." is vacuously true. Section 1.4 made the same point about a universal quantifier over an empty domain.

In an exam, a trivial or vacuous proof is normally the answer to "explain why this is true", not to "prove". If you find yourself writing one for a serious-looking question, re-read it: you have probably misidentified p or q .

Choosing a strategy, and one worked mistake

TEACH

THE DECISION ORDER

- 1 **Write the claim as $p \rightarrow q$.** Name p and name q in plain words. Half of all failed proof attempts are a claim never written down.
- 2 **Try direct.** Replace every word in p by its definition and see how close the algebra takes you to q .
- 3 **If p gives you nothing, negate q .** If $\neg q$ has a usable formula where p did not, do the contrapositive. "Or" in the conclusion is a strong signal, because its negation is an "and".
- 4 **If the claim denies existence, use contradiction.** Words like "no", "never", "cannot", "irrational", "infinitely many" all point here.
- 5 **If p splits, use cases.** Odd or even, positive or negative or zero, and any hypothesis containing "or".
- 6 **Say which one you used, in the first line.** "We prove the contrapositive" costs six words and tells the marker how to read everything that follows.

⚠ TRAP

WHEN THE CONVENTION CHANGES THE ANSWER

Consider the claim "**there is a natural number n for which $n^2 + 3n + 2$ is prime**".

Section 2.1 defined a **prime** as a number with exactly two positive divisors, itself and 1. A whole number above 1 that is not prime is **composite**: it can be written as a product of two smaller whole numbers, each at least 2.

If $\mathbb{N}$ starts at 1, the claim is false, and there is a one-line proof: $n^2 + 3n + 2 = (n + 1)(n + 2)$, a product of two integers both at least 2, so it is composite for every $n \geq 1$. Check: $n = 1$ gives 6, $n = 2$ gives 12, $n = 3$ gives 20.

If $\mathbb{N}$ starts at 0, which is the convention this manual declared in Foundations, the claim is **true**: $n = 0$ gives $0 + 0 + 2 = 2$, and 2 is prime. One witness settles an existence claim, and the whole argument collapses.

The Foundations trap box said "say which convention you are using". This is the case that shows why: the convention is not a matter of taste here, it decides the truth value. **State it in the first line of the answer and you cannot be marked wrong either way.**

Also note the better proof. The factorisation $(n + 1)(n + 2)$ does in one line what a proof by cases on the parity of n does in fifteen. When a polynomial factorises, factorise it.

RECALL AND CHECK

1. Distinguish a lemma from a corollary.
2. Write the definitions of even, odd and "a divides b".
3. What do you assume, and what do you derive, in a proof by contrapositive?
4. What do you assume in a proof by contradiction?
5. How many counterexamples disprove a statement about all x?
6. Prove that the product of two odd integers is odd.
7. Why is "if the door is open and closed then $2 + 2 = 5$ " true?

ANSWERS

1. A lemma is proved **before** and in order to prove a theorem; a corollary follows **after** a theorem with little extra work.
2. m even: $m = 2k$ for some integer k. n odd: $n = 2k + 1$ for some integer k. $a \mid b$: $b = ka$ for some integer k.
3. Assume $\neg q$, derive $\neg p$.
4. Both p and $\neg q$, then derive a statement and its own negation.
5. One.
6. $(2a + 1)(2b + 1) = 2(2ab + a + b) + 1$, which is odd.
7. The hypothesis is false, so the conditional is vacuously true.

FILL IN THE GAP

1. A statement believed true but not yet proved is a _____.
2. To prove $p \rightarrow q$ by contrapositive, assume _____ and derive _____.
3. An integer n is odd when $n =$ _____ for some integer k.
4. A proof by _____ splits the hypothesis into exhaustive alternatives.
5. The negation of "m is even or n is even" is "m is odd _____ n is odd".
6. $\sqrt{2}$ is proved irrational by _____.

ANSWERS

1. conjecture. 2. $\neg q$, $\neg p$. 3. $2k + 1$. 4. cases. 5. and. 6. contradiction.

PART NINE • 9.2

Mathematical Induction

One proof, infinitely many statements. Induction is how you prove something about every natural number without checking them one at a time, and it is the technique behind the tree bound of section 7.2 that the 2024/2025 paper asked for outright.

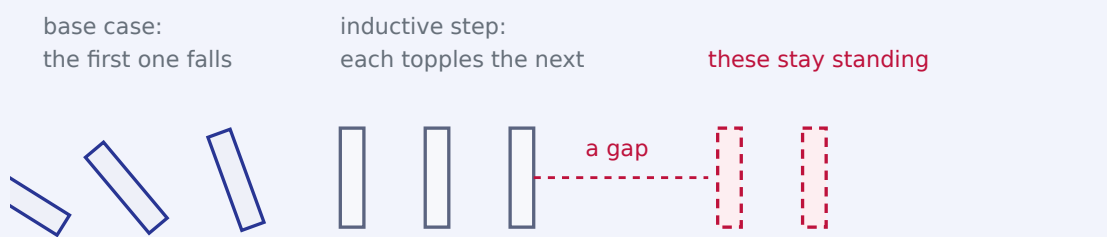
By the end of this section you will be able to state the principle of mathematical induction, lay out the three steps in the order a marker expects, prove a summation formula, a divisibility claim and an inequality by induction, explain why the base case cannot be skipped, and use strong induction where ordinary induction stalls.

Suppose there is an infinite ladder and you know two things:

- ▶ you can reach the **first** rung, and
- ▶ from **any** rung you reach, you can reach the next one.

Can you reach every rung? Yes. And notice you were never told how to reach rung 400 directly. The second fact is a machine that manufactures that answer: reach 1, so reach 2, so reach 3, and the argument runs as far as anyone cares to push it.

The same picture with dominoes standing in a line. Knock the first one over, and space them so that each falling one topples its neighbour. Both conditions are needed. A first domino that never falls leaves the line standing; a gap anywhere in the line stops the chain there and everything past the gap survives.



Both conditions carry weight. Remove either and the chain stops.

THE PRINCIPLE OF MATHEMATICAL INDUCTION

The claim. Let $P(n)$ be a statement about the integer n . If P is true at a starting value, and if the truth of P at any value forces its truth at the next value, then P is true at every value from the start onwards. One check plus one implication buys infinitely many statements.

In symbols.

If $P(a)$ is true, and $P(k) \rightarrow P(k + 1)$ for every $k \geq a$, then $P(n)$ is true for every $n \geq a$.

Conditions. The statement must be indexed by an **integer** that increases in steps of one; there is no induction over the real numbers, because there is no "next" real number. The starting value a is usually 0 or 1 but may be any integer, and the conclusion then only covers $n \geq a$. The inductive step must hold for **every** k from a onwards, not merely for the small ones you tried.

The limiting case. With no base case the principle says nothing at all, however impeccable the inductive step. "If $n = n + 1$ then $n + 1 = n + 2$ " is a perfectly valid inductive step for a statement that is false everywhere; nothing ever gets it started. The base case is not a formality, it is half the proof.

Write these three headings on the page before you write any mathematics. Markers award them separately.

STEP	WHAT YOU WRITE
1. Base case	State $P(a)$ for the smallest a in the claim, and verify it by direct computation . Both sides, evaluated.
2. Inductive hypothesis	Assume $P(k)$ holds for some arbitrary $k \geq a$. Write out what $P(k)$ actually says.
3. Inductive step	Using that assumption, prove $P(k + 1)$. Write out what $P(k + 1)$ says as a target, then reach it.

The whole principle on one line, which is what the three steps above amount to:

$P(a)$, together with $P(k) \rightarrow P(k + 1)$ for every $k \geq a$, gives $P(n)$ for every $n \geq a$

WHERE

$P(n)$ is the **statement** being proved, a predicate in the sense of section 1.4: for each n it is either true or false. It is not a number and you never "calculate $P(n)$ ". **a** is the **base value**, the smallest n the claim covers. **k** is an **arbitrary** value at or above a . It is not a specific number, and it is not n ; using n for both is the most common way to write a proof that says nothing. **$P(k) \rightarrow P(k + 1)$** is the **inductive step**, a conditional, so it is proved by assuming the left side, in the manner of section 9.1.

Close with the conclusion. "By the principle of mathematical induction, $P(n)$ holds for all $n \geq a$." A proof that stops after the algebra has left its last mark on the table.

◆ **Memory hook: check one, assume one, earn the next.** And the assumption is the tool, so every inductive step must use it somewhere; if yours does not, you have proved $P(k + 1)$ outright and did not need induction at all.

A summation formula

WORKED EXAMPLE

METHOD · 1 + 2 + ... + N

GIVEN The claim $P(n)$: $1 + 2 + 3 + \dots + n = n(n + 1) / 2$, for every natural number $n \geq 1$.

FIND A proof by induction on n .

FORMULA Base case, inductive hypothesis, inductive step.

Step 1. Base case, $n = 1$

The left side is the single term 1. The right side is

$$1 \times (1 + 1) / 2 = 2 / 2 = 1$$

Both sides equal 1, so $P(1)$ holds.

Step 2. Inductive hypothesis

Assume $P(k)$ holds for some $k \geq 1$, that is

$$1 + 2 + \dots + k = k(k + 1) / 2$$

Step 3. Inductive step

The target. $P(k + 1)$ says

$$1 + 2 + \dots + k + (k + 1) = (k + 1)(k + 2) / 2$$

Start from the left side and use the hypothesis. The first k terms are exactly what the hypothesis measures, so replace them:

$$\begin{aligned} & 1 + 2 + \dots + k + (k + 1) \\ &= k(k + 1) / 2 + (k + 1) \quad [\text{inductive hypothesis}] \\ &= k(k + 1) / 2 + 2(k + 1) / 2 \quad [\text{common denominator}] \\ &= (k + 1)(k + 2) / 2 \quad [\text{take out the factor } (k + 1)] \end{aligned}$$

which is the target, so $P(k + 1)$ holds.

Conclusion. By the principle of mathematical induction, $1 + 2 + \dots + n = n(n + 1) / 2$ for every $n \geq 1$. $\square$

A check that costs one line. At $n = 100$ the formula gives $100 \times 101 / 2 = 5050$.

Adding one more term, $5050 + 101 = 5151$, and the formula at $n = 101$ gives $101 \times 102 / 2 = 5151$. That agreement is the inductive step happening in front of you at one particular k .

Now redo it: prove $1 + 3 + 5 + \dots + (2n - 1) = n^2$. Base case $n = 1$: left 1, right 1. Step: $k^2 + (2(k + 1) - 1) = k^2 + 2k + 1 = (k + 1)^2$. Sanity check at $n = 4$: $1 + 3 + 5 + 7 = 16 = 4^2$.

A divisibility claim

WORKED EXAMPLE

METHOD · $N^3 + 2N$ IS DIVISIBLE BY 3

- GIVEN** $P(n)$: $n^3 + 2n$ is divisible by 3, for every positive integer n .
FIND A proof by induction.
FORMULA Divisibility is the definition of section 9.1: 3 divides x when $x = 3t$ for some integer t .

Step 1. Base case, $n = 1$

$$1^3 + 2 \times 1 = 1 + 2 = 3$$

and $3 = 3 \times 1$, so $P(1)$ holds.

Step 2. Inductive hypothesis

Assume $P(k)$: $k^3 + 2k$ is divisible by 3, so by the definition there is an integer t with

$$k^3 + 2k = 3t$$

Naming t is the step candidates skip, and without it there is nothing to substitute later.

Step 3. Inductive step

The target. $(k + 1)^3 + 2(k + 1)$ is divisible by 3.

Expand, using $(k + 1)^3 = k^3 + 3k^2 + 3k + 1$:

$$\begin{aligned}(k + 1)^3 + 2(k + 1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= (k^3 + 2k) + (3k^2 + 3k + 3)\end{aligned}$$

Now use the hypothesis on the first bracket, and take 3 out of the second:

$$\begin{aligned}&= 3t + 3(k^2 + k + 1) \\ &= 3(t + k^2 + k + 1)\end{aligned}$$

and $t + k^2 + k + 1$ is an integer, so the total is divisible by 3. $P(k + 1)$ holds.

Conclusion. By induction, $n^3 + 2n$ is divisible by 3 for every positive integer n . $\square$

The move to learn here is the regrouping: split the expanded expression so that the hypothesis appears intact as one bracket, and everything left over is visibly a multiple of 3. Every divisibility induction has that shape.

Check three values: $n = 2$ gives $8 + 4 = 12$, $n = 3$ gives $27 + 6 = 33$, $n = 4$ gives $64 + 8 = 72$. All multiples of 3.

Now redo it: prove $6^n - 1$ is a multiple of 5 for every natural number n . Base $n = 0$: $1 - 1 = 0$, which is 5×0 . Hypothesis: $6^k - 1 = 5j$, so $6^k = 5j + 1$. Then $6^{k+1} - 1 = 6(5j + 1) - 1 = 30j + 5 = 5(6j + 1)$. Check: $n = 1$ gives **5**, $n = 2$ gives **35**, $n = 3$ gives **215**.

An inequality, and where the base case really lives

WORKED EXAMPLE

METHOD · N SQUARED BELOW 2 TO THE N

GIVEN $P(n): n^2 < 2^n$, for every integer $n \geq 5$.

FIND A proof by induction, and an explanation of why the claim starts at 5.

FORMULA Base case at $n = 5$, then the step.

First, why 5. Test the small values and the reason appears:

N	N^2	2^N	IS $N^2 < 2^N$?
1	1	2	yes
2	4	4	no , they are equal
3	9	8	no
4	16	16	no , they are equal
5	25	32	yes
6	36	64	yes

The claim is **false** at $n = 2, 3$ and 4 , and true from 5 onwards. That is why the question says $n \geq 5$, and it is a good illustration that a base case is a genuine check, not a ritual: the same inductive step would run happily from a base of 3 and would then be carrying a false statement.

Step 1. Base case, $n = 5$

$5^2 = 25$ and $2^5 = 32$, and $25 < 32$. $P(5)$ holds.

Step 2. Inductive hypothesis

Assume $P(k)$ for some $k \geq 5$, that is $k^2 < 2^k$.

Step 3. Inductive step

The target. $(k + 1)^2 < 2^{k+1}$.

Work on the left side. $(k + 1)^2 = k^2 + 2k + 1$. Since $k \geq 5$ we have $2k + 1 \leq k^2$, because that inequality holds for every $k \geq 3$. So

$$(k + 1)^2 = k^2 + 2k + 1 \leq k^2 + k^2 = 2k^2$$

Now use the hypothesis: $k^2 < 2^k$, so

$$2k^2 < 2 \times 2^k = 2^{k+1}$$

Chaining the two, $(k + 1)^2 < 2^{k+1}$, which is $P(k + 1)$.

Conclusion. By induction, $n^2 < 2^n$ for every integer $n \geq 5$. $\square$

The technique in inequality inductions is always this: bound the awkward extra term by something the hypothesis can absorb. Here $2k + 1$ was bounded by k^2 , and the side condition $k \geq 3$ that made that legal was stated rather than assumed.

The examined one

WORKED EXAMPLE

PQ 24/25 Q6(B) VII, IN THE INDUCTION TEMPLATE

The 2024/2025 paper asked for a proof that an m -ary tree of height h has at most m^h leaves. That question is quoted as printed and answered in section 7.2, where the tree vocabulary lives. It is repeated here in the three-step layout so you can see the same proof through this section's frame, because the two are the same argument written for two different markers.

GIVEN An m -ary tree of height h , that is a rooted tree in which every vertex has at most m children.

FIND A proof that it has at most m^h leaves.

FORMULA Induction on the **height** h , not on the number of vertices.

Step 1. Base case, $h = 0$

A tree of height 0 is a single vertex, which is both the root and a leaf. It has 1 leaf, and $m^0 = 1$, so the bound holds with equality.

Step 2. Inductive hypothesis

Assume every m -ary tree of height k has at most m^k leaves.

Step 3. Inductive step

Let T be an m -ary tree of height $k + 1$. Remove the root. What remains is at most m subtrees, one per child of the root, and each has height at most k . By the hypothesis each has at most m^k leaves. Every leaf of T lies in one of them, so

$$\text{leaves of } T \leq m \times m^k = m^{k+1}$$

Conclusion. By induction the bound holds at every height. $\square$

Note what the induction was on. Choosing the right quantity to induct on is most of the skill. Inducting on the number of vertices here would be painful, because removing a vertex from an m -ary tree need not leave an m -ary tree of any convenient height. Height was the quantity that made the step easy.

Strong induction

TEACH

WHEN ONE STEP BACK IS NOT ENOUGH

Ordinary induction assumes $P(k)$ and proves $P(k + 1)$: it reaches back exactly one rung. Sometimes the argument needs to reach further, or to reach back to a value it cannot predict.

Strong induction assumes $P(a)$, $P(a + 1)$, ..., $P(k)$ all at once, and proves $P(k + 1)$ from any of them it likes. The principle is the same and is itself provable from ordinary induction, so no new permission is needed.

The standard example: every integer $n \geq 2$ is a product of prime numbers.

Proof. Base case: 2 is itself prime, so it is a product of one prime. Assume every integer from 2 up to k is a product of primes, and consider $k + 1$. Either $k + 1$ is prime, in which case it is a product of one prime and we are done; or it is not, in which case $k + 1 = ab$ for some integers a and b with $2 \leq a \leq k$ and $2 \leq b \leq k$. Both a and b are covered by the hypothesis, so each is a product of primes, and multiplying those products gives one for $k + 1$. $\square$

Why ordinary induction fails here. The factors a and b are not k , and there is no telling in advance where they land. Only an assumption covering **every** smaller value is any use. That is the signal for strong induction: **the step goes back an unpredictable distance**. The same signal appears in section 9.4, where a second-order recurrence needs two previous terms rather than one.

⚠ TRAP

THREE WAYS AN INDUCTION PROOF LOSES MARKS

1. Assuming what is to be proved. Writing " $1 + 2 + \dots + (k + 1) = (k + 1)(k + 2) / 2$ " at the top of step 3 and then manipulating both sides until they agree is circular. Start from **one** side, use the hypothesis, and arrive at the other. If you do work on both sides, say plainly that you are showing two expressions are equal, and never treat the target as available.

2. Never using the hypothesis. An inductive step that does not cite $P(k)$ anywhere has proved $P(k + 1)$ from nothing, which either means induction was unnecessary or, far more often, that a step is missing. **Point at the line where you used it.**

3. A base case that is asserted rather than checked. "Clearly true for $n = 1$ " earns nothing. Evaluate both sides and show they agree. The inequality above is the reason: at $n = 4$ that claim is false, and only an evaluated base case catches it.

RECALL AND CHECK

1. Name the three steps of an induction proof.
2. What exactly do you assume in step 2, and about which value?
3. Why can the base case not be skipped?
4. State the formula for $1 + 2 + \dots + n$ and evaluate it at $n = 100$.
5. What does strong induction assume that ordinary induction does not?
6. Give the signal that tells you strong induction is needed.

ANSWERS

1. Base case, inductive hypothesis, inductive step.
2. That $P(k)$ is true, for an arbitrary k at or above the base value.
3. The inductive step alone only passes truth along; with nothing true to start from it proves nothing, and it will happily carry a false statement forever.
4. $n(n + 1) / 2$, which is $100 \times 101 / 2 = 5050$.
5. Every earlier case, $P(a)$ through $P(k)$, rather than $P(k)$ alone.
6. The step needs a value an unpredictable distance back, as when a number is split into two unknown factors.

FILL IN THE GAP

1. Induction proves a statement for every _____ from the base value onwards.
2. Step 2 assumes $P(\text{_____})$ and step 3 proves $P(\text{_____})$.
3. $1 + 3 + 5 + \dots + (2n - 1) = \text{_____}$.
4. $n^2 < 2^n$ first holds at $n = \text{_____}$.
5. An inductive step that never cites the hypothesis is _____.

ANSWERS

1. integer. 2. $k, k + 1$. 3. n^2 . 4. 5. 5. not an induction proof at all: either it was unnecessary, or a step is missing.

PART NINE • 9.3

Sequences, Series and Progressions

A sequence is an ordered list with a rule behind it. Two rules cover almost everything this course does with them: keep adding the same amount, or keep multiplying by the same amount. Getting the indexing right is where the marks actually go.

By the end of this section you will be able to define a sequence and a term, tell an arithmetic progression from a geometric one, write the general term and the sum of each, read a rule off a printed sequence, use sigma notation and the four standard sums, and state which indexing convention you are using so that an off-by-one cannot cost you.

Sequences

● MUST MEMORISE

SEQUENCE, TERM, FINITE AND INFINITE

A **sequence** is a function from a subset of the integers, usually $\{0, 1, 2, \dots\}$ or $\{1, 2, 3, \dots\}$, to a set S . The image of the integer n is written a_n and is called a **term** of the sequence. The whole sequence is written $\{a_n\}$.

WHERE

n is the **index**, an integer, and it says the position of the term, not its value. a_n is the **term at position n** : one number, not the whole list. The subscript is not multiplication. $\{a_n\}$ in braces is the **sequence itself**, the entire ordered list. S is the set the terms are drawn from, usually the real numbers on this course.

Calling it a function is the useful part. Section 4.1 defined a function as a rule assigning exactly one output to each input; a sequence is that rule with the inputs restricted to integers. So each position has exactly one term, which is why "the third term" is a well-formed phrase.

Finite and infinite. A **finite** sequence has a last term and its **length** is the number of terms. An **infinite** sequence runs on without end.

SEQUENCE	KIND	THE RULE BEHIND IT
2, 4, 6, 8, ..., 20	finite, length 10	add 2 to the previous term
10, 6, 2, -2, ...	infinite	subtract 4 from the previous term
1, 1, 2, 3, 5, 8, 13, ...	infinite	add the previous two terms

A **progression** is a sequence whose terms can be described by a formula. The third row above is the **Fibonacci sequence**, in which each term is the sum of the two before it; section 9.4 solves it properly.

◆ **Memory hook: the index is the seat number, the term is who is sitting in it.**

Arithmetic progressions

● MUST MEMORISE

ARITHMETIC PROGRESSION

An **arithmetic progression** is a sequence in which each term is the previous term plus a fixed amount:

$$a, a + d, a + 2d, a + 3d, \dots$$

The general term and the sum of the first n terms are

$$a_n = a + (n - 1)d \quad \text{and} \quad S_n = n(a + a_n) / 2 = n(2a + (n - 1)d) / 2$$

WHERE

a is the **first term**, the one at position 1. **d** is the **common difference**, the fixed amount added each time; it may be negative, which makes the sequence decrease, and it is found as any term minus the one before it. **n** is the **position** of the term you want, counting the first term as position 1. **a_n** is the **n-th term**. **S_n** is the **sum** of the first n terms, also called the arithmetic **series**. **(n - 1)** is the number of **steps taken** to get from the first term to the n -th, which is one fewer than the number of terms. That subtraction is the single most common error in this topic.

Why the sum formula looks like that. Write the sum forwards and backwards under itself and add column by column. Each column gives $a + a_n$, and there are n columns, so twice the sum is $n(a + a_n)$. It is the first term plus the last, times how many pairs there are. The $n = 1$ to 100 case of section 9.2 is exactly this trick.

◆ **Memory hook: arithmetic adds, and the n -th term has taken n minus one steps.**

△ TRAP

TWO INDEXING CONVENTIONS, ONE SEQUENCE

The definition above starts counting at 1, so $a_1 = a$ and $a_n = a + (n - 1)d$. The official course text writes the same progression as

$$a, a + d, a + 2d, \dots, a + nd$$

which counts from 0, so that its $a_n = a + nd$ and its first term is a_0 . **Neither is wrong.** They describe the same list of numbers and disagree only about what to call the positions.

The danger is mixing them inside one answer, which produces an answer exactly one term out. So do two things:

- ▶ **Say which you are using**, in five words: "counting the first term as a_1 ".
- ▶ **Answer the question in the words it asked.** "Find the 10th term" means the tenth number in the list, whatever you have chosen to call its index. Count the list if you are unsure; ten numbers is not many.

This is the same discipline as the Foundations convention box on whether $\mathbb{N}$ contains 0, and for the same reason: a stated convention cannot be marked wrong.

GIVEN The sequence 7, 12, 17, 22, ...

FIND (a) the 10th term, (b) a general formula for the n -th term, and (c) the sum of the first 10 terms.

FORMULA $a_n = a + (n - 1)d$ and $S_n = n(a + a_n) / 2$, counting the first term as a_1 .

Identify it first. The differences are $12 - 7 = 5$, $17 - 12 = 5$, $22 - 17 = 5$. Constant, so this is an arithmetic progression with $a = 7$ and $d = 5$.

(a) The 10th term

$$a_{10} = a + (10 - 1)d = 7 + 9 \times 5 = 7 + 45 = \mathbf{52}$$

Check by listing, which is cheap and catches an off-by-one at once: 7, 12, 17, 22, 27, 32, 37, 42, 47, **52**. Ten numbers, and the tenth is 52.

(b) The general term

$$a_n = 7 + (n - 1) \times 5 = 7 + 5n - 5 = \mathbf{5n + 2}$$

Test it at both ends: $n = 1$ gives 7, and $n = 10$ gives $50 + 2 = 52$. Both agree.

(c) The sum of the first 10 terms

$$S_{10} = 10 \times (7 + 52) / 2 = 10 \times 59 / 2 = \mathbf{295}$$

Now redo it: for 2, 4, 6, ..., 20, we have $a = 2$, $d = 2$ and 10 terms, so the sum is $10 \times (2 + 20) / 2 = \mathbf{110}$.

Geometric progressions

● MUST MEMORISE

GEOMETRIC PROGRESSION

A **geometric progression** is a sequence in which each term is the previous term times a fixed amount:

$$a, ar, ar^2, ar^3, \dots$$

The general term and the sum of the first n terms are

$$a_n = ar^{n-1} \quad \text{and} \quad S_n = a(r^n - 1) / (r - 1) \quad \text{for } r \neq 1$$

WHERE

a is the **first term**. **r** is the **common ratio**, the fixed multiplier, found as any term divided by the one before it. It may be a fraction, which makes the terms shrink, and it may be negative, which makes the signs alternate. **n - 1** is again the number of **multiplications performed** to reach position n from position 1. **$r \neq 1$** is required because at $r = 1$ the formula divides by zero; that case is just n copies of a , so $S_n = na$.

The infinite sum. If the ratio is small enough that the terms shrink, that is if r lies strictly between -1 and 1 , the infinite series has a finite total:

$$S = a / (1 - r)$$

Outside that range the terms do not shrink and the sum has no finite value.

◆ **Memory hook: arithmetic adds d, geometric times by r.** To tell which one a printed sequence is: take differences, and if they are not constant, take ratios.

GIVEN (a) 4, 8, 16, 32, ... (b) 1, 1/3, 1/9, 1/27, ... (c) 1, 3, 5, 7, 9, ...

FIND The rule and the general term of each, plus one sum.

FORMULA Differences first; if they are not constant, ratios. Then $a_n = a + (n - 1)d$ or $a_n = ar^{n-1}$.

(a) 4, 8, 16, 32

Differences are 4, 8, 16: not constant. **Ratios** are $8 / 4 = 2$, $16 / 8 = 2$, $32 / 16 = 2$: constant. So it is geometric with $a = 4$ and $r = 2$, and

$$a_n = 4 \times 2^{n-1}$$

A neater equivalent form, worth writing beside it: $4 \times 2^{n-1} = 2^2 \times 2^{n-1} = 2^{n+1}$.

Check at $n = 1$: $2^2 = 4$. At $n = 4$: $2^5 = 32$.

The 10th term is $2^{11} = \mathbf{2048}$, and the sum of the first 10 terms is

$$S_{10} = 4(2^{10} - 1) / (2 - 1) = 4 \times 1023 = \mathbf{4092}$$

(b) 1, 1/3, 1/9, 1/27

The denominators are powers of 3, so the ratio is 1/3 and the first term is 1:

$$a_n = (1/3)^{n-1}$$

Here the infinite sum exists, because 1/3 lies between -1 and 1 :

$$S = a / (1 - r) = 1 / (1 - 1/3) = 1 / (2/3) = \mathbf{3/2}$$

So adding infinitely many terms gives 1.5 exactly, not something unbounded. That is worth one sentence of comment in an exam, because it is the point of the formula.

(c) 1, 3, 5, 7, 9

Differences are all 2, so it is arithmetic with $a = 1$, $d = 2$, and

$$a_n = 1 + (n - 1) \times 2 = \mathbf{2n - 1}$$

which is the sequence of odd numbers, exactly the one section 9.2 summed to n^2 .

- GIVEN** The sequence 1, 5, 17, 53, 161, 485.
FIND The rule that generates it.
FORMULA None applies directly: the two standard tests, a constant difference and a constant ratio, both fail, so the rule has to be found by comparing each term with the one before it.

Find the rule behind 1, 5, 17, 53, 161, 485.

Differences: 4, 12, 36, 108, 324. Not constant, so it is not arithmetic. **Ratios:** 5, 3.4, about 3.12, about 3.04, about 3.01. Not constant, so it is not geometric either. But the ratios are creeping towards 3, which is the hint.

Try "times 3 plus something".

TERM	$3 \times \text{PREVIOUS}$	ACTUAL	DIFFERENCE
a_2	$3 \times 1 = 3$	5	2
a_3	$3 \times 5 = 15$	17	2
a_4	$3 \times 17 = 51$	53	2
a_5	$3 \times 53 = 159$	161	2
a_6	$3 \times 161 = 483$	485	2

The extra is 2 every time, so the rule is

$$a_n = 3a_{n-1} + 2, \text{ with } a_1 = 1$$

That is a recurrence relation, a rule that defines each term from earlier ones rather than from n directly. Section 9.4 takes this exact sequence and produces a closed formula for a_n that needs no earlier terms at all.

The general procedure, when a sequence resists both standard shapes:

- 1 Take differences.** Constant means arithmetic.
- 2 Take ratios.** Constant means geometric.
- 3 Take second differences.** Constant means the terms follow a quadratic in n .
- 4 Compare each term with a multiple of its predecessor.** A constant leftover means a recurrence of the form $a_n = ca_{n-1} + b$.
- 5 Try the sum of the previous two.** That is Fibonacci.

Sigma notation and the standard sums

● MUST MEMORISE

FOUR SUMS WORTH KNOWING BY HEART

Summation notation was introduced in Foundations; here are the four closed forms this course expects, each valid for $n \geq 1$.

SUM	CLOSED FORM	CHECK AT $N = 5$
$1 + 2 + \dots + n$	$n(n + 1) / 2$	$5 \times 6 / 2 = 15$
$1^2 + 2^2 + \dots + n^2$	$n(n + 1)(2n + 1) / 6$	$5 \times 6 \times 11 / 6 = 55$
$1^3 + 2^3 + \dots + n^3$	$(n(n + 1) / 2)^2$	$15^2 = 225$
$1 + r + r^2 + \dots + r^n$	$(r^{n+1} - 1) / (r - 1)$	at $r = 2$: $2^6 - 1 = 63$

WHERE

$\sum$ is the **summation sign**: the letter below it is the index, the value below it is where the index starts, the value above it is where it stops, and the expression after it is what to add. **n** is the **last index**, so these sums have n terms when they start at 1 and $n + 1$ terms when they start at 0. The geometric row starts at $r^0 = 1$, which is why its exponent is $n + 1$. **r** is the **common ratio** as above, and $r \neq 1$.

The third one is the square of the first, which is a memorable accident worth using: the sum of the first n cubes equals the square of the sum of the first n numbers.

Two rules that let you take a sum apart, and they are used constantly:

$$\sum (x_k + y_k) = \sum x_k + \sum y_k$$

$$\sum c x_k = c \sum x_k, \text{ for a constant } c$$

◆ **Memory hook: squares over six, cubes are the square of the first sum.** And every one of these four can be proved by induction, which is section 9.2 and a fair exam question.

WORKED EXAMPLE

METHOD · EVALUATING A SUM

GIVEN The sum of k^2 for k running from 1 to 10.

FIND Its value, and the value of the sum of $(3k + 2)$ over the same range.

FORMULA $\sum k^2 = n(n + 1)(2n + 1) / 6$ and $\sum k = n(n + 1) / 2$, with the two splitting rules above.

The squares, at $n = 10$:

$$10 \times 11 \times 21 / 6 = 2310 / 6 = \mathbf{385}$$

The linear one. Split it first, which is where the rules earn their place:

$$\begin{aligned} \sum (3k + 2) &= 3 \sum k + \sum 2 \\ &= 3 \times (10 \times 11 / 2) + 10 \times 2 \\ &= 3 \times 55 + 20 \\ &= 165 + 20 = \mathbf{185} \end{aligned}$$

Note the second term. Summing the constant 2 ten times gives 20, not 2. Forgetting that a constant inside a sum still gets added once per index is the standard slip here.

RECALL AND CHECK

1. Define a sequence, and say what a term is.
2. What distinguishes an arithmetic progression from a geometric one? Give one example of each.
3. Write the n -th term of an arithmetic progression and say what every letter means.
4. Write the sum of the first n terms of a geometric progression, and state its condition.
5. Find the rule behind 4, 8, 16, 32 and give its 10th term.
6. Evaluate the sum of the first 5 cubes.

ANSWERS

1. A function from a subset of the integers to a set; a term a_n is the image of the index n , that is the number in position n .
2. Arithmetic **adds** a constant difference d , geometric **multiplies** by a constant ratio r . For example 7, 12, 17, 22 with $d = 5$, and 4, 8, 16, 32 with $r = 2$.
3. $a_n = a + (n - 1)d$: a is the first term, d the common difference, n the position, and $n - 1$ the number of steps taken.
4. $S_n = a(r^n - 1) / (r - 1)$, valid for $r \neq 1$; at $r = 1$ the sum is na .
5. Geometric, $a = 4$, $r = 2$, so $a_n = 4 \times 2^{n-1} = 2^{n+1}$, and $a_{10} = 2048$.
6. $(5 \times 6 / 2)^2 = 15^2 = 225$.

FILL IN THE GAP

1. In an arithmetic progression the fixed amount added is the common _____.
2. In a geometric progression the fixed multiplier is the common _____.
3. The n -th term of a geometric progression is _____.
4. An infinite geometric series has a finite sum only when r lies between _____ and _____.
5. The sum of the first n squares is _____.
6. 1, 5, 17, 53, 161, 485 follows the rule $a_n =$ _____.

ANSWERS

1. difference. 2. ratio. 3. ar^{n-1} . 4. $-1, 1$. 5. $n(n + 1)(2n + 1) / 6$. 6. $3a_{n-1} + 2$, with $a_1 = 1$.

PART NINE • 9.4

Recurrence Relations

A recurrence defines each term from the terms before it. That is how a recursive program describes its own running time, how the chain letter of section 7.2 grows, and how Fibonacci is written down. Solving one means replacing "look back" with a formula that answers straight away.

By the end of this section you will be able to define a recurrence relation and its order, state initial conditions and say why they are needed, generate terms from a relation, solve a first-order rela-

tion by iteration, solve a homogeneous second-order relation with the characteristic equation in both the distinct-root and repeated-root cases, and recognise a non-homogeneous one.

● MUST MEMORISE

RECURRENCE RELATION, ORDER, INITIAL CONDITIONS

A **recurrence relation** for a sequence $\{a_n\}$ is an equation expressing a_n in terms of one or more of the previous terms:

$$a_n = f(a_{n-1}, a_{n-2}, \dots, a_{n-k})$$

The **initial conditions** are the terms given outright, before the relation takes effect. A sequence is a **solution** of the relation if its terms satisfy it.

WHERE

a_n is the term being defined, and a_{n-1} and its neighbours are the terms already known. f is any rule at all: it need not be linear and need not involve every previous term. k is the **order** of the relation, the number of steps back it reaches. First order looks back one term, second order two. **the initial conditions** are k values supplied separately, and there must be exactly as many of them as the order, because the relation cannot start without them.

Why the initial conditions are not optional. The relation $a_n = 2a_{n-1}$ describes 1, 2, 4, 8, 16, ... and equally describes 5, 10, 20, 40, ... and 0, 0, 0, 0, The relation fixes the **shape**; the initial conditions pick out which sequence of that shape you actually have. A question that gives a recurrence and no initial condition has not specified a sequence.

Solving a recurrence means finding a **closed form**: a formula for a_n in terms of n alone, needing no earlier terms. The point is cost. Computing a_{1000} from a first-order relation takes a thousand steps; from a closed form it takes one.

◆ **Memory hook: the relation gives the shape, the initial conditions choose the sequence, and solving removes the looking back.**

TEACH

READING THE ORDER OFF A RELATION

RELATION	ORDER	WHY
$a_n = 3a_{n-1} + 2$	first	reaches back one term
$F_n = F_{n-1} + F_{n-2}$	second	reaches back two, so it needs two initial values
$T_n = T_{n-1} + T_{n-3}$	third	the furthest reach is three, even though T_{n-2} is absent
$S_n = S_{n/2} + 1$	no finite order	the distance back grows with n , so no fixed k covers it

The order is the furthest reach, not the number of terms mentioned. The third row skips a_{n-2} entirely and is still third order, and it still needs three initial values.

The last row is worth knowing because it is the shape of a divide-and-conquer algorithm: halve the problem, do one unit of work. The binary search that section 11.2 writes in C++ has exactly that recurrence, and it is why halving is so much faster than stepping.

Generating terms, and solving by iteration

WORKED EXAMPLE

METHOD · THE TOWER OF HANOI

Three pegs and n discs of different sizes, stacked smallest on top. Move the whole stack to another peg, one disc at a time, never placing a larger disc on a smaller one. How many moves does it take?

GIVEN H_n = the least number of moves for n discs. Moving n discs means: move the top $n - 1$ to the spare peg, move the largest one across, then move the $n - 1$ back on top.

FIND The recurrence, the first few terms, and a closed form.

FORMULA $H_n = 2H_{n-1} + 1$, with $H_1 = 1$.

Generate the terms before trying to be clever:

N	$H_N = 2H_{N-1} + 1$	VALUE	COMPARE WITH 2^N
1	given	1	$2 - 1$
2	$2 \times 1 + 1$	3	$4 - 1$
3	$2 \times 3 + 1$	7	$8 - 1$
4	$2 \times 7 + 1$	15	$16 - 1$
5	$2 \times 15 + 1$	31	$32 - 1$

The pattern is unmistakable, so the conjecture is

$$H_n = 2^n - 1$$

Now prove it, because a pattern spotted in five rows is a conjecture, not a theorem. Induction, by section 9.2.

Base case, $n = 1$: the formula gives $2^1 - 1 = 1$, which matches $H_1 = 1$.

Hypothesis: $H_k = 2^k - 1$.

Step: $H_{k+1} = 2H_k + 1 = 2(2^k - 1) + 1 = 2^{k+1} - 2 + 1 = 2^{k+1} - 1$, which is the formula at $k + 1$. $\square$

The method just used has a name: solving by iteration. Write out terms until the pattern is visible, conjecture the closed form, then prove it by induction. For a first-order relation it is usually the fastest route, and the induction is what turns a guess into an answer.

- GIVEN** $a_n = 3a_{n-1} + 2$ with $a_1 = 1$, the sequence 1, 5, 17, 53, 161, 485 of section 9.3.
- FIND** A closed form for a_n .
- FORMULA** Iteration, with one substitution that turns the relation into a plain geometric progression.

The trick worth learning. The "+ 2" is what stops this being geometric. Add 1 to both sides and watch it disappear:

$$a_n + 1 = 3a_{n-1} + 3 = 3(a_{n-1} + 1)$$

So the sequence $b_n = a_n + 1$ satisfies $b_n = 3b_{n-1}$, which is geometric with ratio 3 and first term $b_1 = a_1 + 1 = 2$. By section 9.3,

$$b_n = 2 \times 3^{n-1}$$

and therefore

$$a_n = 2 \times 3^{n-1} - 1$$

Check every printed term, which takes four lines and is worth every one:

N	$2 \times 3^{N-1} - 1$	VALUE	THE SEQUENCE SAYS
1	$2 \times 1 - 1$	1	1
2	$2 \times 3 - 1$	5	5
3	$2 \times 9 - 1$	17	17
4	$2 \times 27 - 1$	53	53
5	$2 \times 81 - 1$	161	161
6	$2 \times 243 - 1$	485	485

Note the shape of the trick. Tower of Hanoi was $H_n = 2H_{n-1} + 1$ and the same substitution works there: $H_n + 1 = 2(H_{n-1} + 1)$, giving $H_n + 1 = 2^n$ and $H_n = 2^n - 1$ with no induction needed at all. Any relation $a_n = ca_{n-1} + b$ with $c \neq 1$ yields to it: add $b / (c - 1)$ to both sides.

Linear homogeneous relations with constant coefficients

● MUST MEMORISE

THE CHARACTERISTIC EQUATION METHOD

A relation is **linear** with **constant coefficients** and **homogeneous** of order k when it can be written

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

Its **characteristic equation** is obtained by replacing a_{n-i} with x^{k-i} and moving everything to one side:

$$x^k - c_1 x^{k-1} - \dots - c_k = 0$$

WHERE

linear means every a term appears to the first power on its own: no squares, no products of two terms, no a_n inside a function. **constant coefficients** means each c_i is a fixed number, not something depending on n . **homogeneous** means there is no extra term that does not involve the sequence; a stray "+ 2" makes it non-homogeneous. **x** is an ordinary unknown, and the roots of the characteristic equation are called the **characteristic roots**. **k** is the order, so the characteristic equation has degree k and the general solution carries k constants, fixed by the k initial conditions.

The two cases you will meet at order 2. Solve $x^2 - c_1 x - c_2 = 0$ and look at its roots:

ROOTS	GENERAL SOLUTION
two distinct roots x_1, x_2	$a_n = A x_1^n + B x_2^n$
one repeated root x_1	$a_n = (A + Bn) x_1^n$

Then find A and B by substituting the two initial conditions and solving the resulting pair of simultaneous equations.

Why powers of a root work at all, which is worth one line in an answer: trying $a_n = x^n$ in the relation and dividing through by x^{n-k} gives exactly the characteristic equation. So the guess "a power of something" is not a rabbit out of a hat; the characteristic equation is precisely the condition that makes it work.

◆ **Memory hook: degree equals order, roots give powers, and a repeated root buys an extra n.**

GIVEN $a_n = 5a_{n-1} - 6a_{n-2}$, with $a_0 = 1$ and $a_1 = 4$.

FIND A closed form for a_n .

FORMULA Characteristic equation $x^2 - 5x + 6 = 0$, then $a_n = A x_1^n + B x_2^n$.

Step 1. Write the characteristic equation

$$x^2 - 5x + 6 = 0$$

Watch the signs. The relation subtracts $6a_{n-2}$, so moving it across makes the constant term $+6$.

Step 2. Find the roots

$$(x - 3)(x - 2) = 0, \text{ so } x_1 = 3 \text{ and } x_2 = 2$$

Two real, distinct roots, so this is the first case.

Step 3. Write the general solution

$$a_n = A \times 3^n + B \times 2^n$$

Step 4. Use the initial conditions

$$n = 0: A + B = 1$$

$$n = 1: 3A + 2B = 4$$

From the first, $B = 1 - A$. Substituting into the second: $3A + 2(1 - A) = 4$, so $A + 2 = 4$ and **$A = 2$** , hence **$B = -1$** .

Step 5. The answer

$$a_n = 2 \times 3^n - 2^n$$

Check it against the relation, which is the only way to know the signs came out right:

N	FROM THE RELATION	FROM THE FORMULA
0	given, 1	$2 - 1 = 1$
1	given, 4	$6 - 2 = 4$
2	$5 \times 4 - 6 \times 1 = 14$	$18 - 4 = 14$
3	$5 \times 14 - 6 \times 4 = 46$	$54 - 8 = 46$
4	$5 \times 46 - 6 \times 14 = 146$	$162 - 16 = 146$

GIVEN $a_n = 10a_{n-1} - 25a_{n-2}$, with $a_0 = 3$ and $a_1 = 17$.

FIND A closed form for a_n .

FORMULA Characteristic equation $x^2 - 10x + 25 = 0$, then $a_n = (A + Bn)x_1^n$.

The characteristic equation is $x^2 - 10x + 25 = 0$, that is $(x - 5)^2 = 0$, so there is a **single** root $x_1 = 5$, repeated.

Why the general solution needs the extra n. With only one root there is only one power to build from, so $A \times 5^n$ carries just one constant and cannot satisfy two initial conditions in general. The second independent solution is $n \times 5^n$, so

$$a_n = (A + Bn) \times 5^n$$

Fit the constants.

$$n = 0: A = 3$$

$$n = 1: (A + B) \times 5 = 17, \text{ so } A + B = 17 / 5, \text{ so } B = 17/5 - 3 = 2/5$$

The answer, written twice because the second form is tidier and markers accept either:

$$a_n = (3 + (2/5)n) \times 5^n = 3 \times 5^n + 2n \times 5^{n-1}$$

Check.

N	FROM THE RELATION	FROM THE FORMULA
0	given, 3	$3 \times 1 = 3$
1	given, 17	$15 + 2 = 17$
2	$10 \times 17 - 25 \times 3 = 95$	$75 + 20 = 95$
3	$10 \times 95 - 25 \times 17 = 525$	$375 + 150 = 525$

Do not be alarmed by a fractional constant. A and B are whatever the initial conditions force them to be; $2/5$ is a perfectly good answer, and the check above confirms it produces integers throughout.

The Fibonacci sequence is $F_n = F_{n-1} + F_{n-2}$. The official course text takes $F_0 = 1$ and $F_1 = 1$, giving

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

Other books start $F_0 = 0$, $F_1 = 1$, which shifts every index by one. As always: **say which you are using**.

The characteristic equation is $x^2 - x - 1 = 0$. This one does not factorise into whole numbers, so use the **quadratic formula**: the solutions of $ax^2 + bx + c = 0$ are

$$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$$

Here $a = 1$, $b = -1$ and $c = -1$, so $b^2 - 4ac = 1 + 4 = 5$ and

$$x = (1 \pm \sqrt{5}) / 2$$

The larger of them, about 1.618, is the golden ratio, and it is why Fibonacci numbers grow at roughly 1.618 times per step: the ratio $13 / 8 = 1.625$ and $34 / 21$ is about 1.619.

The point for this course is not the closed form, which is unpleasant, but the two structural facts: the relation is **second order**, so it needs **two** initial conditions, and irrational characteristic roots can produce a sequence of whole numbers. Both are fair short-answer questions.

A relation is **non-homogeneous** when it carries an extra term that does not involve the sequence:

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + f(n)$$

Its solution has two parts added together:

- ▶ the solution of the **associated homogeneous** relation, the same relation with $f(n)$ removed, found by the characteristic equation above;
- ▶ a **particular solution**, any one sequence that satisfies the full relation, found by trying a form that matches $f(n)$: a constant for a constant f , a linear expression for a linear f , a multiple of t^n for $f(n) = t^n$.

You have already done one. $a_n = 3a_{n-1} + 2$ is non-homogeneous with $f(n) = 2$. Its associated homogeneous relation $a_n = 3a_{n-1}$ has solution $A \times 3^n$, and a constant particular solution $a_n = p$ requires $p = 3p + 2$, so $p = -1$. The general solution is $A \times 3^n - 1$, and $a_1 = 1$ gives $3A - 1 = 1$, so $A = 2/3$ and $a_n = (2/3) \times 3^n - 1 = 2 \times 3^{n-1} - 1$, exactly the answer the substitution trick gave.

Two routes, one answer. That agreement is the sort of check worth showing: it costs three lines and demonstrates the method rather than the memory.

One caution on trial solutions. If the form you try is already a solution of the homogeneous relation, it cannot also fix the extra term, and you must multiply the trial by n . That is the same phenomenon as the repeated root above, and for the same reason.

SITUATION	RECURRENCE	SECTION
People at level n of the chain letter	$s_n = 4s_{n-1}, s_0 = 1$	7.2
Leaves of a full m -ary tree by level	$l_n = m l_{n-1}$	7.2
Subsets of an n -element set	$p_n = 2p_{n-1}, p_0 = 1$	2.1
Rows in a truth table with n letters	$r_n = 2r_{n-1}, r_0 = 1$	1.1
Comparisons made by binary search	$T_n = T_{n/2} + 1$	11.2

Four of those five are the same relation with a different letter, and each solves to a power: $4^n, m^n, 2^n, 2^n$. Noticing that a counting problem is a first-order geometric recurrence is often faster than the counting argument, and it is a good sentence to write when a question asks you to justify a formula you already know.

RECALL AND CHECK

1. Define a recurrence relation.
2. What are initial conditions, and why does a relation need them?
3. How many initial conditions does a relation of order k need?
4. What does it mean to solve a recurrence relation?
5. Write the characteristic equation of $a_n = 5a_{n-1} - 6a_{n-2}$ and give its roots.
6. What changes in the general solution when the characteristic equation has a repeated root?
7. Distinguish a homogeneous relation from a non-homogeneous one.

ANSWERS

1. An equation expressing a_n in terms of one or more earlier terms of the same sequence.
2. The terms given outright before the relation applies. Without them the relation fixes only the shape of the sequence, and infinitely many sequences share that shape.
3. Exactly k .
4. Finding a closed form: a formula for a_n in terms of n alone, requiring no earlier terms.
5. $x^2 - 5x + 6 = 0$, with roots 3 and 2.
6. The solution becomes $(A + Bn)x_1^n$: the repeated root buys a factor of n on the second constant.
7. A homogeneous relation involves only terms of the sequence; a non-homogeneous one carries an extra term $f(n)$ that does not.

FILL IN THE GAP

1. The number of steps a relation reaches back is its _____.
2. The Tower of Hanoi with n discs takes _____ moves.
3. The characteristic equation of $a_n = 10a_{n-1} - 25a_{n-2}$ has the single root _____.
4. Fibonacci is a relation of order _____.
5. $a_n = 3a_{n-1} + 2$ with $a_1 = 1$ solves to _____.

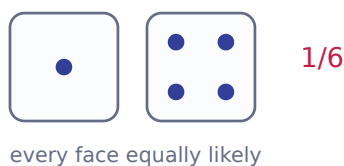
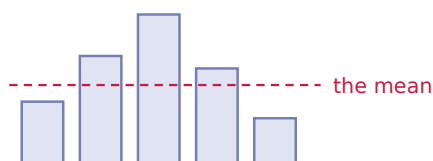
ANSWERS

1. order. 2. $2^n - 1$. 3. 5. 4. two. 5. $2 \times 3^{n-1} - 1$.

PART TEN

Descriptive Statistics and Discrete Probability

Two topics named on the syllabus that no past paper has yet asked about. This part is short by design and complete by obligation: it covers them properly, so that a first appearance cannot cost you, and it shows how probability is simply the counting of Part Five divided through by a total.



Descriptive Statistics

Given a list of numbers, describe it in one or two numbers. Where is the middle, and how spread out is it? Five measures answer both questions, and each is a formula you can compute by hand in under a minute.

By the end of this section you will be able to compute the mean, median, mode, range, variance and standard deviation of a data set, choose between the mean and the median when the data is skewed, work from a frequency table, and say what each measure does and does not tell you.

NAMED ON THE SYLLABUS, MISSING FROM THE COURSE TEXT

The course outline lists a module called Descriptive Statistics. Its unit contains no statistics: it is propositional logic from the first line to the last, the same material this manual teaches in Part One. Probed against the whole text, the words median, variance, standard deviation, quartile, arithmetic mean, frequency distribution and histogram occur **ZERO** times between them. No past paper in the five available sessions has asked a statistics question either. So the risk is small but not nil: the topic is named, and a named topic can appear. This section teaches it from zero and keeps it short, which is the honest proportion.

Measures of the middle

● MUST MEMORISE

MEAN, MEDIAN, MODE

Three answers to "where is the centre", and they can disagree violently on the same data.

MEASURE	HOW TO COMPUTE IT	WHAT IT IS
Mean	add every value, divide by how many there are	the balance point
Median	sort, then take the middle value; with an even count take the mean of the middle two	the halfway mark by position
Mode	the value occurring most often	the most popular value

In symbols, for values $x_1, x_2, \dots, x_n$:

$$\bar{x} = (\sum x_i) / n$$

WHERE

x_i is the **i-th data value**, so the subscript is a position in the list, as with a sequence in section 9.3. **n** is the **number of values** in the set, not the largest of them. $\sum x_i$ is the **total** of all the values, the summation of section 9.3 with i running from 1 to n . $\bar{x}$ is written x with a bar above it and is read "**x bar**". It is the standard symbol for the mean of a sample.

The mode may not be unique, and it may not exist in any useful sense: a set where every value occurs once has no mode, and a set with two equally popular values has two. Say so rather than picking one.

The median needs the data sorted first. Taking the middle of an unsorted list is the most common error in this topic, and it produces a number that means nothing.

◆ **Memory hook: mean is the average, median is the middle, mode is the most.** All three start with m, and so do average, middle and most, in the same order.

Measures of spread

● MUST MEMORISE

RANGE, VARIANCE, STANDARD DEVIATION

The centre alone hides everything. The sets 19, 20, 21 and 0, 20, 40 have the same mean and nothing else in common.

range = largest value – smallest value

$$\sigma^2 = (\sum (x_i - \bar{x})^2) / n$$

$$\sigma = \sqrt{(\sigma^2)}$$

WHERE

$x_i - \bar{x}$ is a **deviation**: how far one value sits from the mean. Deviations are positive above the mean and negative below it. **the square** is there because the deviations always add to exactly zero, so they must be made positive before summing; squaring does that and also weights the far-out values more heavily. σ^2 is the **variance**, the mean of the squared deviations. Its units are the square of the data's units, which is why it is rarely quoted on its own. σ is the **standard deviation**, the square root of the variance, back in the same units as the data. The symbol is a lower-case Greek sigma, and it is not the summation sign, which is the capital. **n** is the number of values.

Dividing by n or by n – 1. The formula above divides by n and describes the data set you have. If the data is a **sample** drawn from a larger population and you want to estimate that population's spread, divide by n – 1 instead and write s^2 and s. Both are correct answers to different questions, so **state which you are computing**; this is the same discipline as the indexing convention in section 9.3.

◆ **Memory hook: square the distances, average them, then take the root to come home.** The root is only there to undo the squaring.

GIVEN Ten test scores: 21, 15, 33, 12, 21, 18, 24, 15, 20, 21.

FIND Mean, median, mode, range, variance and standard deviation.

FORMULA $\bar{x} = (\sum x_i) / n$ and $\sigma^2 = (\sum (x_i - \bar{x})^2) / n$, with $n = 10$.

Step 1. Sort the data, because three of the six measures need it and none of them is harmed by it:

12, 15, 15, 18, 20, 21, 21, 21, 24, 33

Step 2. The mean.

$$\sum x_i = 12 + 15 + 15 + 18 + 20 + 21 + 21 + 21 + 24 + 33 = \mathbf{200}$$

$$\bar{x} = 200 / 10 = \mathbf{20}$$

Step 3. The median. There are 10 values, an even count, so the middle two are the 5th and 6th: 20 and 21.

$$\text{median} = (20 + 21) / 2 = \mathbf{20.5}$$

Step 4. The mode. Counting occurrences: 12 once, 15 twice, 18 once, 20 once, 21 three times, 24 once, 33 once.

$$\text{mode} = \mathbf{21}$$

Step 5. The range.

$$\text{range} = 33 - 12 = \mathbf{21}$$

Step 6. The variance. Tabulate the deviations, because doing this in your head is how arithmetic errors happen.

x_i	$x_i - 20$	$(x_i - 20)^2$
12	-8	64
15	-5	25
15	-5	25
18	-2	4
20	0	0
21	1	1
21	1	1
21	1	1
24	4	16
33	13	169
Total	0	306

Check the middle column adds to zero before going on. It always does, and if yours does not, the mean is wrong and everything after it would be too.

$$\sigma^2 = 306 / 10 = \mathbf{30.6}$$

Step 7. The standard deviation.

$$\sigma = \sqrt{30.6} \approx 5.53$$

If the sample form is wanted instead, divide by $n - 1 = 9$: $s^2 = 306 / 9 = 34$ and $s = \sqrt{34} \approx 5.83$. State which one you have given.

Reading the answer back. Scores centre on about 20 and typically sit within about 5.5 of it. The one score of 33 is more than two standard deviations out, which is what pulls the mean above the median. Saying that sentence is what turns six numbers into a description.

⚠ **TRAP**

WHEN THE MEAN IS THE WRONG ANSWER

In the data above the mean is 20 and the median 20.5. They are close because one outlier is mild. Change the 33 to 133 and watch:

new total = 300, so the mean becomes **30**

the median is unchanged at **20.5**, because the middle two values did not move

The mean has left the data: nine of the ten scores are now below it. **The median is resistant to outliers and the mean is not.** That is the single most useful fact in this section, and the reason incomes and house prices are reported as medians.

What to write in an exam. If a data set has an extreme value, say which measure you are quoting and why. "The median is quoted here because the value 133 distorts the mean" earns the mark that "average = 30" does not.

The mode's own trap: it is the only one of the three that works on data with no numeric order at all. The most common programming language in a class has a mode, but no mean.

GIVEN The number of programming errors found in each of 20 submitted assignments, as a frequency table.

FIND The mean, median and mode.

FORMULA $\bar{x} = (\sum f_i x_i) / (\sum f_i)$, where f_i is the frequency of the value x_i .

ERRORS x_i	FREQUENCY f_i	$f_i x_i$	RUNNING COUNT
0	3	0	3
1	5	5	8
2	7	14	15
3	3	9	18
4	2	8	20
Total	20	36	

The mean.

$$\bar{x} = 36 / 20 = \mathbf{1.8} \text{ errors per assignment}$$

The median. With 20 values the middle two are the 10th and 11th. The running count reaches 8 after the value 1 and 15 after the value 2, so both the 10th and the 11th value are 2.

$$\text{median} = \mathbf{2}$$

The mode is the value with the largest frequency, which is **2**, occurring 7 times.

Note what the frequency column saves you. Writing out all twenty values and adding them would give the same 36; the table is the same arithmetic with the repetitions collected, exactly as the arrangements formula of section 5.2 collects repeated letters.

A mean need not be a possible value. No assignment had 1.8 errors. The mean describes the set, not any member of it, and saying so pre-empts the marker's follow-up question.

RECALL AND CHECK

1. Define the mean, the median and the mode.
2. What must you do to the data before finding the median?
3. Why are deviations squared before they are averaged?
4. What is the difference between the variance and the standard deviation?
5. Which measure of centre resists an outlier?
6. For 12, 15, 15, 18, 20, 21, 21, 21, 24, 33, give the mean, median, mode and range.

ANSWERS

1. Mean: the total divided by the count. Median: the middle value once sorted, or the mean of the middle two when the count is even. Mode: the most frequent value.
2. Sort it.
3. Because the deviations sum to zero, so they must be made positive first; squaring does that and weights the extremes more.
4. The standard deviation is the square root of the variance, so it is in the same units as the data.
5. The median.
6. Mean 20, median 20.5, mode 21, range 21.

FILL IN THE GAP

1. The _____ is the total divided by the count.
2. With an even number of values, the median is the mean of the _____.
3. The sum of all deviations from the mean is always _____.
4. The standard deviation is the _____ of the variance.
5. Range = _____ - _____.

ANSWERS

1. mean. 2. middle two. 3. zero. 4. square root. 5. largest value, smallest value.

PART TEN • 10.2

Discrete Probability

Probability is Part Five with a division at the end. Count the outcomes you want, count all the outcomes there are, divide. Everything else in this section is a rule for handling two events at once, and every one of those rules is a set identity from Part Two wearing a P.

By the end of this section you will be able to define a random experiment, sample space and event; distinguish independent, dependent, mutually exclusive and equally likely events; state the axioms; apply the complement, addition, multiplication and conditional rules; use Bayes' theorem; and recognise which counting rule of Part Five a probability question is really asking for.

IN THE COURSE TEXT, NEVER YET EXAMINED

Discrete probability is a module of the course text and is properly written there, unlike the statistics module. No paper in the five available sessions has asked a probability question. It is taught here in full anyway, and two errors printed in the source module are flagged where they occur, because a reader who memorises them would answer confidently and wrongly.

The vocabulary

● MUST MEMORISE

EXPERIMENT, SAMPLE SPACE, EVENT

TERM	DEFINITION	EXAMPLE: ROLLING ONE DIE
Random experiment	a procedure that can be repeated, whose outcome is not determined in advance and is not affected by the repetition	the roll itself
Outcome	one possible result of the experiment	4
Sample space S	the set of all possible outcomes	{1, 2, 3, 4, 5, 6}
Event E	a subset of the sample space	"an even number" = {2, 4, 6}

A sample space is a set and an event is a subset of it. That one sentence imports the whole of Part Two: the union of two events is the event that at least one happens, the intersection is the event that both happen, and the complement is the event that it does not happen. Nothing new has to be learned about combining events, only about counting them.

For an experiment whose outcomes are all **equally likely**,

$$P(E) = n(E) / n(S)$$

WHERE

P(E) is the **probability** of the event E, a number between 0 and 1 inclusive. **n(E)** is the **number of favourable outcomes**, the cardinality of the event, in the notation of section 2.1. **n(S)** is the **total number of outcomes**, the size of the sample space. **equally likely** is a condition, not decoration: the formula is wrong the moment some outcomes are more probable than others, and a loaded die is the standard counterexample.

◆ **Memory hook: favourable over total, and only when every outcome carries the same weight.**

KIND	MEANING	EXAMPLE
Independent	the outcome of one does not affect the probability of the other	two separate coin tosses
Dependent	the outcome of one changes the probability of the other	drawing two cards without replacing the first
Mutually exclusive	they cannot both happen; their intersection is empty	a single toss landing heads, and landing tails
Equally likely	each has the same probability	the six faces of a fair die

Mutually exclusive in set language is $A \cap B = \emptyset$, which is precisely the disjoint sets of section 2.1. That is why the word **disjoint** is used for the same idea.

"Without replacement" is the phrase that signals dependence. Take a card and put it back, and the second draw faces the same deck; keep it, and the deck has changed. Exam questions turn on that phrase, and it is worth underlining it when you see it.

⚠ TRAP

MUTUALLY EXCLUSIVE IS NOT INDEPENDENT

The official course text writes, under its probability axioms, "if the occurrence of one event is not influenced by another event, they are called mutually exclusive or disjoint". **That sentence defines independence and attaches the wrong name to it.** The two ideas are not merely different; on events that can happen at all they are **opposites**.

	MUTUALLY EXCLUSIVE	INDEPENDENT
Means	they cannot both happen	one happening tells you nothing about the other
In symbols	$P(A \cap B) = 0$	$P(A \cap B) = P(A) \times P(B)$
Knowing A happened	tells you B did not	tells you nothing at all

Why they are opposites. If A and B are mutually exclusive and both have non-zero probability, then learning that A happened tells you B certainly did not: that is the strongest possible influence, so they cannot be independent. Formally, $P(A \cap B) = 0$ while $P(A) \times P(B) > 0$, so the independence equation fails.

The same module prints a second error, in its properties list: "for two non-disjoint events A and B, the probability of the union of two events $P(A \cup B) = P(A) + P(B)$ ". That formula is the one for **disjoint** events. For non-disjoint events the overlap has been counted twice and must come off, exactly as in section 5.3:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

The module's own formula list gives this correct version three paragraphs later, so the two statements contradict each other. Use the one with the subtraction; it is inclusion and exclusion, and it is right.

The axioms and the rules

● MUST MEMORISE

THE AXIOMS AND THE FIVE RULES

The axioms, which every probability obeys:

- ▶ $0 \leq P(E) \leq 1$ for every event E
- ▶ $P(\emptyset) = 0$ for the impossible event, and $P(S) = 1$ for the certain event
- ▶ if $E_1, E_2, \dots, E_n$ are pairwise mutually exclusive, then $P(E_1 \cup \dots \cup E_n) = P(E_1) + \dots + P(E_n)$

The rules, which are the axioms put to work:

RULE	FORMULA	WHEN
Complement	$P(A') = 1 - P(A)$	always
Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	always
Addition, disjoint	$P(A \cup B) = P(A) + P(B)$	only when $A \cap B = \emptyset$
Multiplication	$P(A \cap B) = P(A) \times P(B)$	only when A and B are independent
Conditional	$P(B A) = P(A \cap B) / P(A)$	always, provided $P(A) > 0$

WHERE

A' is the **complement** of A , the event that A does not happen, written with the same prime as the set complement of section 2.2. **A $\cup$ B** is the event that **at least one** of A, B happens, and **A $\cap$ B** is the event that **both** happen. **P(B | A)** is the **conditional probability** of B **given** A : the probability of B once you already know A has happened. The bar is read "given" and is not division. **P(A) > 0** is required because the formula divides by it, and conditioning on something impossible is meaningless.

The general multiplication rule, which works whether or not the events are independent and is just the conditional rule rearranged:

$$P(A \cap B) = P(A) \times P(B | A)$$

and independence is exactly the case where $P(B | A) = P(B)$, which collapses it to the simple form.

◆ **Memory hook: or adds, and multiplies, not subtracts the overlap, given divides.**

PROBABILITY	SETS (PART TWO)	COUNTING (PART FIVE)
$P(A') = 1 - P(A)$	$A' = U - A$	the subtraction rule, 5.1
$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$	inclusion and exclusion, 5.3
$P(A \cap B) = P(A)P(B)$	$n(A \times B) = n(A)n(B)$	the product rule, 5.1

Divide every count in the middle or right column by $n(S)$ and the left column appears.

Probability is not a new subject; it is counting normalised so the total is 1. If a probability question defeats you, answer the counting question instead and divide at the end.

Worked, from the simplest upward

WORKED EXAMPLE

METHOD · COINS, DICE AND CARDS

GIVEN A fair coin, a fair die, and a standard deck of 52 cards.

FIND Several basic probabilities, each by counting the sample space first.

FORMULA $P(E) = n(E) / n(S)$, the outcomes being equally likely in each case.

One coin. $S = \{H, T\}$, so $n(S) = 2$ and $P(\text{heads}) = 1 / 2$.

One die. $S = \{1, 2, 3, 4, 5, 6\}$, so $n(S) = 6$.

$$P(\text{any given face}) = 1/6$$

$$P(\text{even}) = n(\{2, 4, 6\}) / 6 = 3 / 6 = 1/2$$

$$P(\text{odd}) = 3 / 6 = 1/2$$

One card from 52.

$$P(\text{ace}) = 4 / 52 = 1/13$$

$$P(\text{diamond}) = 13 / 52 = 1/4$$

$$P(\text{red queen}) = 2 / 52 = 1/26$$

Three coin tosses, exactly two heads. The sample space is every sequence of three results, so by the product rule of section 5.1 it has $2^3 = 8$ outcomes:

HHH, HHT, HTH, HTT, THH, THT, TTH, TTT

Exactly two heads occurs in HHT, HTH and THH, which is 3 outcomes, so

$$P(\text{exactly two heads}) = 3/8$$

The counting shortcut, for when there are too many outcomes to list: choosing which 2 of the 3 positions hold the heads is $C(3, 2) = 3$, by section 5.2. The same 3, obtained without writing out the sample space, which is the move you need when there are ten tosses rather than three.

The addition rule in action. One card, $P(\text{ace or diamond})$. The events overlap in the ace of diamonds, so

$$P(\text{ace} \cup \text{diamond}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13$$

Adding without subtracting would give $17/52$ and would count the ace of diamonds twice.

GIVEN (a) A jar of 7 red, 3 green and 4 blue marbles, one drawn at random. (b) A committee of 4 chosen from 6 boys and 4 girls.

FIND (a) $P(\text{the marble is not blue})$. (b) $P(\text{the committee contains at least one boy})$.

FORMULA $P(A') = 1 - P(A)$, and $C(n, r)$ from section 5.2 for the counting.

(a) The marbles

The total is $7 + 3 + 4 = 14$ marbles.

Directly: non-blue means red or green, which is $7 + 3 = 10$ marbles, so

$$P(\text{not blue}) = 10 / 14 = 5/7$$

By the complement, which is the same answer by a shorter route:

$$P(\text{blue}) = 4 / 14 = 2/7, \text{ so } P(\text{not blue}) = 1 - 2/7 = 5/7$$

(b) At least one boy

"At least one" is the standard signal to use the complement, because the opposite of "at least one boy" is the single simple case "no boys at all". Section 5.2 answered the counting version of this exact question from the 2025/2026 paper; here it is as a probability.

$$\text{total committees} = C(10, 4) = 210$$

$$\text{committees with no boy} = C(4, 4) = 1$$

$$\text{committees with at least one boy} = 210 - 1 = 209$$

so

$$P(\text{at least one boy}) = 209 / 210$$

Read the answer back. That is about 0.995, which is worth saying: with six boys among ten candidates, a committee of four with no boy at all is nearly impossible, and there is in fact exactly one such committee, the four girls. A probability that close to 1 should agree with your intuition, and if it does not, recheck the counting.

- GIVEN** (a) Half of all teenagers own a cycle, and 30 per cent own both a bike and a cycle. (b) Six good and three defective laptops are mixed up and tested one at a time at random, without replacement.
- FIND** (a) The probability that a teenager owns a bike given that the teenager owns a cycle. (b) The probability that the first two tested are both defective.
- FORMULA** $P(B | A) = P(A \cap B) / P(A)$, and the general multiplication rule $P(A \cap B) = P(A) \times P(B | A)$.

(a) The conditional probability

Let A be "owns a cycle" and B be "owns a bike". We are given $P(A) = 0.5$ and $P(A \cap B) = 0.3$.

$$P(B | A) = P(A \cap B) / P(A) = 0.3 / 0.5 = \mathbf{0.6}$$

So 60 per cent of cycle owners also own a bike.

What conditioning did. The 30 per cent was measured against **all** teenagers; the answer 60 per cent is measured against cycle owners only. Conditioning shrinks the sample space from S to A, and dividing by $P(A)$ is exactly that shrinking. Say that sentence and the formula stops being something to memorise.

(b) Two defectives in a row

There are $6 + 3 = 9$ laptops, of which 3 are defective. Let A be "the first is defective" and B be "the second is defective".

$$P(A) = 3 / 9$$

Now the pool has changed, because the first laptop was not replaced: 8 remain and 2 of them are defective. So

$$P(B | A) = 2 / 8$$

and therefore

$$P(A \cap B) = P(A) \times P(B | A) = (3/9) \times (2/8) = 6 / 72 = \mathbf{1/12}$$

The trap this question sets is using $P(A) \times P(B) = (3/9) \times (3/9) = 1/9$, which treats the draws as independent. They are not, because the first laptop is not put back, and the phrase "tested one by one" is what tells you so.

Cross-check by counting, in the manner of section 5.2: the number of ways to choose 2 defectives from 3 is $C(3, 2) = 3$, and the number of ways to choose any 2 from 9 is $C(9, 2) = 36$, giving $3 / 36 = 1/12$. The two methods agree.

Bayes' theorem

● MUST MEMORISE

BAYES' THEOREM

Conditional probability runs one way; Bayes' theorem turns it round.

$$P(A | B) = P(B | A) \times P(A) / P(B)$$

WHERE

$P(A)$ is the probability of A **before** the evidence, sometimes called the prior. $P(B | A)$ is how likely the evidence B is **when** A holds. $P(B)$ is how likely the evidence is **overall**, and it is usually the part you must compute rather than the part you are given. $P(A | B)$ is the probability of A **after** the evidence, which is what the question wants.

Computing $P(B)$ is the real work. If A and A' are the only two possibilities, then B happens either with A or with A', and those two cases are mutually exclusive, so

$$P(B) = P(B | A) \times P(A) + P(B | A') \times P(A')$$

which is the axiom of adding disjoint cases, applied to a split of the sample space.

◆ **Memory hook: Bayes reverses the bar.** You are given "evidence given cause" and you want "cause given evidence".

GIVEN 60 per cent of a data centre's machines come from supplier A and 40 per cent from supplier B. In the first year 2 per cent of A's machines fail and 5 per cent of B's do. A machine is found to have failed.

FIND The probability that it came from supplier B.

FORMULA $P(B | F) = P(F | B) \times P(B) / P(F)$, with $P(F) = P(F | A)P(A) + P(F | B)P(B)$.

Name everything first, which is where these questions are won or lost:

$$P(A) = 0.6, \quad P(B) = 0.4$$

$$P(F | A) = 0.02, \quad P(F | B) = 0.05$$

Step 1. The overall failure rate.

$$\begin{aligned} P(F) &= 0.02 \times 0.6 + 0.05 \times 0.4 \\ &= 0.012 + 0.020 \\ &= \mathbf{0.032} \end{aligned}$$

Step 2. Reverse the bar.

$$P(B | F) = P(F | B) \times P(B) / P(F) = 0.05 \times 0.4 / 0.032 = 0.020 / 0.032 = \mathbf{0.625}$$

Answer: 62.5 per cent.

Why the answer is interesting. Supplier B provides only 40 per cent of the machines but accounts for 62.5 per cent of the failures, because its failure rate is two and a half times higher. The evidence has moved the probability from 0.4 to 0.625, and that movement is exactly what Bayes' theorem measures.

Check with the complement. $P(A | F)$ should be $0.012 / 0.032 = 0.375$, and $0.625 + 0.375 = 1$. Given that the machine failed, it came from one supplier or the other, so the two conditional probabilities must add to 1. That check costs one line and catches an arithmetic slip immediately.

Now redo it: if the split were 50 per cent each with the same failure rates, $P(F) = 0.02 \times 0.5 + 0.05 \times 0.5 = 0.035$ and $P(B | F) = 0.025 / 0.035$, which is about 0.714.

A **random variable** assigns a number to each outcome of an experiment. It is a function on the sample space, in the sense of section 4.1, and not a variable in the algebraic sense at all.

KIND	TAKES	EXAMPLES
Discrete	countable values, such as 0, 1, 2, ...	the number of heads in three tosses; the number of cars in a car park; the number of correct answers on a test
Continuous	every value in a range	a person's height; the time a reaction takes; a temperature

Discrete is the one this course is about, which is what the word discrete in the course title means: countable, separated values rather than a smooth range. Every structure in this book, from a set to a graph to a tree, is discrete in exactly that sense.

RECALL AND CHECK

1. Define a sample space and an event, and say how they are related.
2. Write the formula for the probability of an event with equally likely outcomes, and state its condition.
3. Distinguish mutually exclusive from independent events.
4. State the addition rule and say when the subtraction term may be dropped.
5. Write the conditional probability formula and say what the bar means.
6. A card is drawn from 52. Give $P(\text{ace})$, $P(\text{diamond})$ and $P(\text{ace or diamond})$.
7. Three good and six bad chips are tested one by one. What is the probability the first two are both good?

ANSWERS

1. The sample space S is the set of all possible outcomes; an event is a subset of S .
2. $P(E) = n(E) / n(S)$, valid only when every outcome is equally likely.
3. Mutually exclusive means they cannot both occur, $P(A \cap B) = 0$. Independent means one gives no information about the other, $P(A \cap B) = P(A)P(B)$. For events of non-zero probability the two are opposites.
4. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; the last term may be dropped only when the events are disjoint, since it is then zero.
5. $P(B | A) = P(A \cap B) / P(A)$; the bar is read "given" and restricts the sample space to A .
6. $1/13$, $1/4$ and $4/13$.
7. $(3/9) \times (2/8) = 1/12$.

FILL IN THE GAP

1. An event is a _____ of the sample space.
2. Every probability lies between _____ and _____.
3. $P(A') =$ _____.
4. Two events are independent when $P(A \cap B) =$ _____.
5. The phrase "without replacement" tells you the events are _____.
6. "At least one" is the signal to use the _____.

ANSWERS

1. subset. 2. 0, 1. 3. $1 - P(A)$. 4. $P(A) \times P(B)$. 5. dependent. 6. complement.

PART ELEVEN

C++ for Discrete Structures

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Enough C++ to Answer the Question

Stub section. Real content follows.

Placeholder paragraph so the section occupies a measurable amount of the page and the layout gate has something real to look at.

The Four Examined Programs

Stub section. Real content follows.

Placeholder paragraph so the section occupies a measurable amount of the page and the layout gate has something real to look at.

REFERENCE

Formulas, Notation and Exam Craft

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Formula and Notation Reference

Stub section. Real content follows.

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Exam Day Strategy

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Trap Index and Cram Sheet

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P A S T P A P E R S

All Five Papers, Solved in Full

Stub divider. Real content follows.

PAST PAPER

2025/2026 Solved in Full

Stub section. Real content follows.

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PAST PAPER

2024/2025 Solved in Full

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PAST PAPER

2023/2024 Solved in Full

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PAST PAPER

2021/2022 Solved in Full

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PAST PAPER

2020/2021 Solved in Full

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MOCK EXAMINATION ONE

Sit This Paper Under Exam Conditions

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Question Paper

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Answers

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MOCK EXAMINATION TWO

Sit This One After the First

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Question Paper

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Answers

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MOCK EXAMINATION THREE

The Hardest of the Three

Stub divider. Real content follows.

Question Paper

Stub section. Real content follows.

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Answers

Stub section. Real content follows.

Placeholder paragraph so the section occupies a measurable amount of the page and the layout gate has something real to look at.